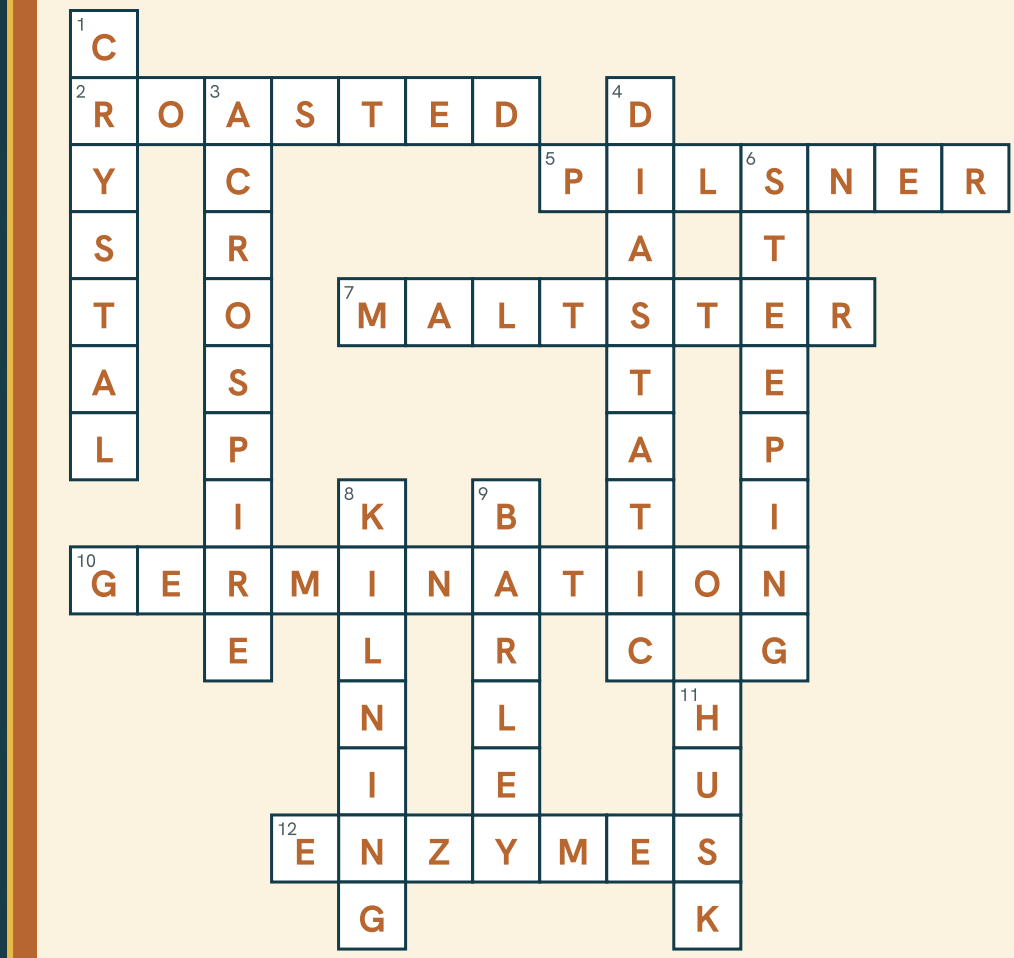
Beer Crosswords

103 puzzles for brewers and beer lovers



Malt, hops, yeast, lagers, sours, the brewhouse, the bar and India's own beer story, plus a puzzle for every Craft Beer School lesson, with answers at the back.

1 player or teams | 15 minutes each | Ages 12+ | Free from Craft Beer School

Craft Beer School
Learn. Brew. Cheer

Beer Crosswords

103 puzzles for brewers and beer lovers

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Craft Beer School, Bengaluru

First edition, October 2026. craftbeerschool.in
Free to print and share for teaching. Please do not sell it.

How to use this book

There are 103 crosswords here. The first 50 cover every corner of brewing in five parts: ingredients, the brewhouse, beer styles, what happens in the glass, and the people, places and history behind it all. Part six is the course companion: 53 more puzzles, one for each lesson of the 18 Craft Beer School courses. Each puzzle has twelve clues and fits on a single page, so you can print just the one you need.

Reading the clues

1. The number in brackets is the length of the answer. Two numbers, such as (3,3), mean two words written together in the grid.
2. A ___ in a clue stands for the missing answer.
3. Spellings are British, temperatures are in °C with °F in brackets and umlauts are dropped in the grid, so Märzen is MARZEN.

In the classroom or the taproom

1. Hand out the course companion puzzle at the end of each lesson as a ten-minute recap.
2. Race in teams: the first team with a full, correct grid wins the round.
3. Read one clue aloud at a tasting and let the table shout the answer.

Answers

Every grid is solved at the back of the book, six to a page, with the page number of its puzzle. Try not to peek.

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PART ONE

Ingredients

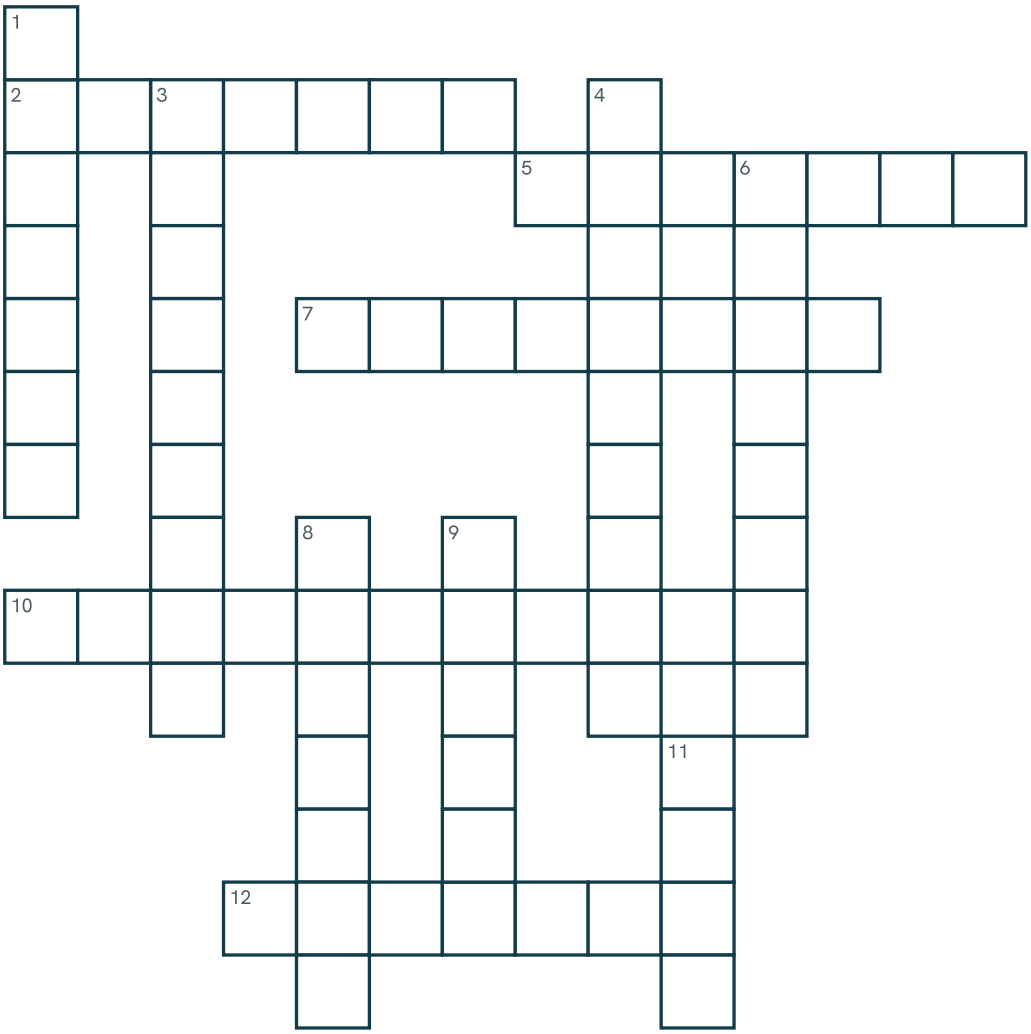
Malt, hops, yeast and water, and the chemistry that binds them.

Malt basics, Malt varieties, Hops, Hop varieties, Hop farming, Yeast, Brewing water, Adjuncts, Brewing chemistry, Recipe design

Crossword 1

Malt basics

From field to maltings: the grain behind nearly every beer.



Across

2 ___ barley gives dry stout its coffee bite (7)

5 Very pale base malt named after a Czech lager (7)

7 Person who turns grain into malt (8)

10 Sprouting stage when the grain builds its enzymes (11)

12 Amylases are examples of these malt proteins (7)

Down

1 Stewed malt with glassy sugars, also called caramel malt (7)

3 Shoot that grows under the husk during malting (9)

4 ___ power: a malt's ability to convert starch (9)

6 First maltings step: soaking the grain in water (8)

8 Hot-air drying that stops growth and sets colour (7)

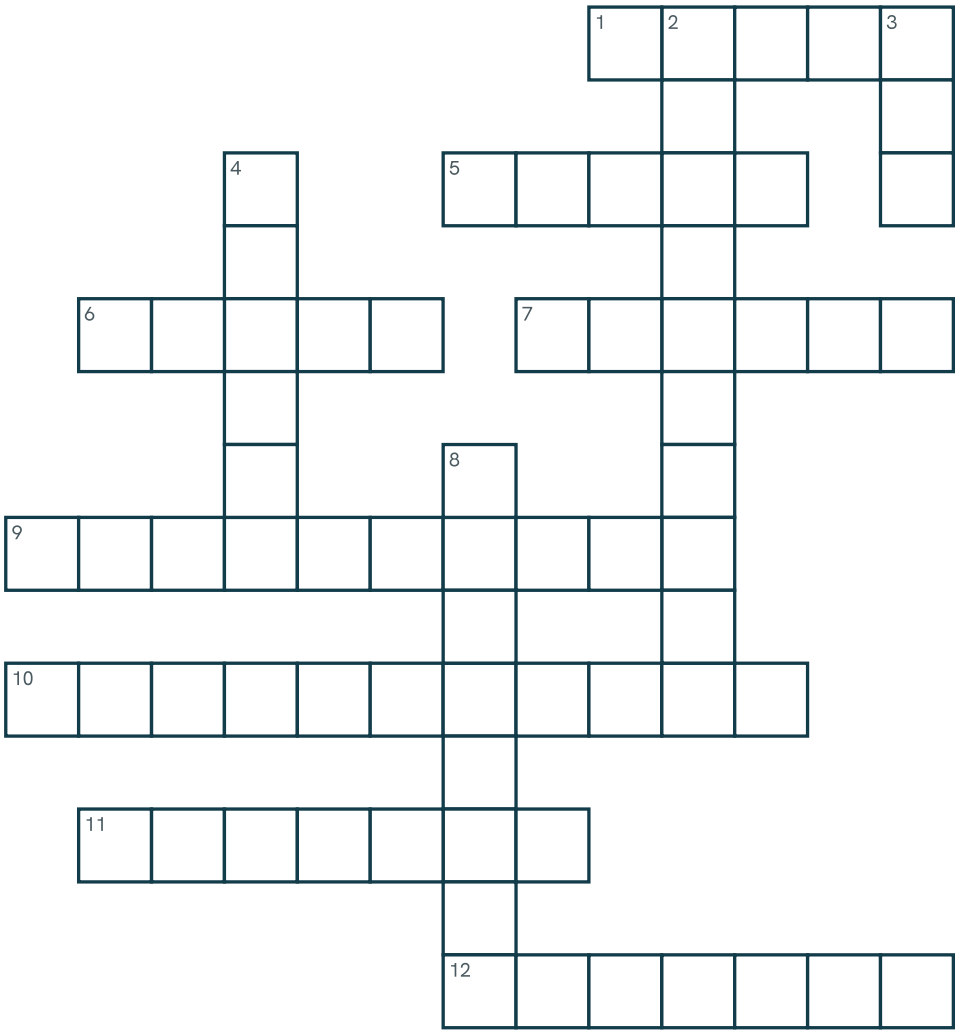
9 Cereal grain behind most brewing malt (6)

11 Outer layer that forms the filter bed in the mash tun (4)

Crossword 2

Malt varieties

Specialty malts and what they bring.



Across

1 British malt between pale and brown (5)

5 Ancient wheat that can be malted (5)

6 Old porter malt (5)

7 Toasty base malt from Bavaria (6)

9 Malt coated in lactic acid to lower mash pH (10)

10 Very dark malt patented in 1817 (5,6)

11 Toasted Belgian malt (7)

12 ___ B, a dark Belgian crystal malt (7)

Down

2 Malt with rich red colour and honey notes (10)

3 Spicy grain malted for Roggenbier (3)

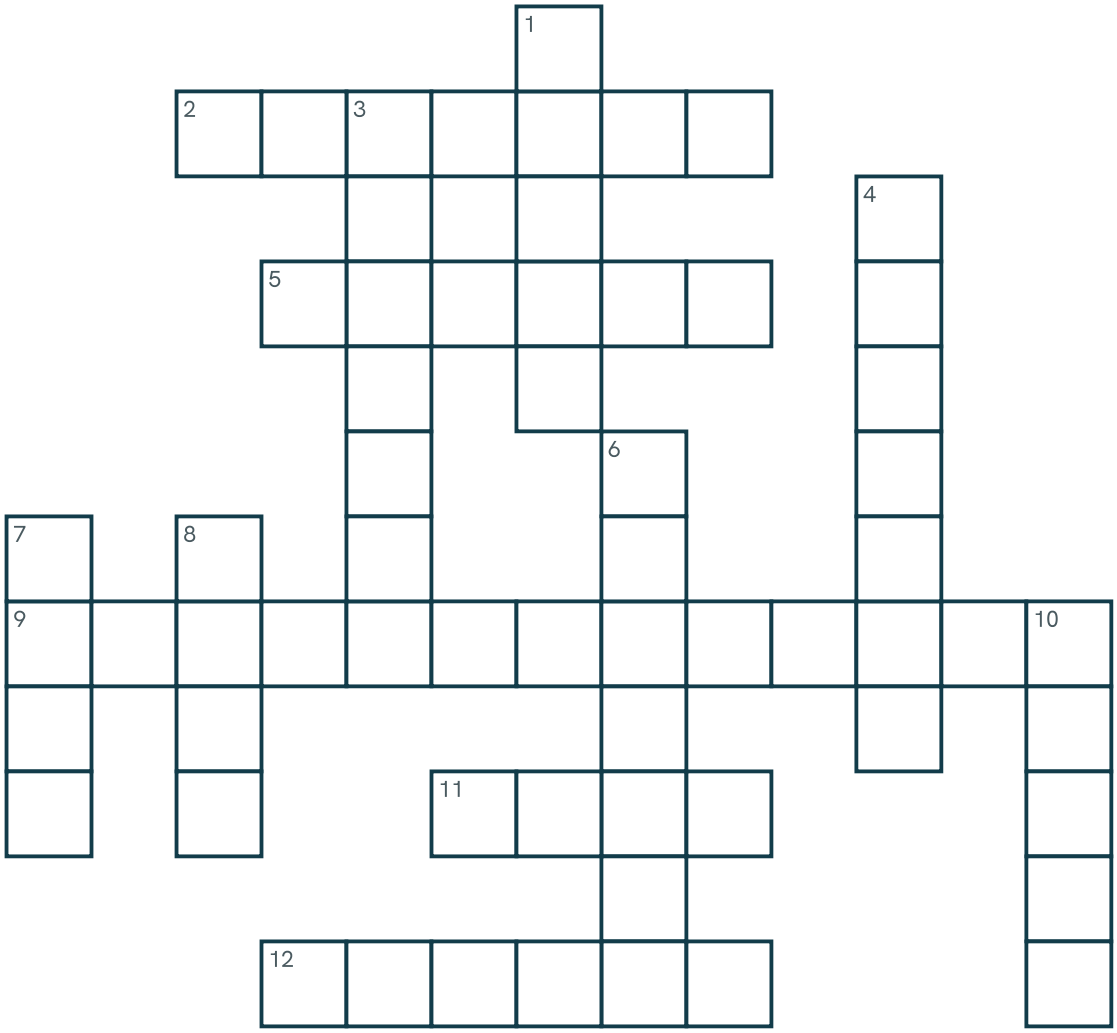
4 Beechwood-___ malt for Rauchbier (6)

8 Dextrin malt for foam and body (8)

Crossword 3

Hops

The green cone that brings bitterness, aroma and a lot of argument.



Across

2 Genus of the hop plant (7)

5 To add hops cold, once fermentation is under way (3,3)

9 Boil reaction that makes hop acids bitter and soluble (13)

11 Noble hop named after the Czech town of Žatec (4)

12 Most common hop form: milled and pressed (6)

Down

1 ___ acids, the source of bitterness (5)

3 Most plentiful hop oil, green and resinous (7)

4 Yellow glands in the cone that hold resins and oils (7)

6 US variety released in 1972 that shaped American pale ale (7)

7 Climbing stem of the hop, which twists rather than clings (4)

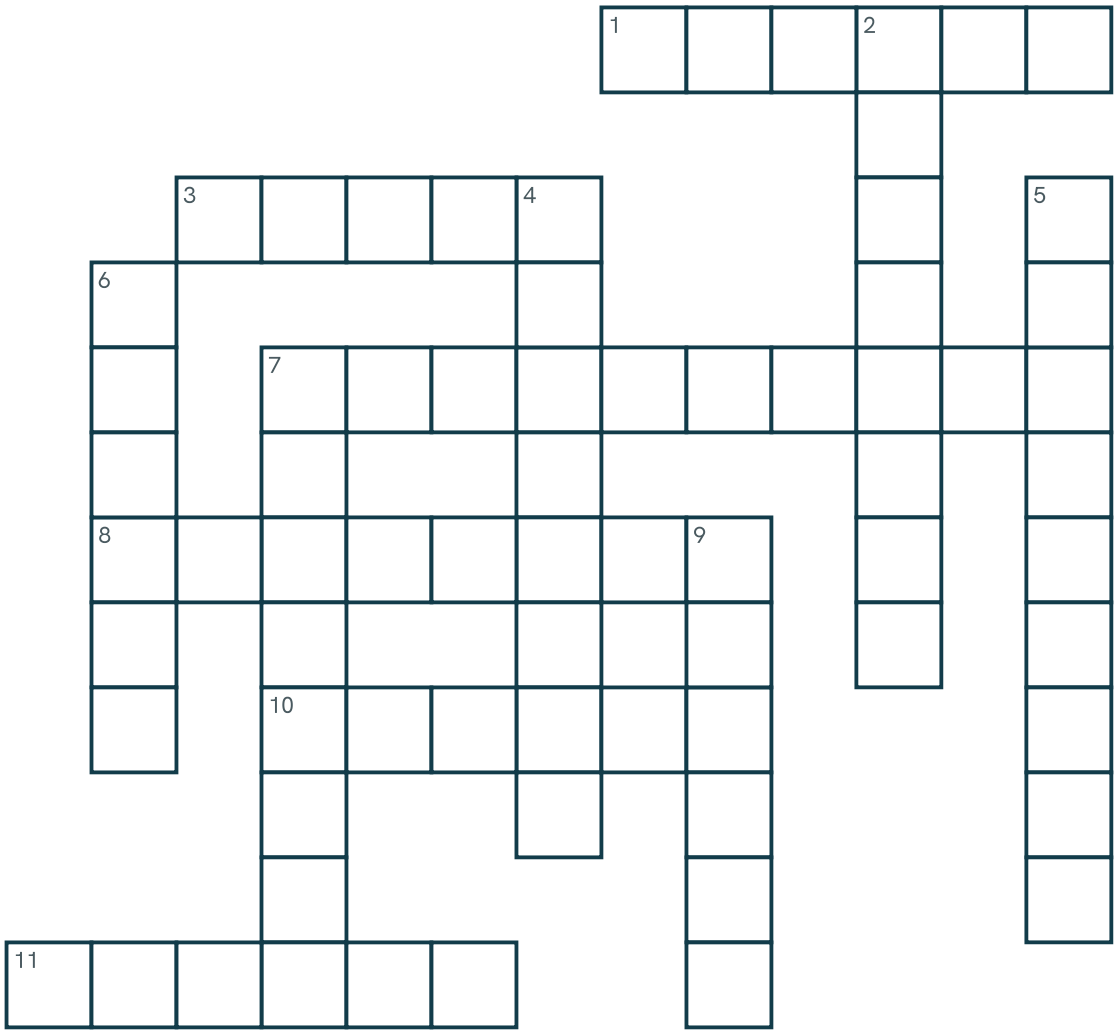
8 Female flower used in brewing (4)

10 ___ hops: the classic German and Czech aroma varieties (5)

Crossword 4

Hop varieties

Name the hop from its home and its smell.



Across

1 Australian hop with a passion fruit punch (6)

3 Noble hop town south of Nuremberg (5)

7 US hop nicknamed Super Cascade (10)

8 East Kent ___ (8)

10 High-alpha bittering hop (6)

11 ___ Sauvin, a New Zealand hop with white grape notes (6)

Down

2 Orange-forward US hop (8)

4 Noble hop town near Lake Constance (8)

5 Bavarian hop region, the largest in the world (9)

6 Earthy English hop from Kent, introduced in 1875 (6)

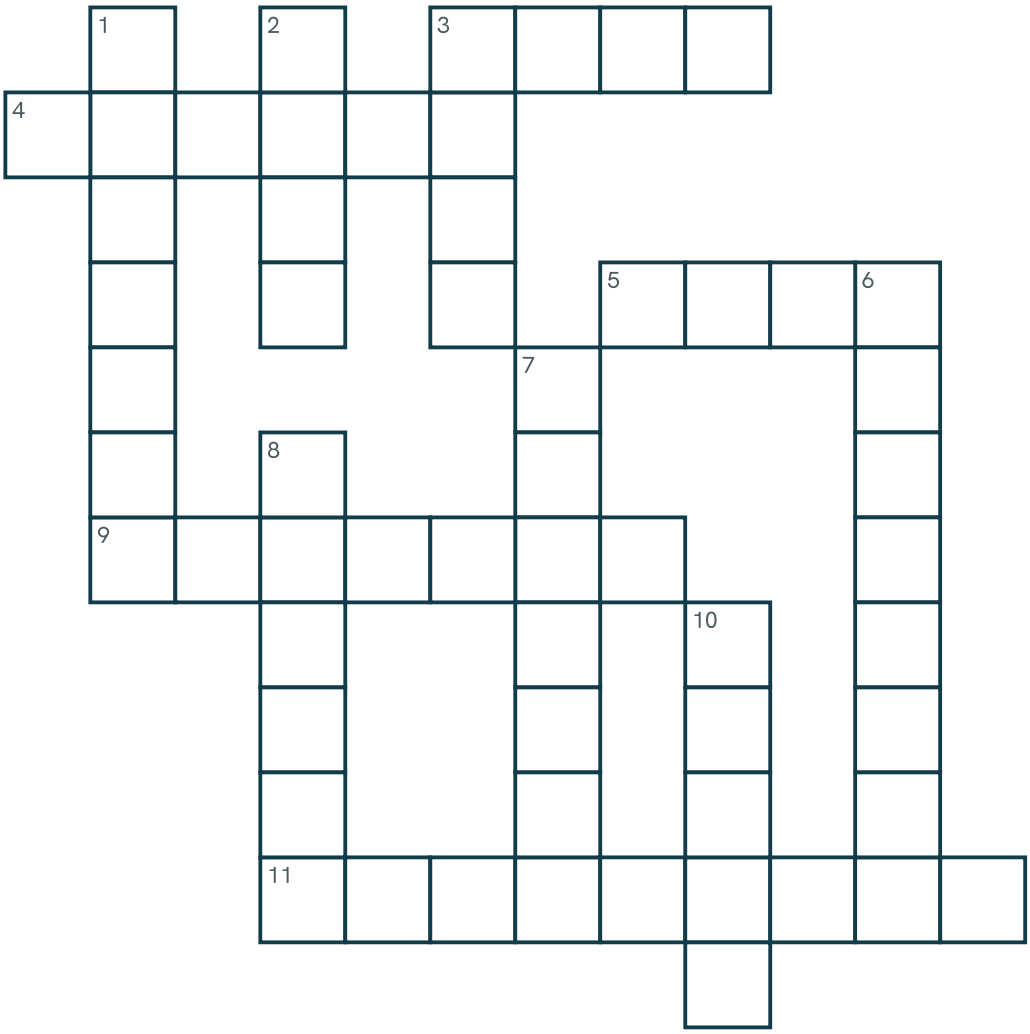
7 Pungent high-alpha US hop, the C in CTZ (8)

9 US hop with pine and passion fruit (6)

Crossword 5

Hop farming

From rhizome to bale.



Across

3 Early flower stage before the cone forms (4)

4 Washington valley that grows most US hops (6)

5 Kent building with a hop kiln (4)

9 Wire frame the hops climb (7)

11 Hops come back every year, so they are this (9)

Down

1 Late August to September in the north (7)

2 Where hops are dried (4)

3 Pressed block of dried hops (4)

6 Australian island hop region (8)

7 Root cutting used to plant hops (7)

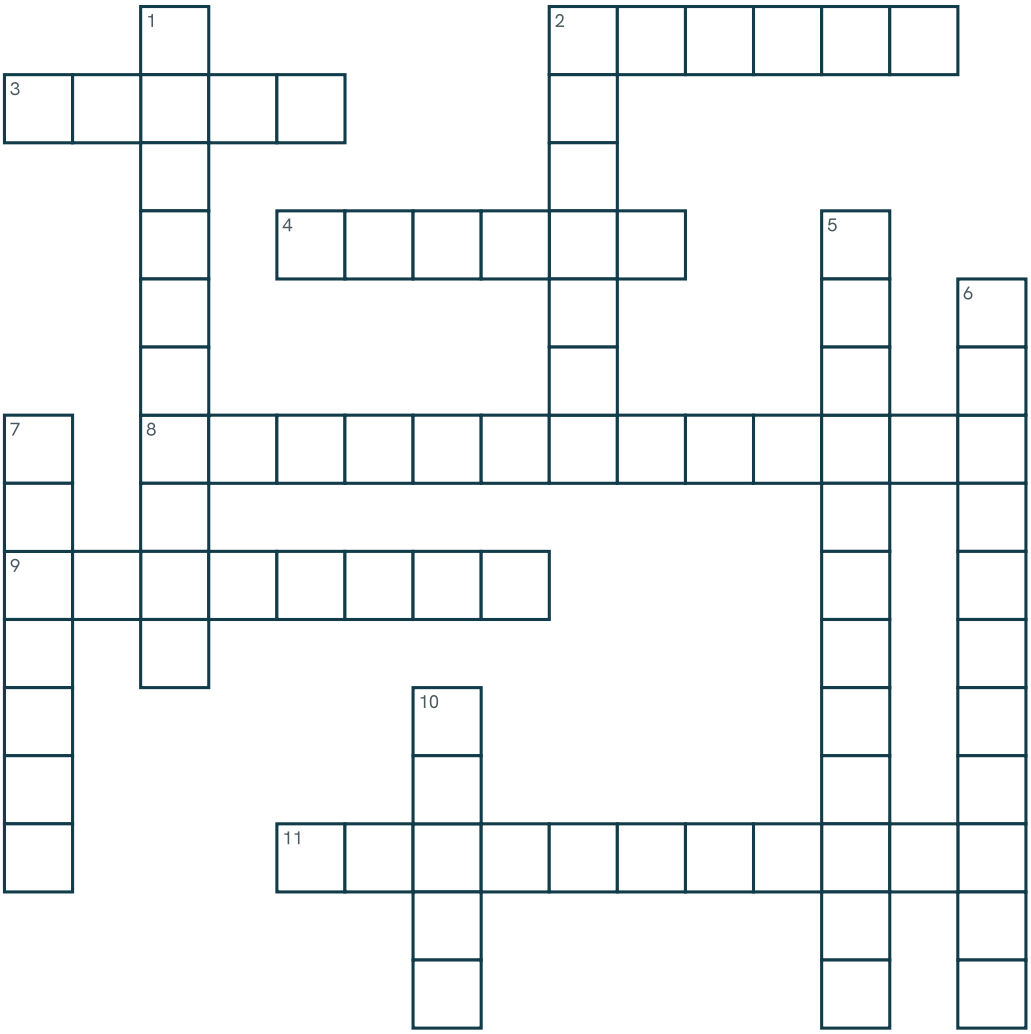
8 Fresh, undried hops used within a day of picking (3,3)

10 ___ mildew, a serious hop disease (5)

Crossword 6

Yeast

Brewers make wort. Yeast makes beer.



Across

2 Thick yeast harvested from the cone for repitching (6)

3 Norwegian farmhouse yeast happy at 30°C (86°F) and above (5)

4 Fruity compounds made by yeast (6)

8 Genus of brewer's yeast (13)

9 Buttery compound yeast cleans up during a warm rest (8)

11 How much of the sugar the yeast eats, as a percentage (11)

Down

1 Species name of ale yeast (10)

2 Small batch of wort that grows yeast before brew day (7)

5 Yeast clumping together and settling out (12)

6 Species name of lager yeast (11)

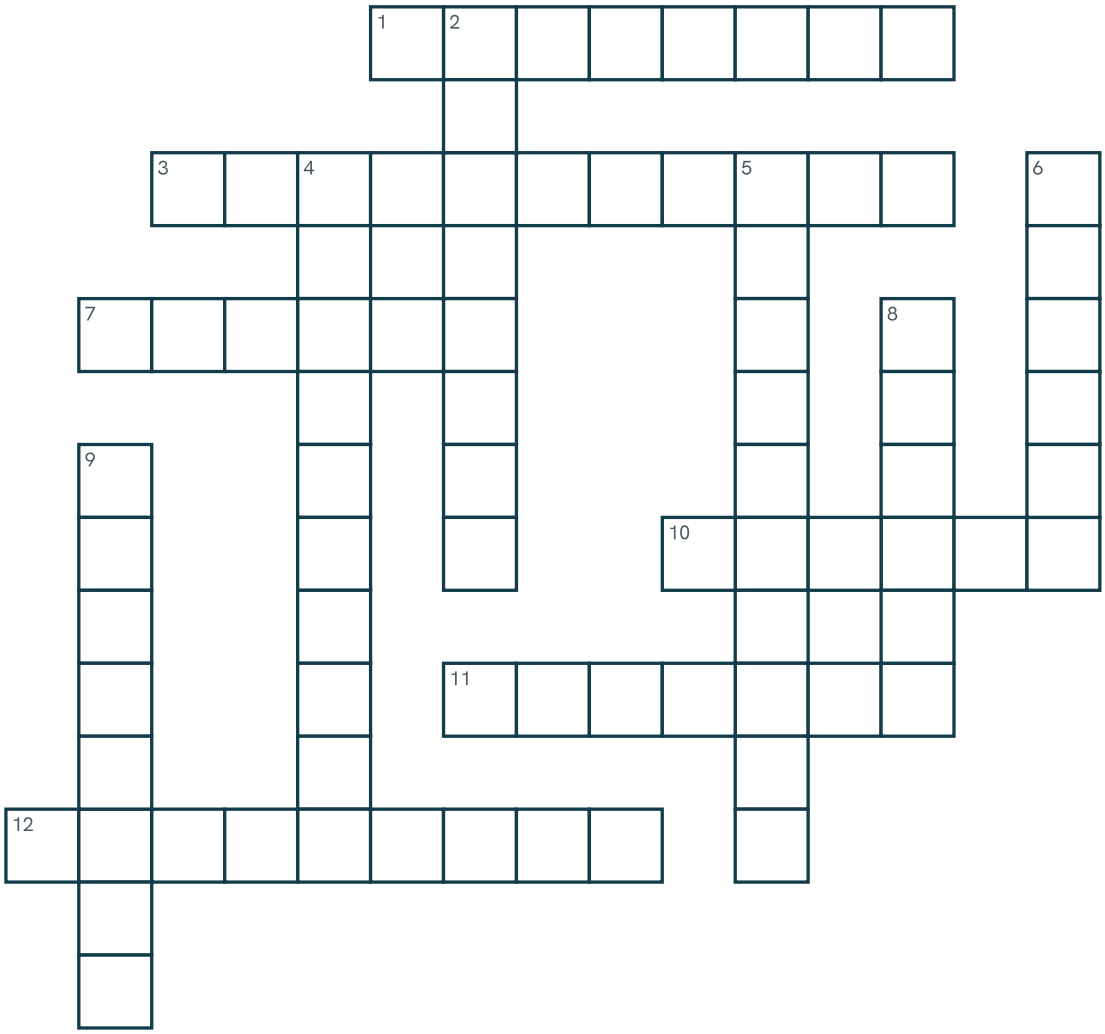
7 How yeast cells reproduce (7)

10 To add yeast to cooled wort (5)

Crossword 7

Brewing water

Beer is more than nine parts water, so the water matters.



Across

1 Ion that rounds out maltiness (8)

3 Alkaline ion that pushes mash pH up (11)

7 City whose carbonate-rich water suited dark stout (6)

10 Czech city with very soft water and a famous pale lager (6)

11 Ion that helps enzymes, clarity and yeast health (7)

12 Yeast nutrient ion that tastes bitter in excess (9)

Down

2 Measure of dissolved calcium and magnesium (8)

4 Mains-water disinfectant that carbon filters remove (10)

5 Water's resistance to a fall in pH (10)

6 English town famed for hard, sulphate-rich water (6)

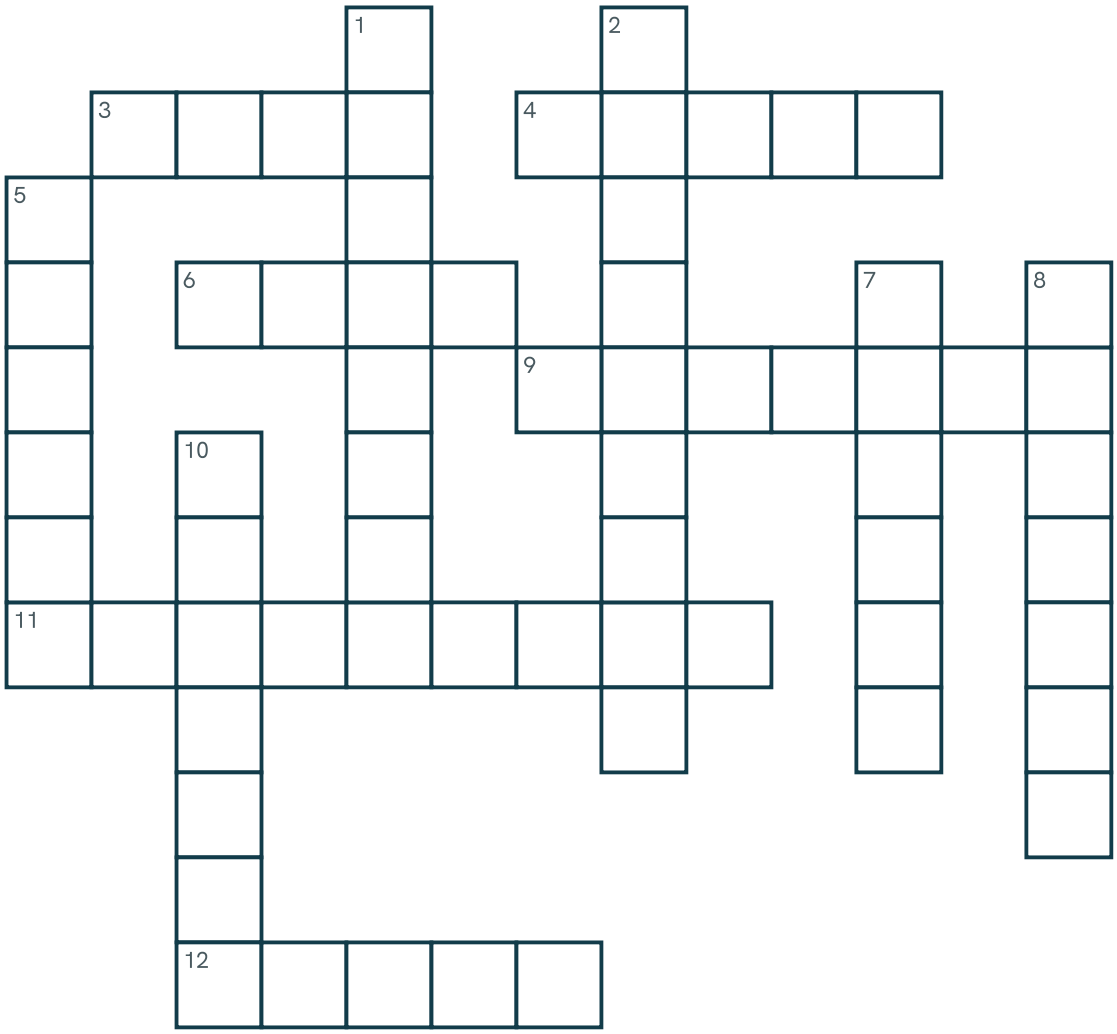
8 Calcium sulphate, added to sharpen hop bitterness (6)

9 Ion that makes bitterness feel drier (8)

Crossword 8

Adjuncts

Everything in the mash or kettle that is not barley malt.



Across

3 Adjunct in many Asian lagers (4)

4 Sweet adjunct, and the base of mead (5)

6 Grain for a silky body (4)

9 Bean found in many pastry stouts (7)

11 Puffed with heat, like some wheat (9)

12 Corn by its British name (5)

Down

1 Corn sugar, often used for priming (8)

2 Spice in witbier (9)

5 Small grain used in Himalayan and African brews (6)

7 ___ barley: rolled unmalted grain (6)

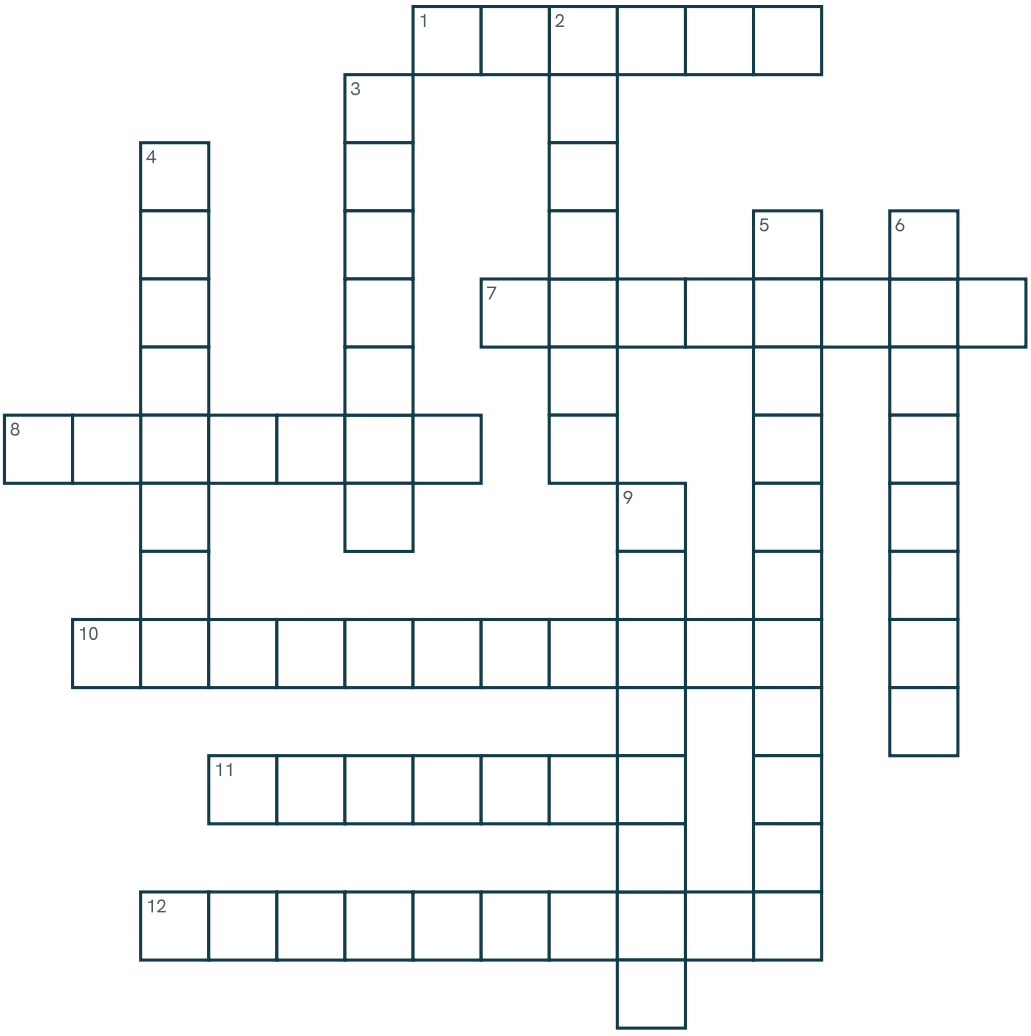
8 Indian cane sugar for strong ales (7)

10 Gluten-free grain used in African brewing (7)

Crossword 9

Brewing chemistry

The molecules behind the flavour.



Across

1 Malt phosphate acts as one in the mash (6)

7 Yeast's energy store (8)

8 Main sugar in wort (7)

10 Three-glucose sugar that lager yeast can use (11)

11 Enzyme family that breaks down starch (7)

12 Splitting a molecule with water (10)

Down

2 Acid that becomes clove in weissbier (7)

3 Simplest sugar (7)

4 ___ acids, the bitter compounds in beer (3,5)

5 Tannins from husks and hops (11)

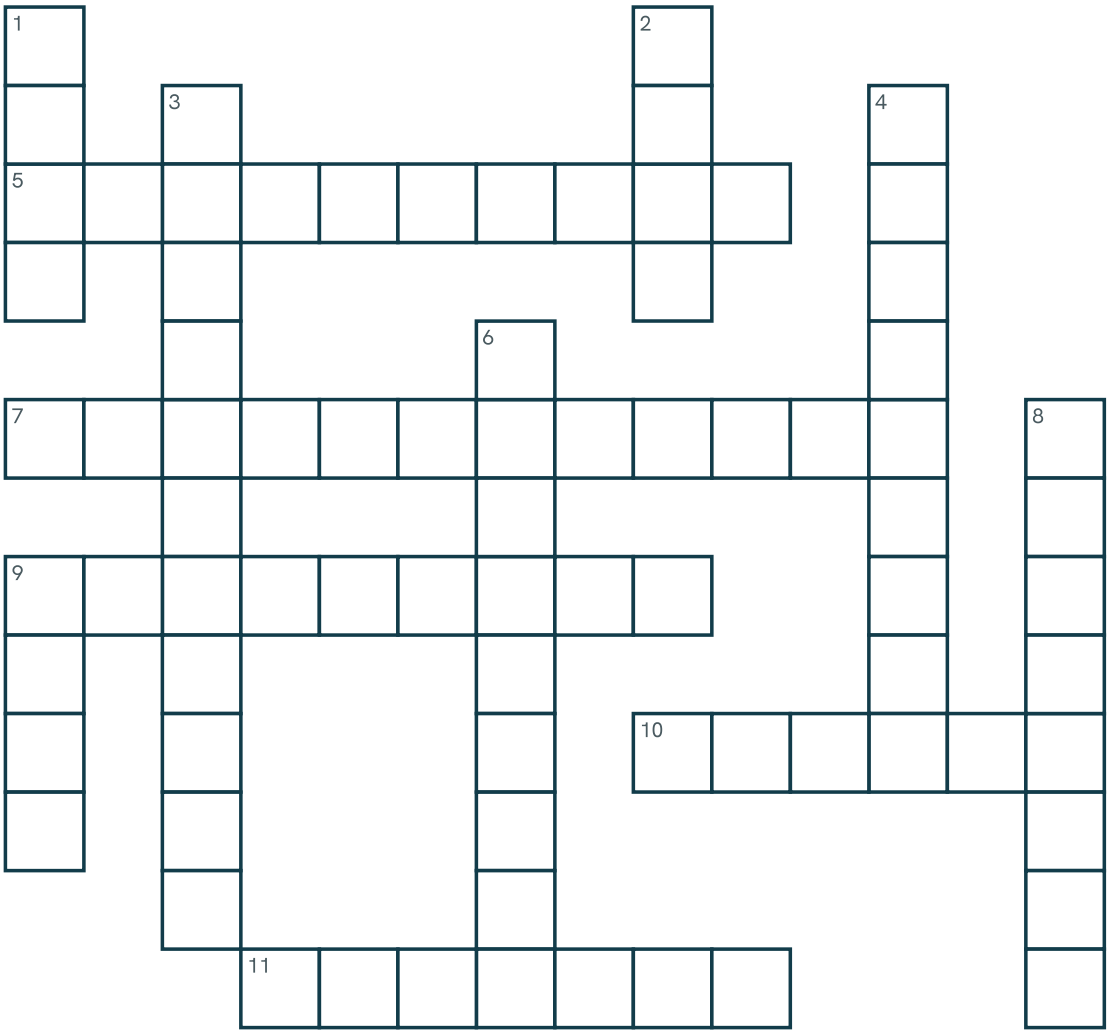
6 Unfermentable chains that give body (8)

9 Enzyme that breaks down protein (8)

Crossword 10

Recipe design

Building a beer on paper before brew day.



Across

5 Small-share malts that add character (10)

7 Mineral recipe for the brewing liquor (5,7)

9 Volume the recipe makes (5,4)

10 Aim for a ___ original gravity (6)

11 Adjusting a recipe to another size (7)

Down

1 ___ malt: most of the grist (4)

2 ___ hop: added near the end of the boil (4)

3 The culture you choose to ferment with (5,6)

4 Early-boil hop addition (9)

6 List of malts in a recipe (5,4)

8 Raise it for more body (4,4)

9 BU:GU ratio of bitterness to gravity (4)



PART TWO

The brewhouse

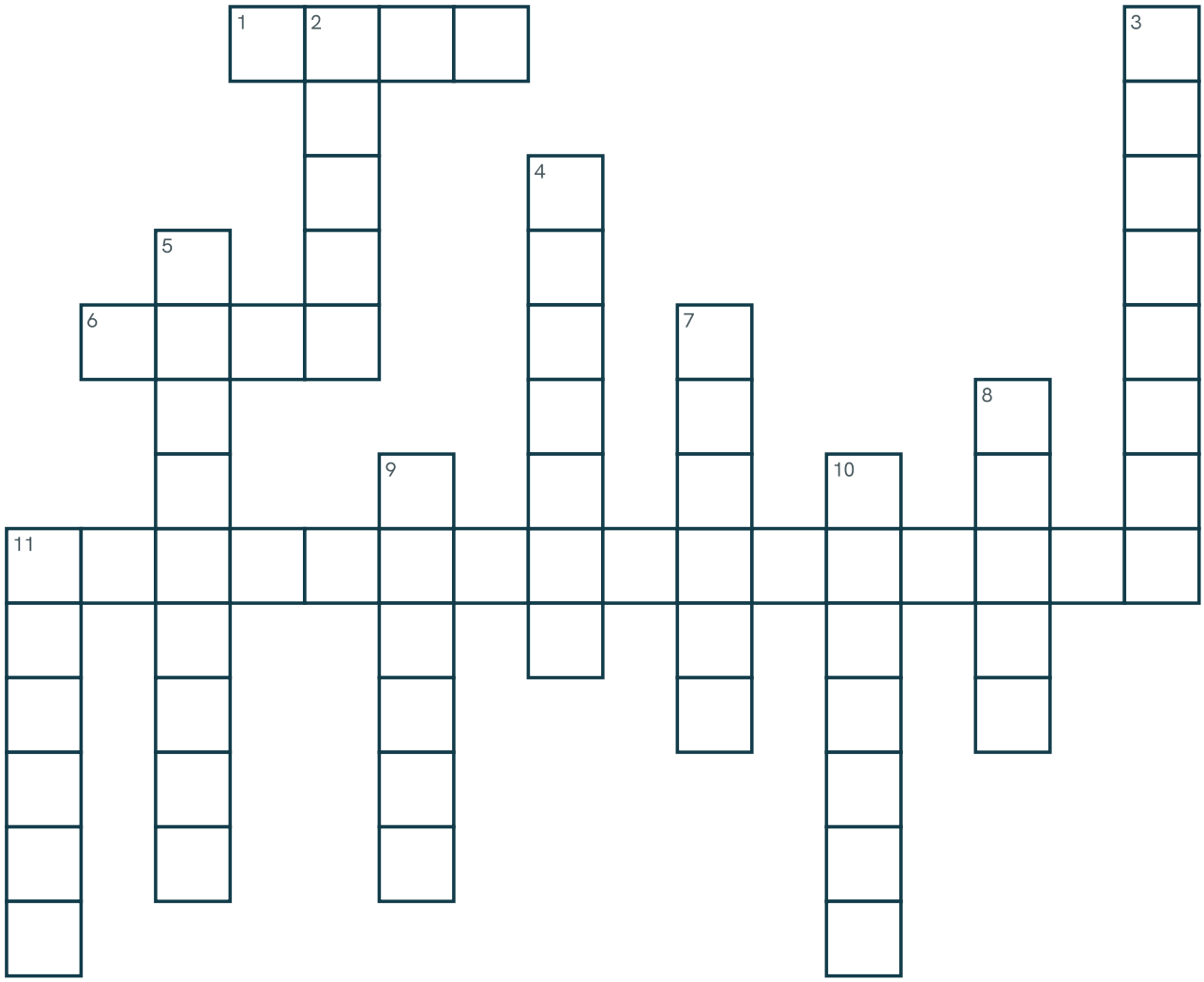
From mash tun to bright tank, and the plant that keeps it running.

Mashing, The boil, Brew day in order, Brewhouse kit, Fermentation, Cellar and conditioning, Cleaning and hygiene, Utilities and the plant, Brewing numbers, Quality control

Crossword 11

Mashing

Where hot water and crushed malt turn starch into sugar.



Across

1 Mix of crushed malt and hot water (4)

6 ___ amylase, busiest around 60 to 65°C (140 to 149°F) (4)

11 The starch-to-sugar rest (16)

Down

2 ___ amylase, busiest near 70°C (158°F) and good for body (5)

3 Mash method held at a single temperature (8)

4 To mix grist and water at the start of the mash (5,2)

5 Mash method that boils a portion and returns it (9)

7 Test that turns blue-black if starch remains (6)

8 Milled malt before it meets water (5)

9 To separate sweet wort from the grain bed (6)

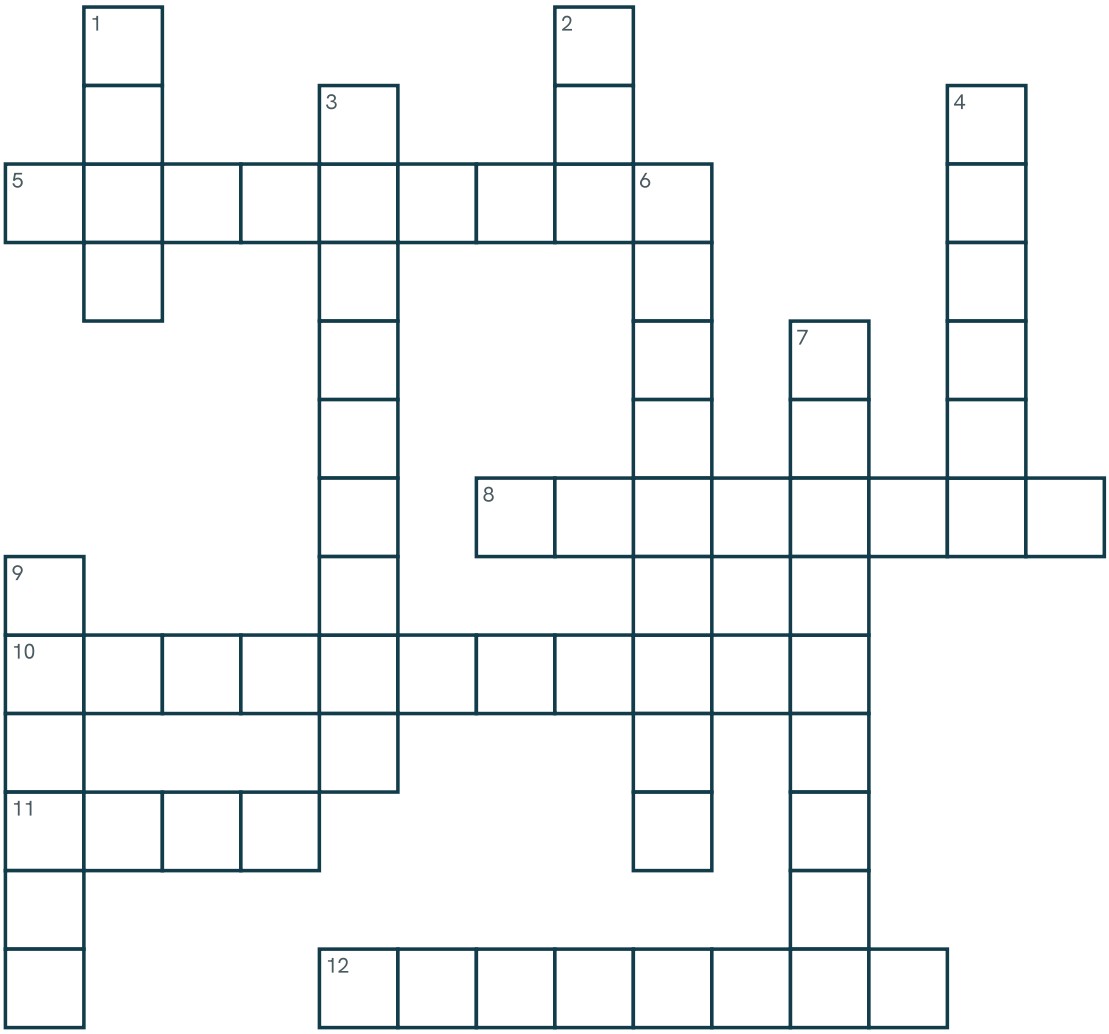
10 Raise to about 76°C (169°F) to stop the enzymes (4,3)

11 To rinse the grain bed with hot water (6)

Crossword 12

The boil

An hour or so of rolling heat that sets bitterness and stability.



Across

5 Seaweed fining added late in the boil (5,4)

8 Browning reaction between sugars and amino acids (8)

10 Loss of water as steam during the boil (11)

11 Sludge of protein and hop matter (4)

12 Proteins clumping early in the boil (3,5)

Down

1 Sweet liquid boiled with hops (4)

2 Sulphur compound with a cooked-corn smell driven off by boiling (3)

3 Spin after the boil that piles trub into a cone (9)

4 Traditional British name for the kettle (6)

6 One reason to boil: to ___ the wort (9)

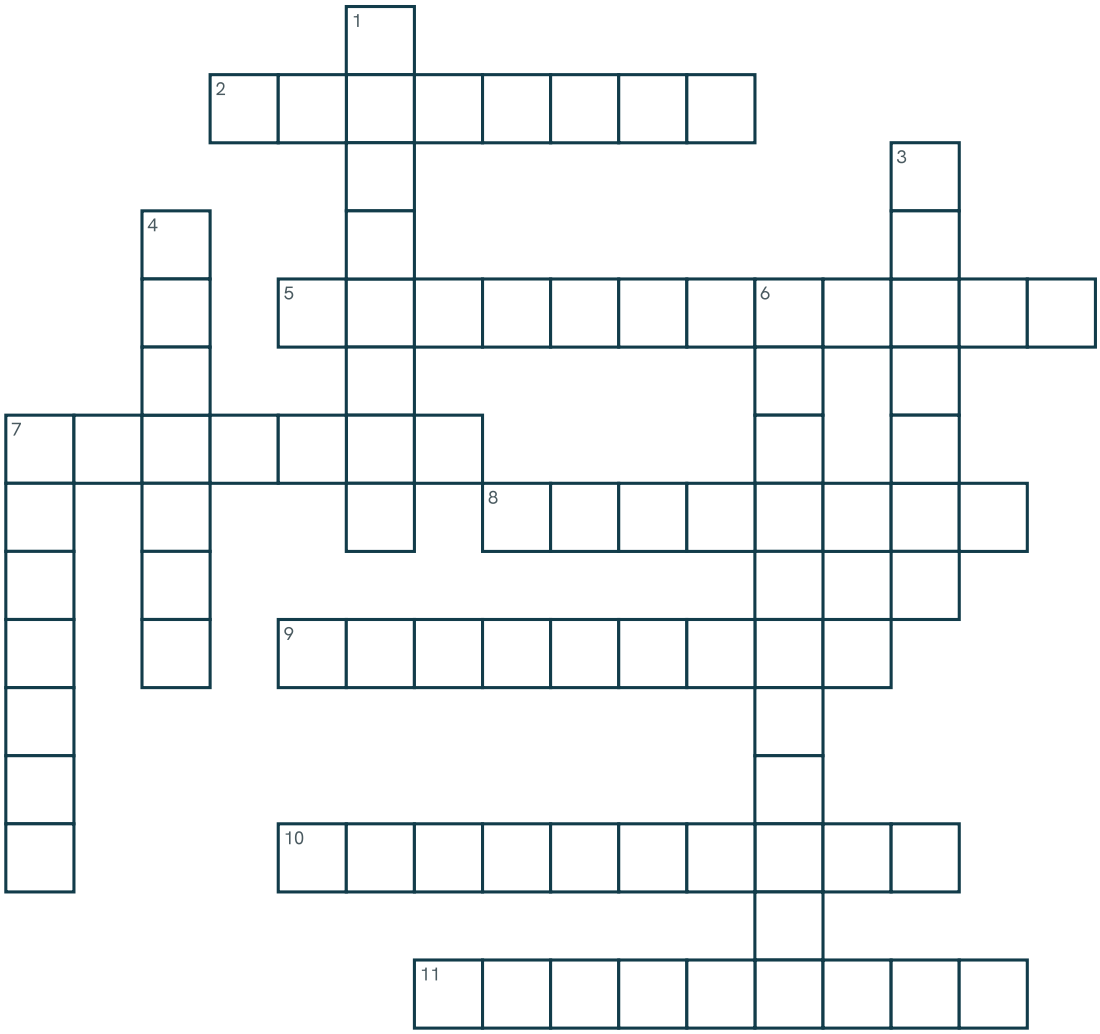
7 Bundle of heating tubes inside some kettles (9)

9 Vessel where wort boils (6)

Crossword 13

Brew day in order

Twelve steps from sack to tank.



Across

2 Step 8: cool the wort fast (8)

5 Step 7: spin to gather the trub (12)

7 Step 1: crack the malt (7)

8 Step 5: rinse the bed with hot water (8)

9 Step 4: run the wort off the grain (9)

10 Step 11: let the yeast work (10)

11 Step 12: into keg, can or bottle (9)

Down

1 Step 10: add the yeast (8)

3 Step 6: roll the wort with hops (7)

4 Step 3: recirculate until the wort runs clear (7)

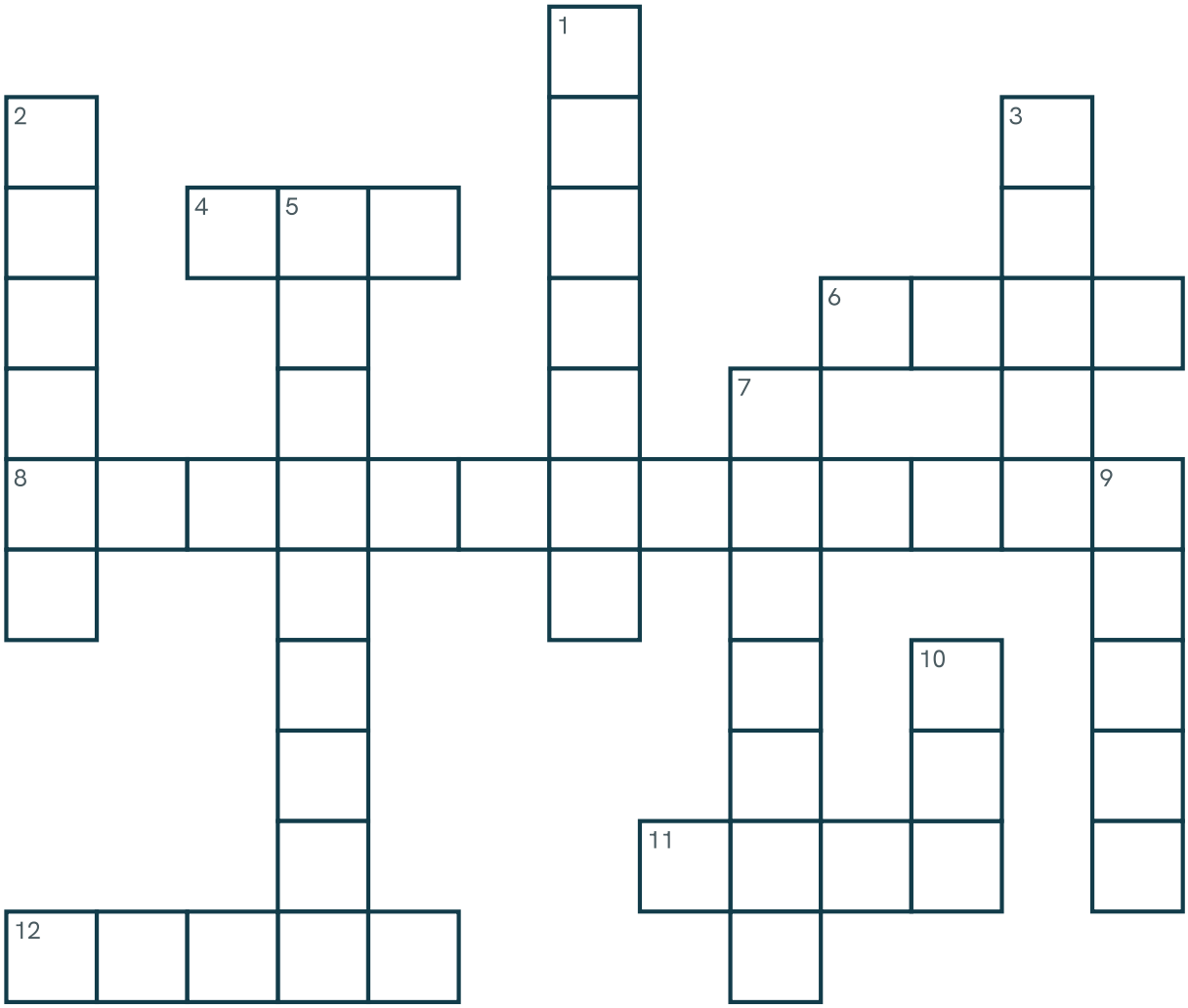
6 Step 9: give the yeast some air (11)

7 Step 2: hot water meets grist (7)

Crossword 14

Brewhouse kit

Name the steel on the brewhouse floor.



Across

4 Hot liquor tank, by its letters (3)

6 Crushes the malt (4)

8 Plate unit that cools wort on its way to the fermenter (4,9)

11 Moves wort and water around the brewhouse (4)

12 Small vessel that collects runoff (5)

Down

1 Hop-filled vessel the hot wort passes through (3,4)

2 ___ tank: holds finished beer (6)

3 Butterfly ___ (5)

5 Vessel with a false bottom for wort separation (6,3)

7 Vessel for mashing (4,3)

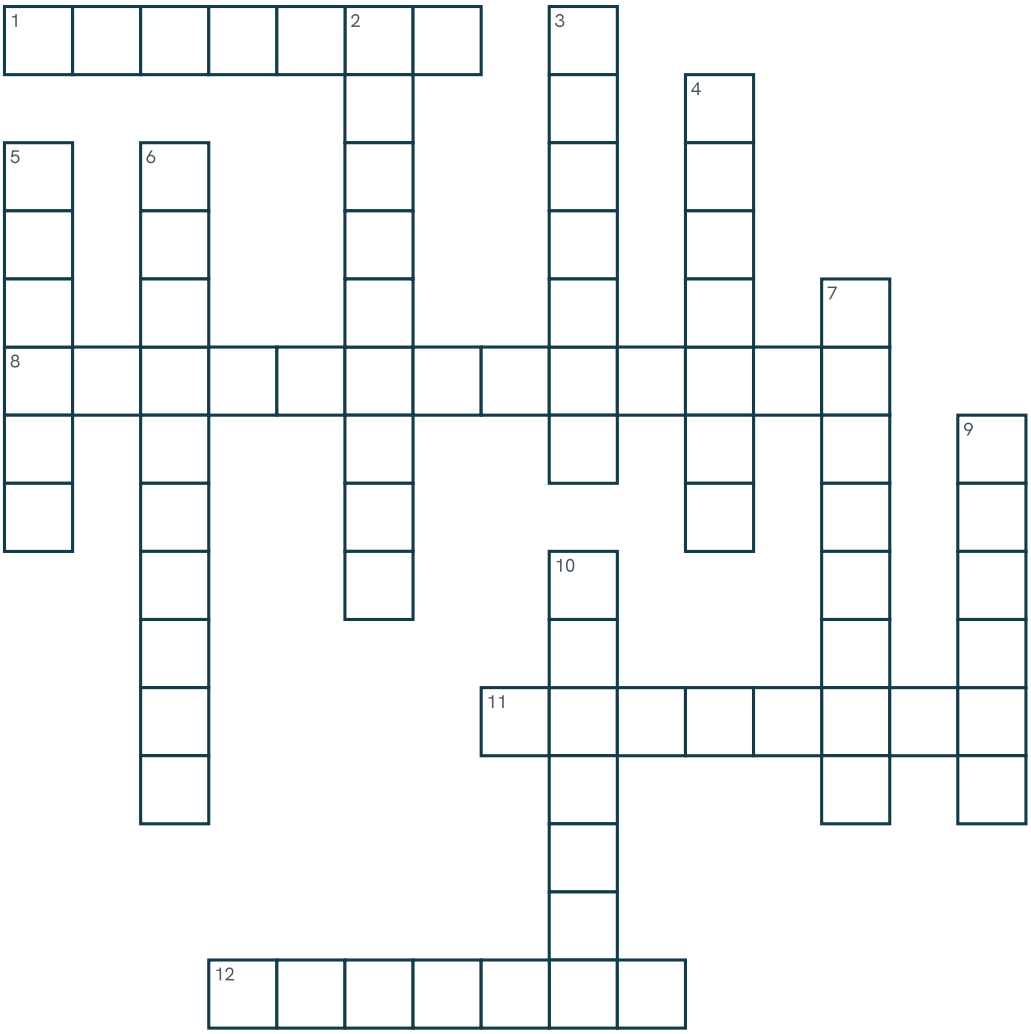
9 Lauter tun knives that cut the grain bed (5)

10 Clean in place, by its letters (3)

Crossword 15

Fermentation

Sugar in, alcohol and carbon dioxide out.



Across

1 ___ tube: lets foam and gas escape a busy fermenter (4,3)

8 Gas made alongside alcohol (6,7)

11 Quiet start while yeast adapts to the wort (3,5)

12 Conical tank for fermenting and conditioning in one (7)

Down

2 Vessel where wort becomes beer (9)

3 The alcohol in beer (7)

4 Density reading tracked every day (7)

5 Coolant run through tank jackets (6)

6 Floating gauge for specific gravity (10)

7 ___ gravity: when the reading stops falling (8)

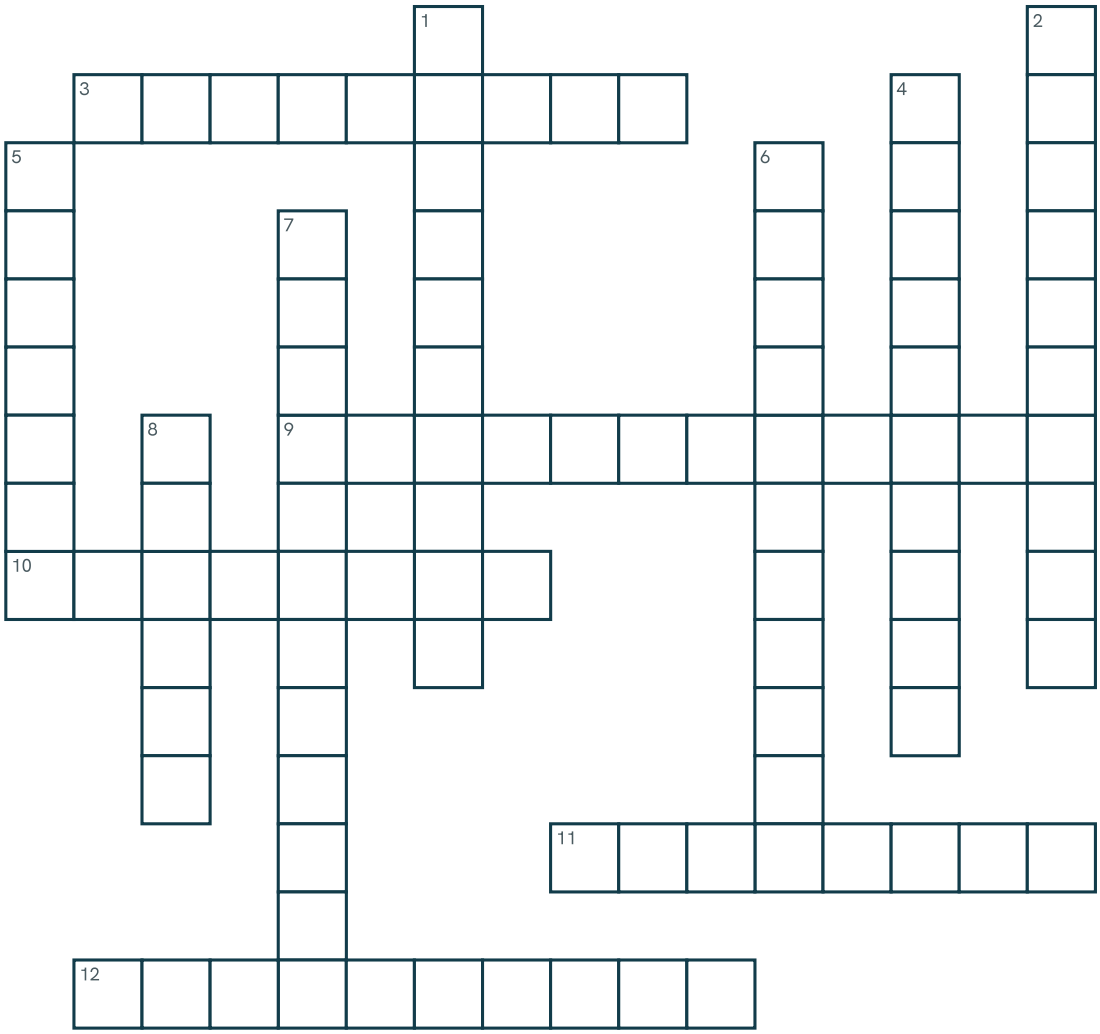
9 Yeast needs it at pitching, finished beer hates it (6)

10 Rocky foam head on fermenting beer (7)

Crossword 16

Cellar and conditioning

What happens after the yeast has done the hard work.



Across

3 Drop the temperature to settle yeast (4,5)

9 Maturing beer after fermentation (12)

10 Fining made from collagen (8)

11 Capping a tank to keep natural CO2 (8)

12 Removing yeast and haze (10)

Down

1 Adding fermenting wort to carbonate (10)

2 Diatomaceous earth used as a filter aid (10)

4 Spins yeast out of beer (10)

5 Adding sugar for bottle conditioning (7)

6 Fizz (11)

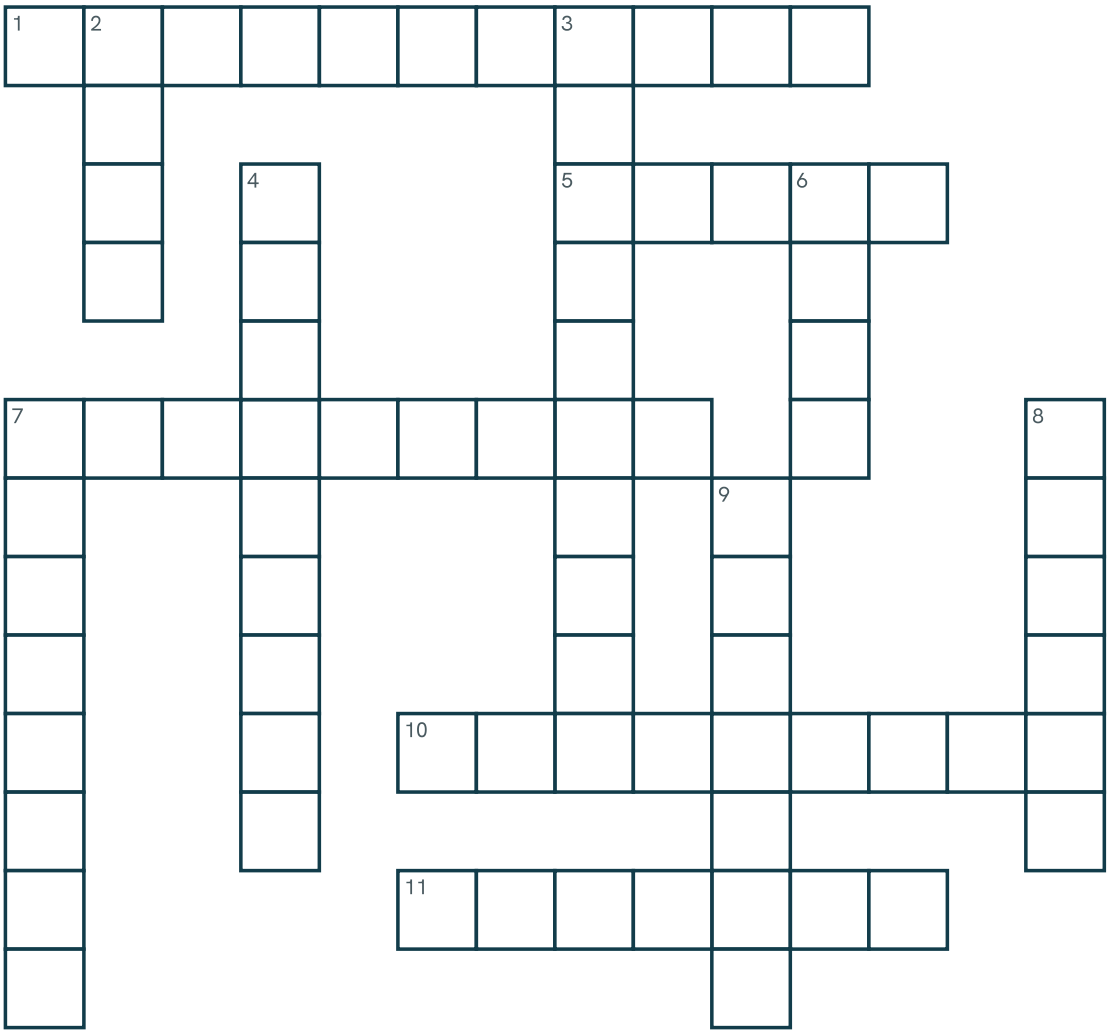
7 Warm pause before the crash (8,4)

8 Gel that removes haze proteins (6)

Crossword 17

Cleaning and hygiene

Brewing is mostly cleaning. Here is the vocabulary.



Across

1 Acid treatment that protects stainless steel (11)

5 Water step between chemicals (5)

7 Head inside a tank that sprays cleaning fluid (5,4)

10 Calcium oxalate scale on tank walls (9)

11 Slimy layer of microbes (7)

Down

2 Cleaner that removes beerstone (4)

3 CIP needs enough flow to create this (10)

4 ___ acid, a popular no-rinse sanitiser (9)

6 ATP ___: a quick hygiene test (4)

7 To bring microbes down to a safe level (8)

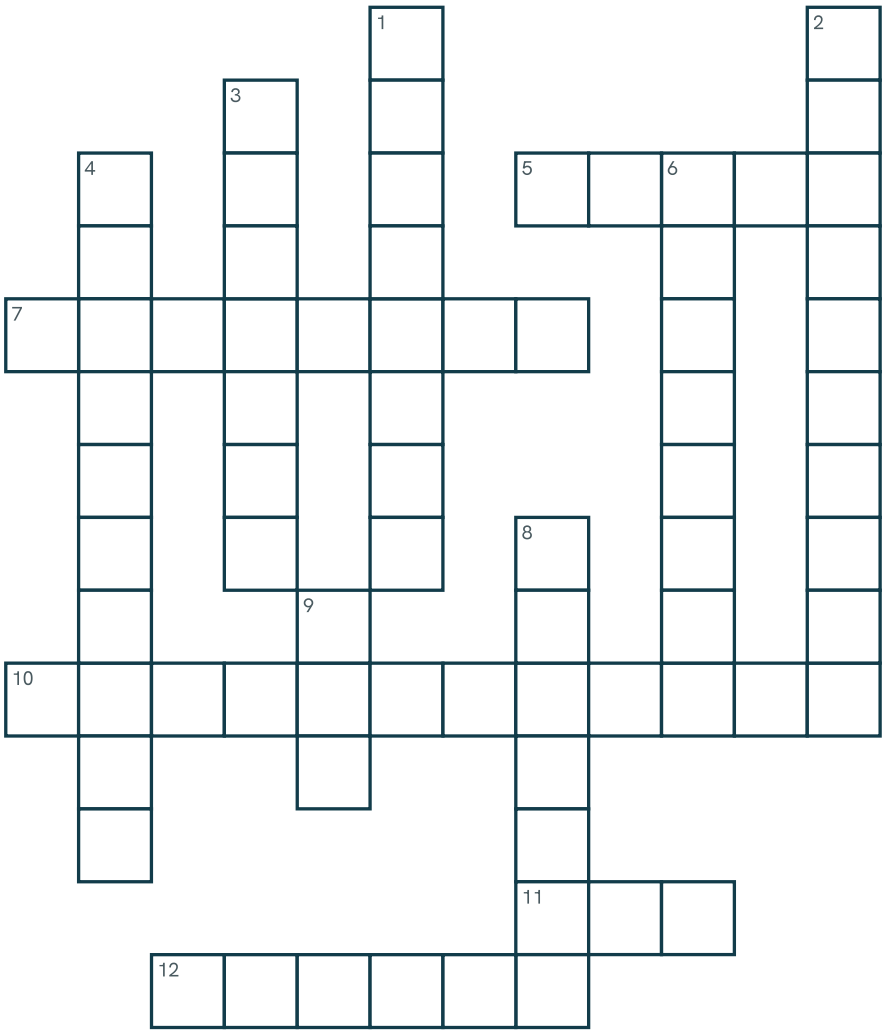
8 Seal that can hide bacteria (6)

9 Alkaline cleaner for hot CIP (7)

Crossword 18

Utilities and the plant

Steam, cold and water: what keeps a brewery running.



Across

5 Common heat source for a kettle (5)

7 CO2 ___: capturing fermentation gas (8)

10 Removes chlorine from water (6,6)

11 Effluent treatment plant, by its letters (3)

12 Makes the steam (6)

Down

1 Removes hardness by ion exchange (8)

2 Machine at the heart of refrigeration (10)

3 Refrigerant in large plants (7)

4 Mash leftover fed to cattle (5,5)

6 Waste water leaving the brewery (8)

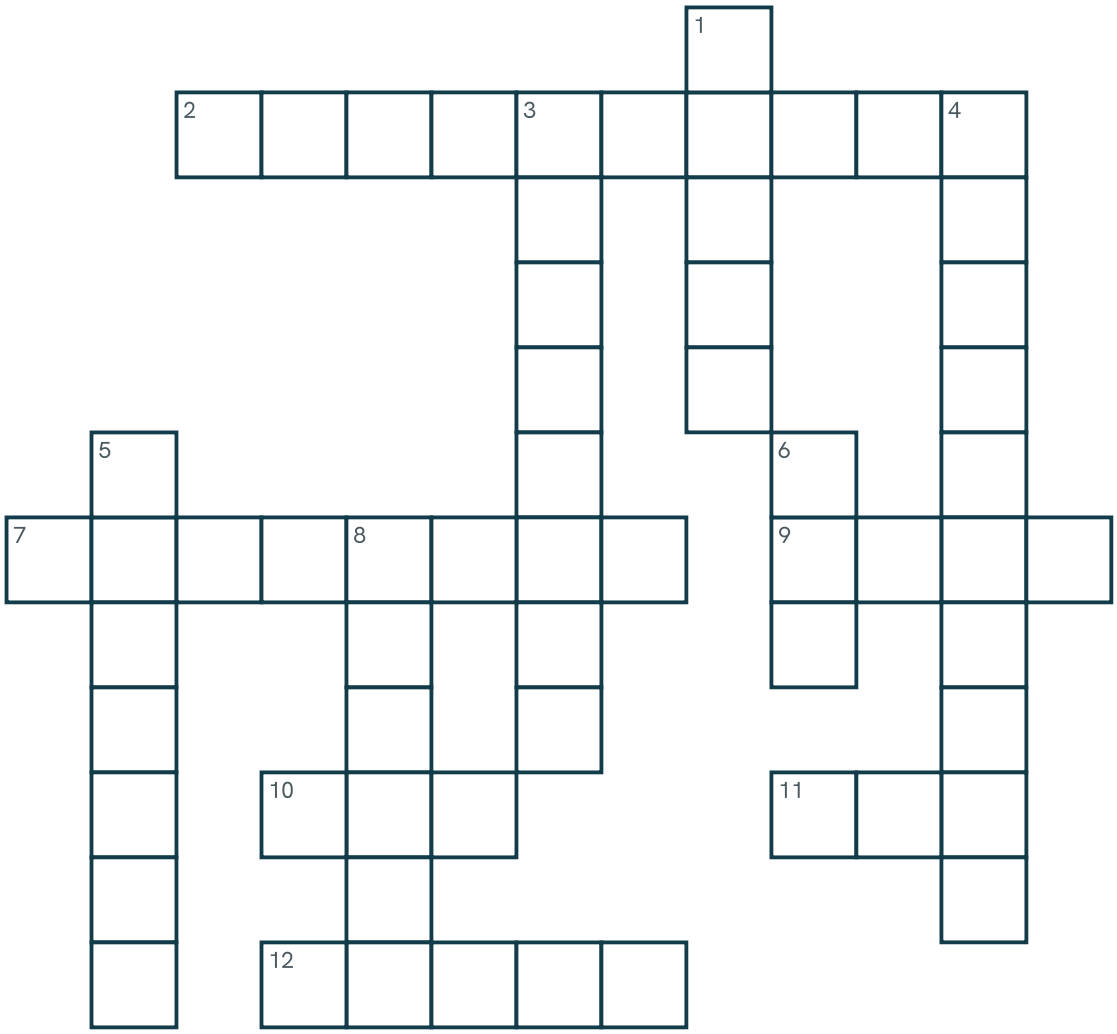
8 Makes the cold for glycol (7)

9 Chemical oxygen demand, by its letters (3)

Crossword 19

Brewing numbers

The units on every brew sheet.



Across

2 One hundred litres (10)

7 Older colour scale still used for malt (8)

9 Scale read on a refractometer (4)

10 American colour scale, by its letters (3)

11 European colour scale, by its letters (3)

12 Degrees ___: sugar by weight (5)

Down

1 ___ gravity, measured after fermentation (5)

3 ___ gravity, measured before fermentation (8)

4 Brewhouse ___: extract won against extract possible (10)

5 Carbonation is measured in ___ of CO2 (7)

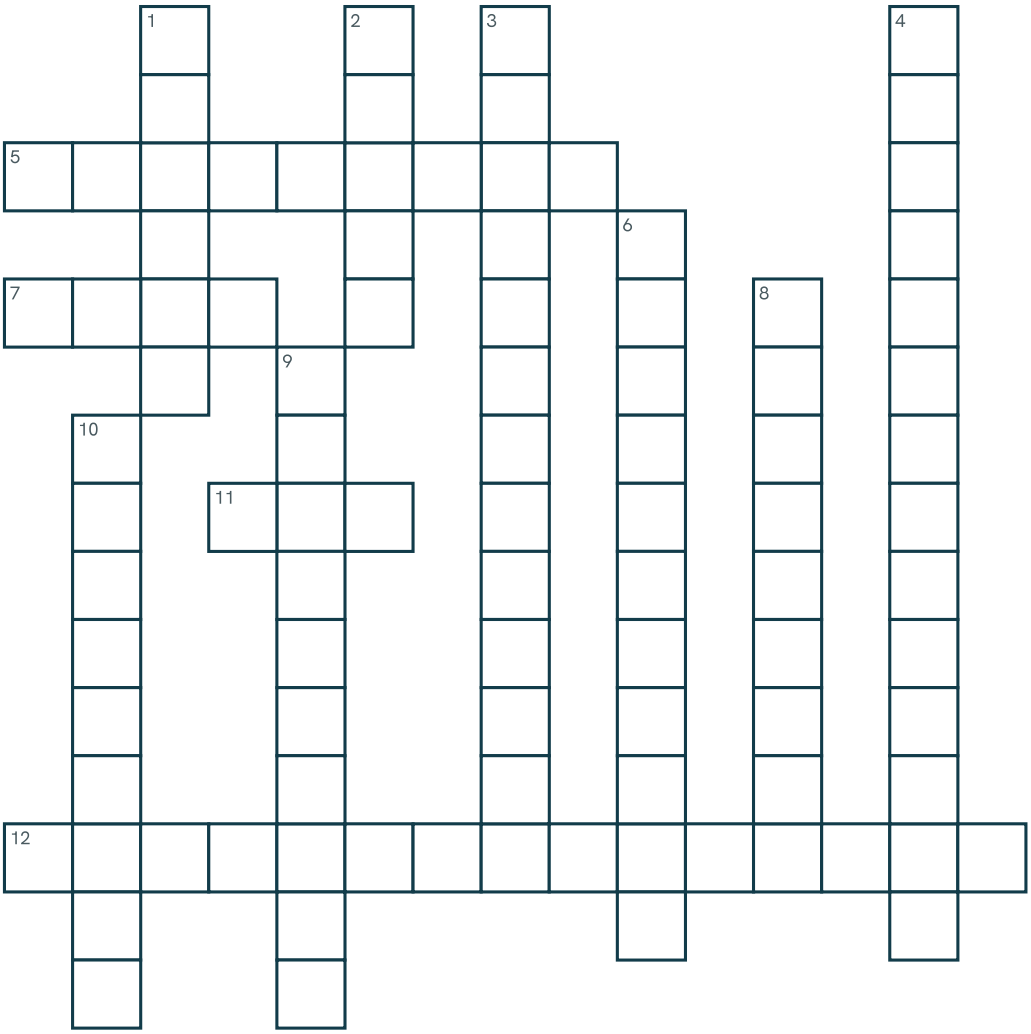
6 Alcohol by volume, by its letters (3)

8 US measure of about 117 litres (6)

Crossword 20

Quality control

How a brewery knows the beer is right.



Across

5 Haze measurement (9)

7 Target range for a product, for short (4)

11 DNA test that finds spoilers, by its letters (3)

12 Measured in parts per billion after packaging (9,6)

Down

1 ___ ageing: warm storage to predict staling (6)

2 Systematic check of a process (5)

3 Stain that marks dead yeast (9,4)

4 Counting chamber for cells (14)

6 Checking an instrument against a standard (11)

8 Share of live yeast cells (9)

9 Tool for counting yeast cells (10)

10 How long beer stays fresh (5,4)



PART THREE

Beer styles

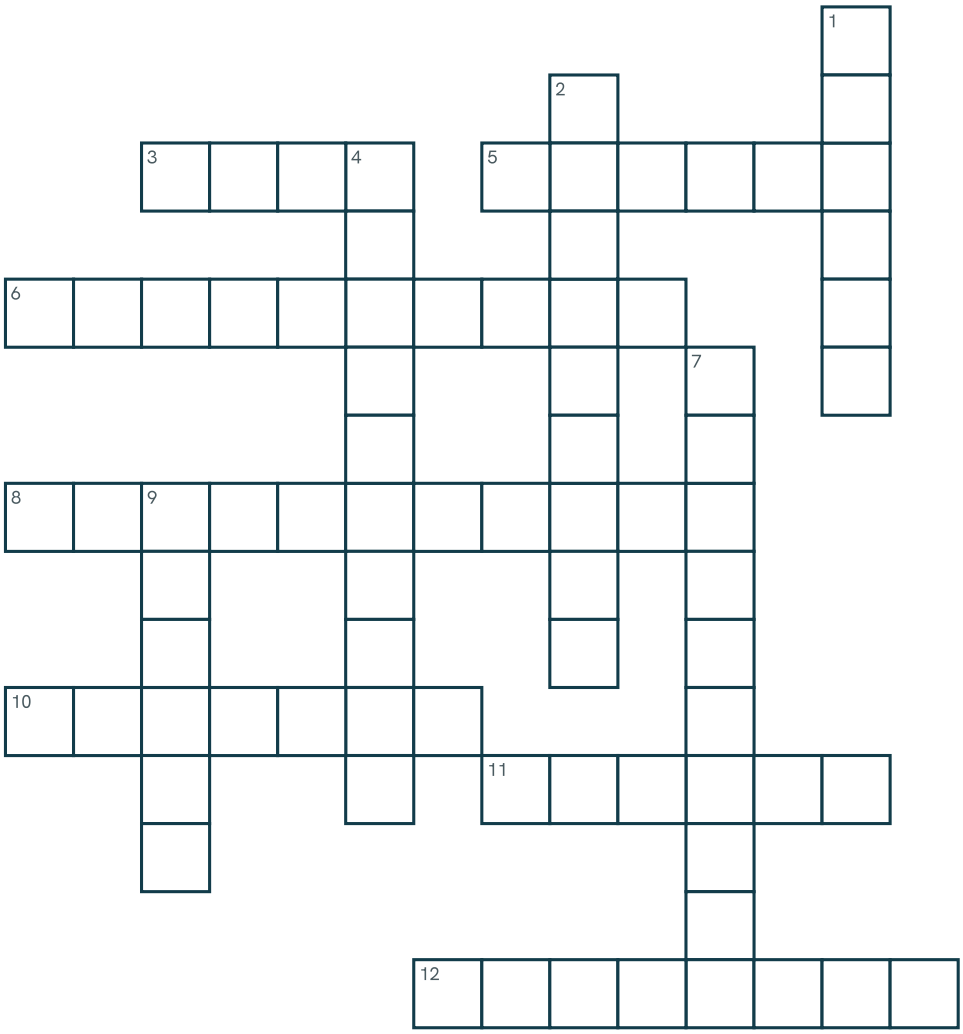
Lagers, ales, sours and everything brewed in between.

Lagers, British ales, Belgian beers, Wheat beers, Stouts and porters, India pale ale, Sour beers, Farmhouse traditions, Low and no alcohol, Barrel ageing

Crossword 21

Lagers

Cold, slow and patient: the lager family.



Across

3 Strong German lager linked to the town of Einbeck (4)

5 Amber Oktoberfest lager (no umlaut needed) (6)

6 Stronger bock, a style started by Munich's Paulaner monks (10)

8 German black lager (11)

10 Pale lager first brewed in 1842 (7)

11 Amber lager city that also names a malt (6)

12 Cold storage after fermentation (8)

Down

1 Dark Munich lager (6)

2 Smoked lager from Bamberg (9)

4 Unfiltered cellar lager from Franconia (10)

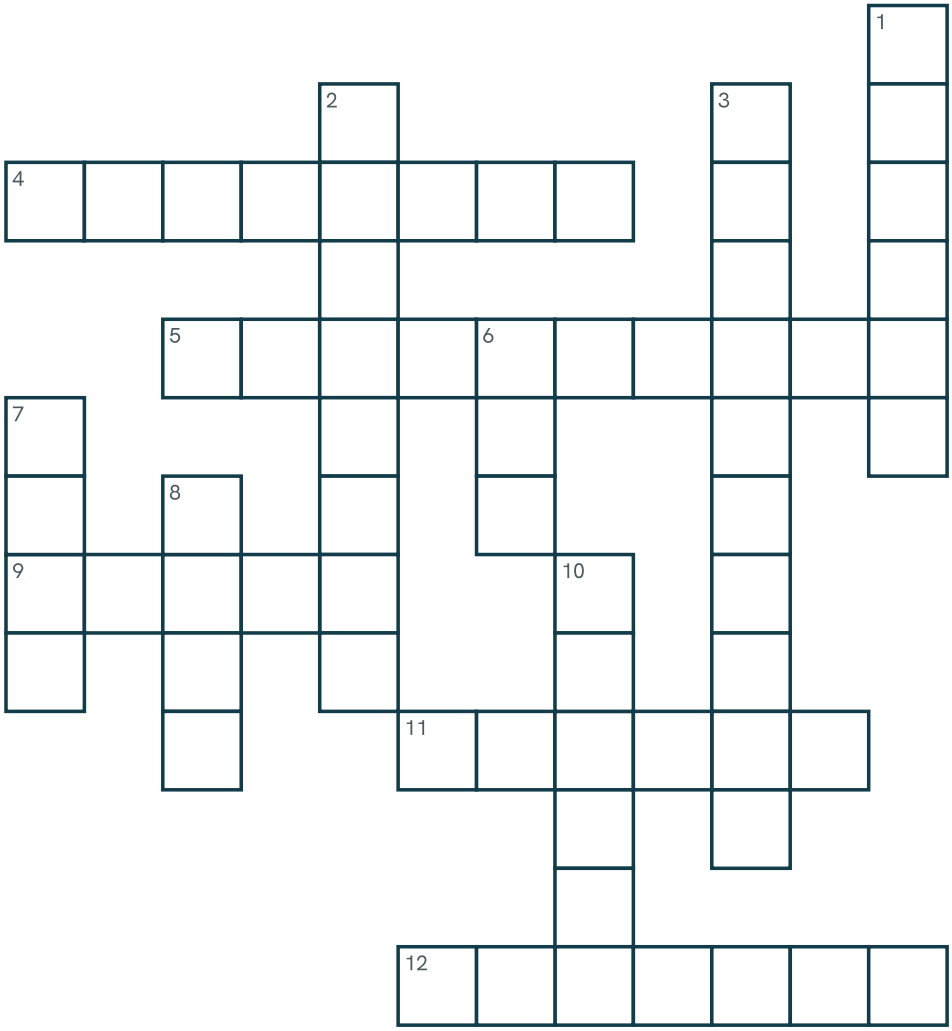
7 Export-strength pale lager from a Ruhr city (10)

9 Pale everyday lager of Munich (6)

Crossword 22

British ales

Pints, pumps and the pub.



Across

4 Beer engine on the bar (4,4)

5 Strong ale that ages like wine (10)

9 Wooden peg that vents a cask (5)

11 Dark ale named after London's street workers (6)

12 Isinglass is one of these clarifiers (7)

Down

1 Session ale of the English pub (6)

2 Nozzle that gives a northern pint its tight head (8)

3 English heritage malting barley (5,5)

6 Extra special bitter, by its letters (3)

7 Vessel where unfiltered real ale conditions (4)

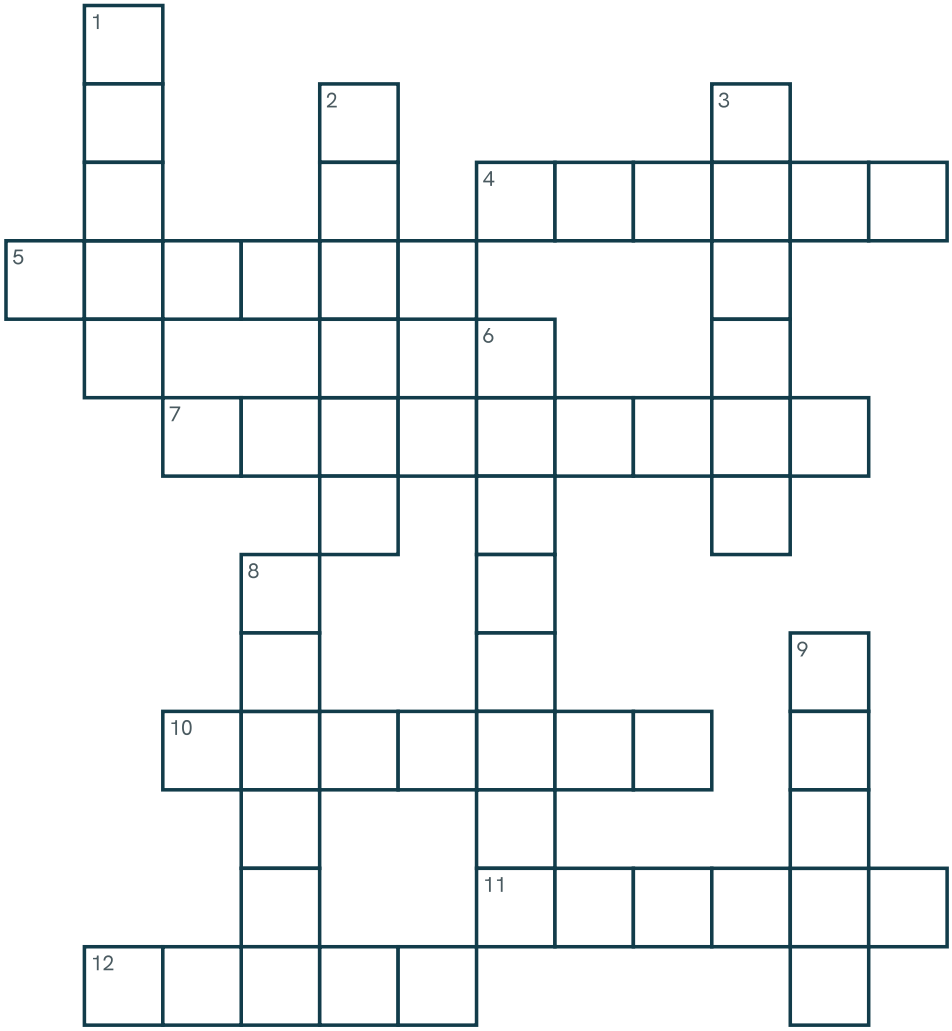
8 Low-strength English ale, often dark (4)

10 Cask holding nine imperial gallons (6)

Crossword 23

Belgian beers

Abbeys, farmhouses and the wild air of the Senne valley.



Across

4 Blend of young and old lambic (6)

5 Spontaneously fermented beer from around Brussels (6)

7 Strongest of the dark abbey styles (9)

10 Wheat beer spiced with coriander and orange peel (7)

11 Strong golden abbey ale (6)

12 Belgian ___ sugar (5)

Down

1 Trappist abbey whose single beer is dry hopped and finished with Brett (5)

2 Trappist brewery near the French border, known for its Blue (6)

3 Dark, raisiny abbey ale (6)

6 Beer brewed under a monastery's supervision (8)

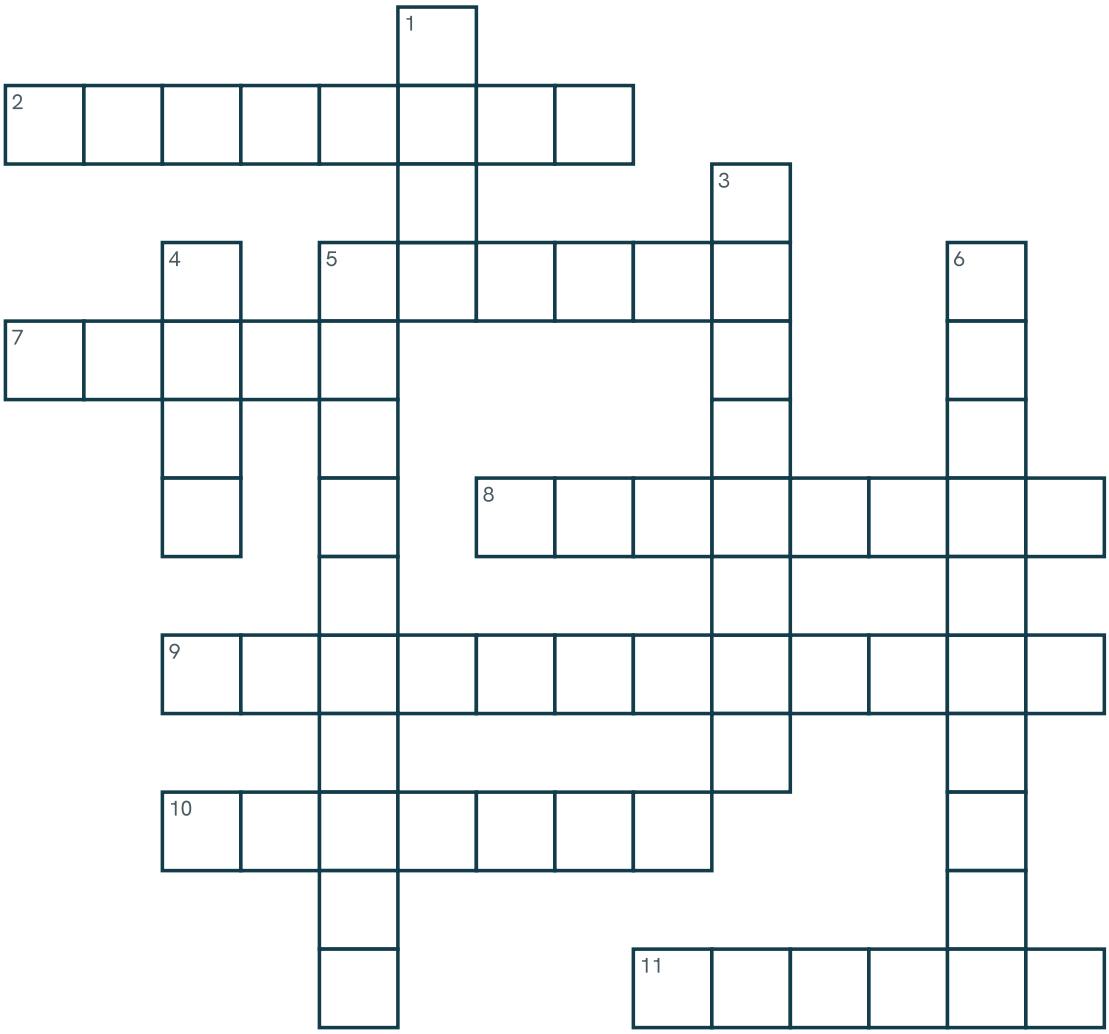
8 Farmhouse ale from Wallonia (6)

9 Lambic with sour cherries (5)

Crossword 24

Wheat beers

Banana, clove, haze and a tall glass.



Across

2 Filtered weissbier (8)

5 German for wheat (6)

7 Spice note from 4-vinyl guaiacol in weissbier (5)

8 ___ Weisse, a tart wheat beer of low strength (8)

9 Dark Bavarian wheat beer (12)

10 Wheat brings more of this than barley, so more haze and foam (7)

11 Isoamyl acetate gives this fruit note (6)

Down

1 Cloudiness from yeast and protein (4)

3 Witbier uses this kind of raw wheat (8)

4 Salty sour with coriander, from Goslar and Leipzig (4)

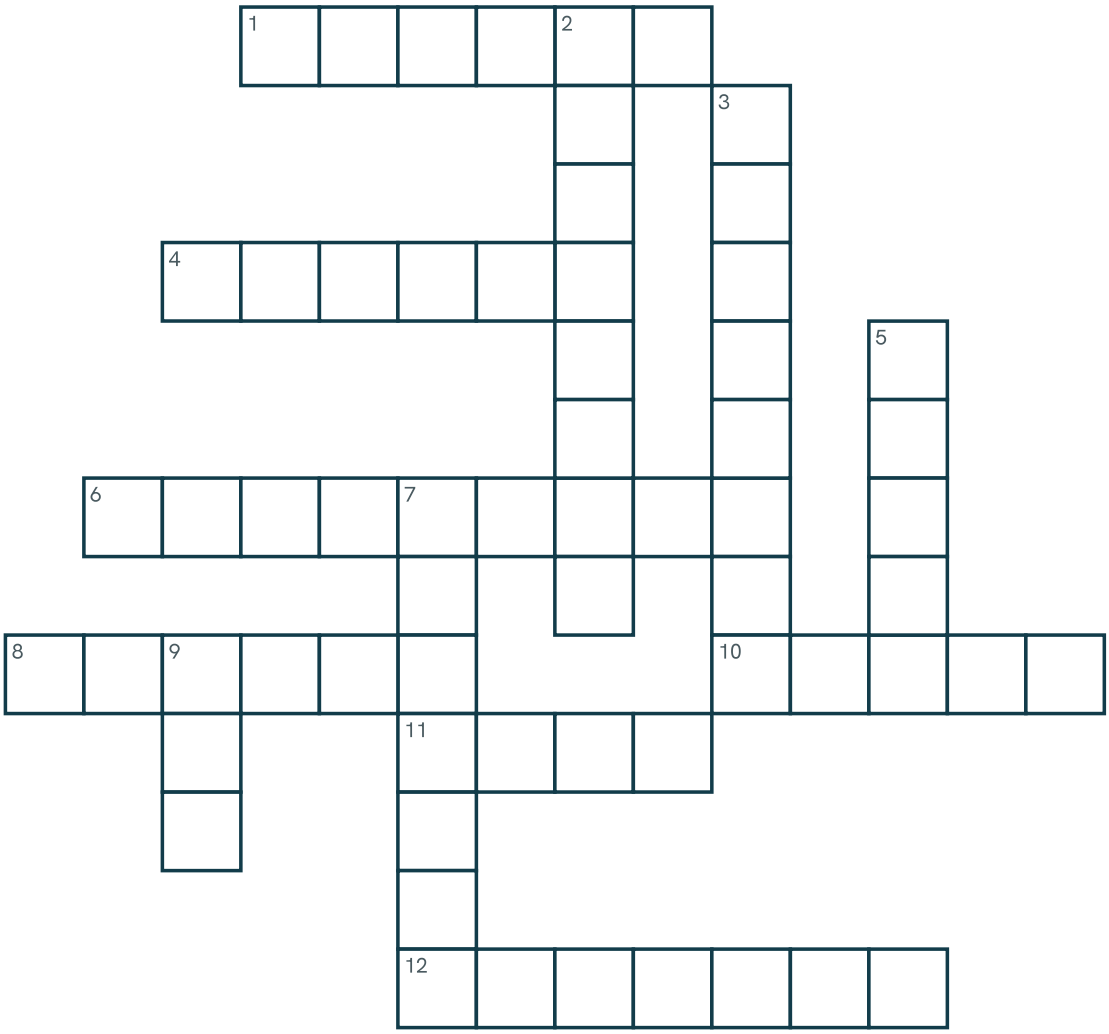
5 Strong wheat beer (10)

6 Bavarian wheat beer served with its yeast (10)

Crossword 25

Stouts and porters

Roast, coffee, chocolate and a creamy head.



Across

1 ___ porter, a dark beer fermented with lager yeast (6)

4 Flavour roasted barley often suggests (6)

6 Dark malt that adds cocoa notes (9)

8 Nitrogen capsule inside a can (6)

10 Dark beer whose name once just meant strong (5)

11 ___ stout, sweetened with lactose (4)

12 Milk sugar that yeast cannot ferment (7)

Down

2 ___ stout, brewed strong for the Russian market (8)

3 Stout first brewed at St James's Gate, Dublin (8)

5 Pour that gives a creamy, cascading head (5)

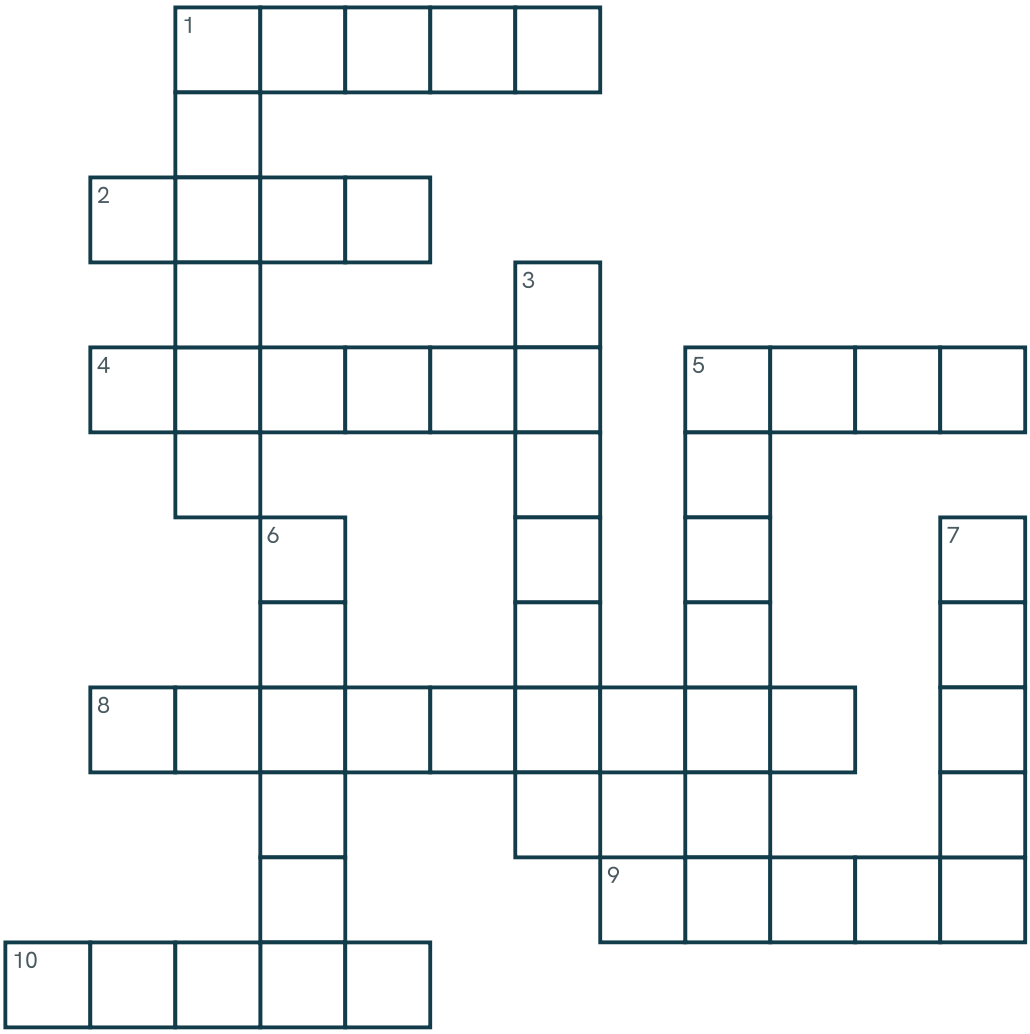
7 Grain that adds silky body to a stout (7)

9 Irish ___ stout (3)

Crossword 26

India pale ale

The style that sailed east and came back as a craze.



Across

1 ___ IPA, dark in the glass but hoppy (5)

2 Bone-dry IPA made with an added enzyme (4)

4 ___ IPA, stronger and hoppier (6)

5 New England IPA is often called this (4)

8 Clear, bitter American IPA (4,5)

9 The I in IPA (5)

10 Soft, low-bitterness IPAs are often called this (5)

Down

1 Trent town that made IPA its own (6)

3 ___ IPA, low in strength (7)

5 George ___, London brewer whose pale ale shipped east in the 1700s (7)

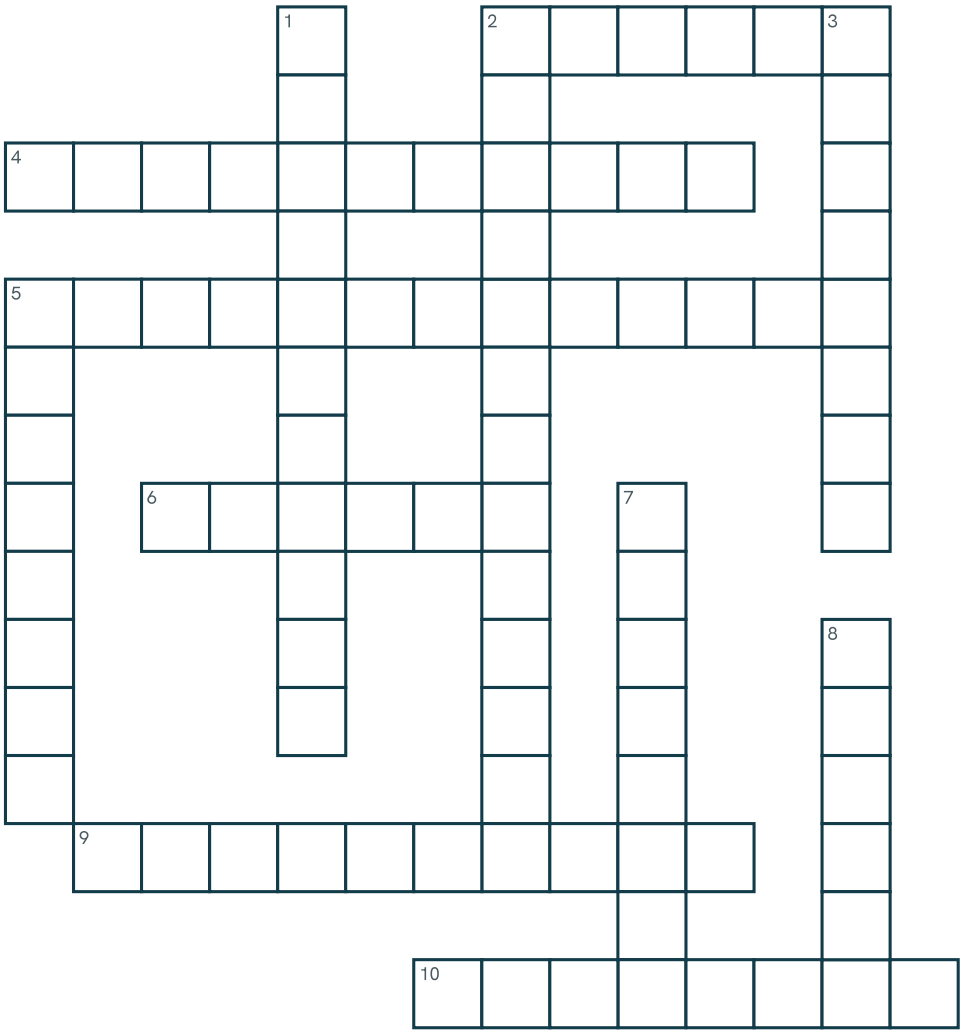
6 US hop bred from Simcoe, with berry and pine notes (6)

7 US hop released in 2008, known for lime and tropical fruit (5)

Crossword 27

Sour beers

Acid, wood and patience.



Across

2 Clean, yoghurt-like acid (6)

4 Acid-making bacteria often paired with Brett (11)

5 Wild yeast with farmyard and fruity notes (13)

6 Vinegar acid (6)

9 Quick souring before the boil (6,4)

10 ___ red ale, a vinous Belgian sour (8)

Down

1 Fermentation with no yeast added by the brewer (11)

2 Bacteria that make lactic acid (13)

3 Shallow open vessel for spontaneous inoculation (8)

5 Mixing barrels of different ages (8)

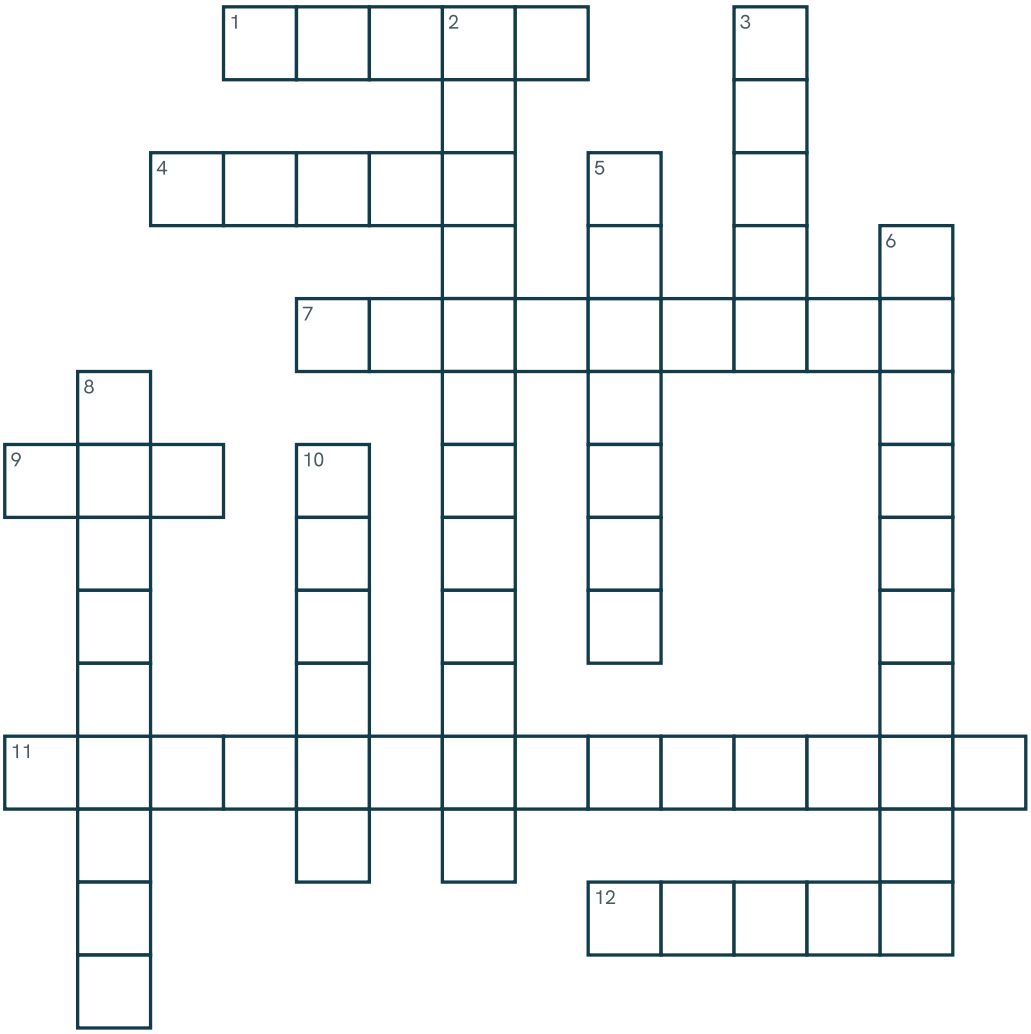
7 Flemish brown sour (3,5)

8 Large wooden vat for ageing sour beer (6)

Crossword 28

Farmhouse traditions

Old beers brewed on farms and still alive.



Across

1 Gruit was made from these (5)

4 Main character of grodziskie (5)

7 German stone beer (9)

9 ___ ale: wort that is never boiled (3)

11 Smoky farmhouse ale of a Swedish island (14)

12 Steinbier is boiled with a hot ___ (5)

Down

2 French ale made for keeping (5,2,5)

3 Finnish beer filtered through juniper (5)

5 Branches used in a sahti filter bed (7)

6 Smoky Polish wheat beer (10)

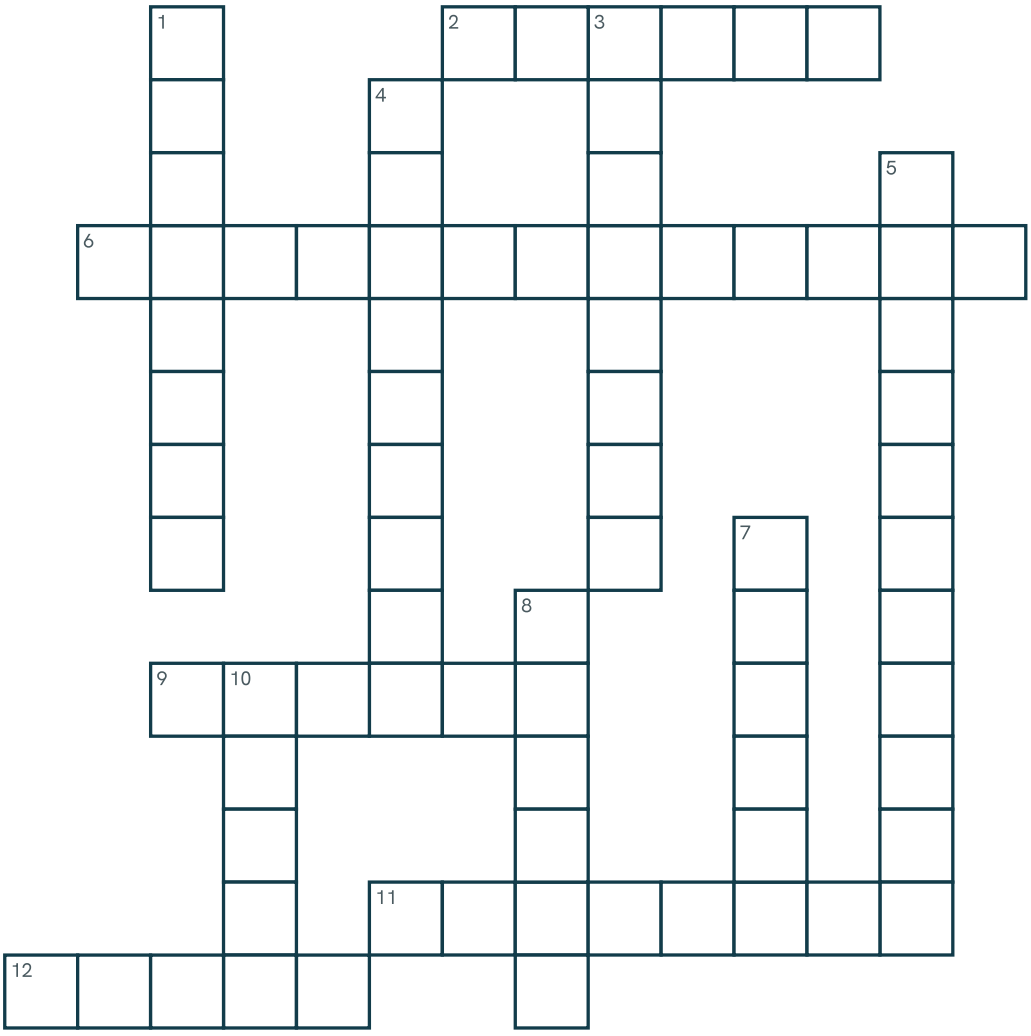
8 Saison and bière de garde are this kind of ale (9)

10 Home of kveik (6)

Crossword 29

Low and no alcohol

Lighter beers and how they are made.



Across

2 ___ distillation strips alcohol at low temperature (6)

6 With the alcohol removed after fermentation (13)

9 Barley protein some drinkers must avoid (6)

11 Reverse osmosis pushes beer through one (8)

12 Common flaw of alcohol-free beer: tasting too ___ (5)

Down

1 ___ fermentation: stopped early on purpose (8)

3 Energy on the nutrition panel (8)

4 Gluten-free seed brewed like a grain (9)

5 Easy to drink several of (11)

7 Andean seed used in some gluten-free beers (6)

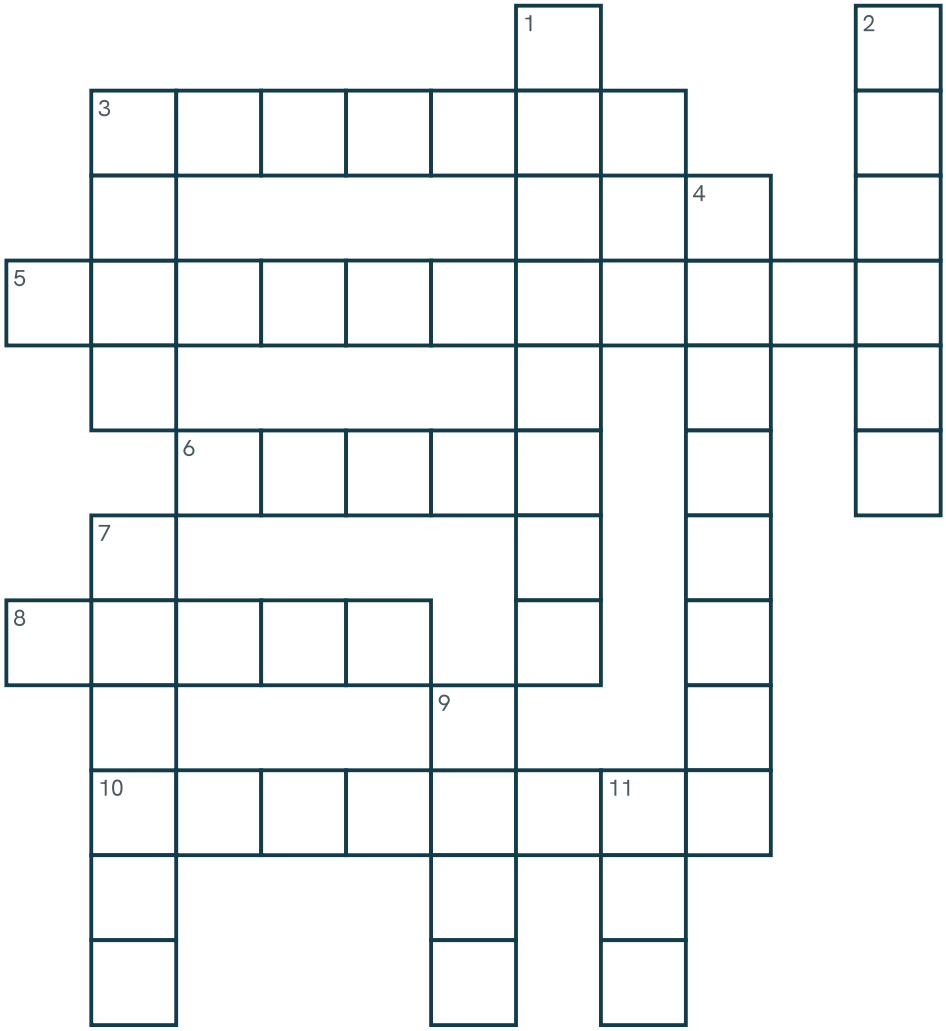
8 Some brewers add one to break down gluten (6)

10 Low-calorie lager type (5)

Crossword 30

Barrel ageing

Wood, spirit and time.



Across

3 US whiskey whose barrels age many stouts (7)

5 Liquid lost to evaporation (6,5)

6 Plank of a barrel (5)

8 Gentler heating of barrel wood (5)

10 Large cask, often about 500 litres (8)

Down

1 Cask of about 250 litres (8)

2 Fractional blending system (6)

3 Barrel stopper (4)

4 Oak compound with vanilla flavour (8)

7 Barrel maker (6)

9 Burnt inner layer of a barrel (4)

11 Wood of most barrels (3)



PART FOUR

In the glass

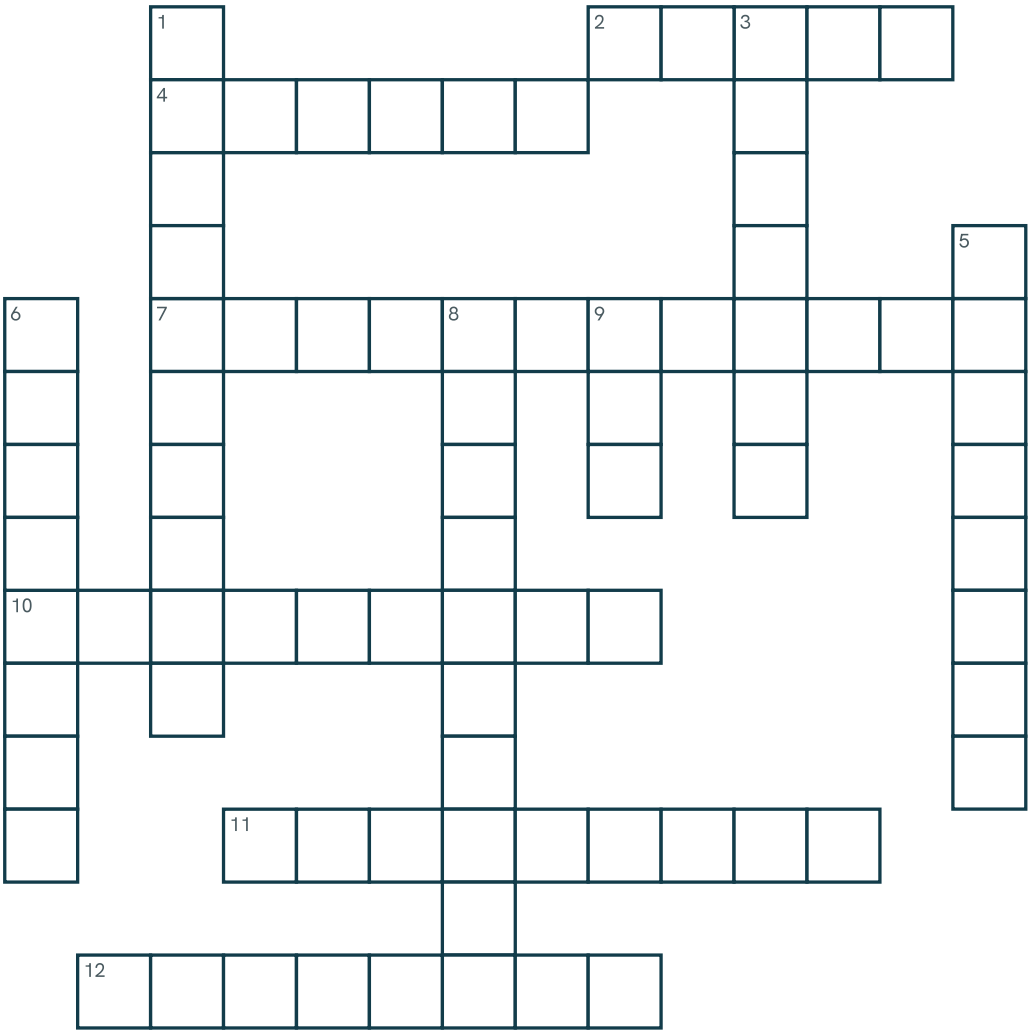
Tasting, faults, serving and the food on the table.

Off-flavours, Tasting and sensory, Colour and clarity, Foam and fizz, Glassware, Draught and serving, Kegs, casks and bottles, Packaging, Beer and food, Judging and competitions

Crossword 31

Off-flavours

Name the fault, then find the cause.



Across

2 Hot, solvent-like alcohols (5)

4 Aroma of light-struck beer (6)

7 Green apple or fresh-cut pumpkin (12)

10 Wet cardboard comes from this (9)

11 Contamination by wild microbes (9)

12 Butter or butterscotch (8)

Down

1 Cheesy, sweaty smell from old hops (10)

3 Rotten egg, often from lager yeast (7)

5 Taste of coins or blood (8)

6 Plastic or medicinal, often from chlorine (8)

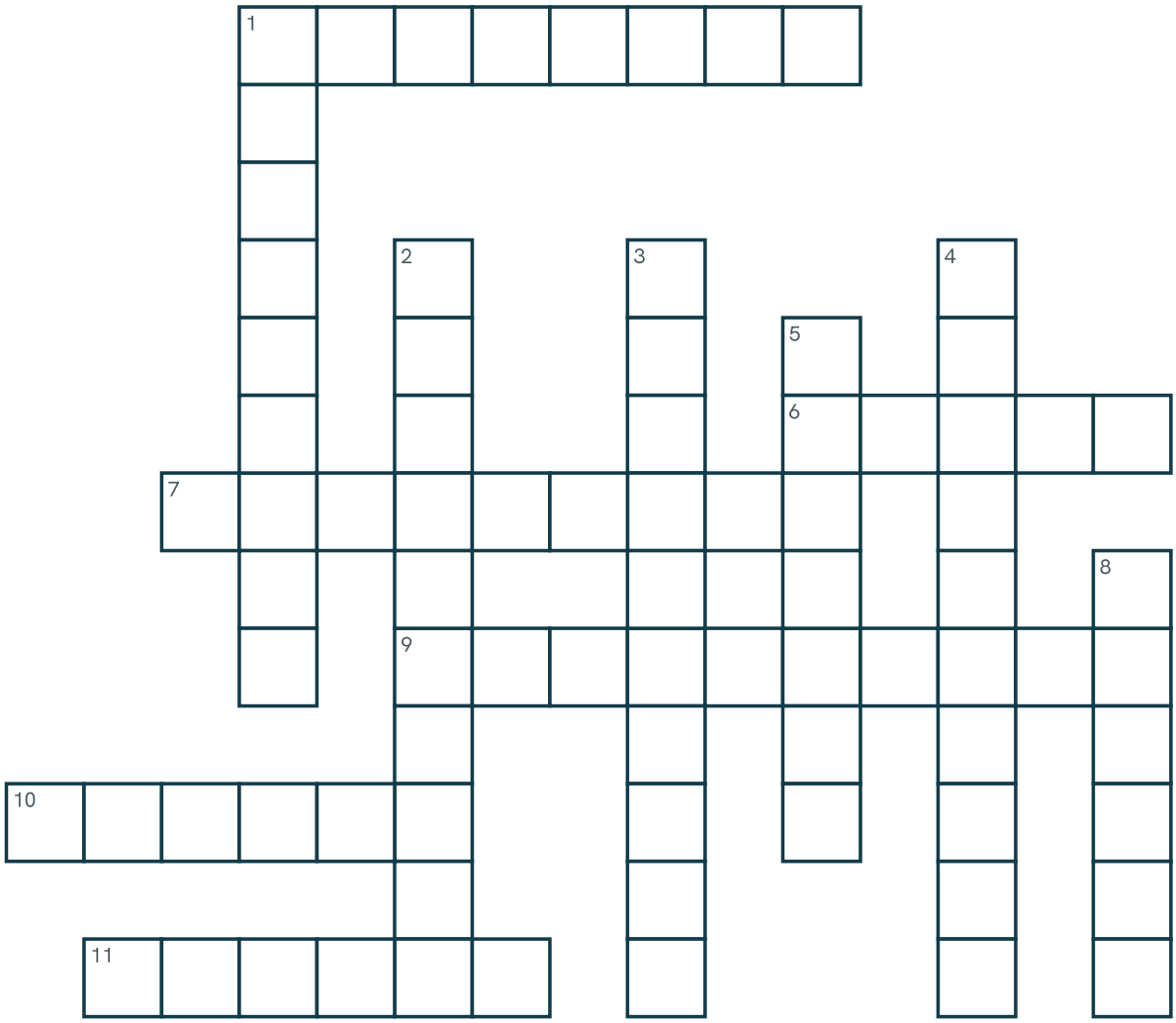
8 Puckering dryness, like stewed tea (10)

9 Cooked sweetcorn (3)

Crossword 32

Tasting and sensory

How to taste like a judge.



Across

1 Test with three samples, one of them different (8)

6 What you smell first (5)

7 Body, carbonation and texture (9)

9 Smelling through the back of the mouth (10)

10 Your sense of taste (6)

11 Flavour that lingers after you swallow (6)

Down

1 Lowest level you can detect (9)

2 Taste from hop acids (10)

3 Colour, clarity and head (10)

4 Judging form (10)

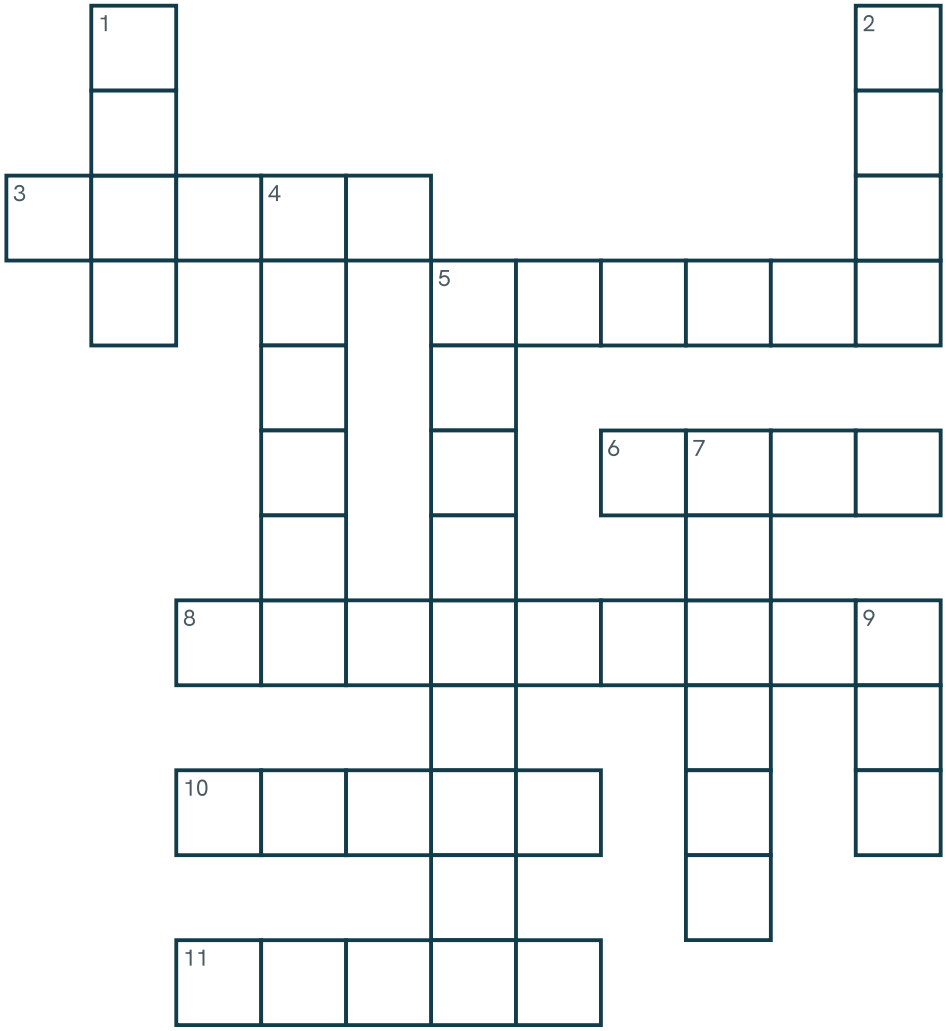
5 Malt set against hops (7)

8 A set of small tasters (6)

Crossword 33

Colour and clarity

What your eyes tell you before the first sip.



Across

3 Colour of a Schwarzbier (5)

5 Unfiltered look (6)

6 Colour of a Helles (4)

8 Top grade of clarity (9)

10 Palest beer colour, like dried grass (5)

11 Colour of a Märzen (5)

Down

1 ___ ale, named for its colour (4)

2 Red glint in a porter held to the light (4)

4 Colour of many bitters (6)

5 Cloudiness that appears cold and clears when warm (5,4)

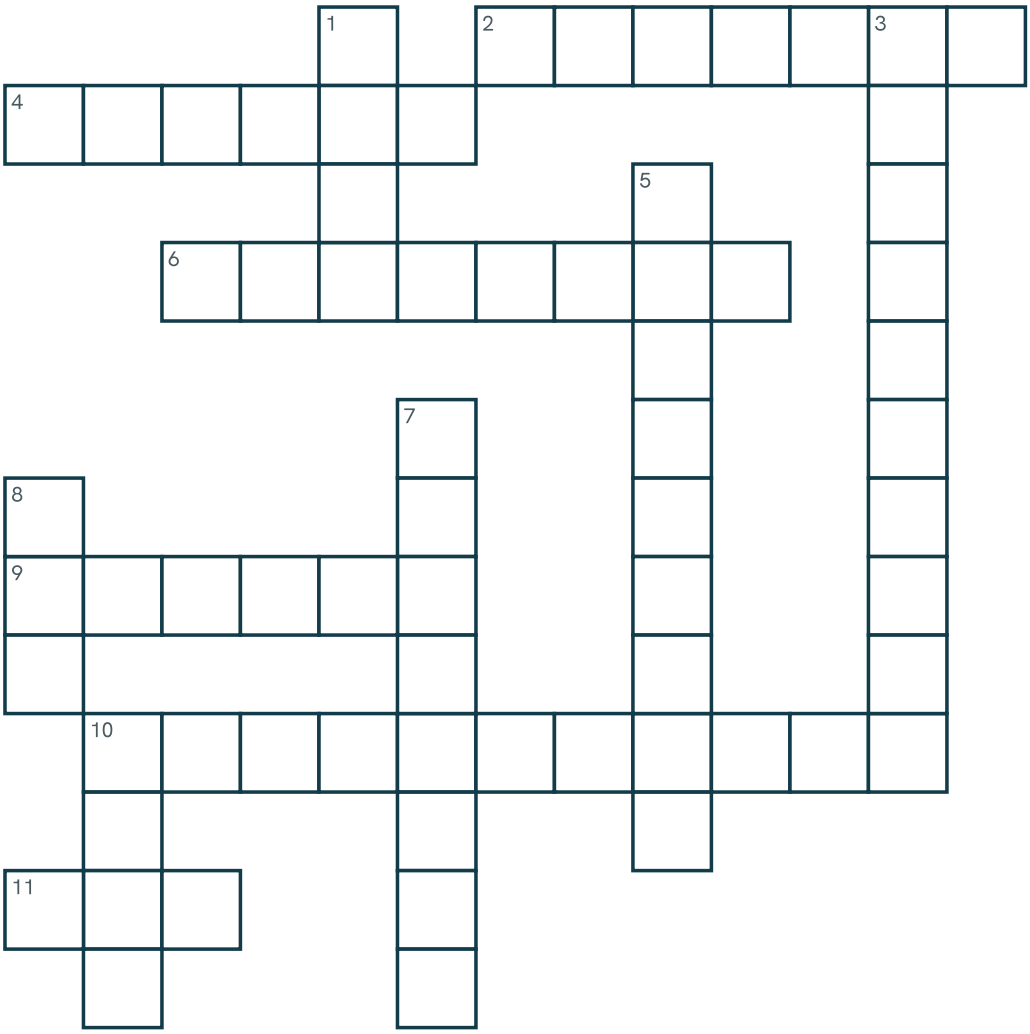
7 Letting no light through (6)

9 Colour of a stout's head (3)

Crossword 34

Foam and fizz

The science of bubbles.



Across

2 Foaming out on opening, sometimes from mouldy grain (7)

4 Rising pocket of gas (6)

6 Gas that makes tiny bubbles (8)

9 Ultrasonic base that sets off a nitro pour (6)

10 Fungal protein that causes gushing (11)

11 Metric unit of pressure (3)

Down

1 With no fizz (4)

3 Bubbles forming at a rough spot (10)

5 How long the head lasts (9)

7 ___ acid, the bite of CO2 (8)

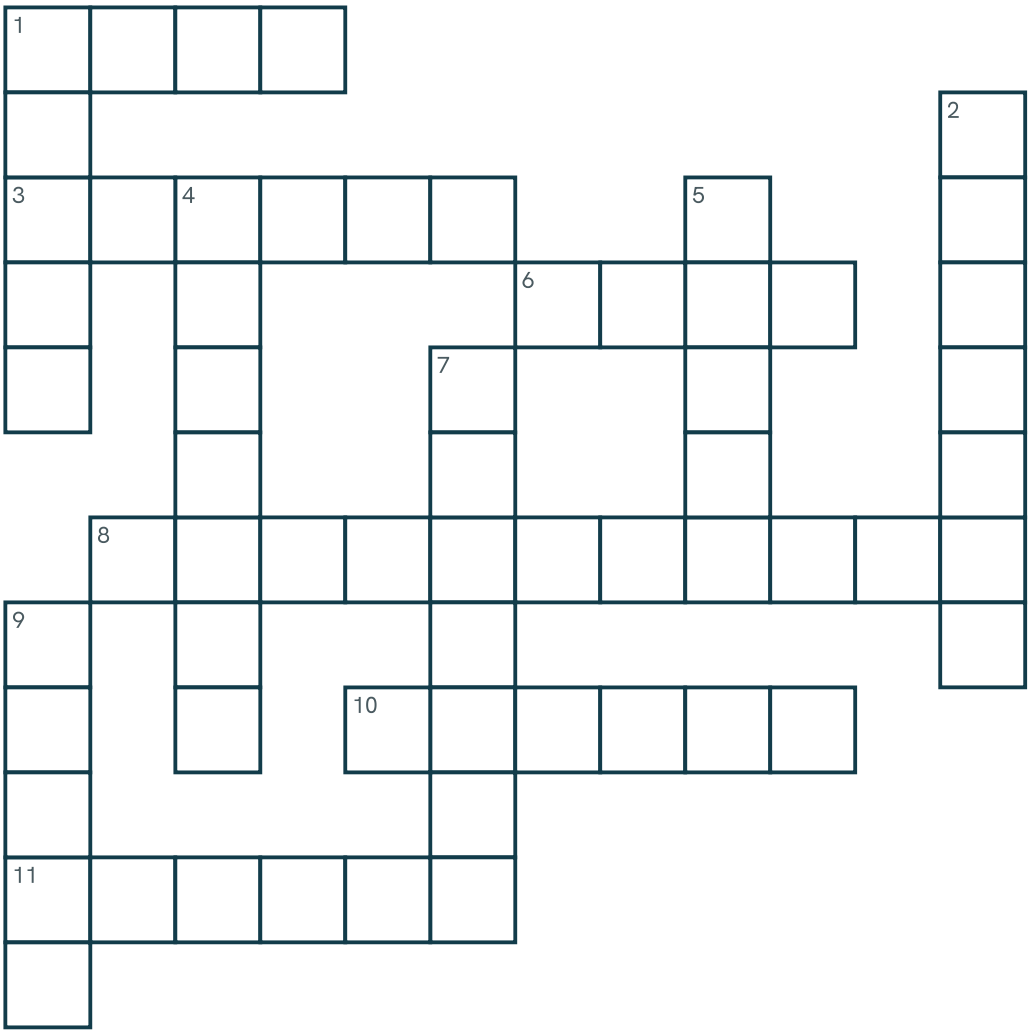
8 Pressure unit in North America (3)

10 Foam on top of a beer (4)

Crossword 35

Glassware

The right glass, and why it matters.



Across

1 Modern stemmed tasting glass from Italy (4)

3 Rings of foam left on the glass (6)

6 Glass shaped like footwear (4)

8 German glass that tapers at the top (11)

10 Tall, thin glass for Kölsch (6)

11 Small glass in a flight (6)

Down

1 Stemmed glass that holds aroma, good for strong ales (5)

2 Mug with a handle (7)

4 Wide goblet used for abbey ales (7)

5 Pint glass with a bulge near the top (5)

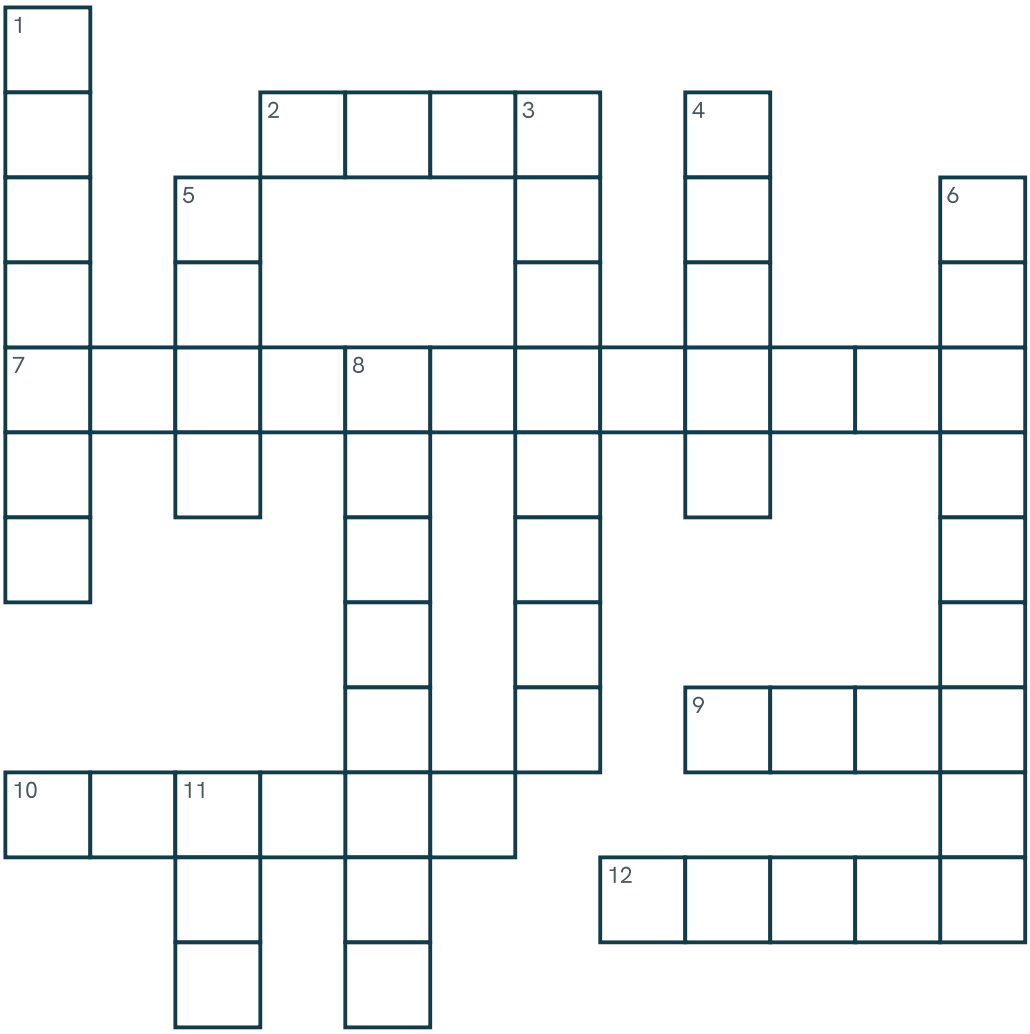
7 Balloon glass for barleywine (7)

9 Narrow glass for fruit lambic (5)

Crossword 36

Draught and serving

From keg to glass without the foam disaster.



Across

2 Too much of this means waste (4)

7 Job due every one to two weeks (4,8)

9 Bar-top fitting that holds the taps (4)

10 Chilled bundle of beer lines (6)

12 Tube inside a keg (5)

Down

1 Fits onto the keg spear (7)

3 Blend of nitrogen and carbon dioxide (5,3)

4 Tube that runs through the font (5)

5 Standard pour (4)

6 Sets the gas pressure (9)

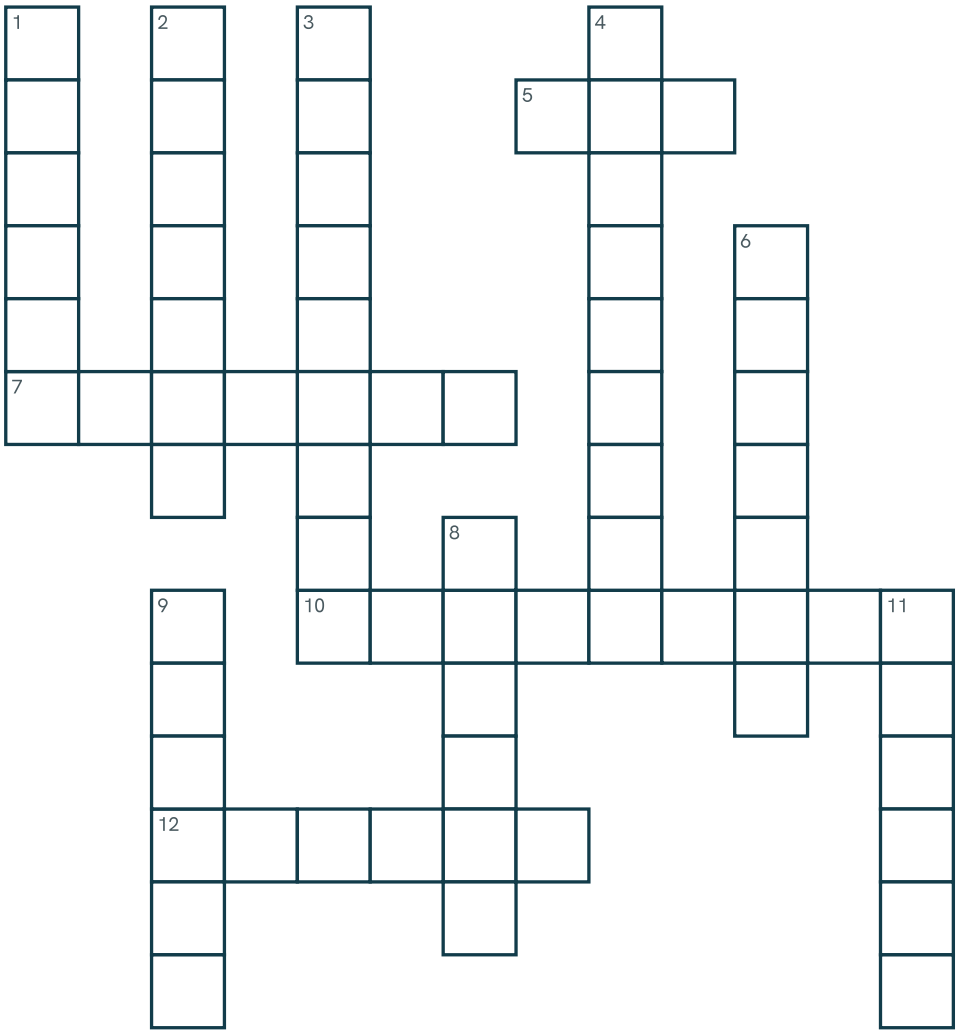
8 Walk-in chiller for kegs (4,4)

11 Faucet on the bar (3)

Crossword 37

Kegs, casks and bottles

Every container a beer can live in.



Across

5 Smallest cask, four and a half imperial gallons (3)

7 Refillable jug from the taproom (7)

10 Steel used for most kegs (9)

12 650 ml bottle (6)

Down

1 One-way bag-in-ball keg (6)

2 Sixteen-ounce can (7)

3 Soda keg loved by homebrewers (9)

4 Cask of 18 imperial gallons (9)

6 Can filled and sealed at the bar (7)

8 Common keg fitting (6)

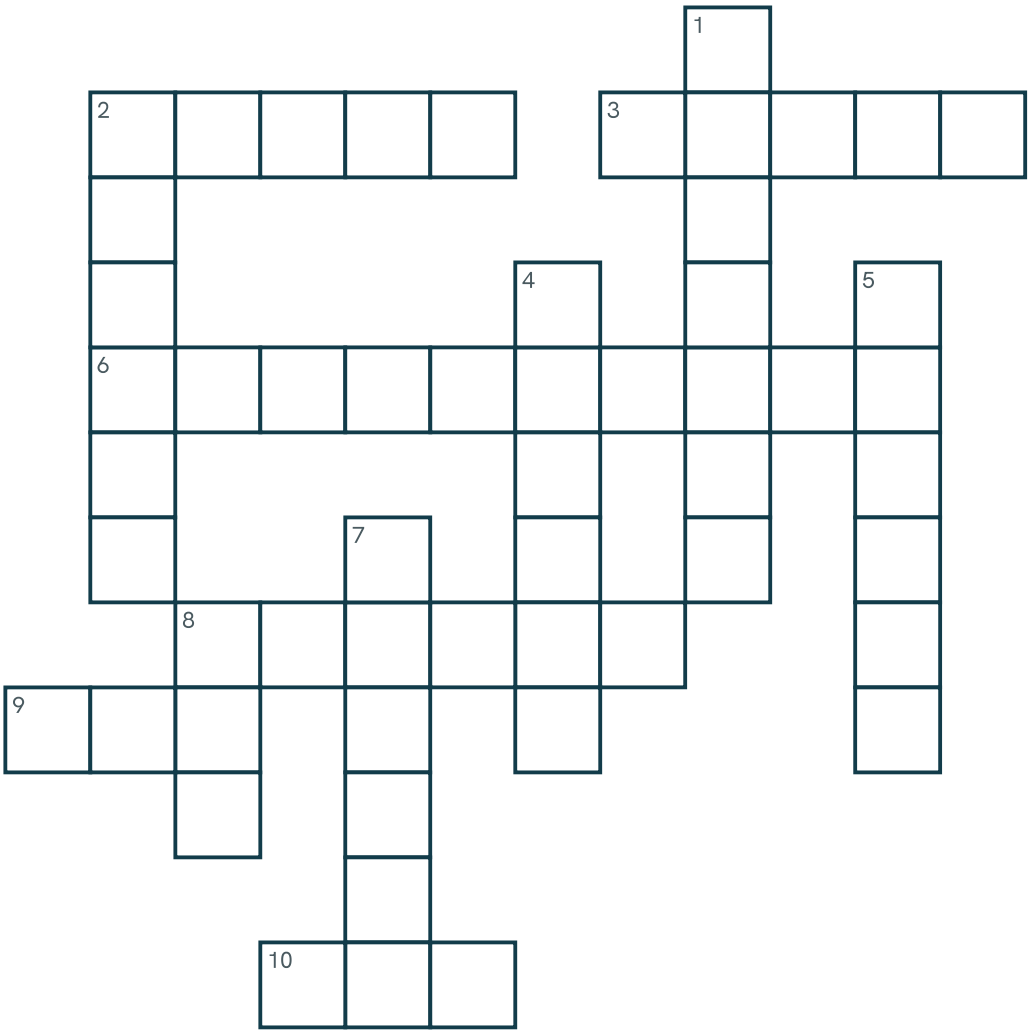
9 Short, squat bottle (6)

11 US sixth-barrel keg (6)

Crossword 38

Packaging

Getting beer into something you can sell.



Across

2 Bottle cap (5)

3 Paper or film on a bottle (5)

6 To heat-treat beer for stability (10)

8 Machine that fills bottles or cans (6)

9 Total package oxygen, by its letters (3)

10 Steel container for draught beer (3)

Down

1 Packing beer into aluminium (7)

2 Machine that crimps on the crowns (6)

4 ___ pasteuriser (6)

5 Machine that closes the lid on a can (6)

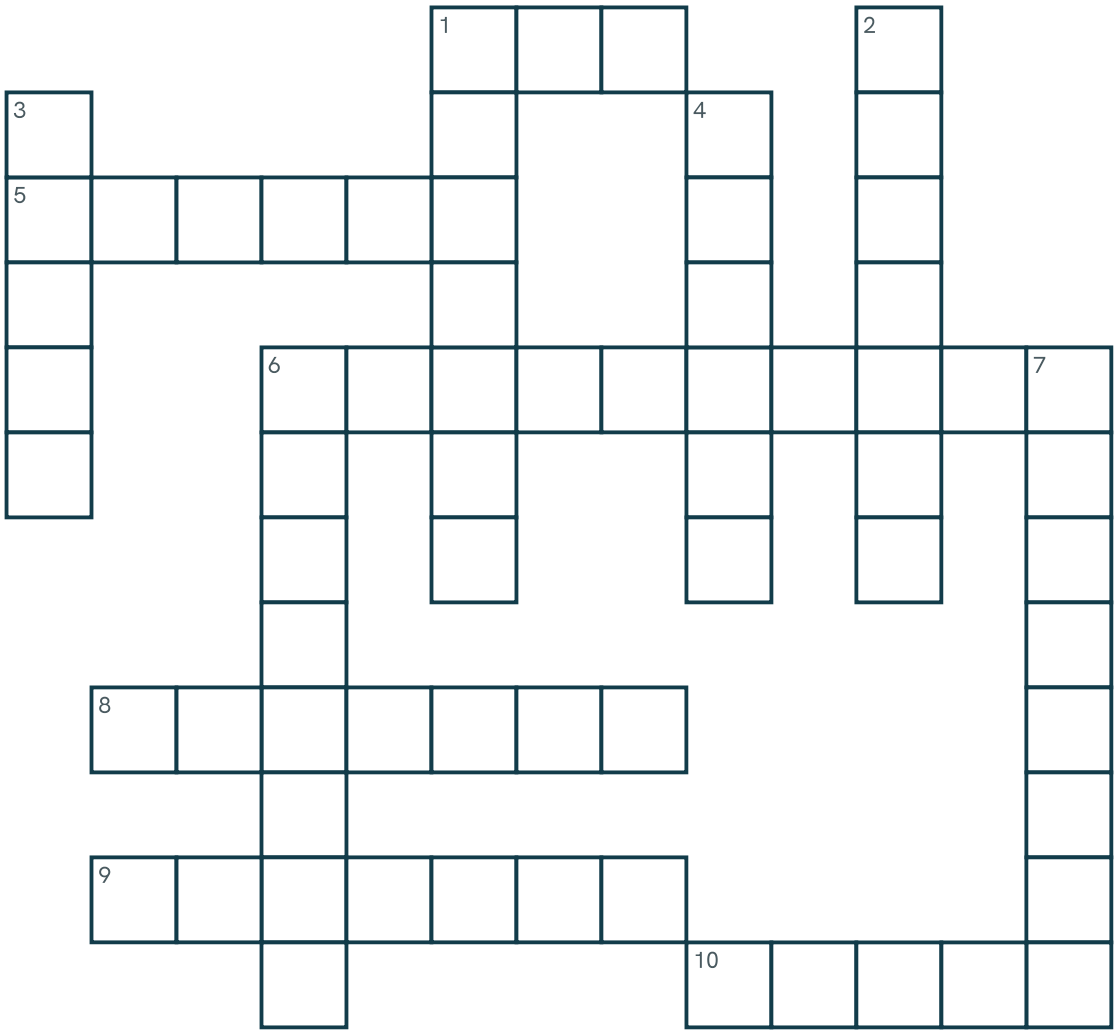
7 Shrink ___: printed wrap for a can (6)

8 Foam on beer detector, by its letters (3)

Crossword 39

Beer and food

Pairing rules, from the dinner table.



Across

1 Carbonation and bitterness ___ through fat (3)

5 Indian cheese to try with a witbier (6)

6 Pairing by shared flavours (10)

8 Spiced rice dish that suits a crisp lager (7)

9 Imperial stout can stand in for one (7)

10 Savoury fifth taste (5)

Down

1 Maillard notes in amber ale echo this (7)

2 Classic match for dry stout (7)

3 Hop bitterness can amplify chilli ___ (5)

4 Washed-rind ___ with a Belgian ale (6)

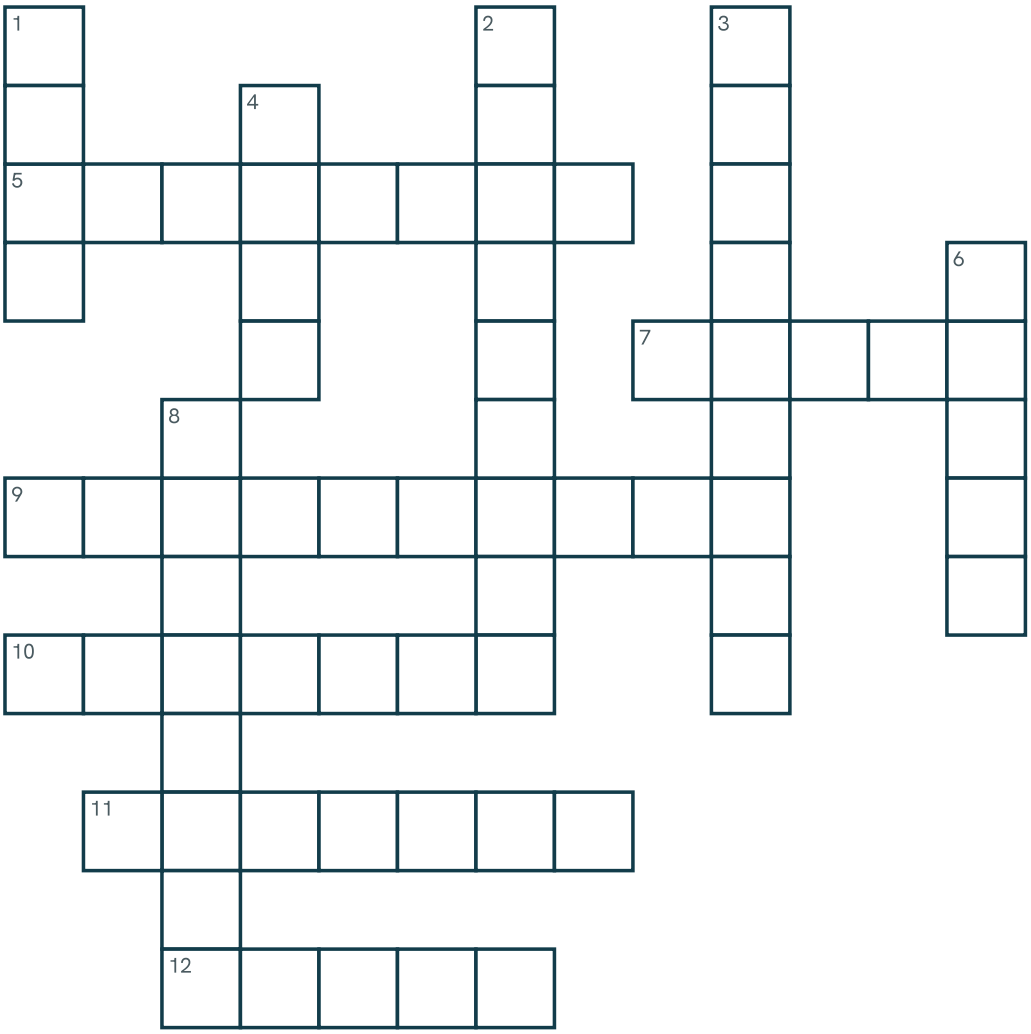
6 Pairing by opposites (8)

7 Clay-oven dishes that suit malty or smoky beers (8)

Crossword 40

Judging and competitions

How beers win medals.



Across

5 Style group in a competition (8)

7 Gold, silver or bronze (5)

9 Style document the judges use (10)

10 Volunteer who pours for the judges (7)

11 How well a beer matches its guidelines (2,5)

12 One beer sent in (5)

Down

1 US beer judge programme, by its letters (4)

2 ___ Beer Server, a Cicerone level (9)

3 Agreed score between judges (9)

4 ___ of show (4)

6 Tasting without labels (5)

8 Beer service certification (8)



PART FIVE

People, places and history

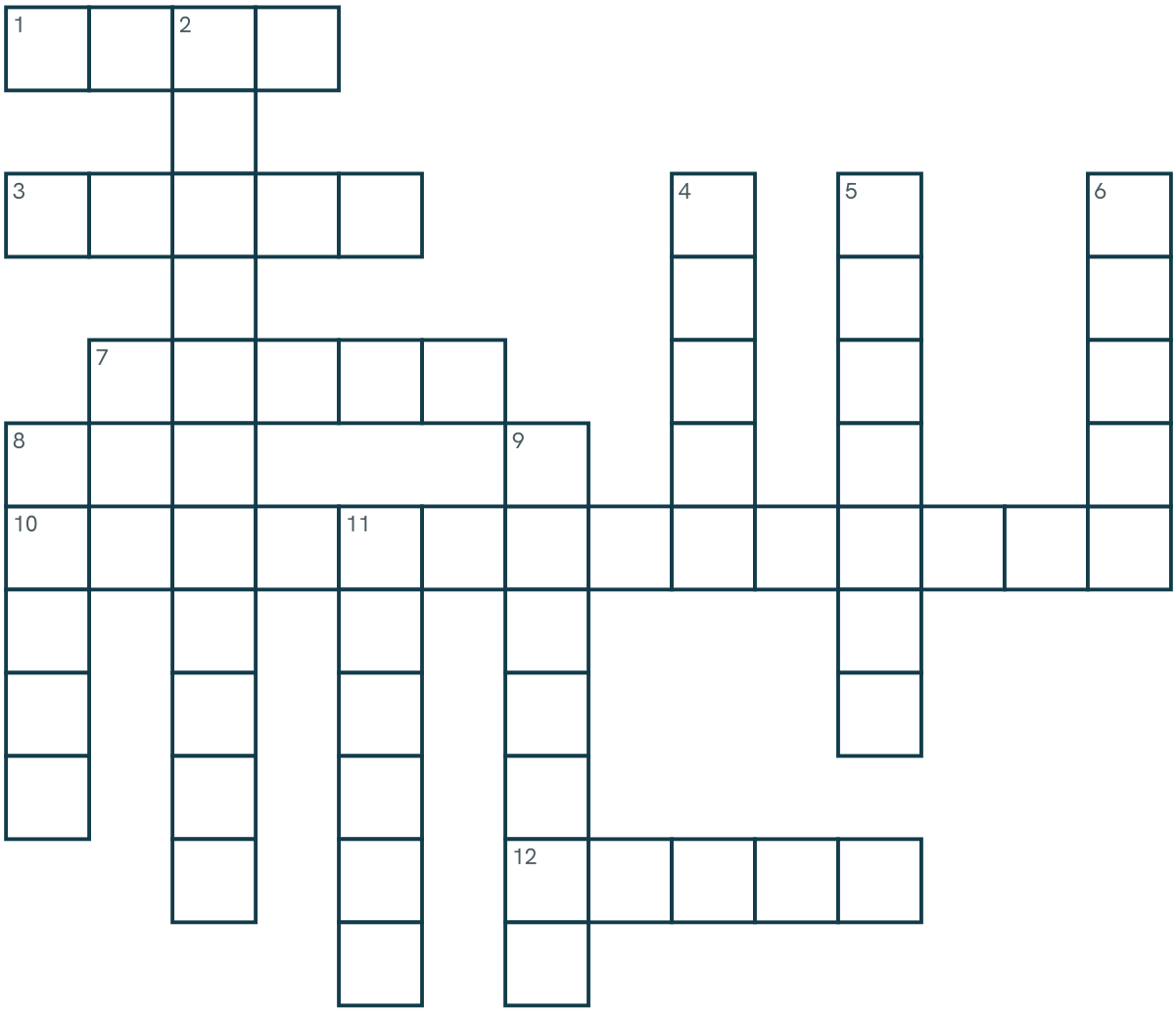
Seven thousand years of beer, from Sumer to Bengaluru.

Beer history, Beer in India, Famous breweries, American craft, German beer words, Festivals and culture, Other fermented drinks, The business of beer, Homebrewing, Schools and careers

Crossword 41

Beer history

Seven thousand years in twelve clues.



Across

1 Plant that replaced gruit (4)

3 Josef ___, brewer of the first Pilsner in 1842 (5)

7 Carl von ___, whose refrigeration changed lagering (5)

10 Bavarian purity law of 1516 (14)

12 Ancient land with a goddess of beer (5)

Down

2 US ban on alcohol from 1920 to 1933 (11)

4 Abbey brewers of medieval Europe (5)

5 Scientist who showed that yeast drives fermentation (7)

6 Land where pyramid builders were paid partly in beer (5)

8 Herb mix used before hops (5)

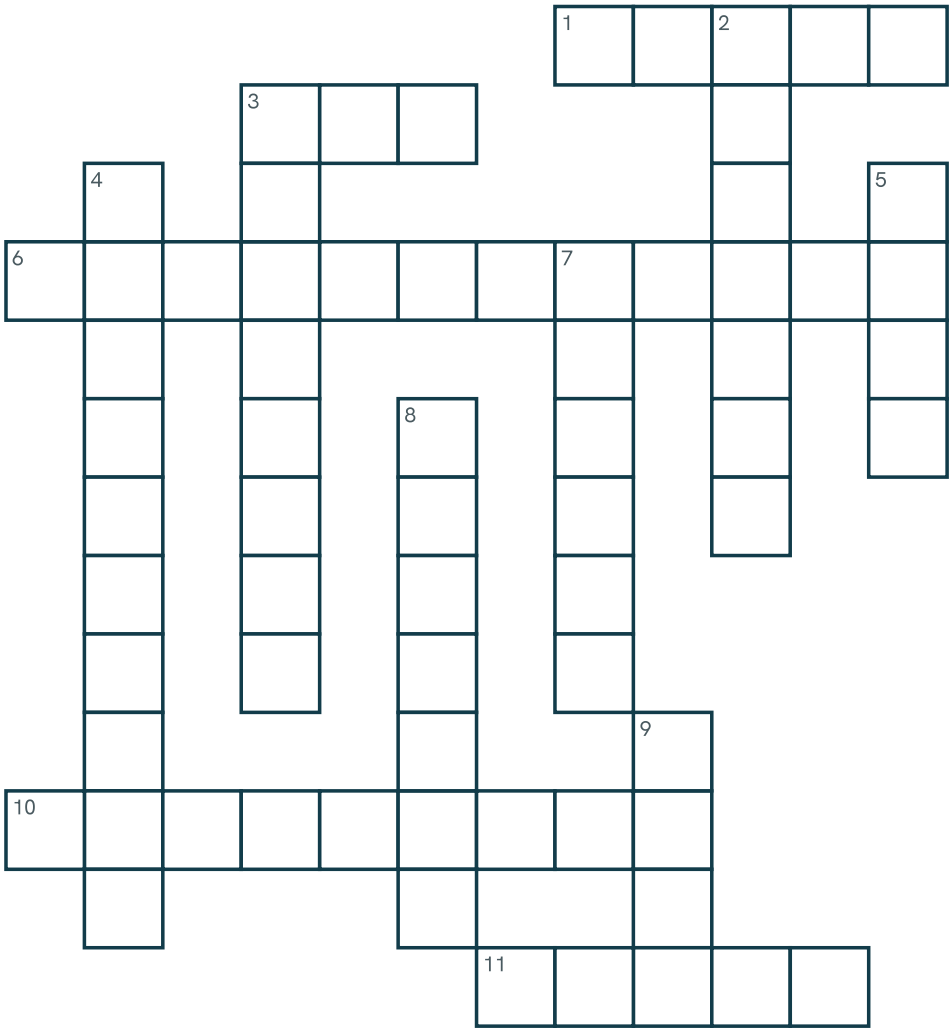
9 Sumerian goddess of beer (7)

11 Emil Christian ___, who isolated pure lager yeast in 1883 (6)

Crossword 42

Beer in India

From a hill-station brewery to the brewpub boom.



Across

1 Himachal town where Mohan Meakin brews today (5)

3 Coastal state famous for beach shacks and cold beer (3)

6 Small brewery, often attached to a pub (12)

10 City often called India's pub capital (9)

11 Brewpub menus in India often lead with a ___ beer (5)

Down

2 Permission every brewpub needs (7)

3 Haryana city where brewpubs boomed in the 2010s (8)

4 India's best-selling lager brand (10)

5 Edward ___, who founded that brewery (4)

7 State duty on alcohol (6)

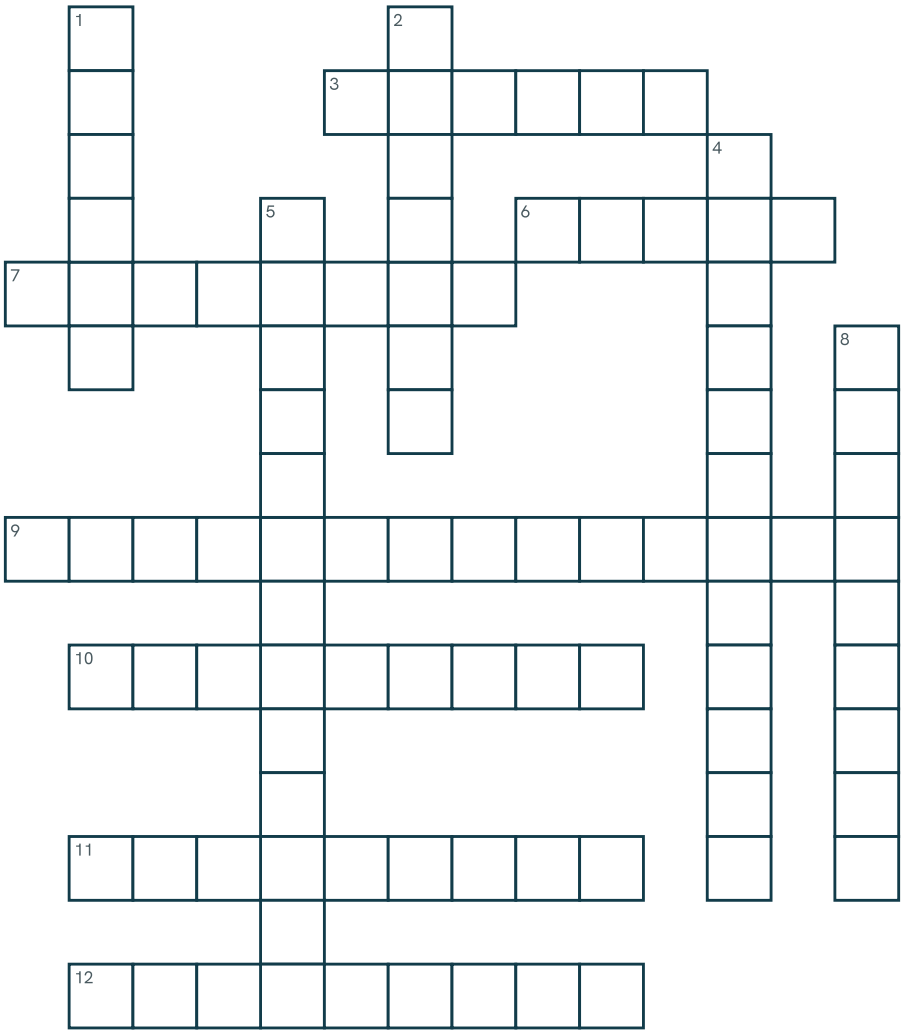
8 Himachal town with one of Asia's oldest breweries, from the 1820s (7)

9 Maharashtra city with a growing microbrewery scene (4)

Crossword 43

Famous breweries

Names every brewer should know.



Across

3 Brewery in České Budějovice (6)

6 Belgian strong golden ale named after the devil (5)

7 Dutch brewer with a red star (8)

9 Plzeň brewery of the original pale lager (7,7)

10 Brussels lambic brewery (9)

11 Trappist abbey in the Ardennes (9)

12 Weissbier brewery founded by Georg ___ in 1872 (9)

Down

1 Munich brewery of Gabriel Sedlmayr (6)

2 Chiswick brewer of the original ESB (7)

4 Trappist abbey whose beer is sold mainly at its gate (12)

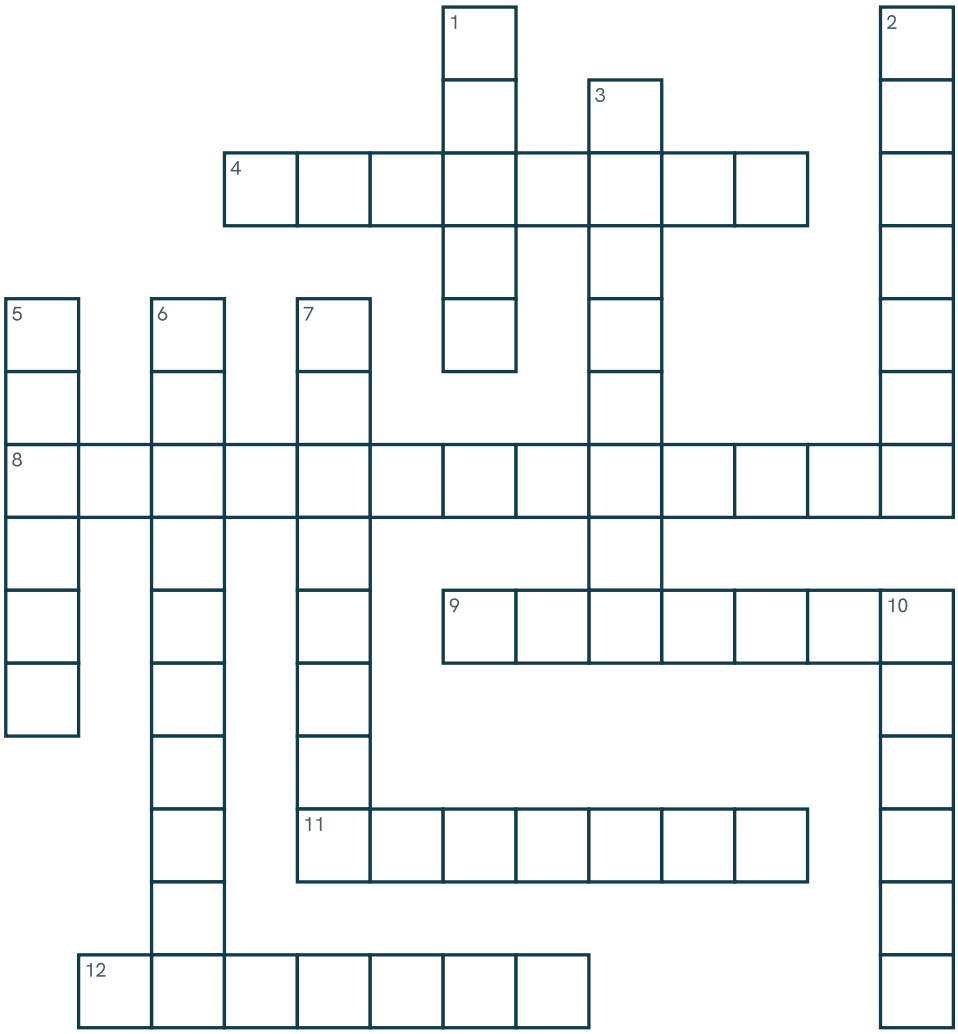
5 Bavarian brewery with roots in 1040, often called the oldest (13)

8 Copenhagen brewer whose lab isolated lager yeast (9)

Crossword 44

American craft

How a homebrew hobby became an industry.



Across

4 Hobby made legal at federal level in the US in 1978 (8)

8 Beer brewed by two breweries together (13)

9 Pub that brews its own beer (7)

11 Cascade-forward style that launched US craft (4,3)

12 Bar at the brewery (7)

Down

1 Anchor's trademark name for California common (5)

2 Autumn seasonal with spice (7)

3 Light US hybrid, fermented warm then cold (5,3)

5 San Francisco brewery revived by Fritz Maytag (6)

6 ___ common: lager yeast fermented warm (10)

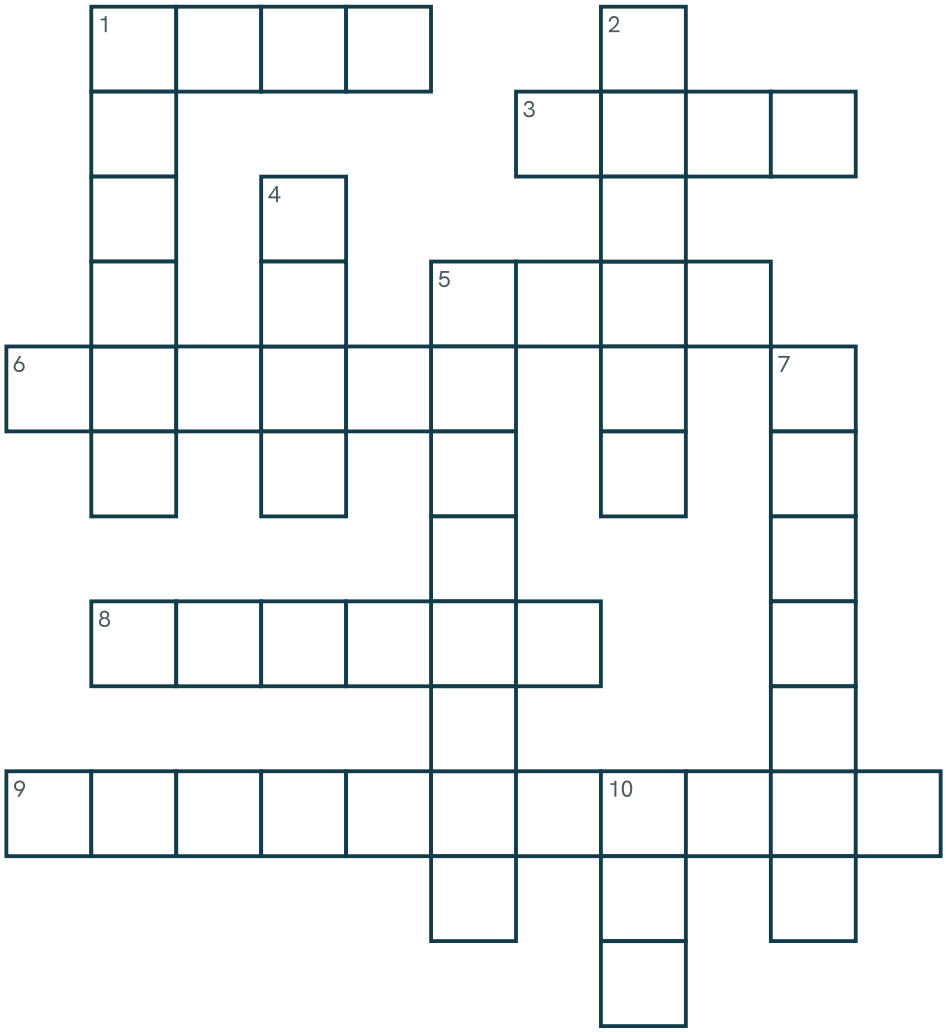
7 A brewery's core beer (8)

10 Easy-drinking golden ale (6)

Crossword 45

German beer words

A brewer's German for the brewhouse.



Across

1 Yeast (4)

3 Cask or keg (4)

5 Beer (4)

6 Cellar (6)

8 Water (6)

9 Head brewer (11)

Down

1 Hops (6)

2 To store (6)

4 Malt (4)

5 Brewery (8)

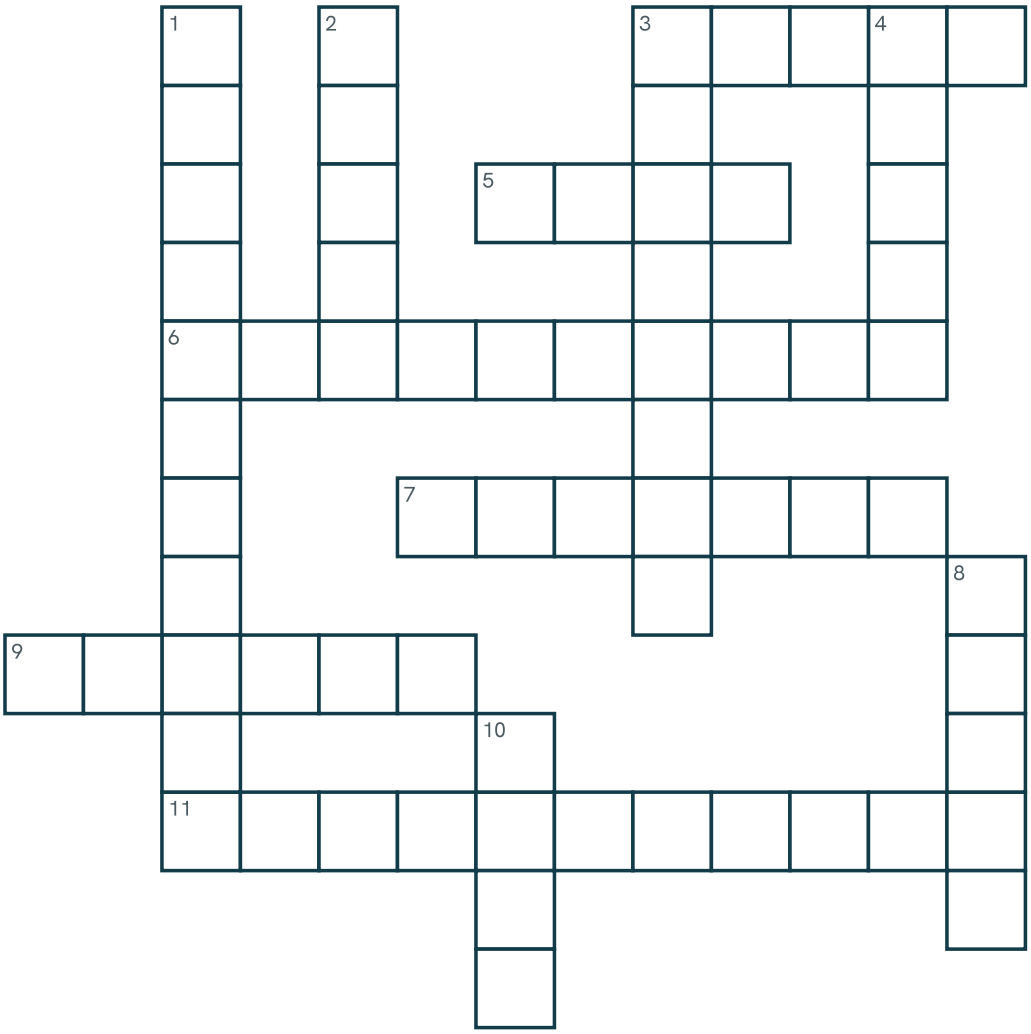
7 Sampling tap on a tank (7)

10 One brew, a single batch (3)

Crossword 46

Festivals and culture

Raise a glass.



Across

3 German for cheers (5)

5 Great American Beer Festival in Denver, by its letters (4)

6 Outdoor drinking space under chestnut trees (4,6)

7 German toast to your health (3,4)

9 The English toast (6)

11 Night a guest brewery holds a bar's lines (3,8)

Down

1 Munich festival that starts in September (11)

2 French for cheers (no accent needed) (5)

3 Tour of several bars in one night (3,5)

4 Stoneware mug (5)

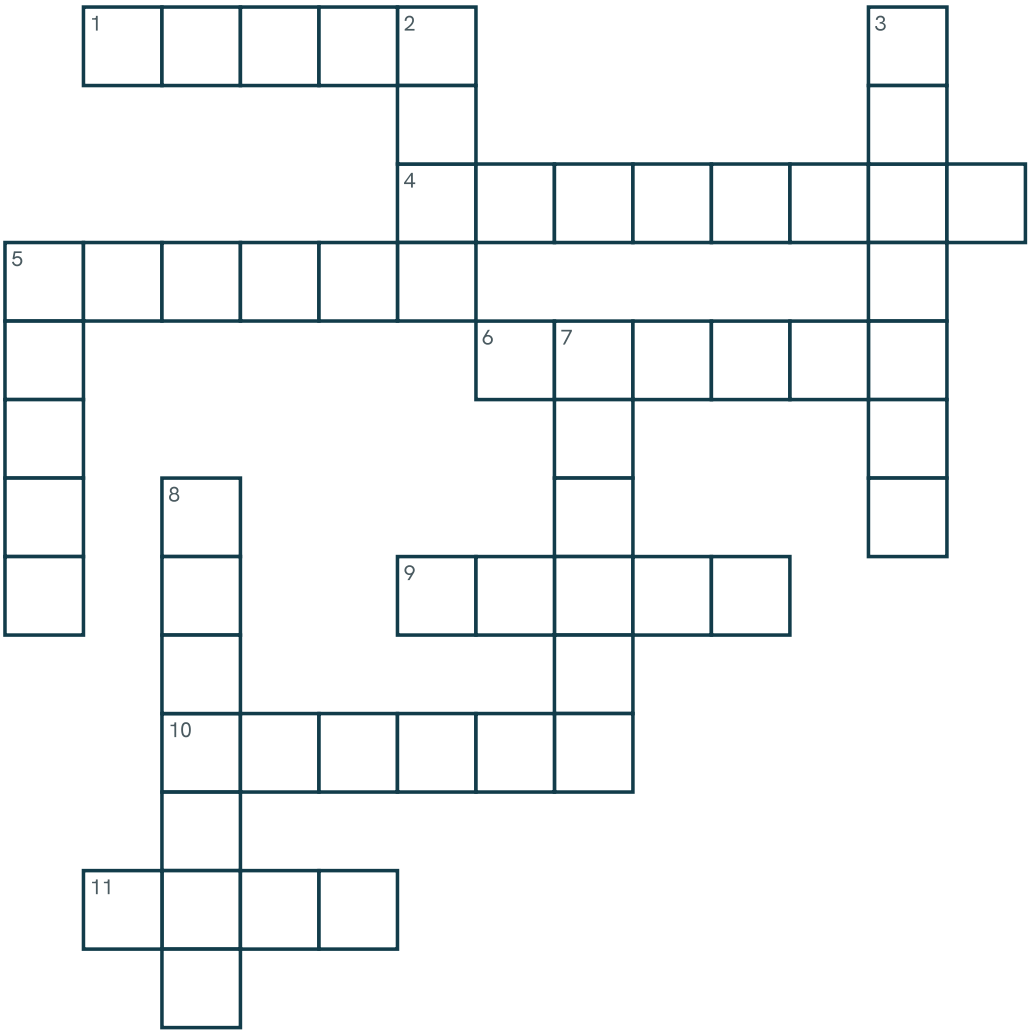
8 UK real ale campaign founded in 1971 (5)

10 One-litre mug at Oktoberfest (4)

Crossword 47

Other fermented drinks

Beer's cousins, from India and around the world.



Across

1 Slavic drink brewed from bread (5)

4 Fermented sweet tea (8)

5 Fermented agave sap from Mexico (6)

6 Andean corn beer (6)

9 Fermented apple juice (5)

10 Millet drink of Sikkim and Nepal, sipped through hot water (6)

11 Honey wine (4)

Down

2 Japanese rice brew (4)

3 Himalayan brew of millet or barley (7)

5 Fermented pear juice (5)

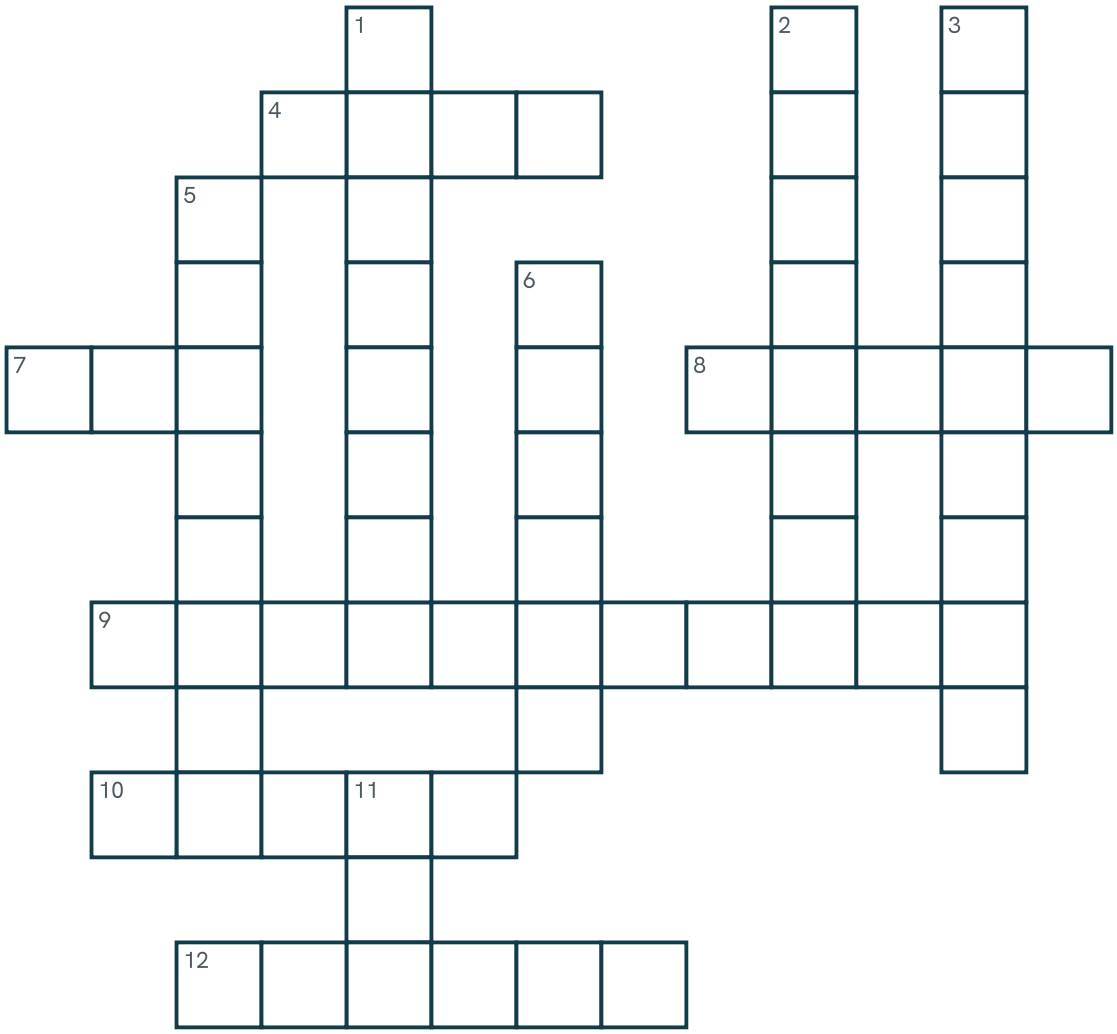
7 Rice beer of Jharkhand and Odisha (6)

8 Hard ___: alcoholic fizzy water (7)

Crossword 48

The business of beer

Turning good beer into a going concern.



Across

4 Cost of goods sold, by its letters (4)

7 Maximum retail price printed on Indian packs (3)

8 What drinkers remember (5)

9 Middle tier between brewer and bar (11)

10 ___ brewer: one with no brewery of their own (5)

12 First release of a beer (6)

Down

1 Sales prediction (8)

2 ___ brewing: renting another brewery's kit (8)

3 Stock on hand (9)

5 The most beer a plant can make (8)

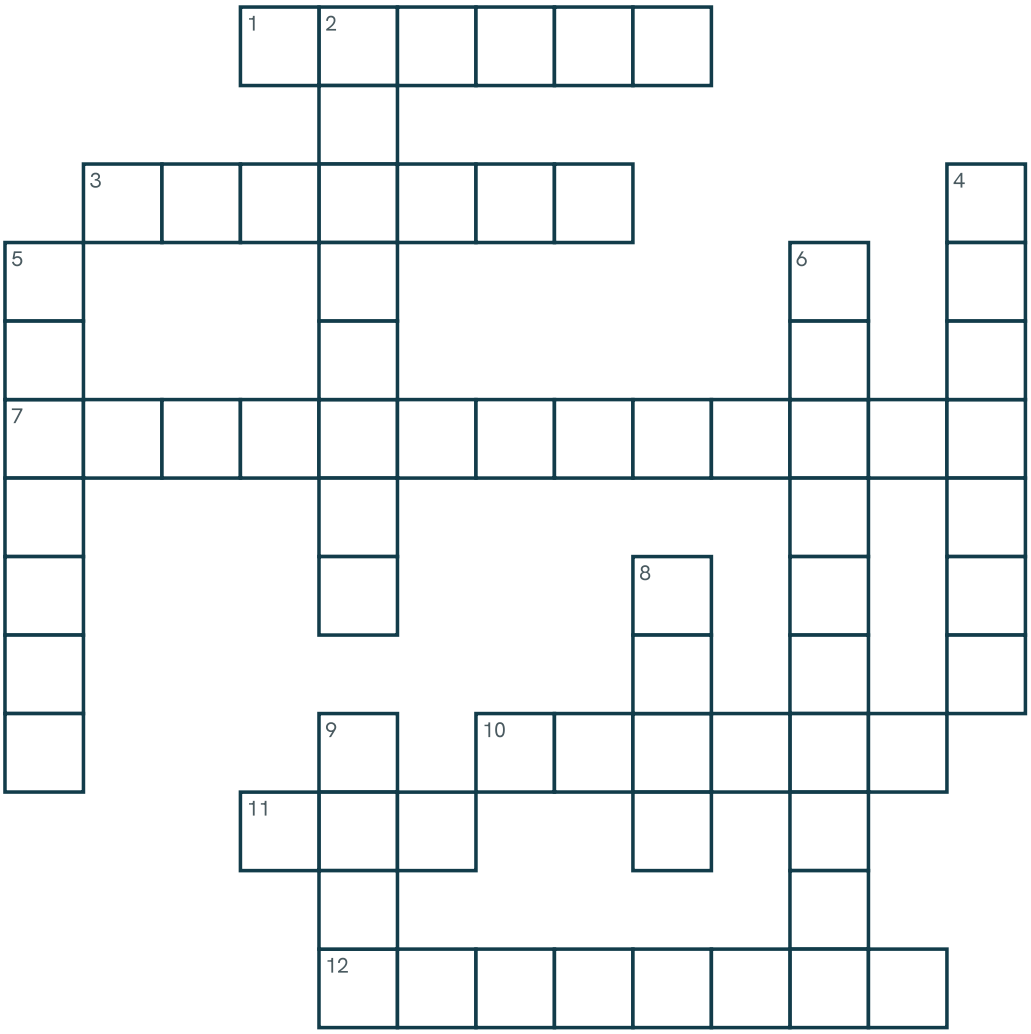
6 Profit left on each pint (6)

11 Stock keeping unit, by its letters (3)

Crossword 49

Homebrewing

The words you meet on your first kit.



Across

1 Glass fermenter (6)

3 Immersion ___ cools the wort (7)

7 Needs only a drop of wort (13)

10 List of grain, hops and yeast (6)

11 Beginner's starter pack (3)

12 Packaging at home (8)

Down

2 Brewing from malted grain only (3,5)

4 Malt in syrup or powder (7)

5 Valve that lets gas out of a fermenter (7)

6 Tool for racking beer (10)

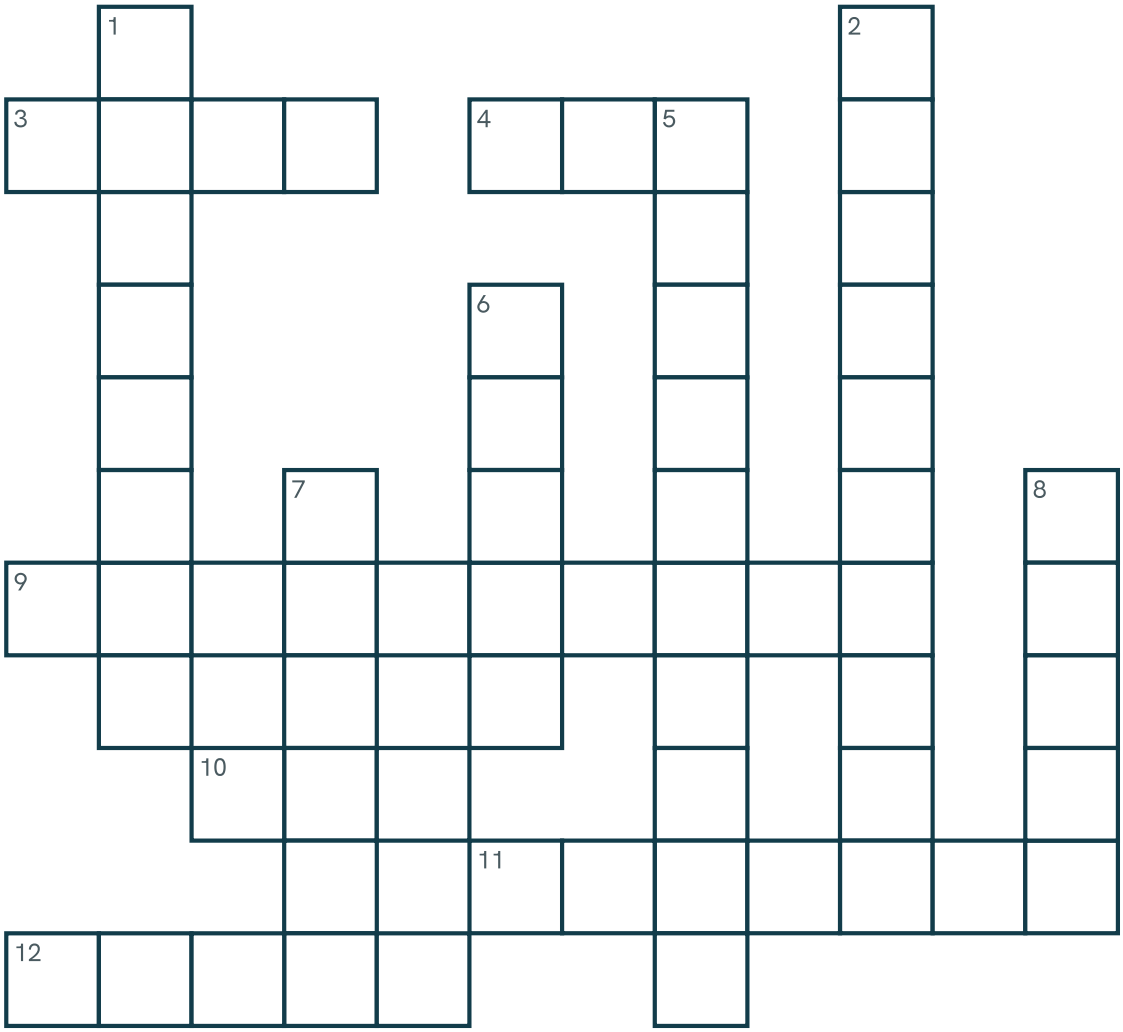
8 To move beer off the yeast (4)

9 Brew in a bag, by its letters (4)

Crossword 50

Schools and careers

Where brewers learn and where they go next.



Across

3 The point of it all (4)

4 Berlin brewing research and teaching institute, by its letters (3)

9 Edinburgh university with a brewing degree (6,4)

10 Institute of Brewing and Distilling, by its letters (3)

11 Brewing academy near Munich (7)

12 Litres out for every kilo in (5)

Down

1 Person who keeps the cellar (8)

2 Learner in the brewhouse (10)

5 Head of brewing (10)

6 What lean brewing cuts (5)

7 Chicago brewing school founded in 1872 (6)

8 University of California campus with a brewing programme (5)



PART SIX

Course companion

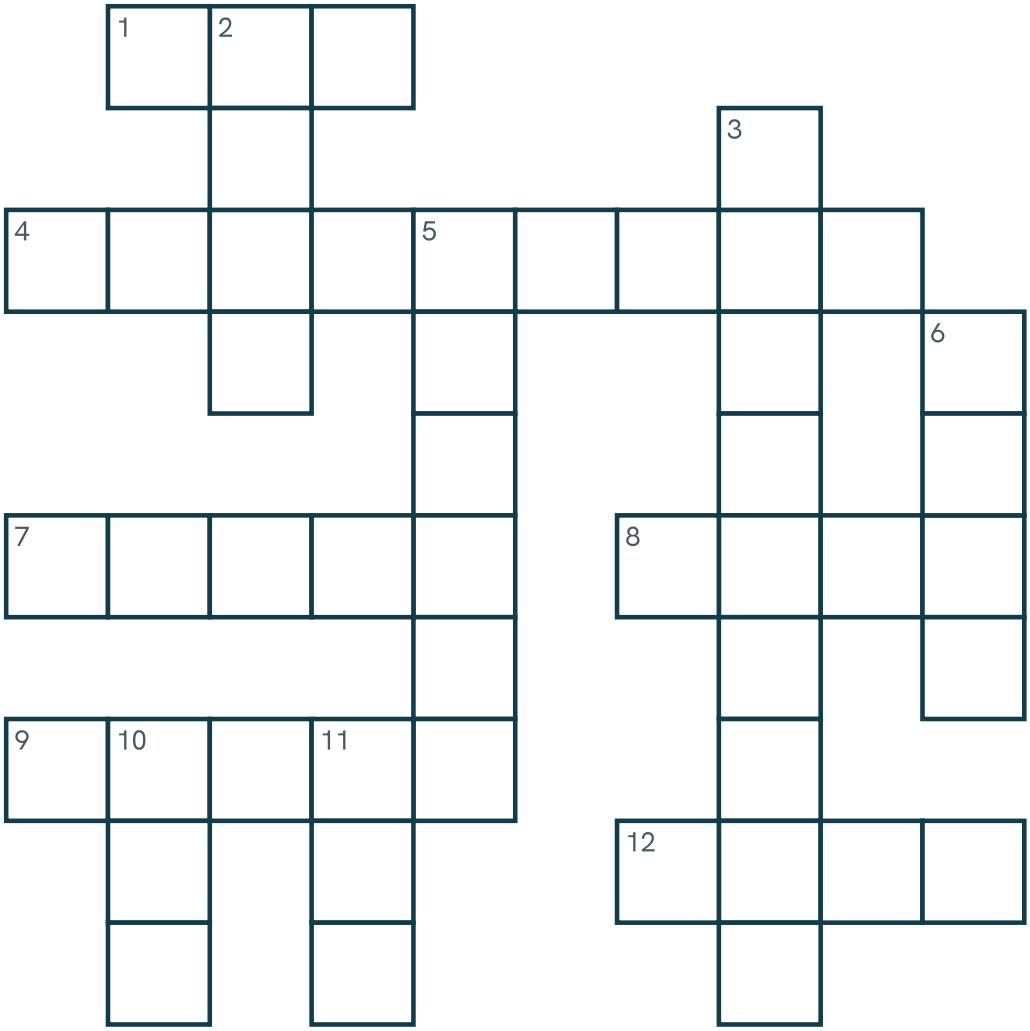
One crossword for every lesson of every Craft Beer School course, to use as a recap.

Brewing Fundamentals, Advanced Brewing Science, Brewery Business Management, Style Specialisation, Beer Branding & Packaging, Sensory Evaluation, AI for Craft Breweries, Digital Transformation Basics for Brewing, Power BI and Tableau for Breweries, ESG in Craft Brewing, Safety in Brewing, Craft Distilling, Craft Gin Making, RTD Drinks: Alcoholic and Non-Alcoholic, Hop Water, Hard Seltzer, Kombucha, Non-Alcoholic Beer

Crossword 51. Brewing Fundamentals, lesson 1

Grain to glass

The four ingredients and the journey from field to pint.



Across

1 Bitterness on the label, by its letters (3)

4 Spin that leaves the trub in a cone (9)

7 Turns sugar into alcohol and CO2 (5)

8 First brewhouse stop: crack the grain (4)

9 The biggest ingredient by volume (5)

12 Ingredient for bitterness and aroma (4)

Down

2 Kettle stage where the hops go in (4)

3 To mature beer after fermentation (9)

5 To separate wort from the spent grain (6)

6 Grain that brings sugar, colour and body (4)

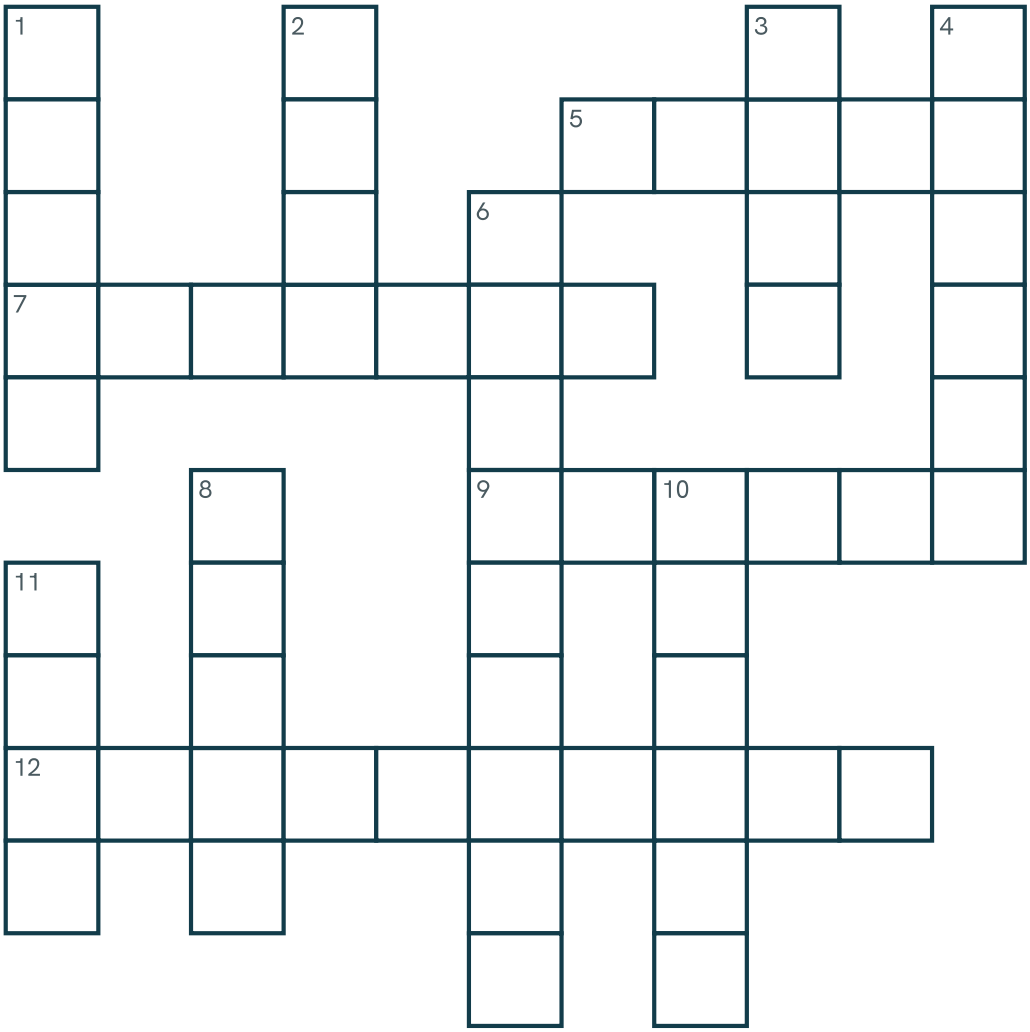
10 Strength on the label, by its letters (3)

11 European colour scale, by its letters (3)

Crossword 52. Brewing Fundamentals, lesson 2

Malt, water and the mash

Malting, enzymes and the mash temperatures that decide body.



Across

5 Milled malt ready for the mash (5)

7 Amylases are examples of these (7)

9 It should sit at 5.2 to 5.6 (4,2)

12 Malt used in small amounts for colour and flavour (10)

Down

1 Mashing at 62 to 65°C (144 to 149°F) makes the beer this (5)

2 Mashing at 68 to 70°C (154 to 158°F) gives more of this (4)

3 Dryer that stops growth and sets colour (4)

4 What the enzymes turn into sugar (6)

6 To sprout, building enzymes (9)

8 First malting step: soak the grain (5)

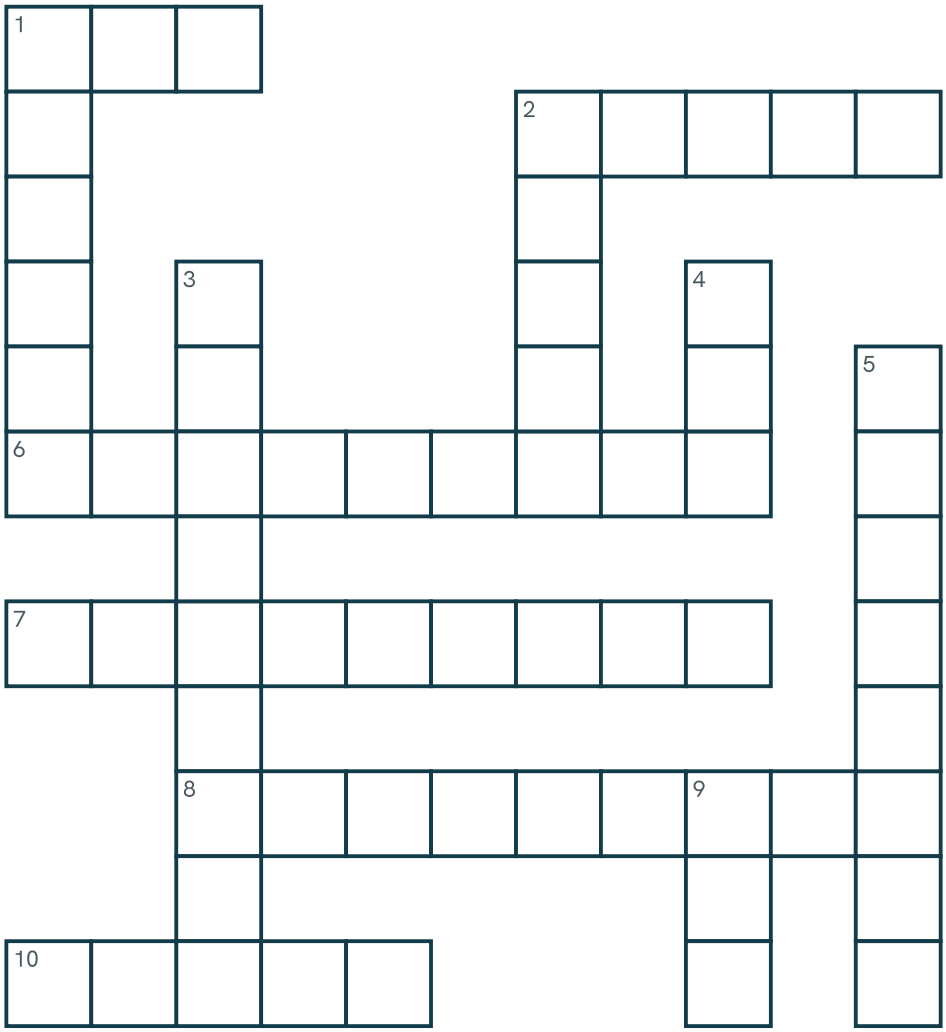
10 ___ water: the hot water that meets the grist (6)

11 ___ malt: most of the grist (4)

Crossword 53. Brewing Fundamentals, lesson 3

Hops, boil and yeast

Bitterness, the boil and the yeast that finishes the job.



Across

1 Cooked-corn compound the boil drives off (3)

2 Late hop addition (5)

6 Yeast cells added per millilitre of wort (5,4)

7 Boil job: kill the microbes (9)

8 Boil job: make hop acids bitter and soluble (9)

10 Yeast that works cold, around 8 to 12°C (46 to 54°F) (5)

Down

1 Hop addition made cold, after the boil (3,3)

2 ___ acids: they need the boil to turn bitter (5)

3 Early hop addition (9)

4 Yeast that works warm, often 18 to 22°C (64 to 72°F) (3)

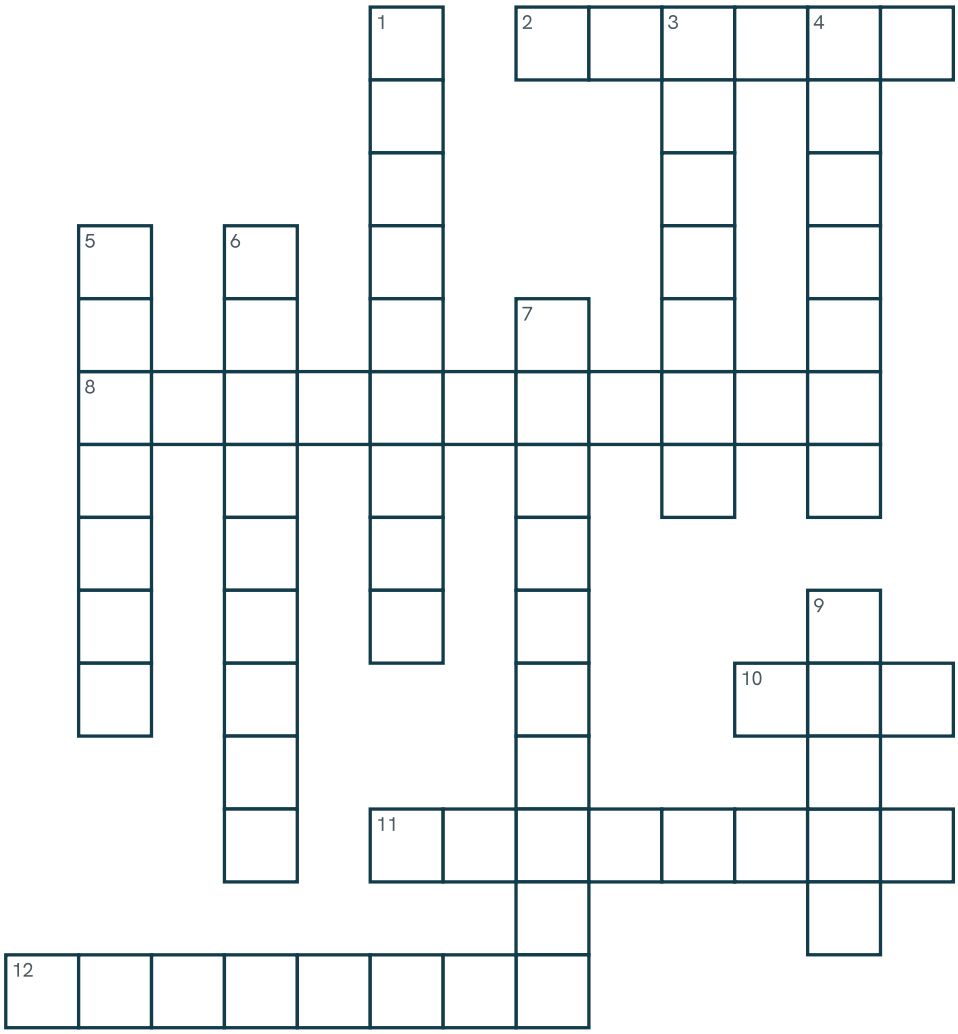
5 Proteins that clump early in the boil (3,5)

9 Bitterness figure you calculate for your recipe (3)

Crossword 54. Brewing Fundamentals, lesson 4

Sanitation, kit and your first recipe

Clean, sanitise, measure and package your first beer.



Across

2 What you present to the cohort at the end (6)

8 Fizz from dissolved CO2 (11)

10 Homebrew ___: a small all-in-one setup (3)

11 ___ gravity, read before fermentation (8)

12 Removing the soil you can see (8)

Down

1 A microbrewery's set of hot-side vessels (9)

3 Alkaline brewery cleaner (7)

4 Adding sugar to carbonate in the bottle (7)

5 OG minus FG, times 131.25, estimates this by volume (7)

6 ___ acid, a no-rinse sanitiser (9)

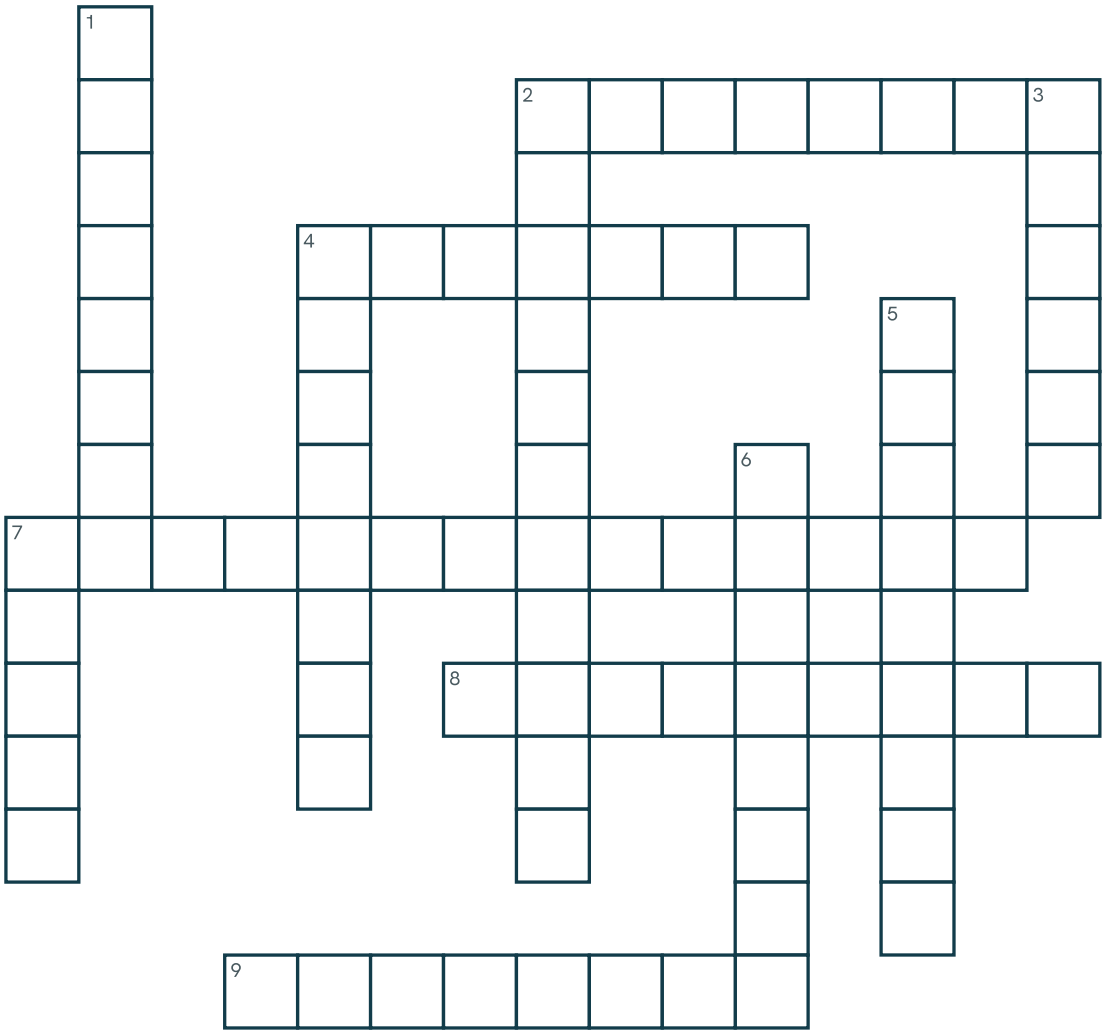
7 Cutting microbes down to a safe level (10)

9 ___ gravity, read after fermentation (5)

Crossword 55. Advanced Brewing Science, lesson 1

Brewing water chemistry

Ions, alkalinity and treating Indian water for two styles.



Across

2 Mixing RO and raw water to hit a target (8)

4 Ion that helps enzymes, yeast and clarity (7)

7 Membrane treatment that strips minerals (7,7)

8 Yeast nutrient ion, bitter in excess (9)

9 ___ alkalinity: what is left after calcium and magnesium act (8)

Down

1 Ion that makes bitterness drier (8)

2 Ion that pushes mash pH up (11)

3 Salt added to raise sulphate (6)

4 Ion that rounds out malt (8)

5 City supply, often chlorinated (9)

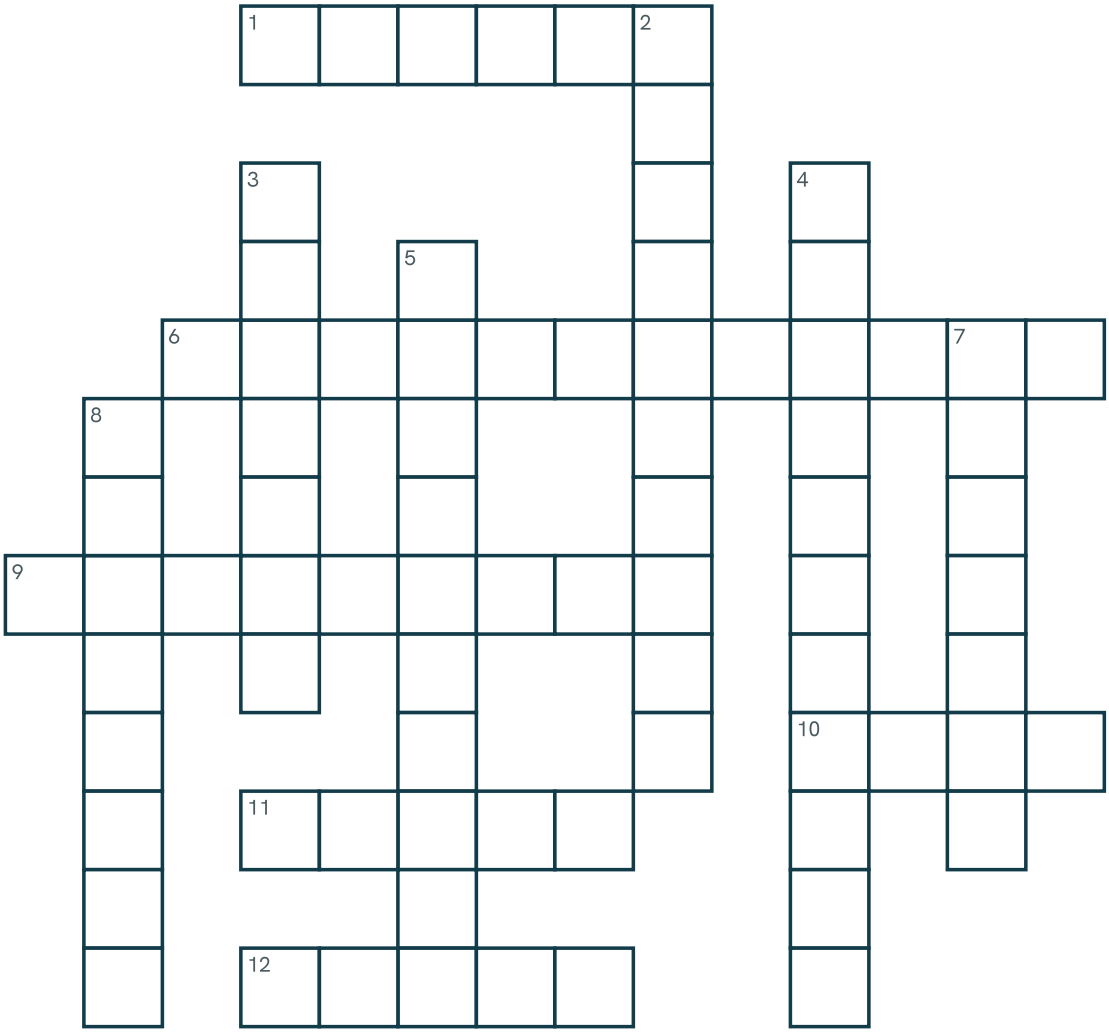
6 Groundwater source in India, often hard (8)

7 Sulphate to chloride ___ (5)

Crossword 56. Advanced Brewing Science, lesson 2

Mash chemistry and advanced mashing

Step mashes, adjuncts and the sums behind efficiency.



Across

1 To rinse the grain bed (6)

6 Vessel that cooks adjuncts before the mash (6,6)

9 Boil part of the mash and return it (9)

10 Adjunct that must be cooked first (4)

11 Corn adjunct (5)

12 Lauter ___: the extract collected in the kettle (5)

Down

2 Brewhouse ___: actual extract over potential (10)

3 Record you calculate efficiency from (4,3)

4 Low-temperature rest around 50°C (122°F) (7,4)

5 What cooking does to adjunct starch (10)

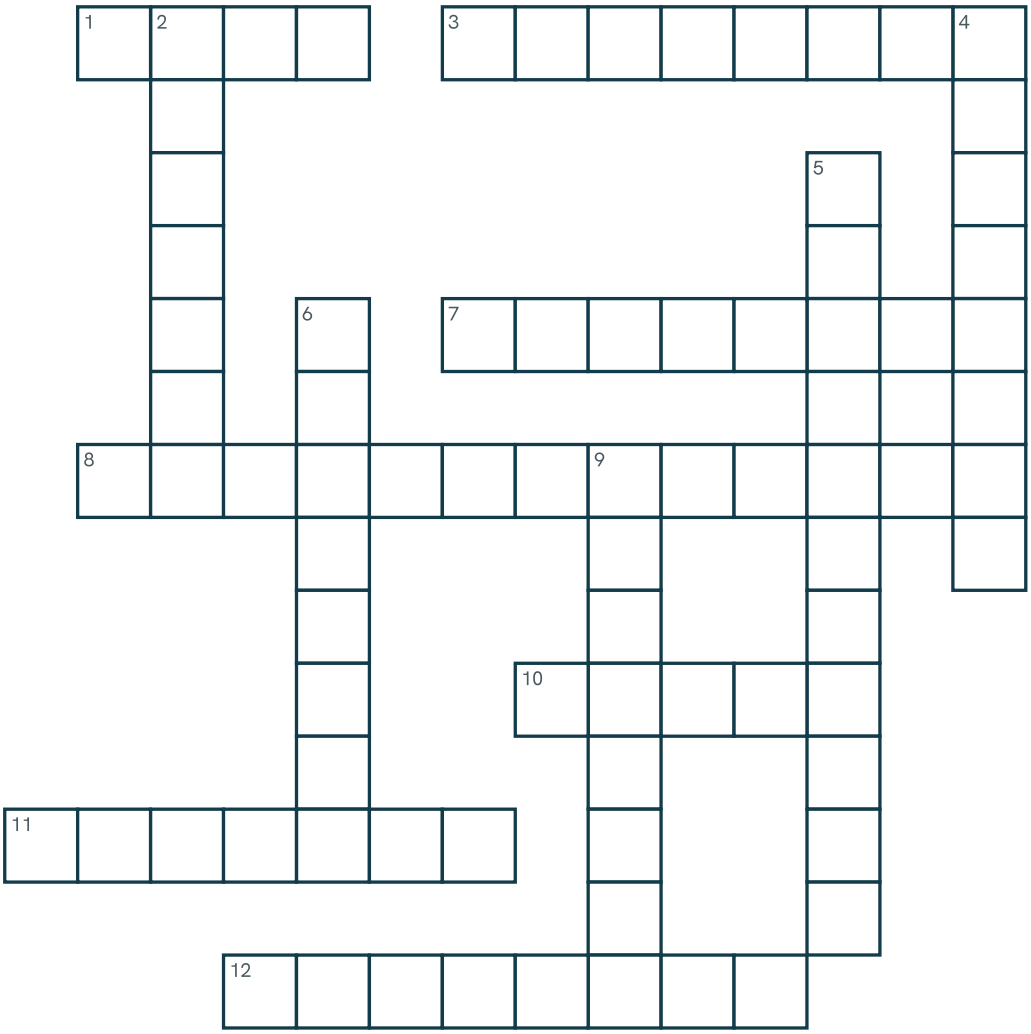
7 Sugar won from the grist (7)

8 Mash that climbs through several rests (4,4)

Crossword 57. Advanced Brewing Science, lesson 3

Hop chemistry

Acids, oils and what yeast does to them.



Across

1 ___ acids: barely bitter in the kettle (4)

3 Floral, citrus hop oil (8)

7 Rose-like hop oil (8)

8 Heat change that makes hop acids bitter (13)

10 Changes some hop compounds during fermentation (5)

11 Green, resinous hop oil (7)

12 Refermentation set off by dry hopping (3,5)

Down

2 Hop ___ break dextrins and cause hop creep (7)

4 Main beta acid (8)

5 Share of alpha acid that ends up in the beer (11)

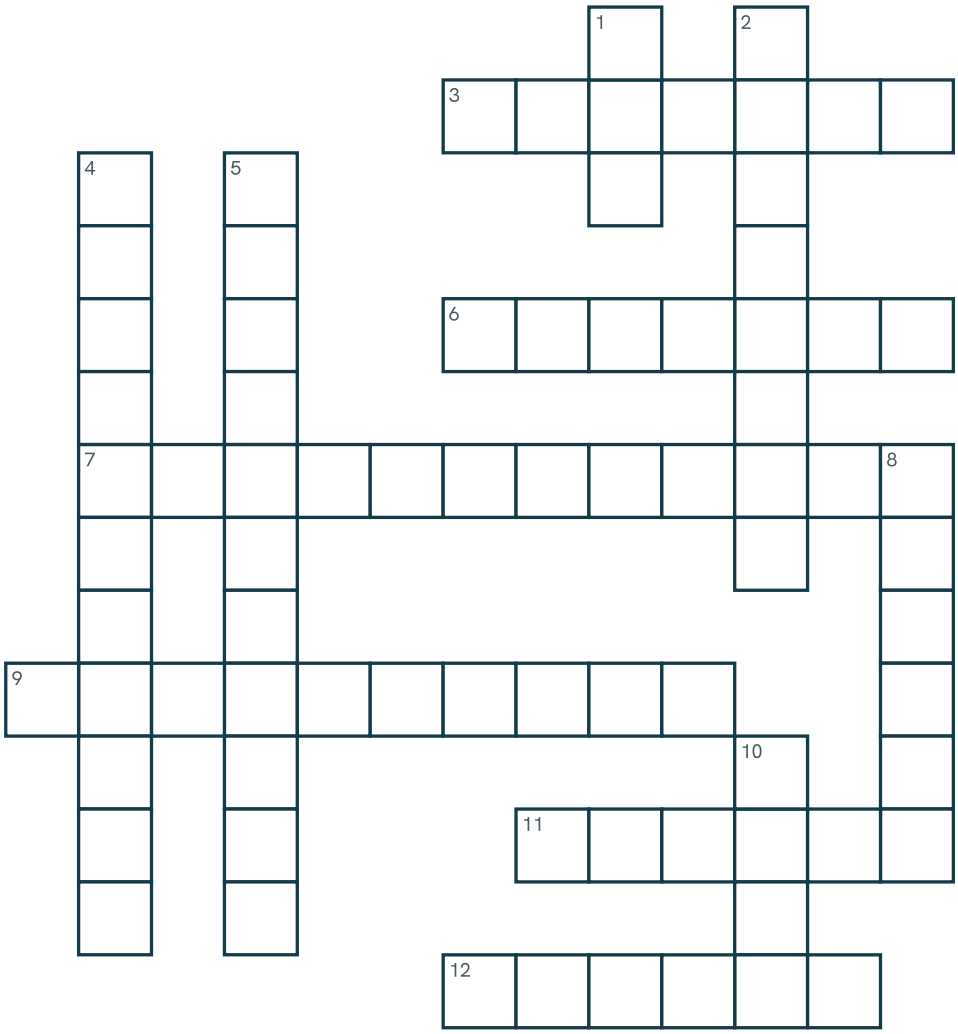
6 Main alpha acid (8)

9 Hop ___: when each addition goes in (8)

Crossword 58. Advanced Brewing Science, lesson 4

Yeast and fermentation microbiology

Phases, flavours and faults you can read from a log.



Across

3 Reading you plot against time (7)

6 To collect yeast from the cone (7)

7 Green apple in young beer (12)

9 Phase when growth stops (10)

11 Hot alcohols from warm, fast fermentation (6)

12 Phase of rapid multiplication (6)

Down

1 First phase, when yeast adapts (3)

2 Butter or butterscotch (8)

4 Growing up a fresh yeast crop (11)

5 Dissolving oxygen at pitching (11)

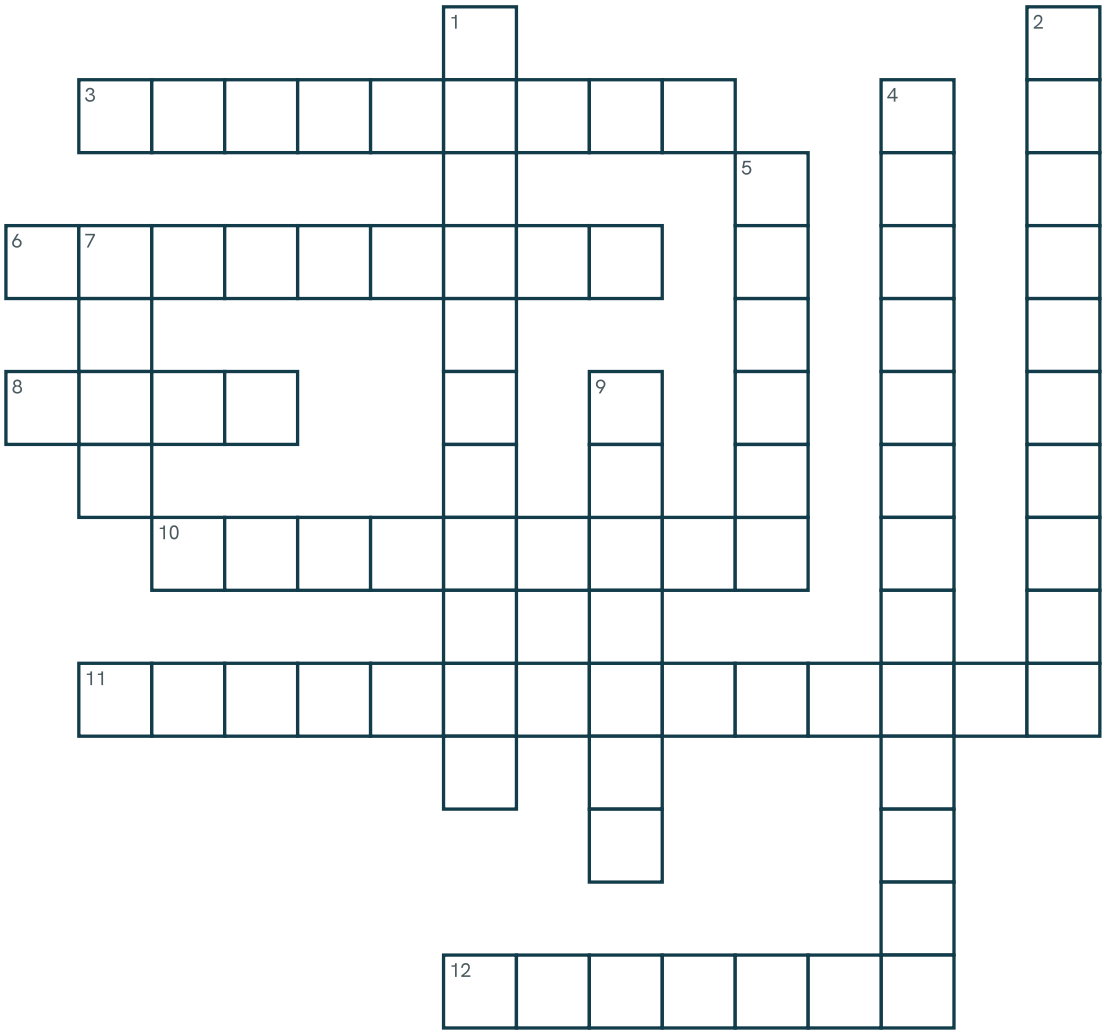
8 Fruity compounds that rise with temperature (6)

10 Diacetyl ___: a warm hold near the end (4)

Crossword 59. Advanced Brewing Science, lesson 5

Contamination, stability and shelf life

Spoilers, staling and keeping beer fresh in the heat.



Across

3 Non-brewing yeast that over-attenuates (4,5)

6 How long a beer stays fresh (5,4)

8 Cloudiness in finished beer (4)

10 Keeping beer chilled all the way to the bar (4,5)

11 Heat treatment dosed in units (14)

12 Clarifying agents (7)

Down

1 Bacteria that can make diacetyl and rope (11)

2 Removes yeast and haze (10)

4 Bacteria that make lactic acid (13)

5 Pickup of this stales beer (6)

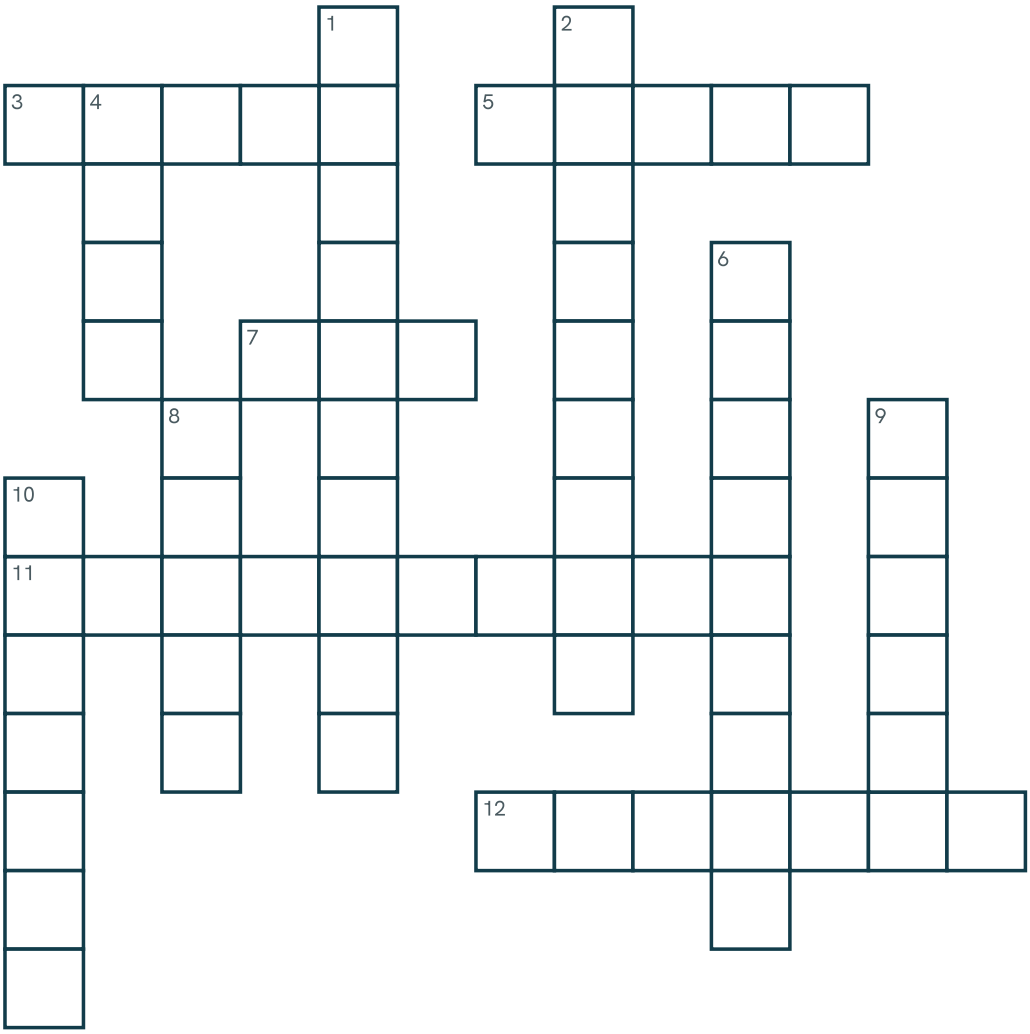
7 Indian summer enemy on the supply chain (4)

9 Cardboard ageing (7)

Crossword 60. Advanced Brewing Science, lesson 6

Quality assurance and control

A small QC lab, specs and food safety.



Across

3 Indian food safety regulator, by its letters (5)

5 Hazard analysis food safety system, by its letters (5)

7 Standard operating procedure, by its letters (3)

11 Floating gauge for gravity (10)

12 ___ test: warm storage to predict stability (7)

Down

1 Tool for checking yeast (10)

2 To check a meter against a standard (9)

4 Target range, for short (4)

6 Allowed deviation from spec (9)

8 Systematic check of a process (5)

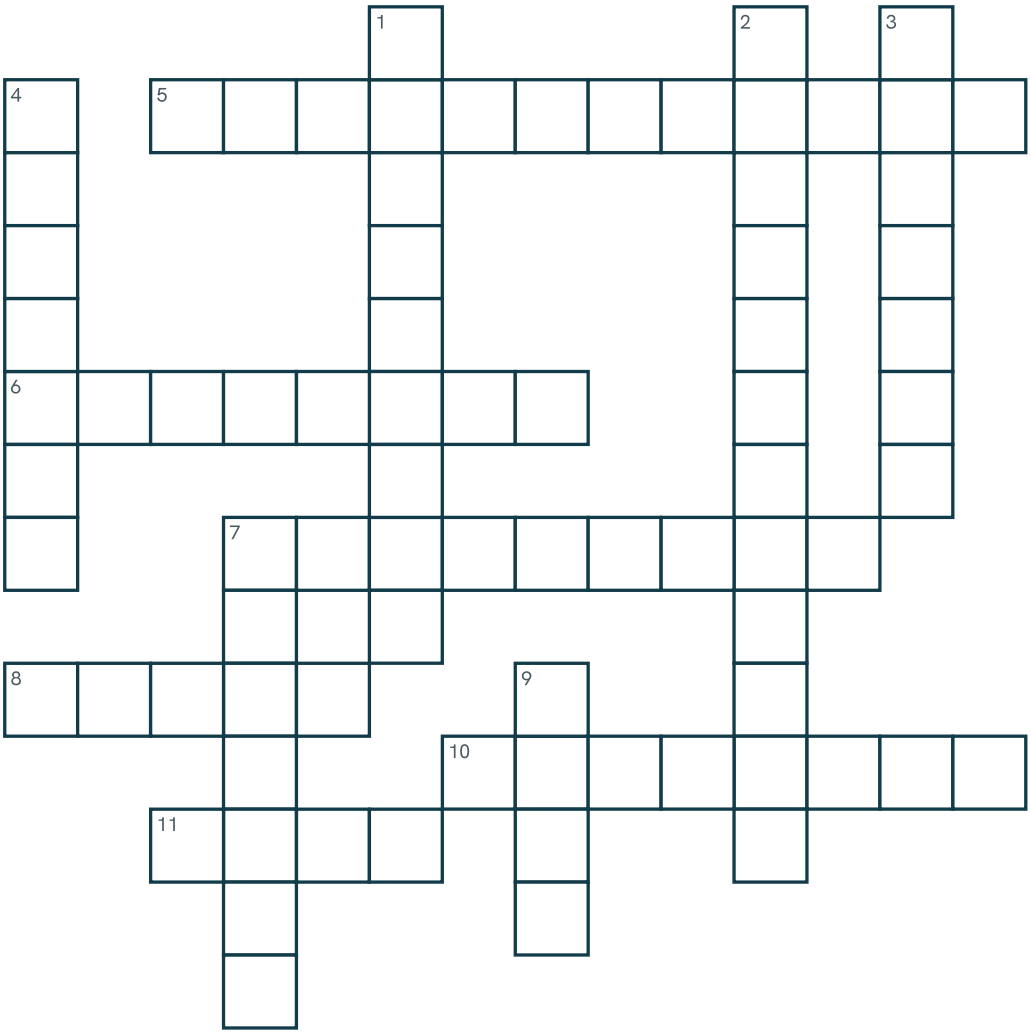
9 ___ sample: packaged beer kept for later checks (6)

10 Instrument for acidity (2,5)

Crossword 61. Brewery Business Management, lesson 1

Plan and finance

Choosing a model, costing it and finding the money.



Across

5 The number that sets your price (4,3,5)

6 Backer who takes equity (8)

7 Point where sales cover costs (5,4)

8 Money spent on kit up front (5)

10 ___ brewing: making beer in someone else's plant (8)

11 Running costs (4)

Down

1 Capex item: power, water, steam and cooling (9)

2 Small-scale brewery (12)

3 Setting what a pint sells for (7)

4 Money to get started (7)

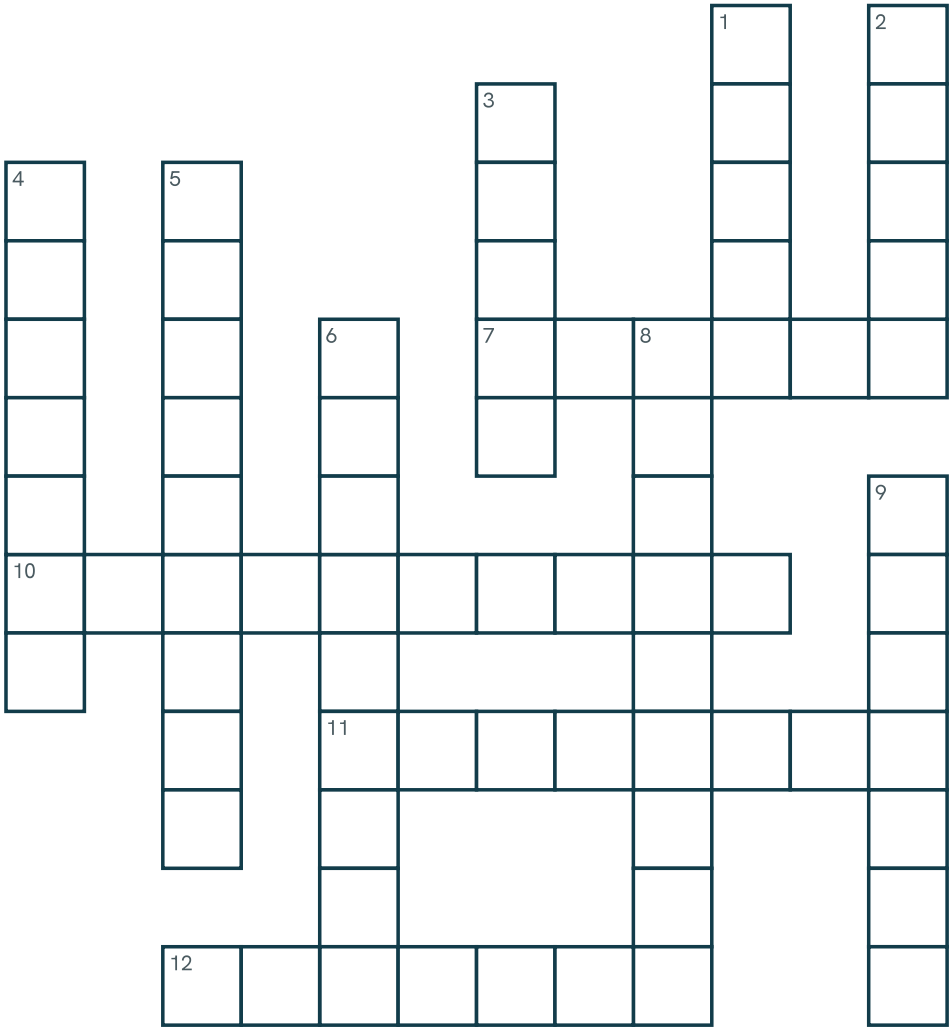
7 Pub that brews for its own bar (7)

9 Debt route to funding (4)

Crossword 62. Brewery Business Management, lesson 2

Licensing and regulation in India

The permissions every Indian brewery needs.



Across

7 Duty set state by state (6)

10 Staying within the rules (10)

11 Realistic plan for approvals (8)

12 Permission to operate from that board (7)

Down

1 Food safety regulator, by its letters (5)

2 Level of government that sets most alcohol rules in India (5)

3 ___ registration before sale (5)

4 Permission to brew and sell (7)

5 Protects your brand name (9)

6 ___ control board (9)

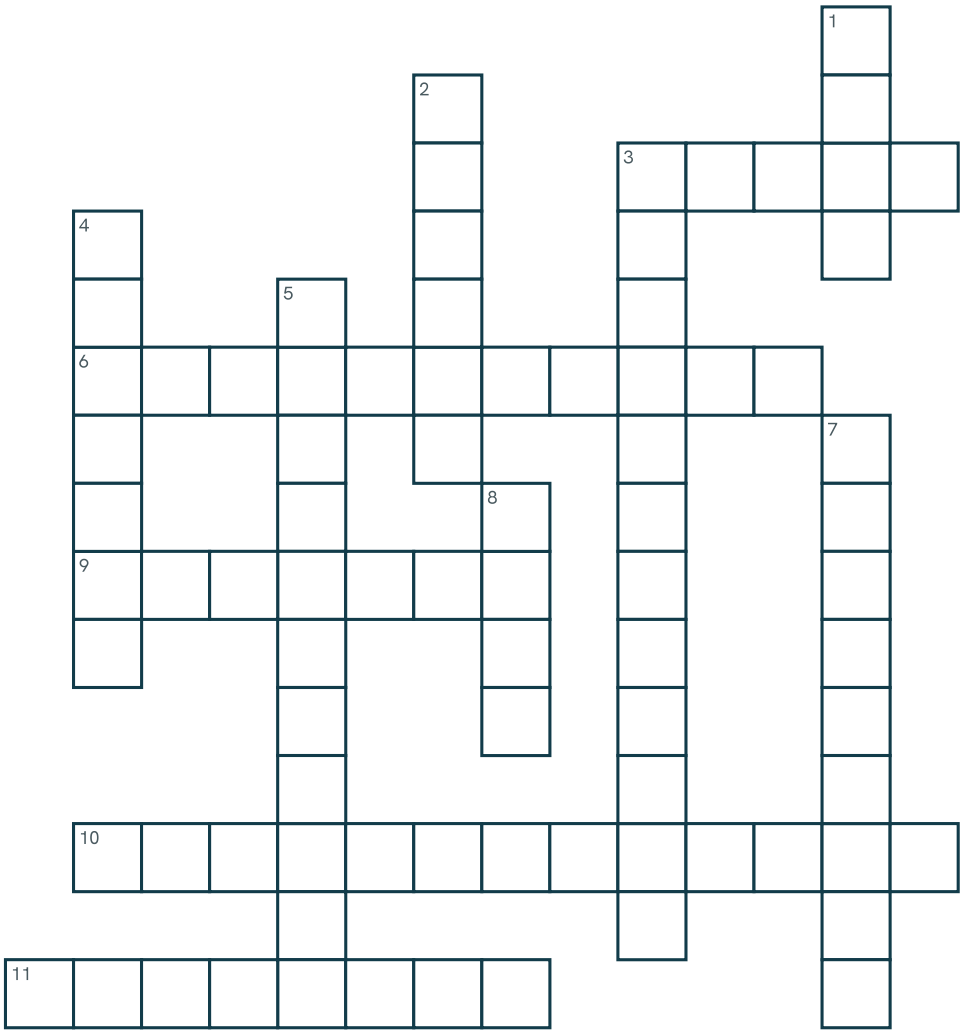
8 Homework: a list of every licence you need (9)

9 Licences need periodic ___ (7)

Crossword 63. Brewery Business Management, lesson 3

Market, brand and distribution

Positioning, channels and the keg or can question.



Across

3 What drinkers remember (5)

6 Where your beer sits in the drinker's mind (11)

9 Bars and restaurants (2,5)

10 How beer reaches the drinker (5,2,6)

11 Shops selling packaged beer (3,5)

Down

1 Light, stackable packs that block light (4)

2 Profit on each pint (6)

3 One-page homework for the cohort (8,4)

4 Bar at the brewery, often the best margin (7)

5 Middle link to bars and shops (11)

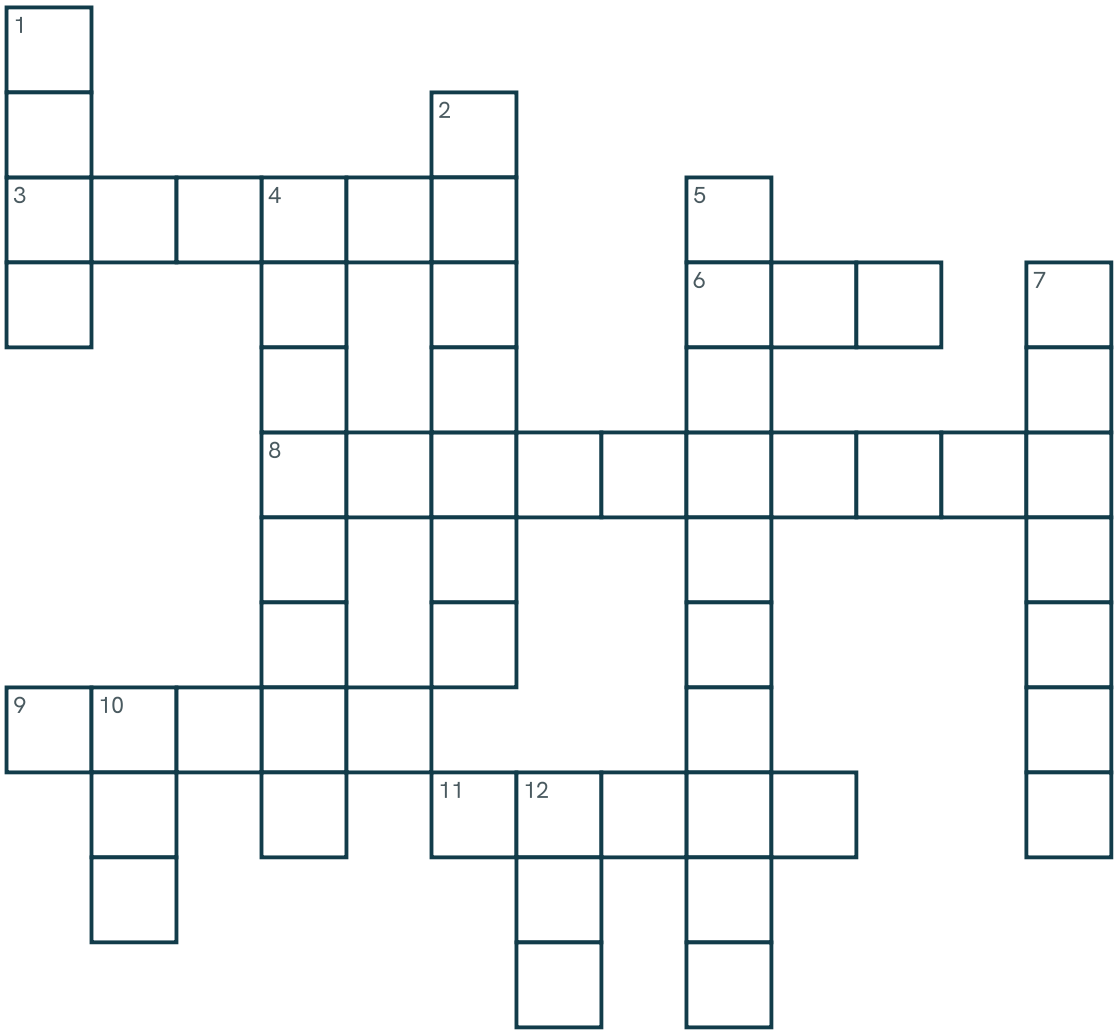
7 Why local kegs beat long trips (9)

8 Draught containers, cheaper per litre (4)

Crossword 64. Style Specialisation, lesson 1

How styles work

Guidelines, ranges and the history behind them.



Across

3 Range given in SRM or EBC (6)

6 Strength-based ___ shaped British ales (3)

8 Style documents judges use (10)

9 ___ gravity range (5)

11 Burton's hard ___ shaped pale ale (5)

Down

1 Beer judge programme, by its letters (4)

2 Homework: ___ one style (7)

4 ___ gravity range (8)

5 Page that lists the ranges (5,5)

7 It shaped many styles (7)

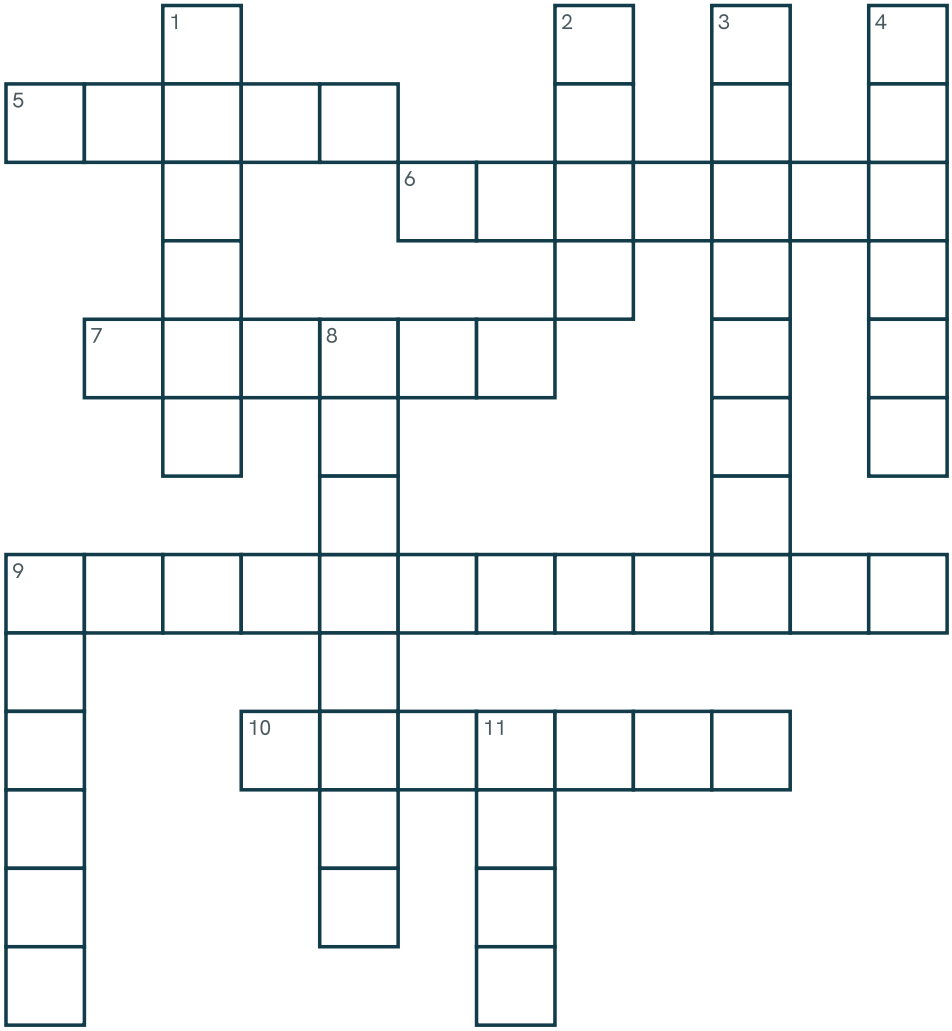
10 Bitterness range, by its letters (3)

12 Strength range, by its letters (3)

Crossword 65. Style Specialisation, lesson 2

Lagers

From Helles to Dunkel, with the cold patience they need.



Across

5 Word for a clean lager finish (5)

6 Aroma lager yeast can give off early (7)

7 Pale Munich lager (6)

9 Warm-up near the end of fermentation (8,4)

10 Hoppy pale lager (7)

Down

1 City where Pilsner began (6)

2 Lager ferments like this, around 8 to 12°C (46 to 54°F) (4)

3 Fermentation ___ for the homework (8)

4 Amber festival lager (no umlaut) (6)

8 Weeks of cold storage (8)

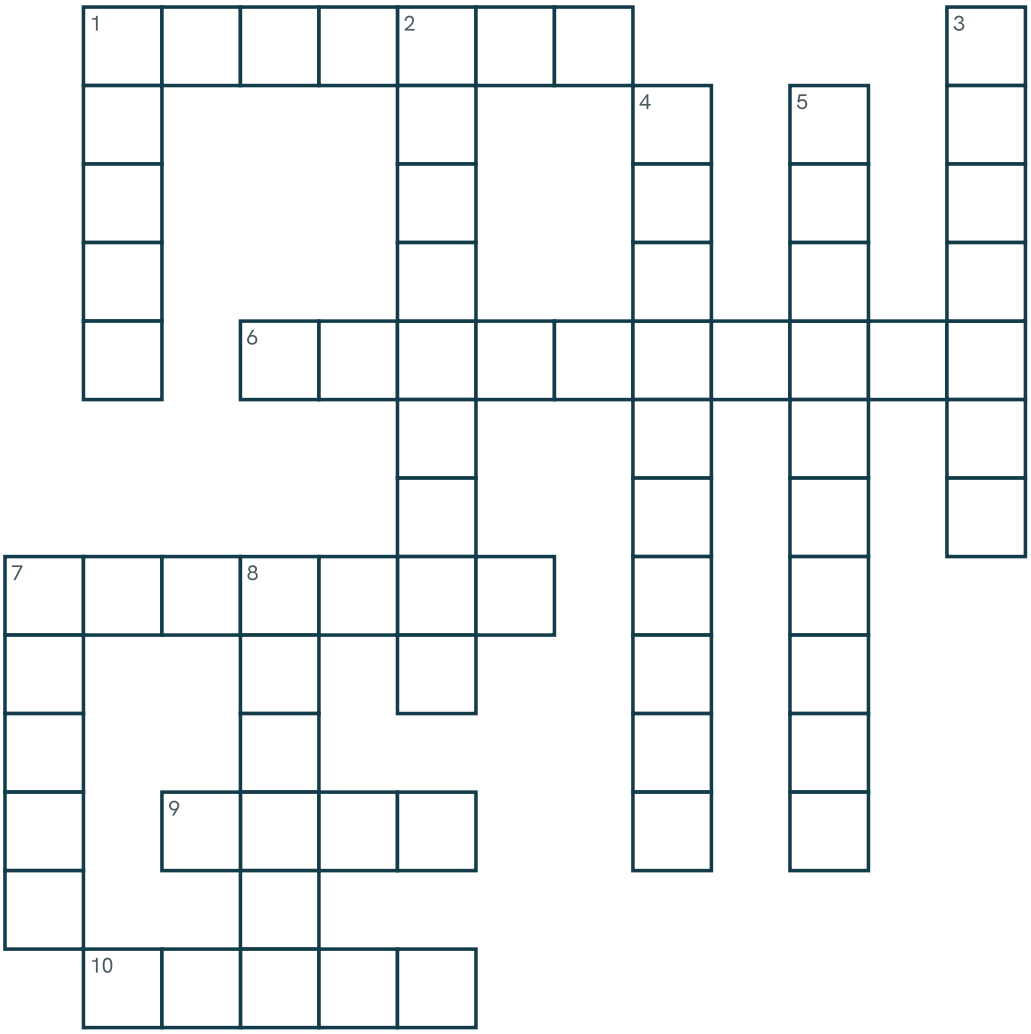
9 Dark Munich lager (6)

11 Czech hop in a classic Pilsner (4)

Crossword 66. Style Specialisation, lesson 3

Wheat beers

Clove, banana and spice in the right measure.



Across

1 Bitter orange often used in witbier (7)

6 Citrus spice in witbier (6,4)

7 Belgian spiced wheat beer (7)

9 Wheat protein gives this (4)

10 Choice that sets clove against banana (5)

Down

1 Phenol from 4-vinyl guaiacol (5)

2 Seed spice in witbier (9)

3 ___ acid rest, near 45°C (113°F) (7)

4 Bavarian wheat beer with its yeast (10)

5 To add less yeast and lift the banana (10)

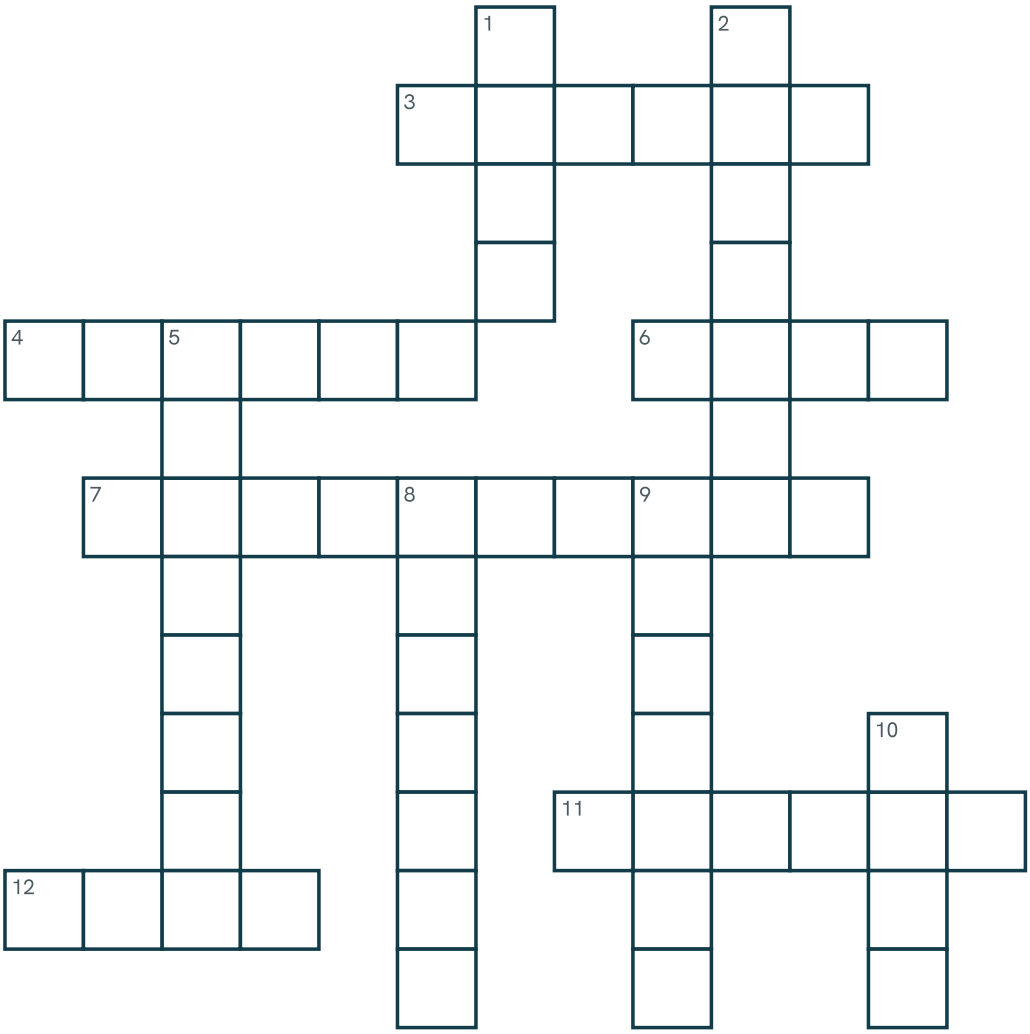
7 Half or more of the grist (5)

8 Isoamyl acetate note (6)

Crossword 67. Style Specialisation, lesson 4

British ales

Bitters, milds, porters and the cask.



Across

3 London dark ale (6)

4 Earthy English hop (6)

6 Low-strength ale, often dark (4)

7 Settling well, like most English yeast (10)

11 Pub ale of England (6)

12 Where real ale conditions (4)

Down

1 Hard to keep at under 4% ABV (4)

2 Low-strength beer brewed for drinking a few in one sitting (7)

5 East Kent ___ (8)

8 Malt that adds caramel and body (7)

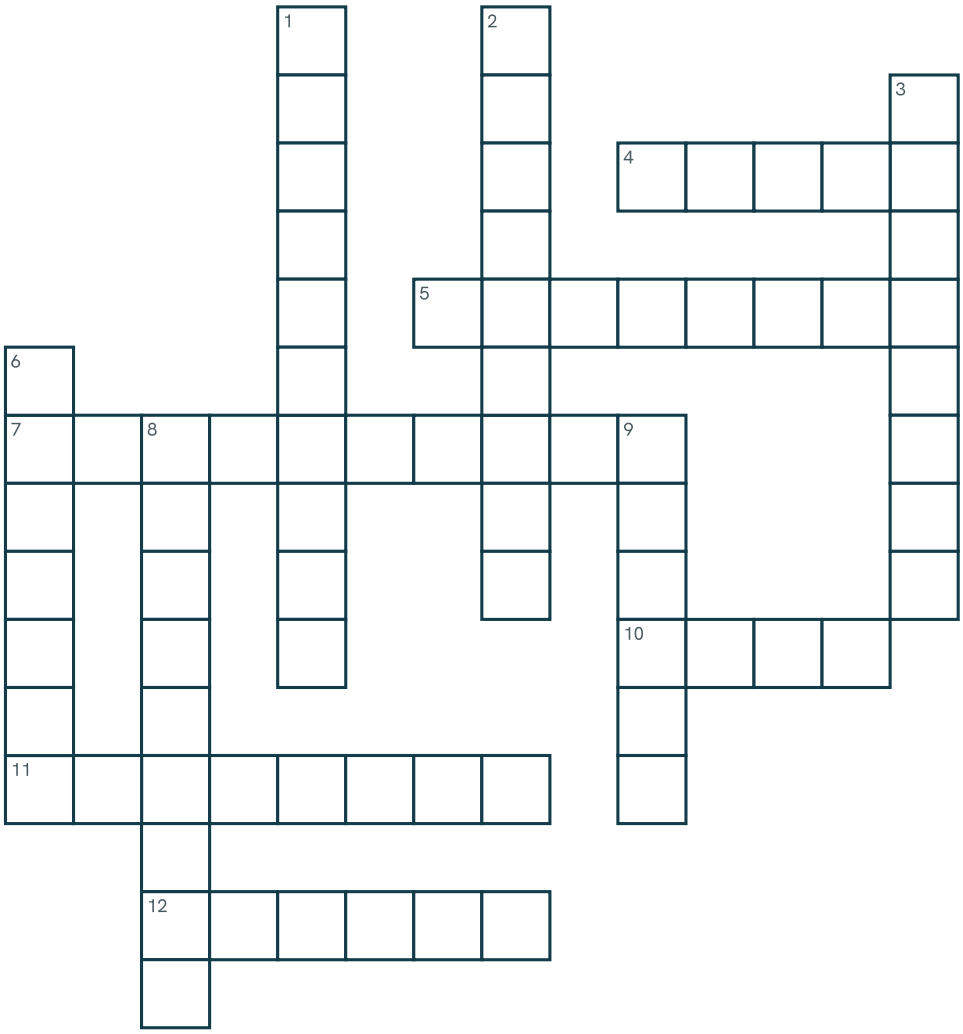
9 ___ yeast: fruity and quick to settle (7)

10 ___ bitter: the homework style (4)

Crossword 68. Style Specialisation, lesson 5

IPAs

English, West Coast and New England.



Across

4 Popular hop with lime and tropical fruit (5)

5 Hop ___ (8)

7 Hazy, soft US IPA (3,7)

10 The look of a New England IPA (4)

11 Extra fermentation from dry hops (3,5)

12 Enemy of hop aroma (6)

Down

1 West Coast has more of it (10)

2 Clear, bitter US IPA (4,5)

3 The same ___ for both homework recipes (4,4)

6 The original IPA, earthier (7)

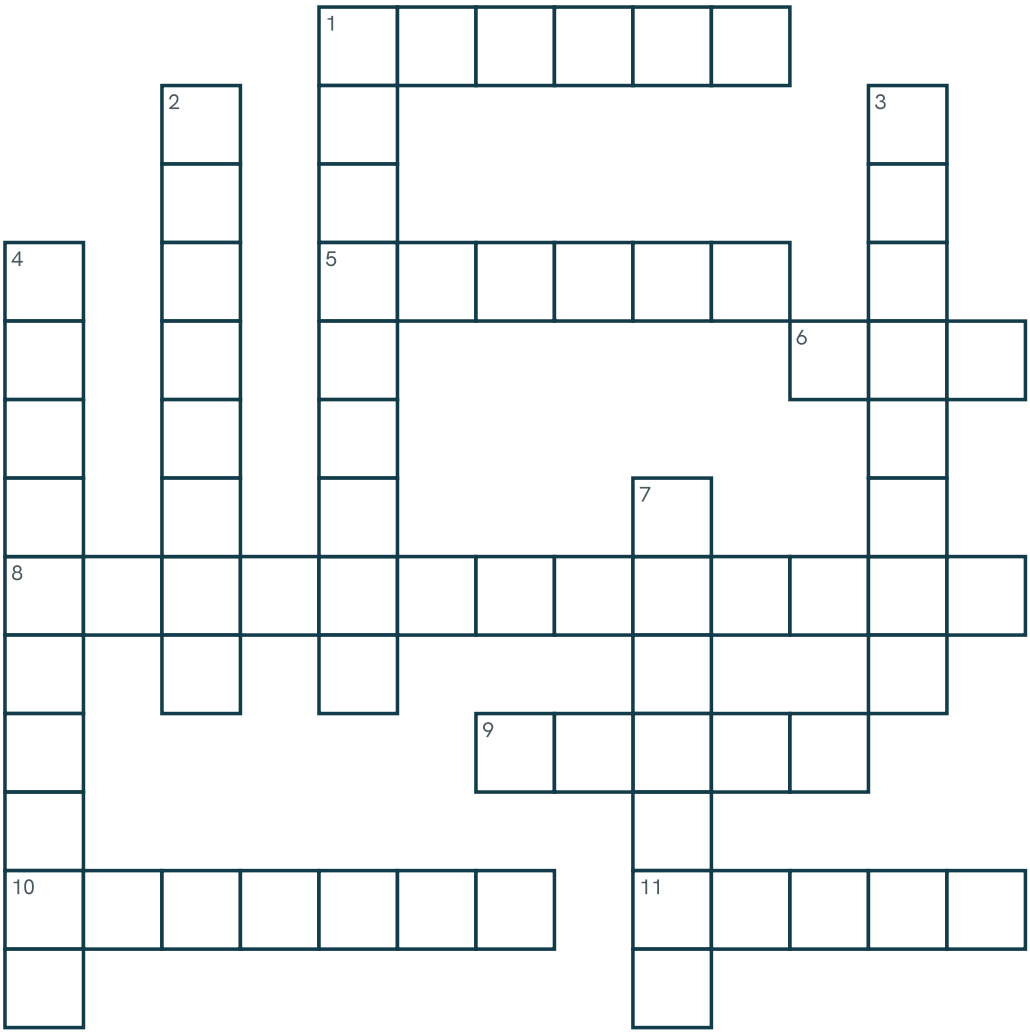
8 Hot-side aroma stage (9)

9 Cold hop addition (3,3)

Crossword 69. Style Specialisation, lesson 6

Stouts and dark beers

Roast, sweetness and the nitro pour.



Across

1 The head a nitro pour gives (6)

5 Roast flavour note (6)

6 Irish ___ stout (3)

8 Unmalted grain behind stout's bite (7,6)

9 Creamy pour (5)

10 Grain for silky body (7)

11 ___ stout, with unfermented sugar left in (5)

Down

1 Dark malt with cocoa notes (9)

2 Strongest stout family (8)

3 Gas in that pour (8)

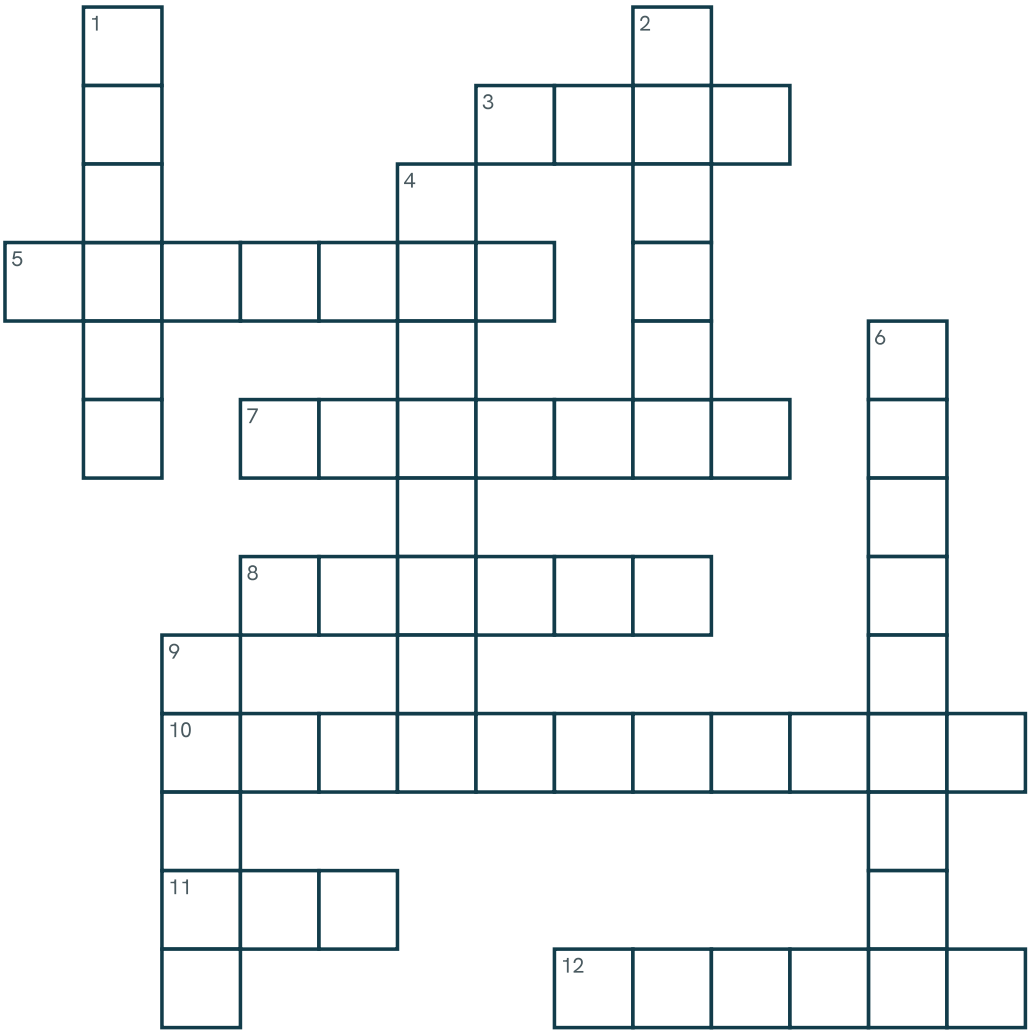
4 Plate inside a stout tap (10)

7 Milk sugar in sweet stout (7)

Crossword 70. Style Specialisation, lesson 7

Belgian ales

Saison, dubbel and tripel, fermented warm.



Across

3 How Belgian yeast is fermented (4)

5 Classic saison note (7)

7 Spicy, peppery yeast compounds (7)

8 Farmhouse ale (6)

10 Saison yeast is famous for high ___ (11)

11 A saison should finish very ___ (3)

12 Dark abbey ale (6)

Down

1 Fruity notes from warm fermentation (6)

2 Strong golden abbey ale (6)

4 Letting temperature climb on its own (4,4)

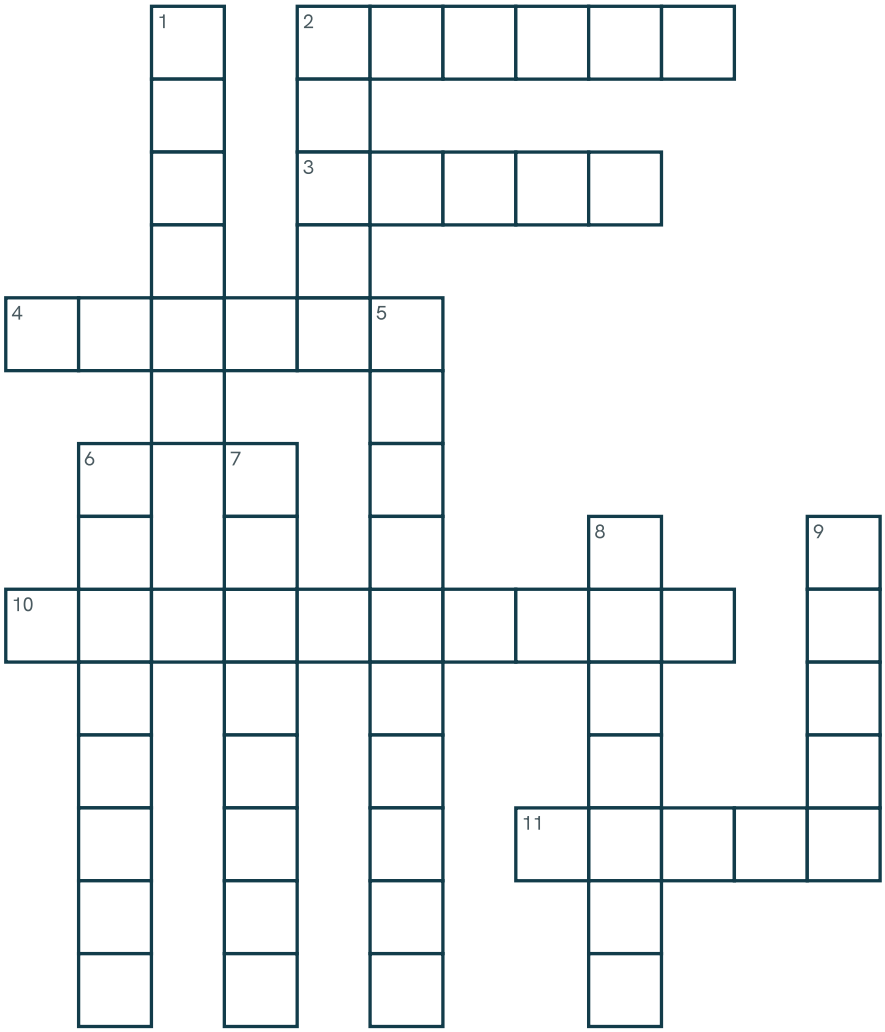
6 Saison's family (9)

9 Belgian ___ sugar (5)

Crossword 71. Style Specialisation, lesson 8

Sours and competition brewing

Souring methods and how beers are judged.



Across

2 Wooden home for a mixed ferment (6)

3 One beer sent in (5)

4 BJCP scores run to 50 of these (6)

10 Quick souring before the boil (6,4)

11 ___ fermentation: yeast and bacteria together (5)

Down

1 Clean souring acid (6)

2 Wild yeast, for short (5)

5 Form the judges fill in (10)

6 What the sheets give the brewer (8)

7 Style group you enter (8)

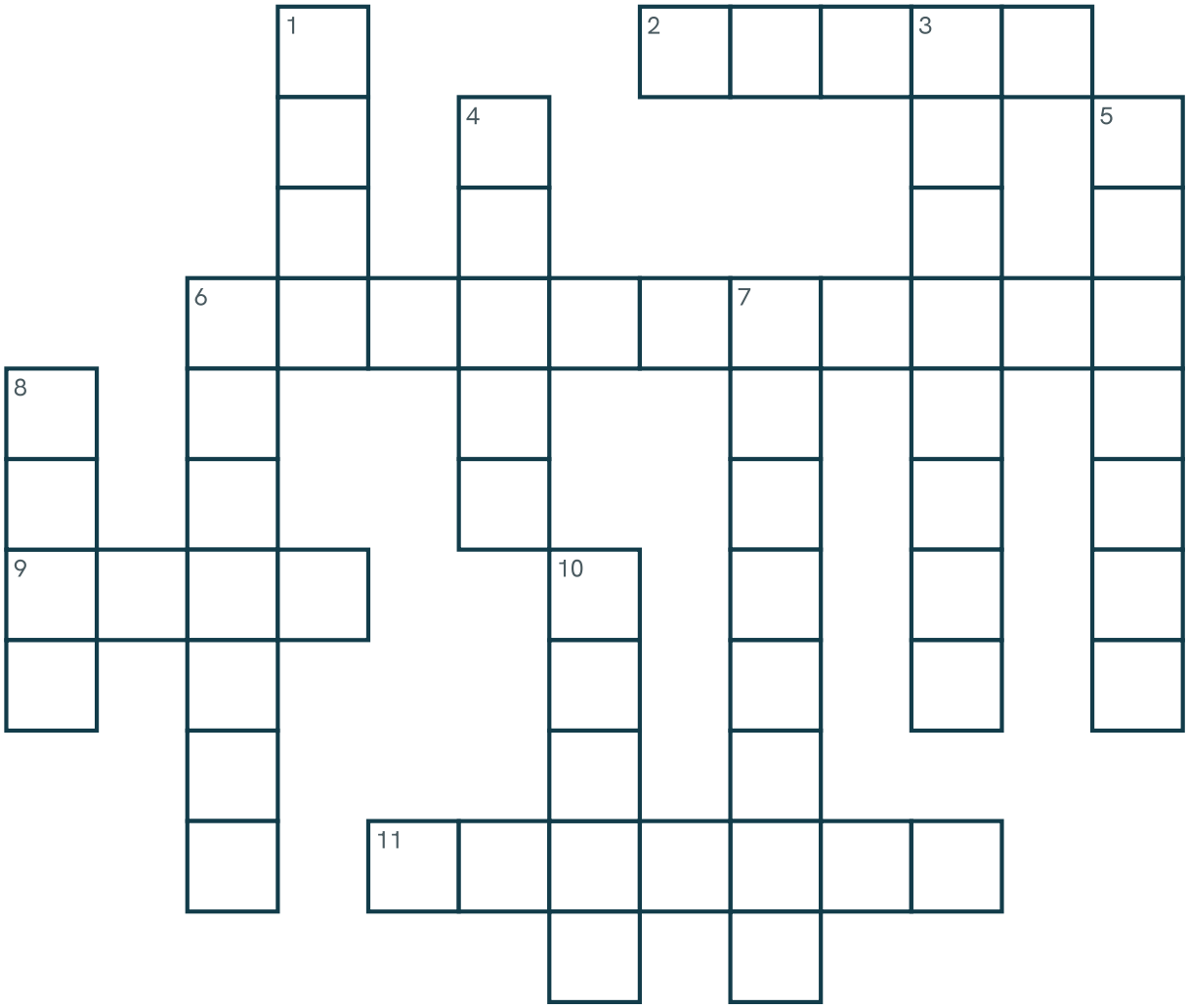
8 Scoring beers against a style (7)

9 Judged with no labels (5)

Crossword 72. Beer Branding & Packaging, lesson 1

Brand identity

Who drinks your beer, what you call it and how it speaks.



Across

2 Indian craft brands do well when they feel this: rooted in their own city (5)

6 Where a brand sits in the drinker's mind against its rivals (11)

9 The first thing a drinker says when ordering (4)

11 Imagined typical drinker, given a name and a life (7)

Down

1 Mark that sits on every tap handle and label (4)

3 The drinkers a brand is built for (8)

4 One-page document that sets out a brand before any design starts (5)

5 Short line that sums up the brand (7)

6 What a brand commits to deliver in every glass (7)

7 When and where a beer gets drunk: a match, a dinner, a weekend (8)

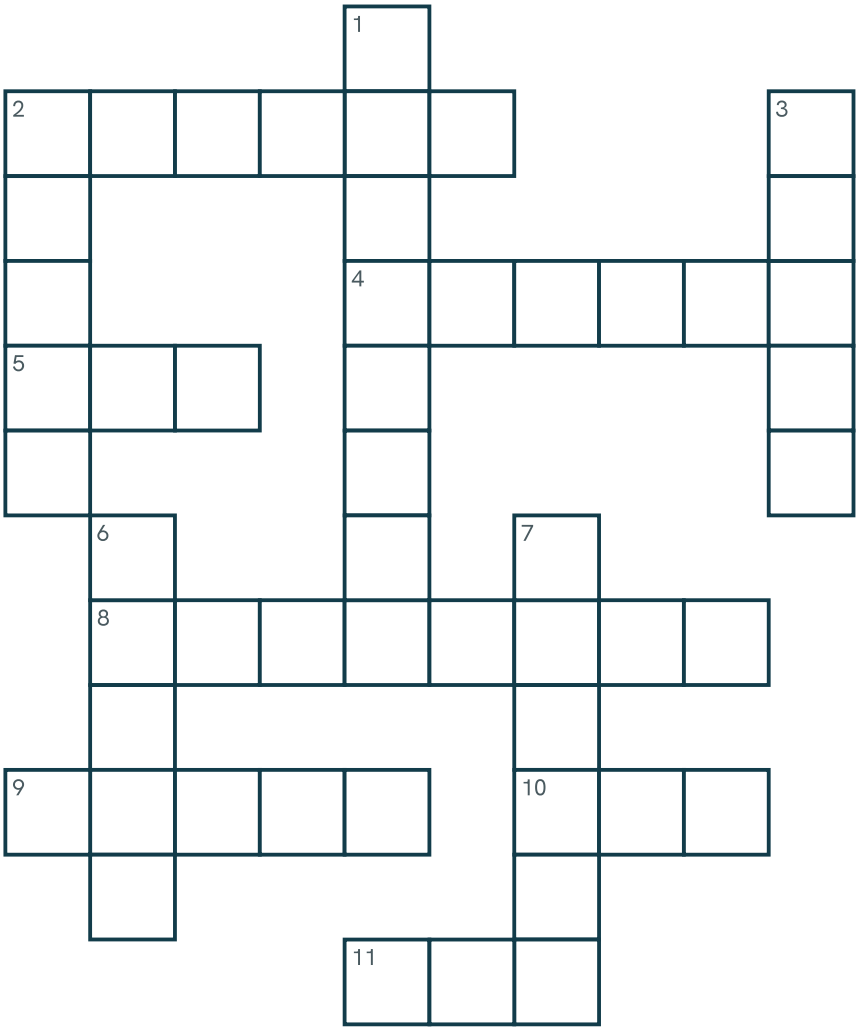
8 ___ of voice: how a brand sounds in every line it writes (4)

10 The tale behind a brand that a bartender can retell (5)

Crossword 73. Beer Branding & Packaging, lesson 2

Packaging that sells

Can, bottle or keg plus a label that wins the shelf.



Across

2 Glass pack, often amber to keep light out (6)

4 State department whose label approval every beer needs (6)

5 Light, fast-chilling pack that blocks all light (3)

8 What a good label has in a crowded fridge (8)

9 Printed face of the pack on the shelf (5)

10 Steel container for draught in bars and taprooms (3)

11 Maximum retail price, printed on every Indian pack (3)

Down

1 Barley and wheat must be declared as one (8)

2 ___ number: printed so a complaint can be traced (5)

3 Where a pack has seconds to catch the eye (5)

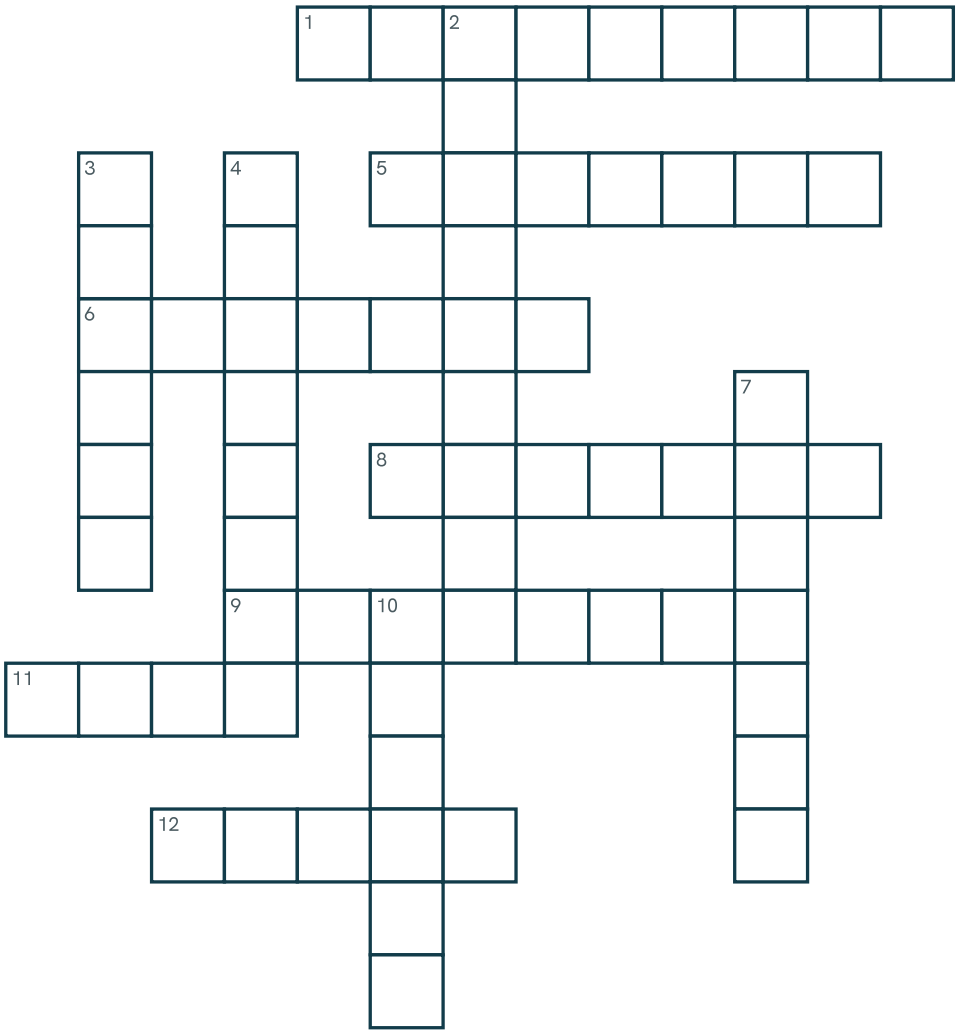
6 Food Safety and Standards Authority of India, by its letters (5)

7 Trial version of a label to check against the rules (6)

Crossword 74. Beer Branding & Packaging, lesson 3

Launch and promotion

Getting a new beer in front of drinkers, within India's rules.



Across

1 Photo-led social platform most craft brands lead on (9)

5 Early tasting for trade and regulars before launch day (7)

6 Content maker a brand may work with on social media (7)

8 Tag that groups posts around a launch (7)

9 Plan of posts and events across the months (8)

11 Slide presentation of a brand for feedback (4)

12 Quiz night, food pairing or tap takeover, for example (5)

Down

2 ___ advertising: promoting a brand through other products, as alcohol brands in India often do (9)

3 ___ media: where most craft launches now play out (6)

4 What the cohort gives on your brand deck (8)

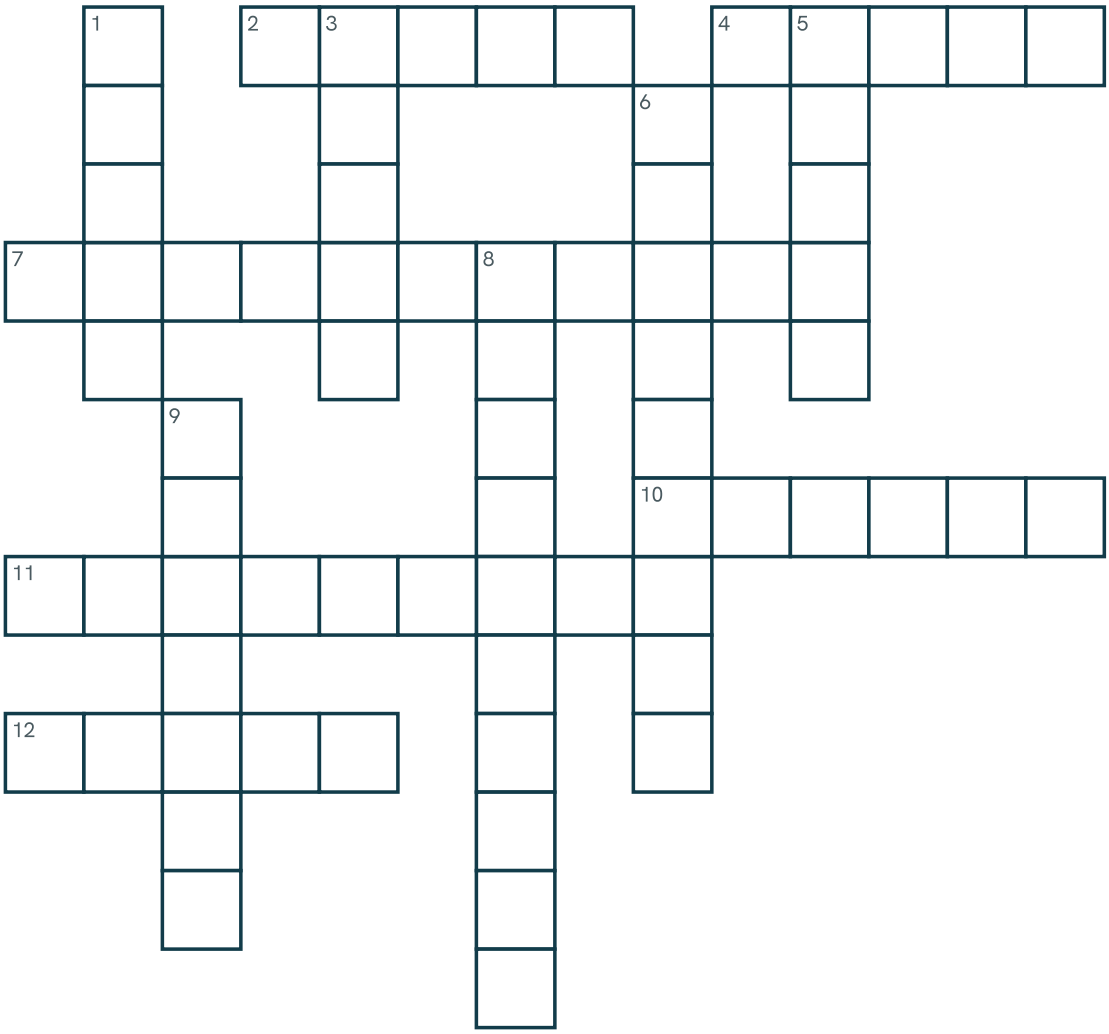
7 Bar at the brewery, the natural home for events (7)

10 First release of a beer to the public (6)

Crossword 75. Sensory Evaluation, lesson 1

How tasting works

The senses, the glass and the order you taste in.



Across

2 Short draws of air over the glass (5)

4 Move the beer to lift aroma (5)

7 Serving ___: too cold mutes aroma (11)

10 A set of small samples tasted together (6)

11 A tulip or a nonic, for example (9)

12 What the nose picks up first (5)

Down

1 Taste light to strong, mild to hoppy: the tasting ___ (5)

3 Written description of each beer (5)

5 Flavour ___: a circular chart of descriptors (5)

6 Body, carbonation and texture (9)

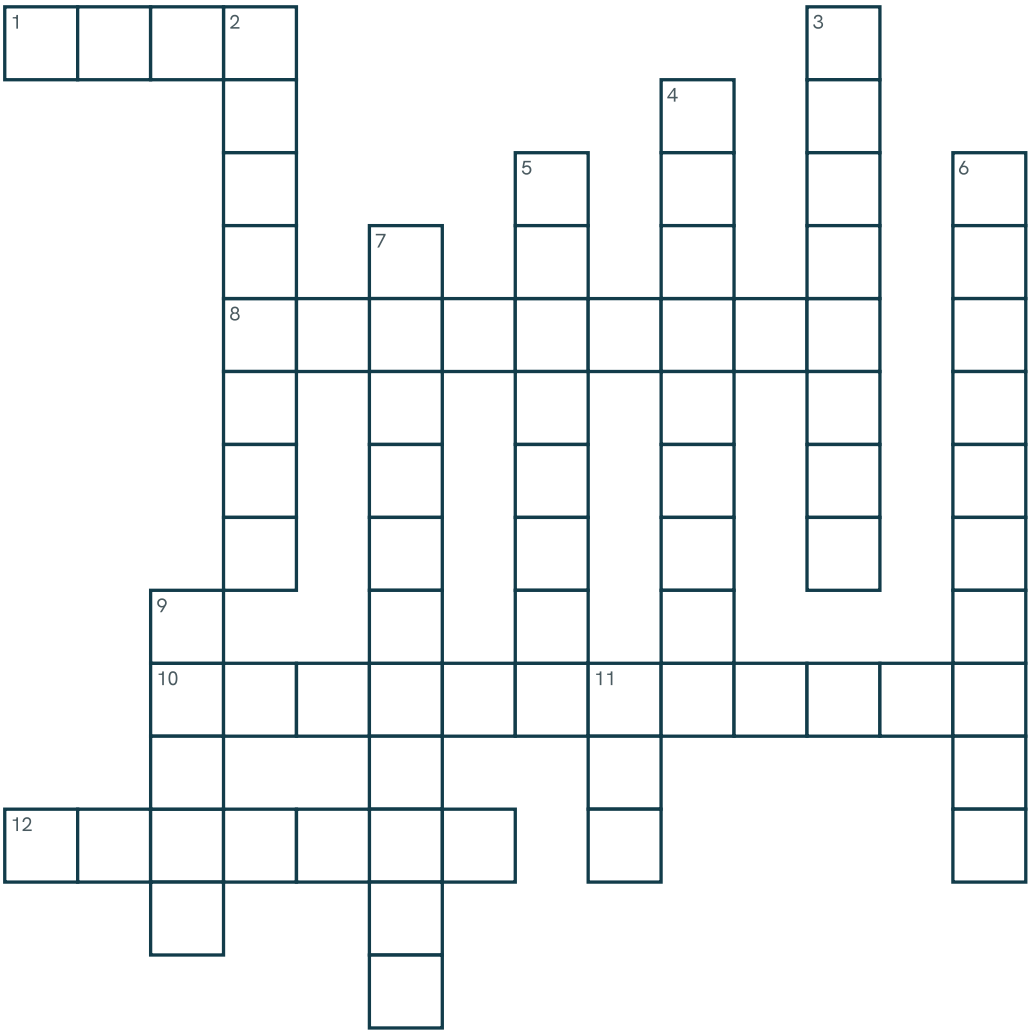
8 Colour, clarity and head (10)

9 Taste and smell working together in the mouth (7)

Crossword 76. Sensory Evaluation, lesson 2

Off-flavours and scoring

Name the fault, find the cause, score the beer.



Across

1 US judge programme whose scoresheets the course follows, by its letters (4)

8 Cause of wet cardboard and stale flavour (9)

10 Green apple from young beer (12)

12 Rotten egg, common in young lager (7)

Down

2 Plastic or clove-like fault, often wild yeast or chlorine (8)

3 Test with three samples, one different (8)

4 Compare your scores with an expert's to ___ your palate (9)

5 Butter or butterscotch from yeast (8)

6 Form judges fill in for each beer (10)

7 Skunky fault from sunlight on hop compounds (11)

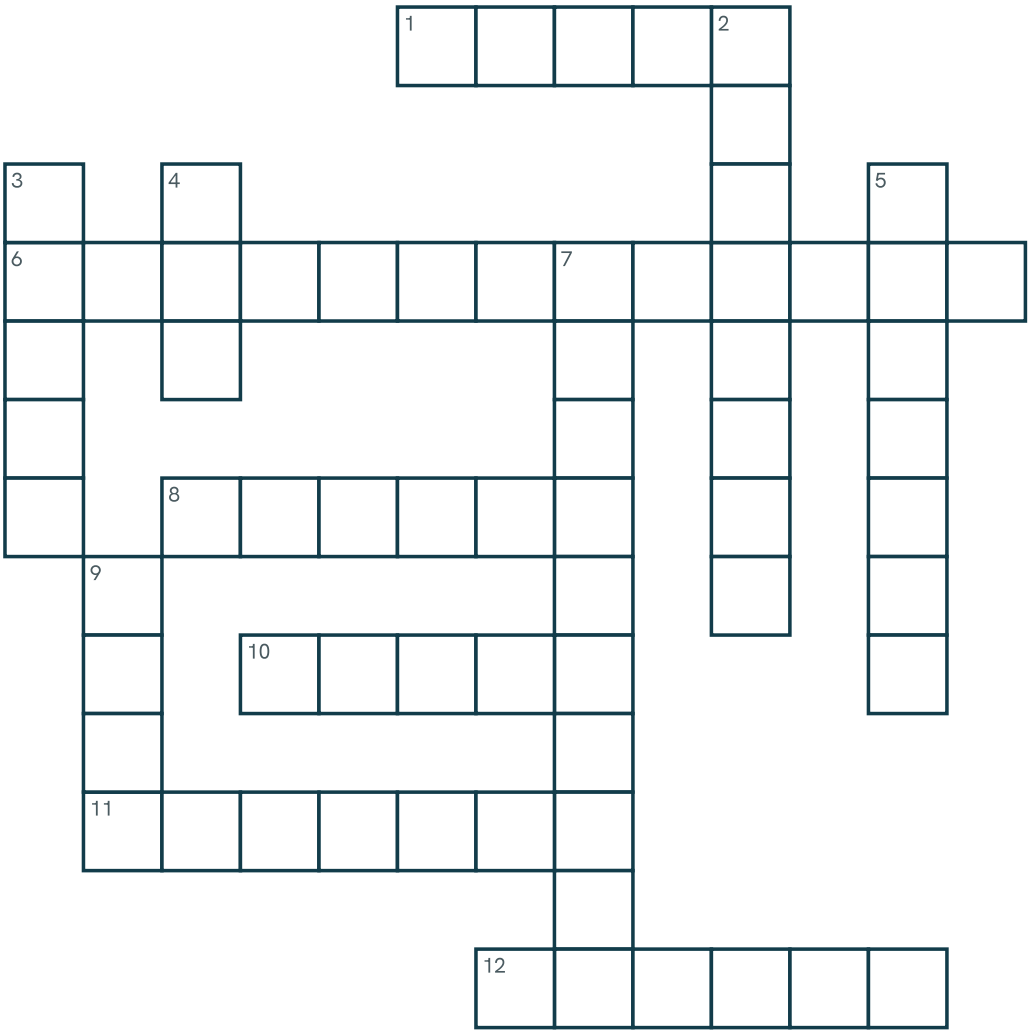
9 Flavour that should not be there (5)

11 Cooked corn, by its letters (3)

Crossword 77. AI for Craft Breweries, lesson 1

GenAI for your brewery's words

Using AI to write tap notes, labels and posts that sound like you.



Across

1 A good prompt is a proper ___: style, ABV, ingredients and voice (5)

6 When AI states something false with confidence (13)

8 Chunks of text a language model reads and writes (6)

10 Your brewery's way of speaking, which AI must learn (5)

11 Board or menu of beers on draught (7)

12 The instructions you give an AI tool (6)

Down

2 Diwali or Holi, for example, in a content plan (8)

3 A real one of your beer often beats an AI image (5)

4 Large language model, by its letters (3)

5 Rainy season to plan a month of Indian posts around (7)

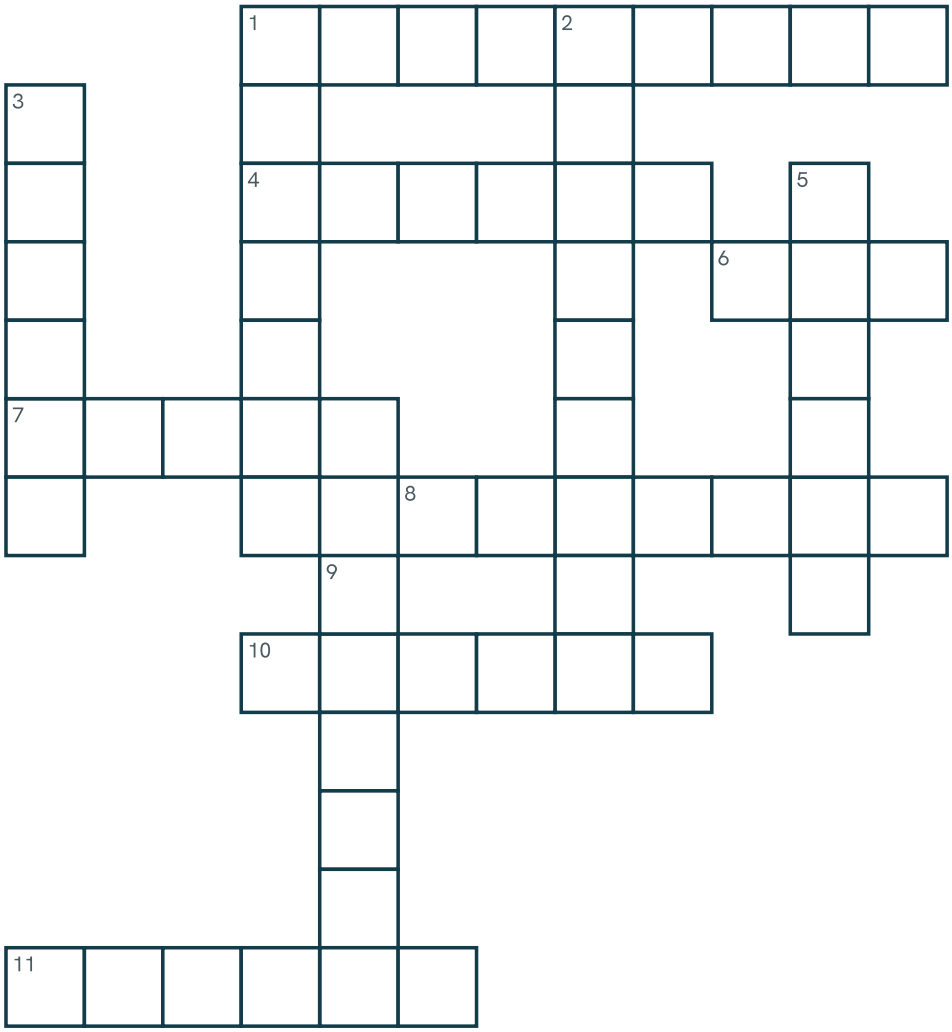
7 Regular email to regulars (10)

9 What every AI draft needs before it goes out (4)

Crossword 78. AI for Craft Breweries, lesson 2

Repetitive tasks and responsible AI

Hand off the dull work and keep the facts straight.



Across

1 What every ABV, award and ingredient needs before publishing (9)

4 A human must ___ anything AI produces (6)

6 Standard operating procedure, by its letters (3)

7 Drafted answer to a review or common question (5)

8 What customer data deserves (7)

10 Beer copy must make no ___ claims (6)

11 Image ___: who owns a photo you post (6)

Down

1 Spreadsheet calculation AI can help write (7)

2 List of steps a tool can draft for review (9)

3 Never aim alcohol content at these (6)

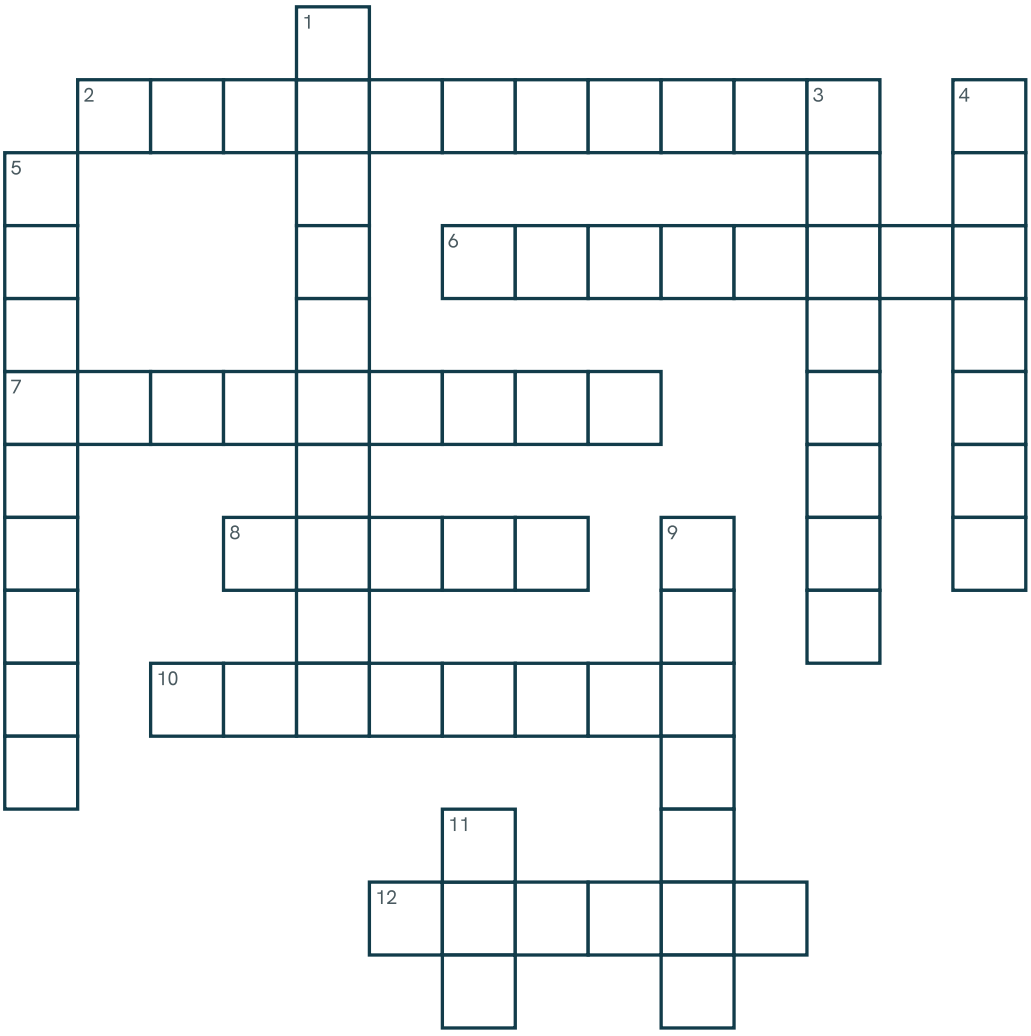
5 One-page AI ___ for your brewery (6)

9 Weekly summary from brew logs and sales notes (6)

Crossword 79. Digital Transformation Basics for Brewing, lesson 1

From paper to data

Moving brew sheets and stock off paper.



Across

2 Shared grid where a small brewery starts (11)

6 Brewery management ___: the step beyond spreadsheets (8)

7 When one comes in, batch codes matter most (9)

8 Malt, hops and finished beer on hand (5)

10 Records that prove CIP was done (8)

12 ___ log: daily gravity and temperature (6)

Down

1 100 litres; a small brewery might make 10 per brew (10)

3 Following kegs, batches and stock (8)

4 Record of every brew, one row per batch (7)

5 Number that traces a pint back to its brew (9)

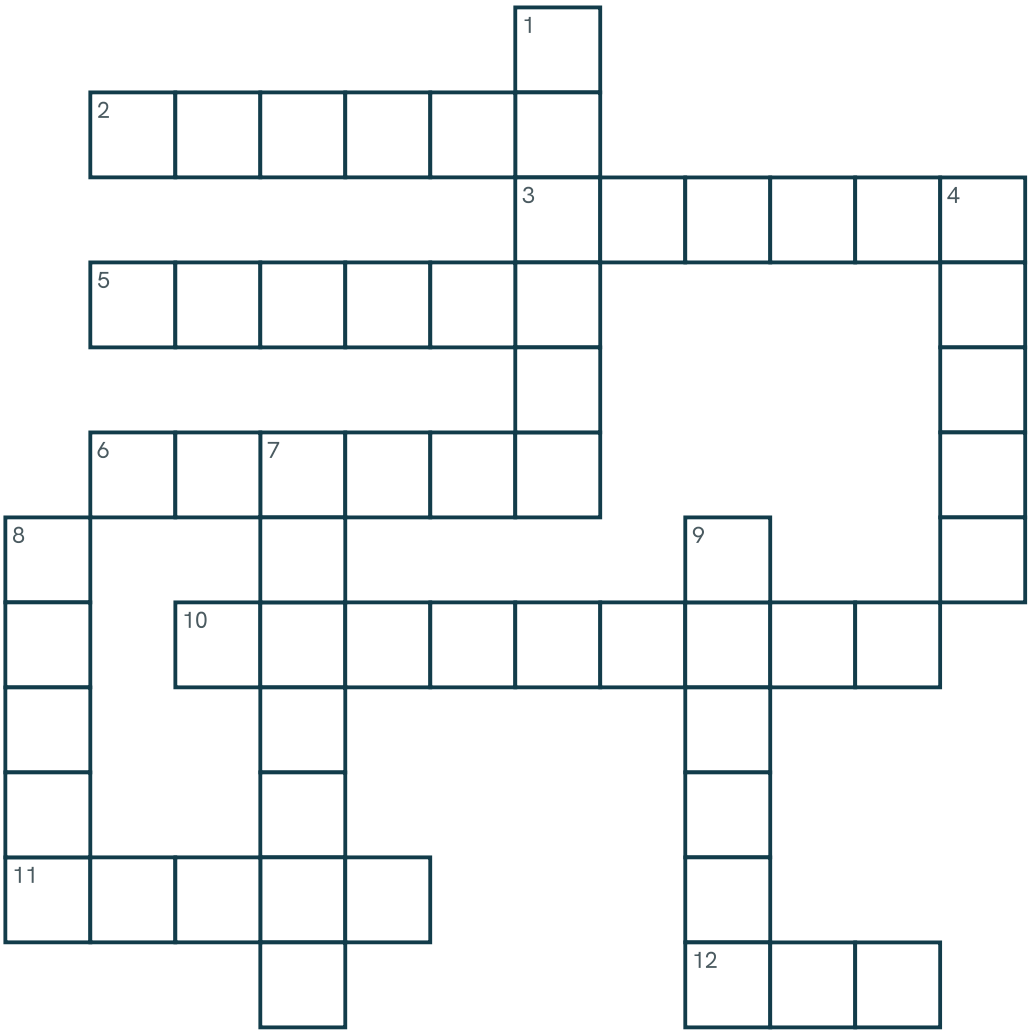
9 ___ transformation: changing how work is done with data (7)

11 Item that needs tracking out and back (3)

Crossword 80. Digital Transformation Basics for Brewing, lesson 2

Sensors, dashboards and your plan

Loggers, dashboards and a 90-day change plan.



Across

2 Litres brewed or sold (6)

3 Number of days in the change plan (6)

5 Who can edit what (6)

6 Records readings at set intervals (6)

10 One screen of the numbers that matter (9)

11 Single source of ___: one master copy (5)

12 Point of sale, by its letters (3)

Down

1 Device that measures temperature or gravity (6)

4 Beer out compared with what you should have got (5)

7 Reading that shows fermentation progress (7)

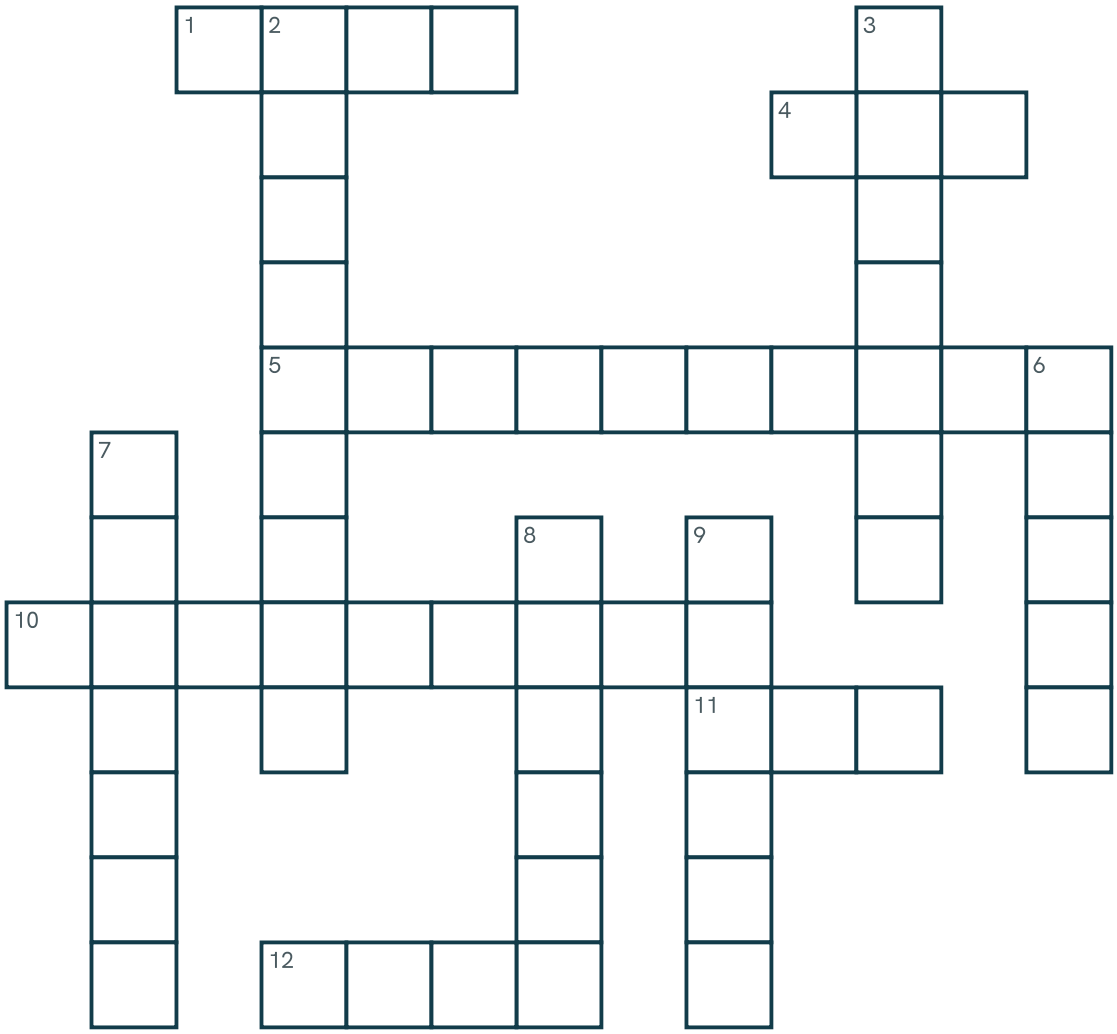
8 What makes a change stick (5)

9 Copy of data kept safe elsewhere (6)

Crossword 81. Power BI and Tableau for Breweries, lesson 1

The brewery data model

Getting brew log data clean enough to chart.



Across

1 Beer that never reaches a glass (4)

4 Formula language for Power BI measures (3)

5 Power BI's tool for cleaning data (10)

10 Key for joining brew, cellar and sales data (9)

11 One per batch in a clean brew log (3)

12 Linking two tables on a shared key (4)

Down

2 Tank ___: how long vessels are full (9)

3 BI tool with calculated fields (7)

6 Brewhouse ___: extract won against expected (5)

7 A calculation such as yield or loss (7)

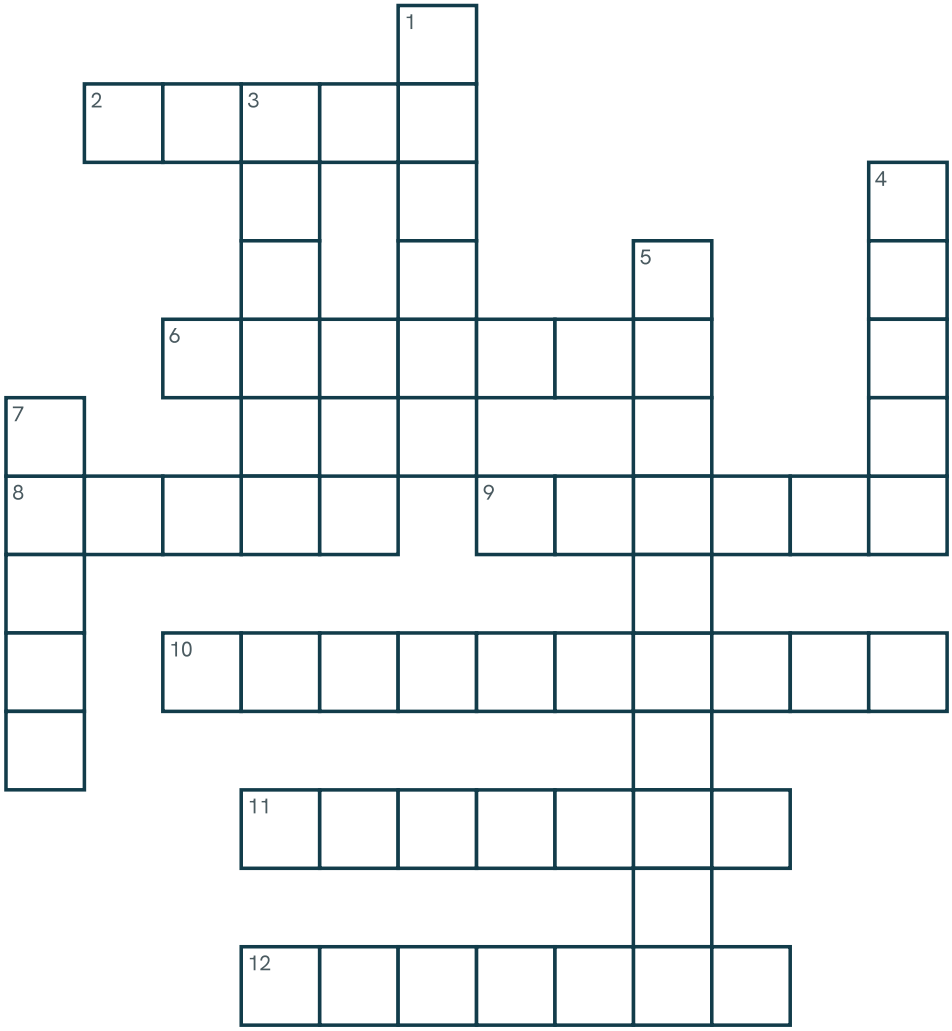
8 One unit per ___, no mixing (6)

9 Cells to avoid in any data sheet (6)

Crossword 82. Power BI and Tableau for Breweries, lesson 2

Dashboards people open

One page, five numbers and a weekly habit.



Across

2 Getting a report to the team (5)

6 Cost that differs between Power BI and Tableau (7)

8 Direction a quality number moves over time (5)

9 Litres brewed or packaged (6)

10 Time beer spends in a vessel (10)

11 ___ limits: the band a stable process stays inside (7)

12 Updating data without retyping it (7)

Down

1 Ten-minute weekly look at the numbers (6)

3 What a moving number should trigger (6)

4 Fermentation ___: gravity against time (5)

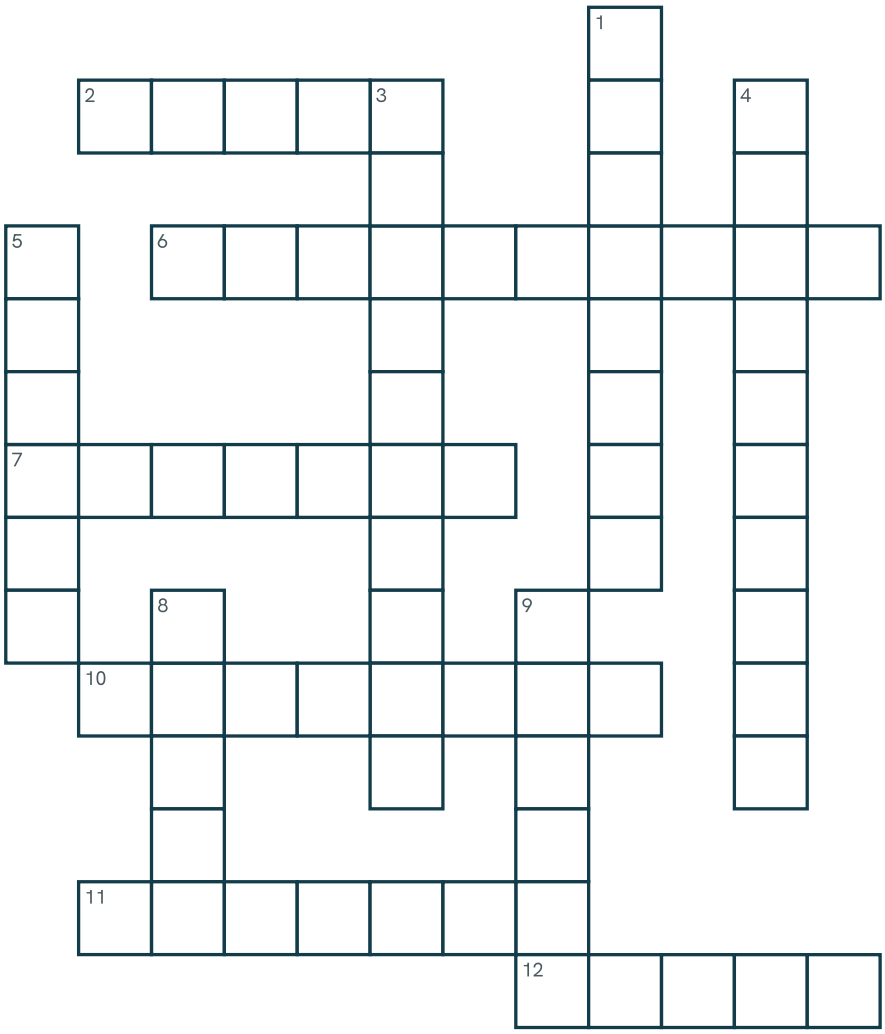
5 Beer sold out of distributors and bars (10)

7 What surprises you when sales are not next to production (5)

Crossword 83. ESG in Craft Brewing, lesson 1

The E: water, energy and waste

Water, energy and waste per litre of beer.



Across

2 The resource used most per litre of beer (5)

6 Effluent that needs a pollution control consent (10)

7 Permission from a state pollution board (7)

10 First measurement to improve from (8)

11 Where the rest often goes (7)

12 Surplus ___: can become feed or food (5)

Down

1 Where many brewery litres go (8)

3 Glass bottle or keg that comes back (10)

4 Mash leftover that can feed cattle (10)

5 Coolant that runs the cellar chillers (6)

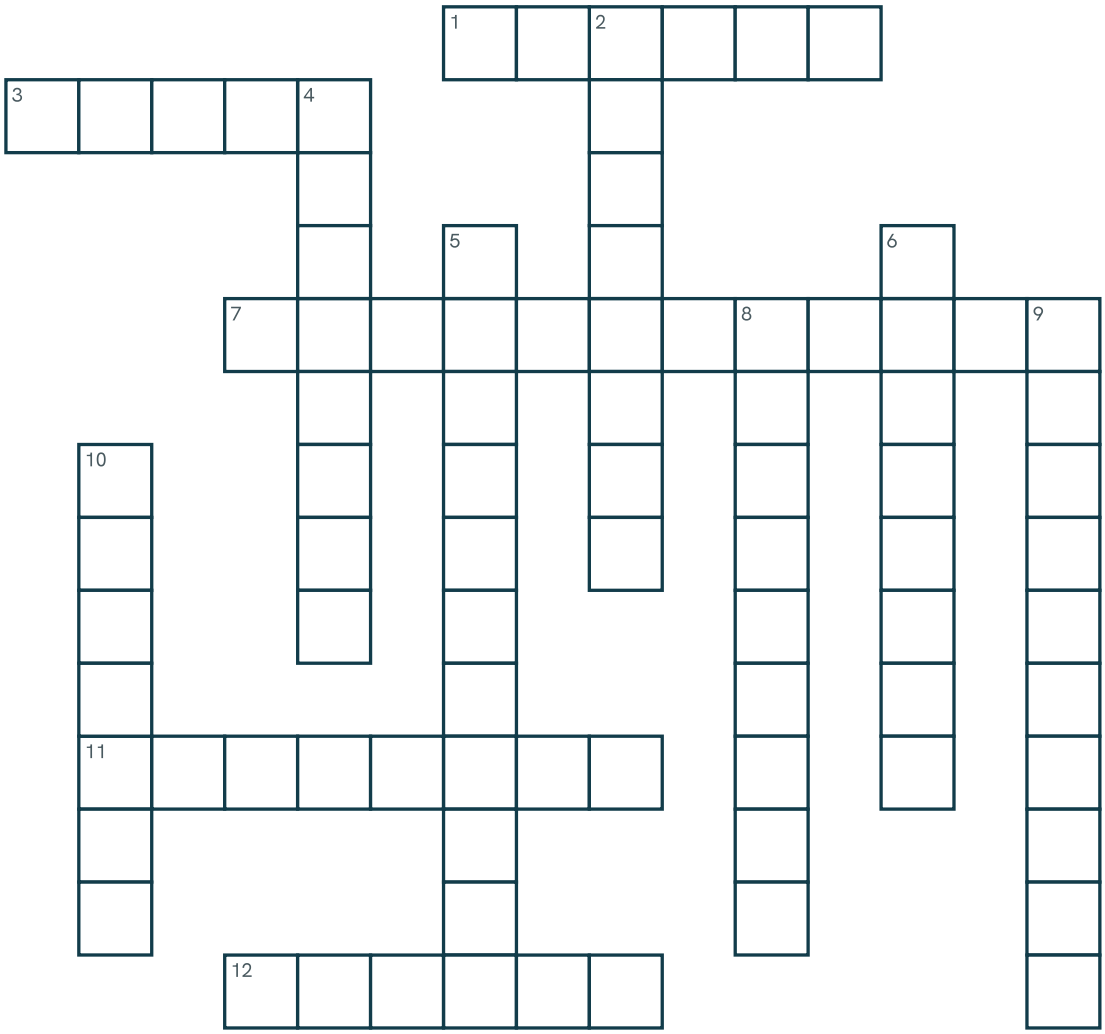
8 Water to beer ___: litres used per litre made (5)

9 Heat and power per litre of beer (6)

Crossword 84. ESG in Craft Brewing, lesson 2

The S and the G, and telling the story

People, governance and honest reporting.



Across

1 Staff ___: the first social measure (6)

3 Clean ___: honest accounts (5)

7 Overclaiming environmental credentials (12)

11 Skills that keep staff safe and growing (8)

12 What every ESG post should be (6)

Down

2 Decent pay and hours for staff (8)

4 Local ___: buying from nearby suppliers (8)

5 ___ drinking: a message every brewery owes (11)

6 Permits a compliance calendar keeps current (8)

8 One-page ESG summary updated every quarter (9)

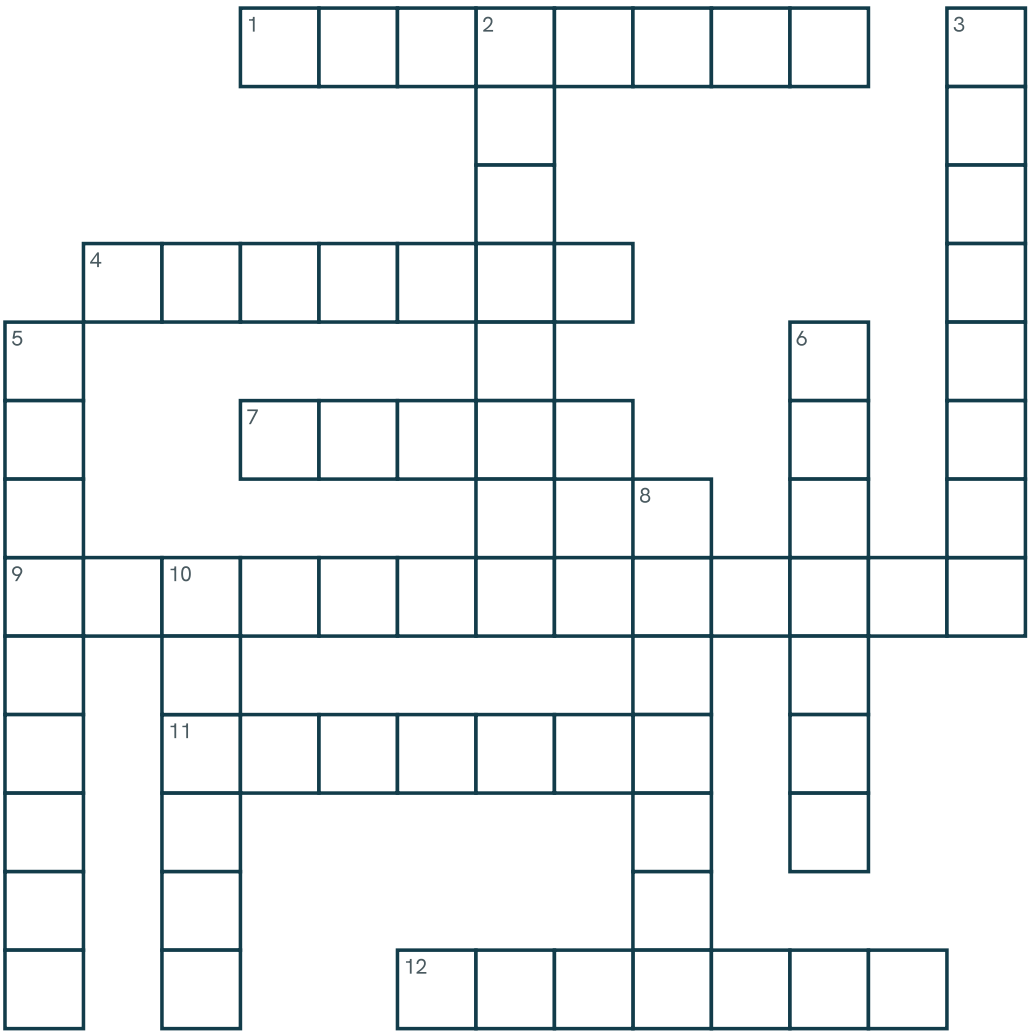
9 The G in ESG (10)

10 How often to update the scorecard (7)

Crossword 85. Safety in Brewing, lesson 1

The hazards and how to work around them

The everyday dangers on a brewery floor.



Across

1 ___ space: never climb into a tank (8)

4 Station to rinse eyes after a splash (7)

7 Scalds from hot wort and ___ (5)

9 Gas from fermentation that collects low down (13)

11 Bend the knees when ___ malt sacks (7)

12 Alkaline cleaner that burns skin and eyes (7)

Down

2 Immediate care before help arrives (8)

3 ___ vessel: a tank built to hold gas (8)

5 Keep these away from water (9)

6 Gas ___: alarm for cellars and cold rooms (7)

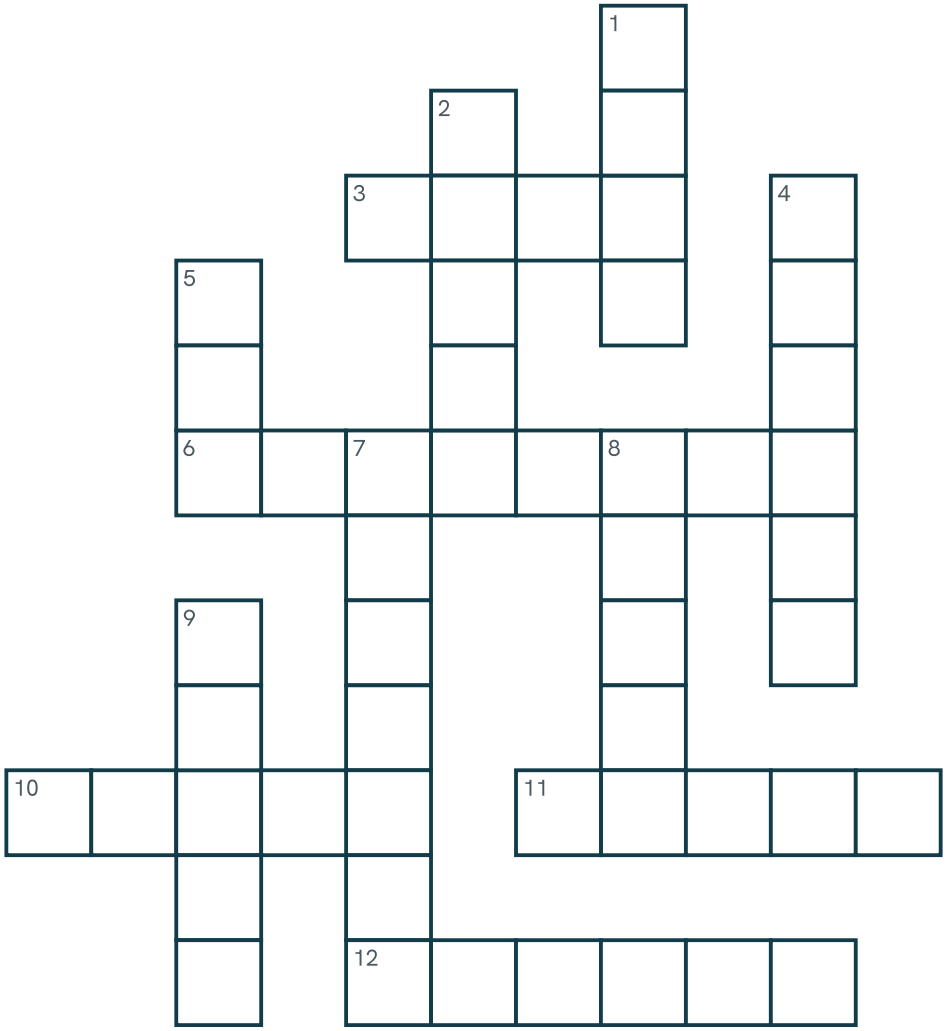
8 Eye protection for chemical work (7)

10 ___ valve: lets out excess pressure (6)

Crossword 86. Craft Distilling, lesson 1

Wash, spirits and the law

From the fermenter to the licence: where every spirit starts.



Across

3 Ingredient a beer has and a wash leaves out (4)

6 Sticky sugar-cane by-product behind most rum (8)

10 Malted or unmalted cereal for whisky and vodka (5)

11 Base for brandy and eau de vie (5)

12 State department that licenses a distillery (6)

Down

1 Fermented liquid that goes into the still (4)

2 Neutral spirit, often filtered through charcoal (5)

4 Grain spirit aged in oak (6)

5 Spirit made from cane juice or molasses (3)

7 What distilling in India needs, separate from a brewing one (7)

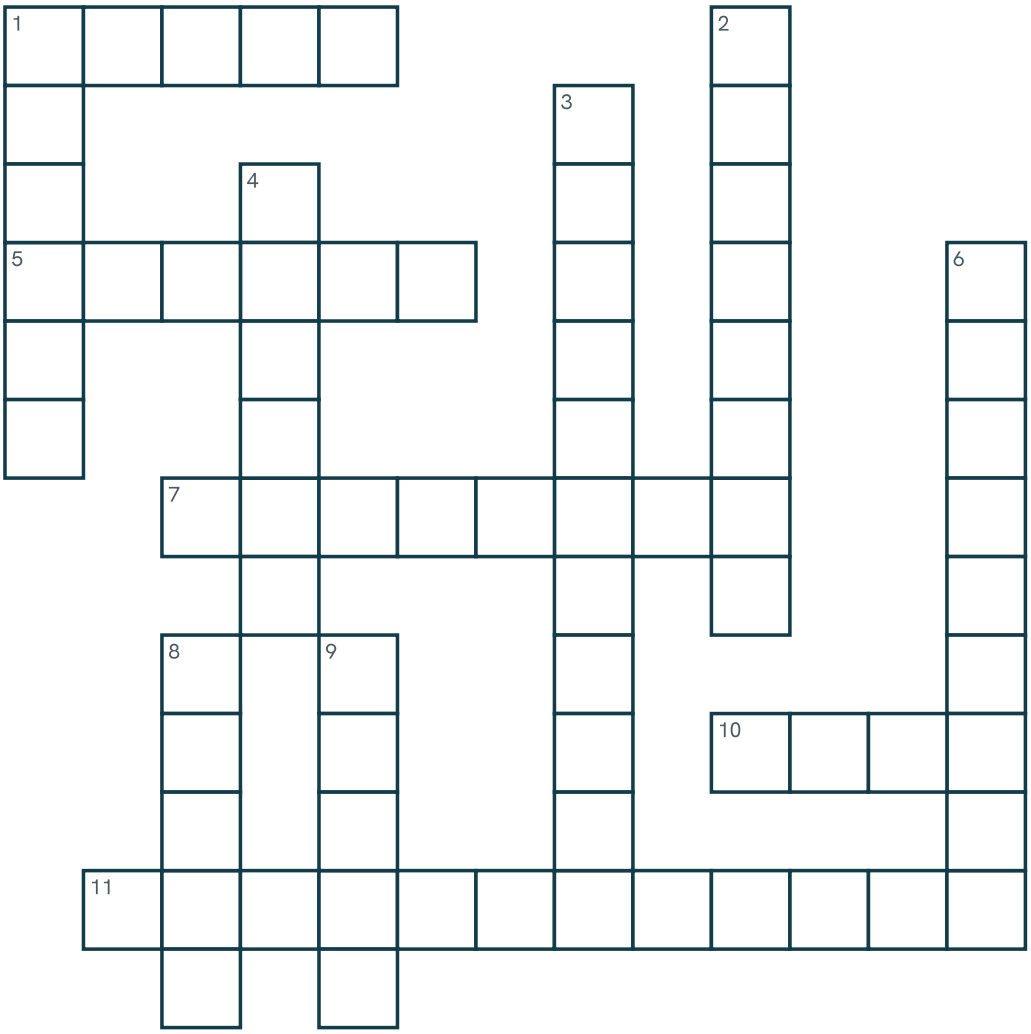
8 Simple fermentable for a quick, clean wash (5)

9 Rules on spirits vary from one ___ to the next (5)

Crossword 87. Craft Distilling, lesson 2

Stills and the run

Copper, heat and the cuts that make or ruin a spirit.



Across

1 First, volatile fraction of the run, high in acetone (5)

5 Vapour condensing and falling back for more purity (6)

7 Toxic alcohol that comes off early in the run (8)

10 Decisions on where heads, hearts and tails divide (4)

11 Floating gauge for the strength of a spirit (12)

Down

1 The middle cut that becomes the spirit (6)

2 Batch still that keeps more character in the spirit (3,5)

3 Continuous still built for high strength (6,5)

4 Trays inside a column that each act as a small still (6)

6 Cools vapour back into liquid (9)

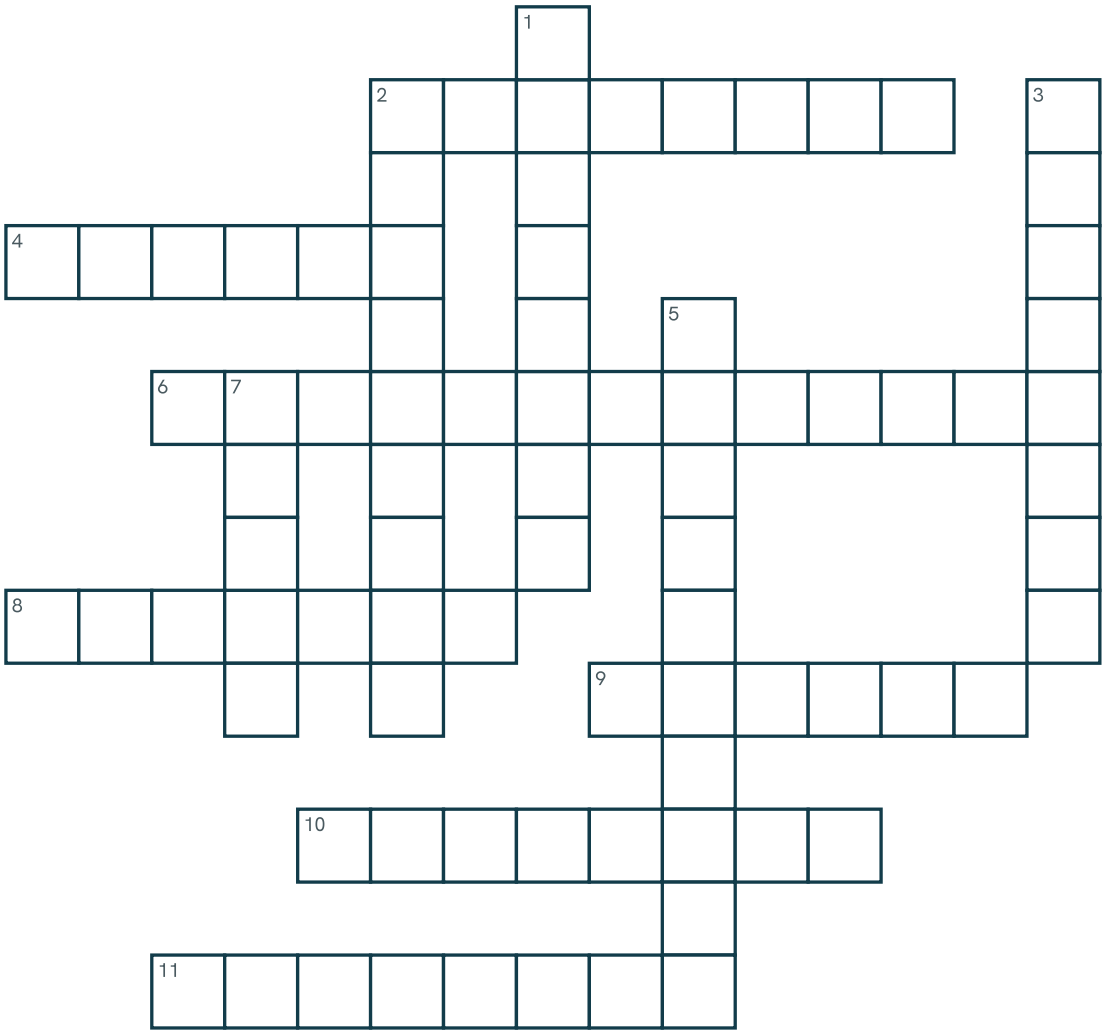
8 Last fraction, oily and full of fusel alcohols (5)

9 US measure of strength, double the ABV (5)

Crossword 88. Craft Distilling, lesson 3

Gin and botanicals

Juniper first, then everything else in the botanical bill.



Across

2 Indian pod that adds a cooling spice (8)

4 ___ infusion: botanicals sit in a basket above the spirit (6)

6 The recipe of everything that flavours a gin (9,4)

8 Berry that must lead in every gin (7)

9 Holds botanicals in the vapour path (6)

10 Root that binds the other botanicals (8)

11 Adding water to bring spirit to strength (8)

Down

1 Cutting spirit down to its bottling strength (8)

2 Seed that brings citrus and spice (9)

3 ___ strength: the ABV on the label (8)

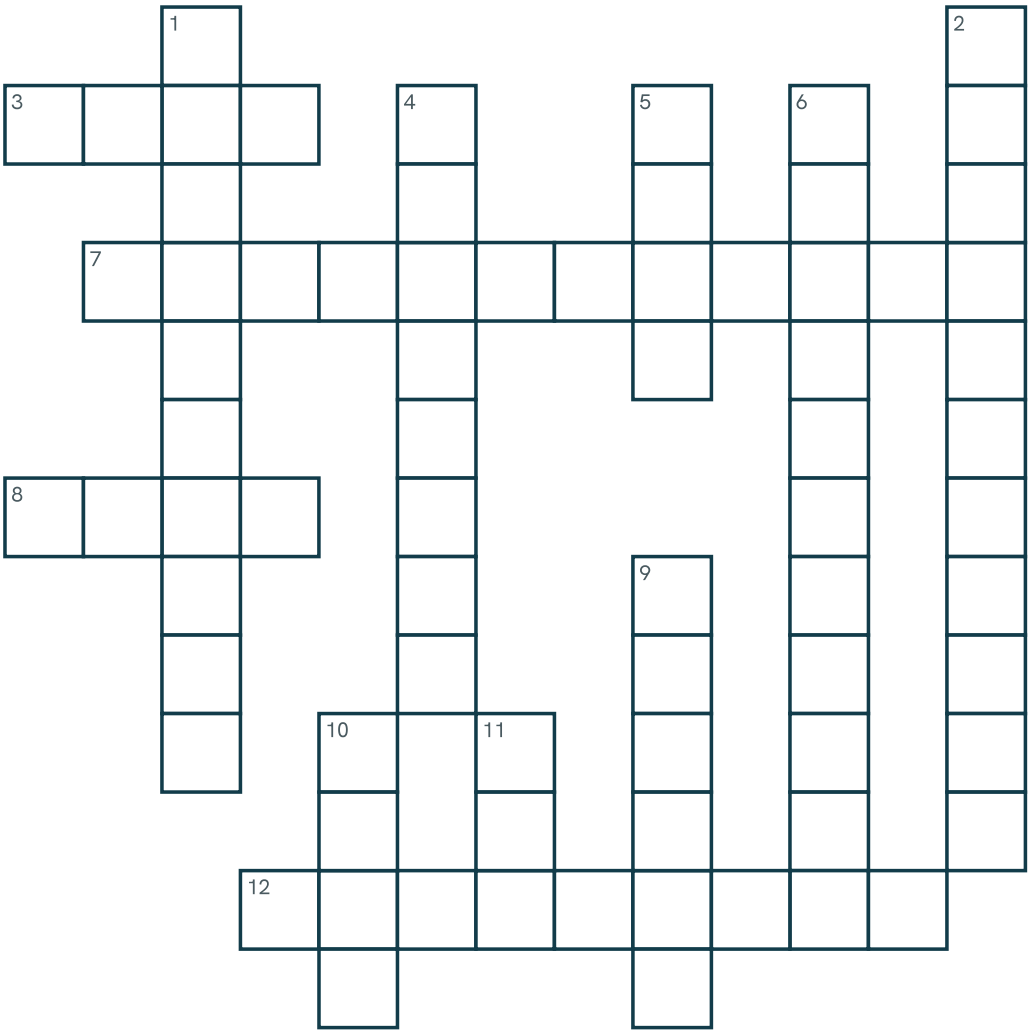
5 Steeping botanicals in spirit before the run (10)

7 Iris root used as a fixative (5)

Crossword 89. Craft Distilling, lesson 4

Ageing, costs and getting to market

Oak, heat, duty and the first year of a small distillery.



Across

3 Why maturation runs faster in India (4)

7 One-page argument for your first spirit (8,4)

8 Tax that can be the biggest line in a bottle's cost (4)

12 Glass, closure and label in the cost sheet (9)

Down

1 Time in wood that rounds a spirit (10)

2 Spirit lost to evaporation in the cask (6,5)

4 Oak compound with a vanilla flavour (8)

5 Wooden vessel for maturing spirit (4)

6 Losses climb in a hot warehouse through this (11)

9 Fuel and power to run the still (6)

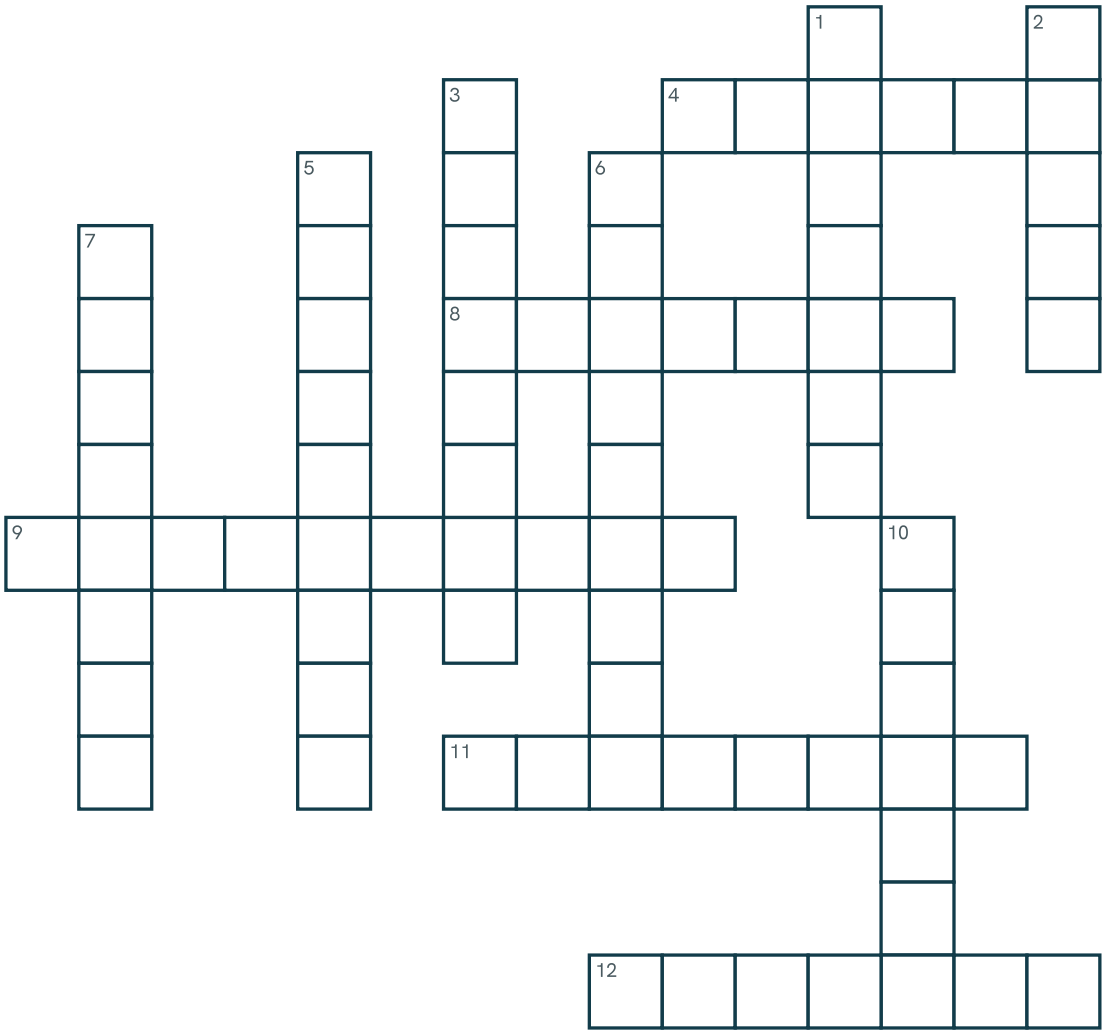
10 Burnt layer inside a cask that filters and colours (4)

11 Wood used for nearly all casks (3)

Crossword 90. Craft Gin Making, lesson 1

What gin is, and the law

What makes a gin a gin and what the rules say.



Across

4 Base liquid of every gin (6)

8 Comparing three gins side by side (7)

9 Plants that flavour a gin (10)

11 To run base spirit again with botanicals (8)

12 ___ spirit: the clean base for gin (7)

Down

1 Permission you need before you distil (7)

2 Gin rules vary by this in India (5)

3 ___ distilling: making gin on someone else's still (8)

5 Gin with nothing but water added after distilling (6,3)

6 ___ gin: botanicals redistilled with spirit (9)

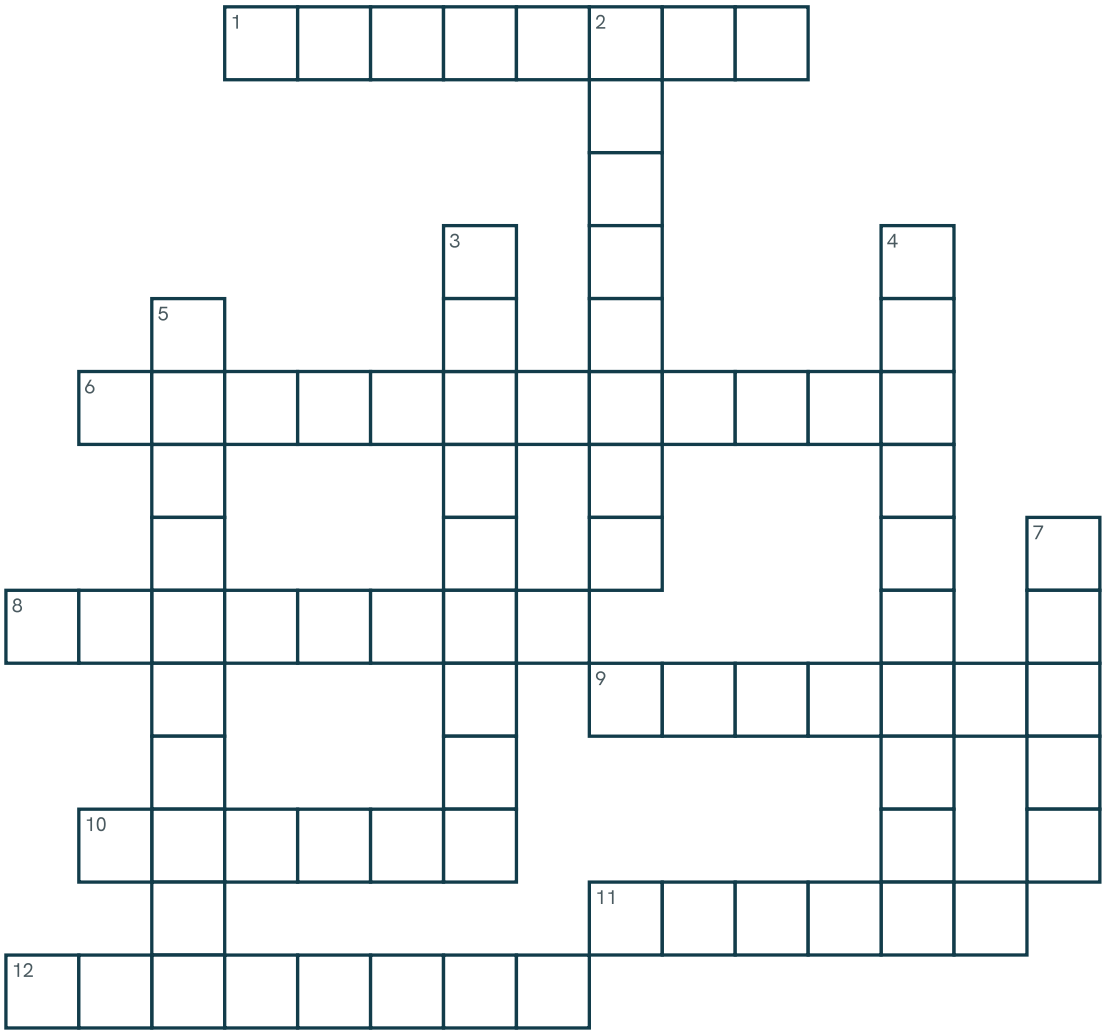
7 Gin made by mixing flavourings into spirit, no still needed (8)

10 Flavour that has to lead (7)

Crossword 91. Craft Gin Making, lesson 2

Botanicals

The classic five and the Indian twist.



Across

1 Green pod from Kerala (8)

6 Rack that holds botanicals above the spirit (6,6)

8 Combining separate distillates into one gin (8)

9 Berry that must still lead (7)

10 Warm rhizome from Indian kitchens (6)

11 ___ botanical distillates, blended afterwards (6)

12 Earthy root that holds the flavours together (8)

Down

2 To steep botanicals in spirit (8)

3 Seed with a citrus, spicy note (9)

4 Dried orange or lemon zest (6,4)

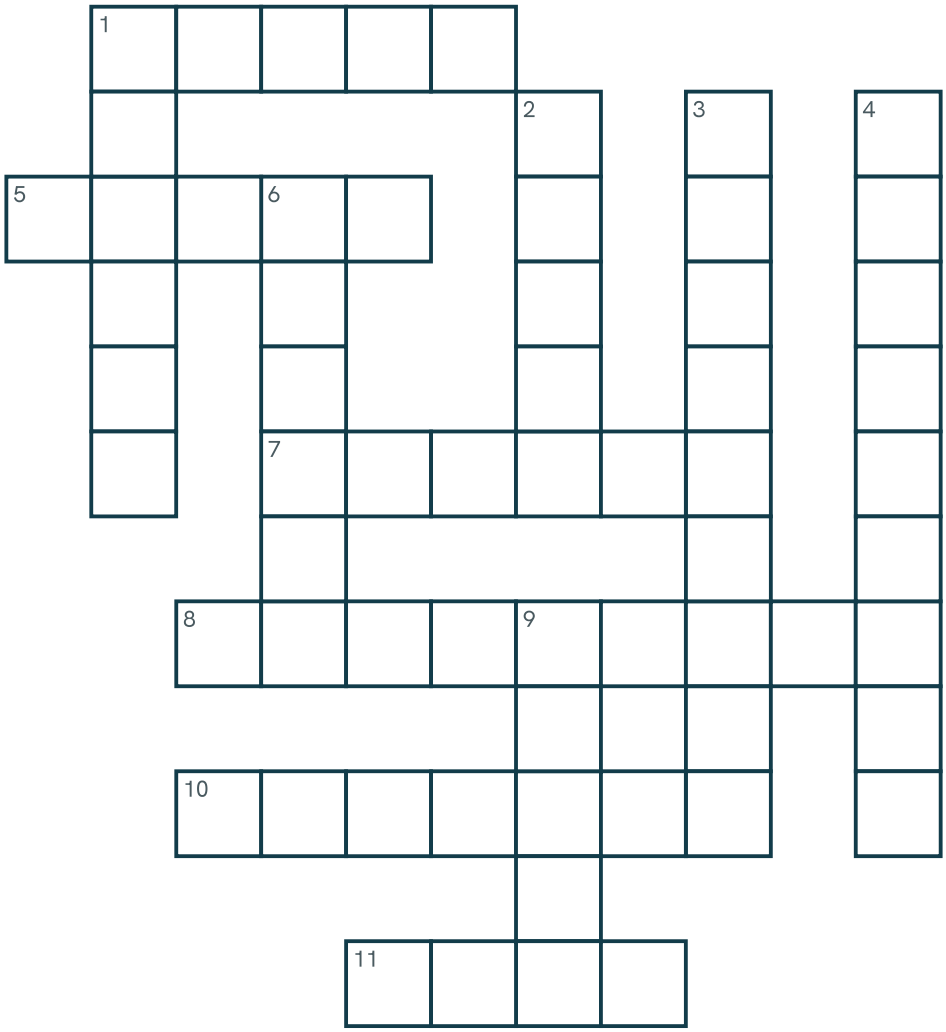
5 Hill tea used in some Indian gins (10)

7 Violet-scented root, a classic fixative (5)

Crossword 92. Craft Gin Making, lesson 3

The run and the finish

Charging the still, making the cuts and finishing the gin.



Across

1 Sharp early fraction set aside (5)

5 Late fraction, heavy and oily (5)

7 To fill the still for a run (6)

8 Bringing gin down to bottling strength (9)

10 Letting gin settle before bottling (7)

11 Botanical compounds that can cloud a gin (4)

Down

1 Middle cut where the best flavour sits (6)

2 What you add to reduce strength (5)

3 Passing spirit through a medium (9)

4 Where you switch between fractions (3,6)

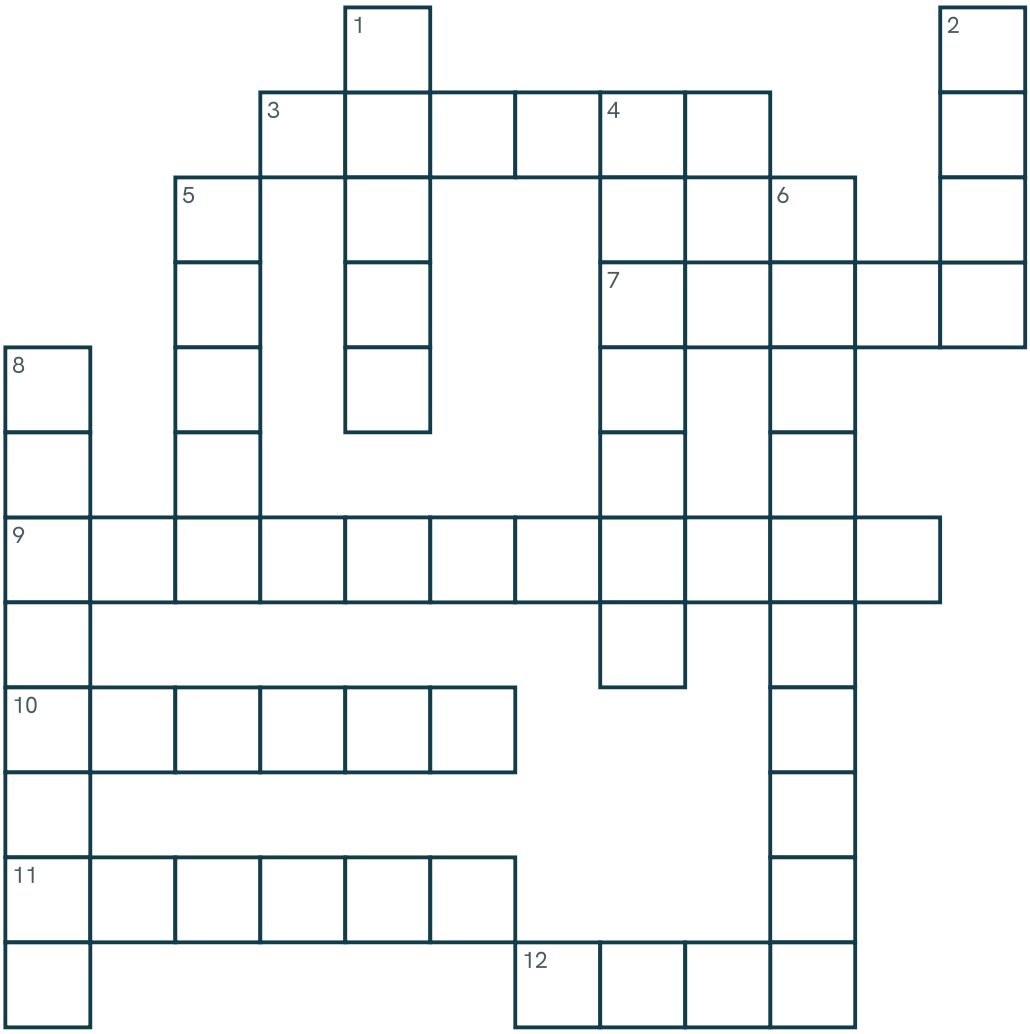
6 Cloudiness when oils drop out of the spirit (6)

9 ___ filtering removes haze-forming compounds (5)

Crossword 93. Craft Gin Making, lesson 4

Brand, costing and market

Getting a gin from the still to the bar.



Across

3 First release to the market (6)

7 What makes a drinker remember your brand (5)

9 Where your gin sits against imported brands (11)

10 Shops that sell bottles to take home (6)

11 Department that approves your label (6)

12 On-trade outlets for a new gin (4)

Down

1 Needs registering with the excise department (5)

2 Tax that weighs on every bottle (4)

4 Working out what each bottle costs (7)

5 Often the biggest packaging cost (5)

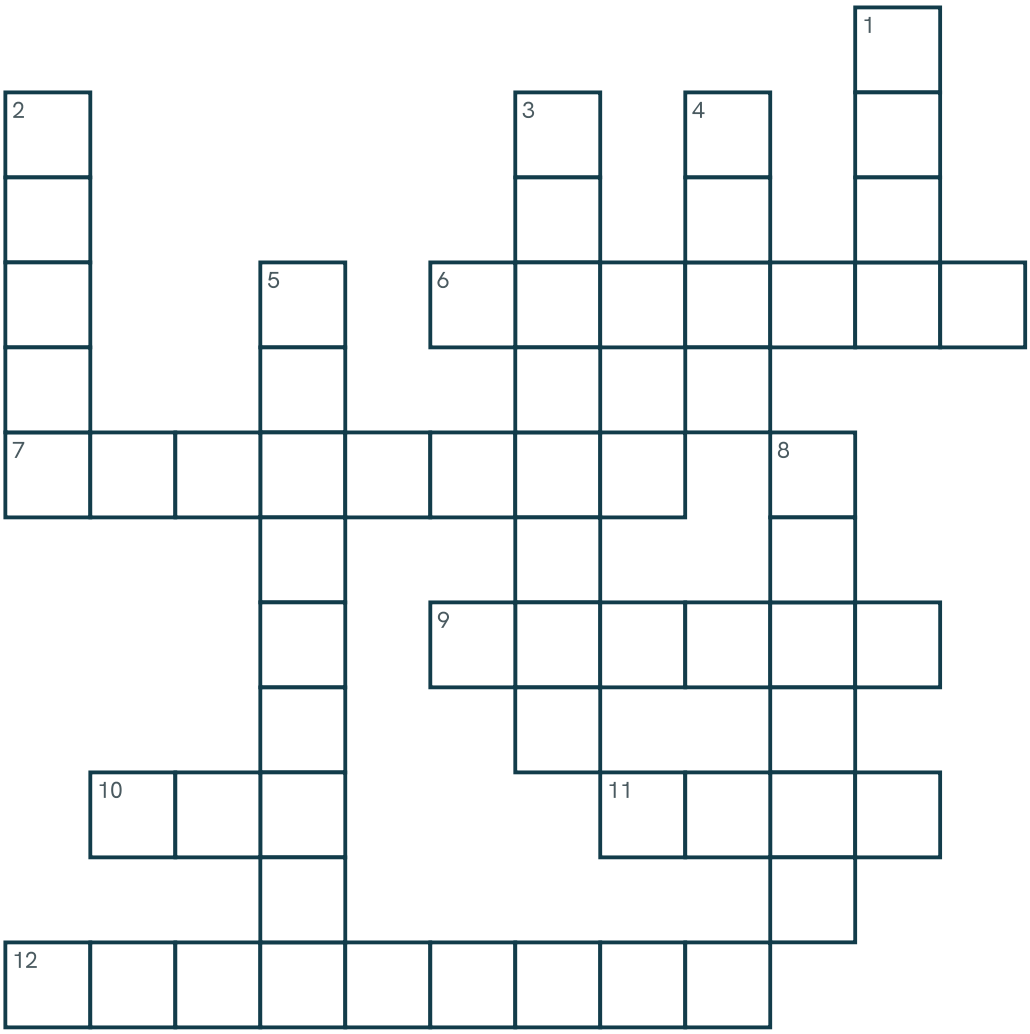
6 Ingredient line beside the spirit (10)

8 Gins made abroad that set the price ceiling (8)

Crossword 94. RTD Drinks: Alcoholic and Non-Alcoholic, lesson 1

Formulation

Balancing sweet, sour and strong in a ready-to-drink can.



Across

6 The other side, measured not guessed (7)

7 Concentrated flavours added in small doses (8)

9 ___ acid, the commonest sour note in soft drinks (6)

10 Base for an iced RTD (3)

11 Scale for sugar content (4)

12 One side of the balance, measured in Brix (9)

Down

1 Brewed base used in some alcoholic RTDs (4)

2 Fruit liquid that adds body and colour (5)

3 Alcohol-free mixed drink (8)

4 Craft ___: a carbonated soft drink (4)

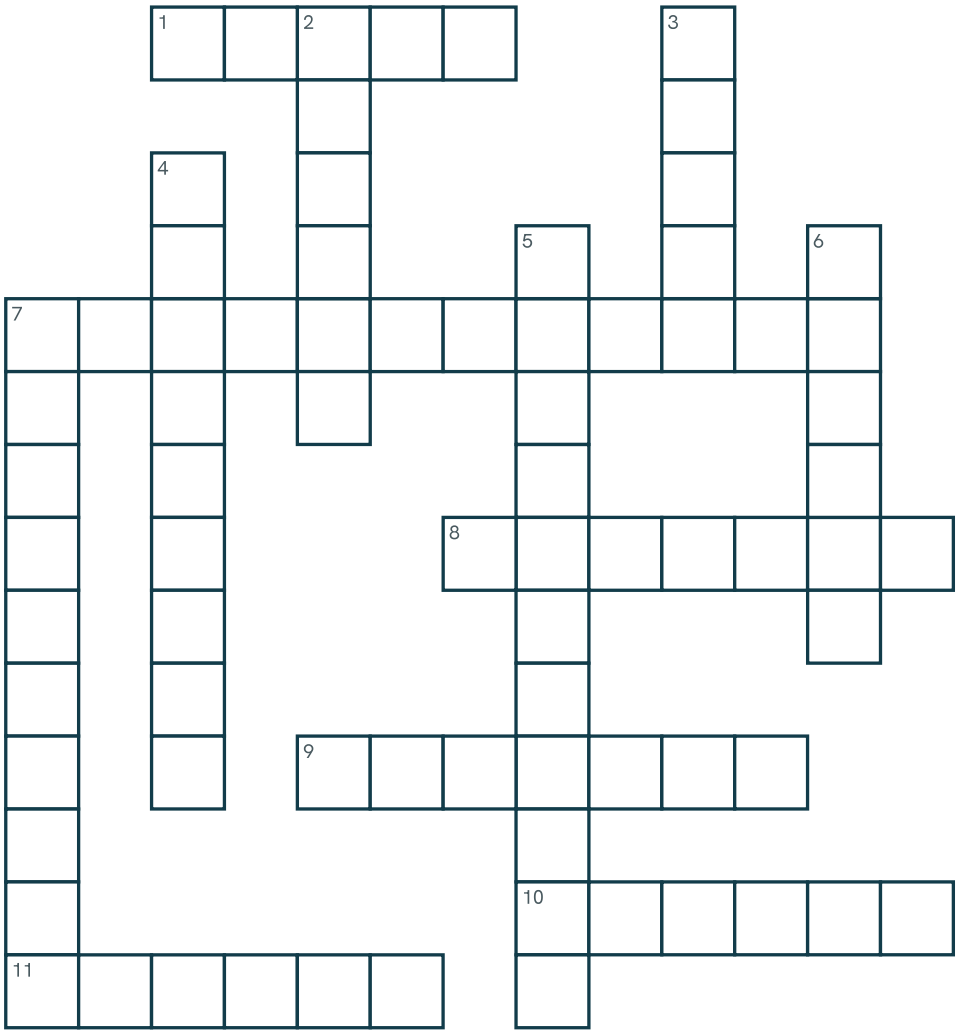
5 To design the recipe on paper (9)

8 Alcoholic base such as vodka or rum (6)

Crossword 95. RTD Drinks: Alcoholic and Non-Alcoholic, lesson 2

Production, shelf life and the rules

Fizz, safety and the paperwork before launch.



Across

1 Must meet the regulator's rules (5)

7 Additive such as sorbate or benzoate (12)

8 Unit for dissolved CO2 (7)

9 Checking samples over weeks (7)

10 Gas to keep out of the can (6)

11 Department that rules alcoholic RTDs (6)

Down

2 Small-scale shelf-life tests need a tight one (6)

3 Food regulator for non-alcoholic RTDs, by its letters (5)

4 How long the drink stays good (5,4)

5 Fizz, which changes how sweet a drink tastes (11)

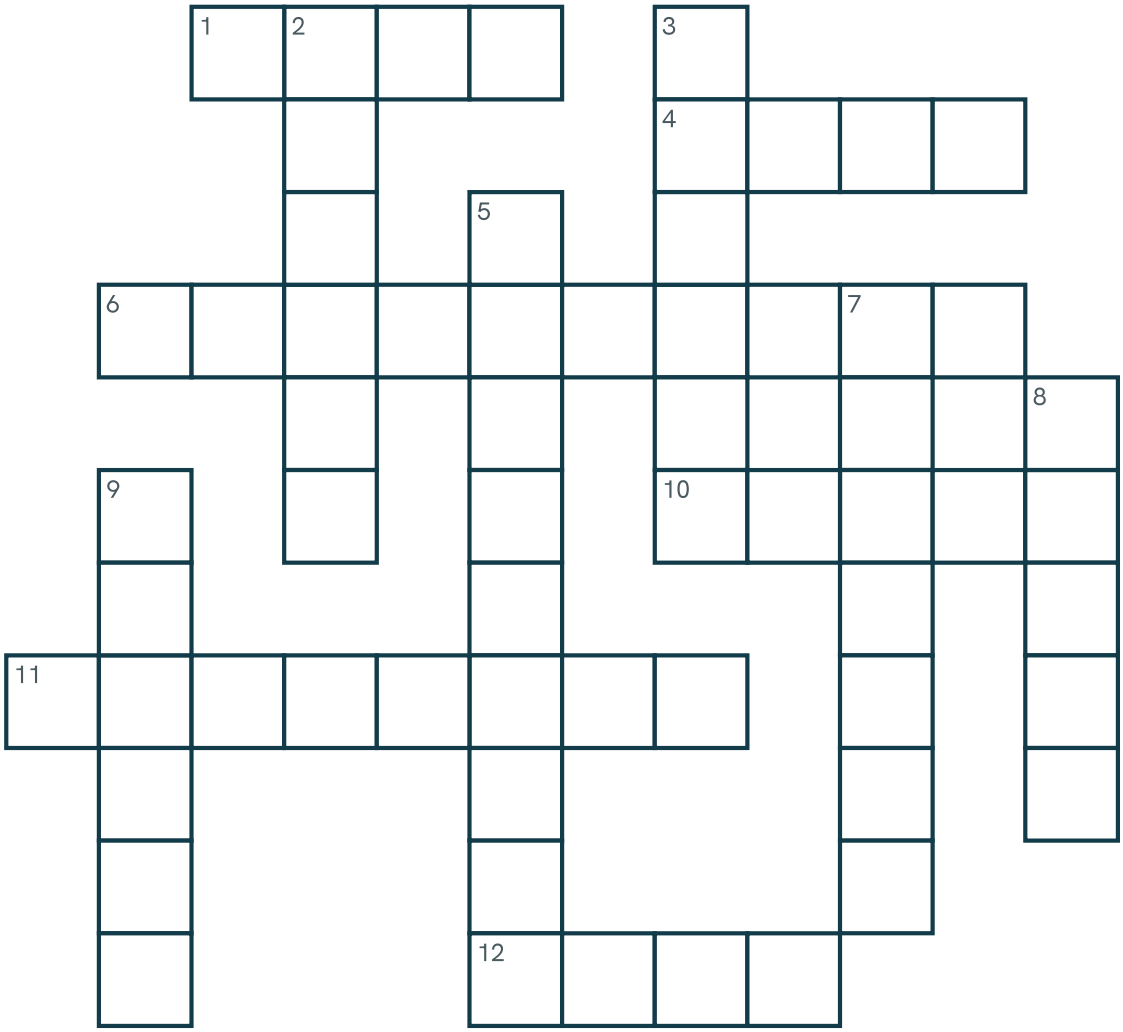
6 Machine that closes the can (6)

7 To heat-treat a drink for safety (10)

Crossword 96. Hop Water, lesson 1

Making hop water

Water, hops and bubbles, with nowhere to hide.



Across

1 Often added in small amounts for brightness (4)

4 Volatile compounds that carry hop aroma (4)

6 Fizzy, like every hop water (10)

10 Hop for a lime and tropical hop water (5)

11 Carbonated water flavoured with hops (3,5)

12 Resinous aroma from some hops (4)

Down

2 Aroma family of many US hops (6)

3 Hop with berry and tropical notes (6)

5 Soaking hops in cold water (4,5)

7 Concentrated hop product for dosing (7)

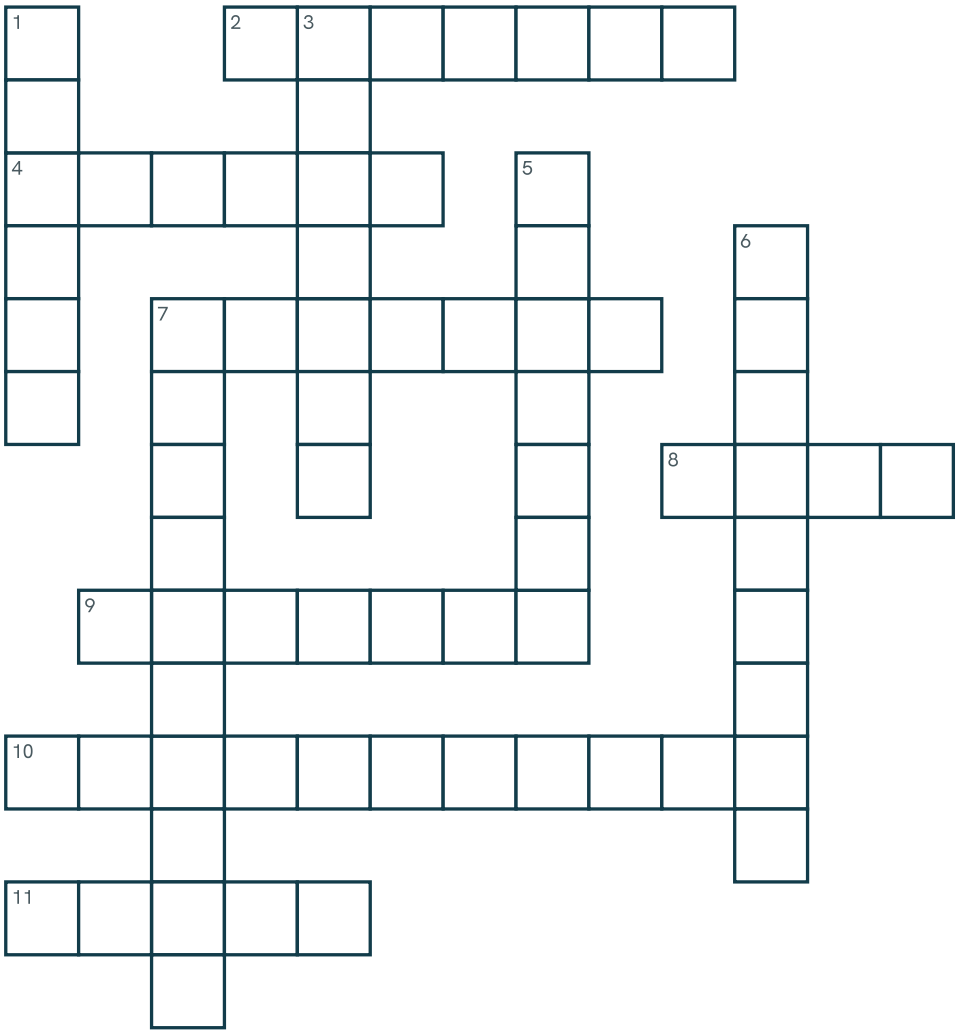
8 Its quality matters most, as nothing else hides it (5)

9 Aroma of many European hops (6)

Crossword 97. Hop Water, lesson 2

Stability and selling it

Safety, fizz and a price for a drink with no alcohol.



Across

2 Where many breweries sell it first (7)

4 Acid with a sharper, fruit-like edge (6)

7 Setting what the drink sells for (7)

8 ___ treatment protects a drink with no alcohol (4)

9 Filling kegs on brewery kit (7)

10 Fizz set in the keg or can (11)

11 Labelling regulator for hop water, by its letters (5)

Down

1 Acid that gives a soft sourness (6)

3 To lower pH for a safer, brighter drink (7)

5 Packing hop water into aluminium (7)

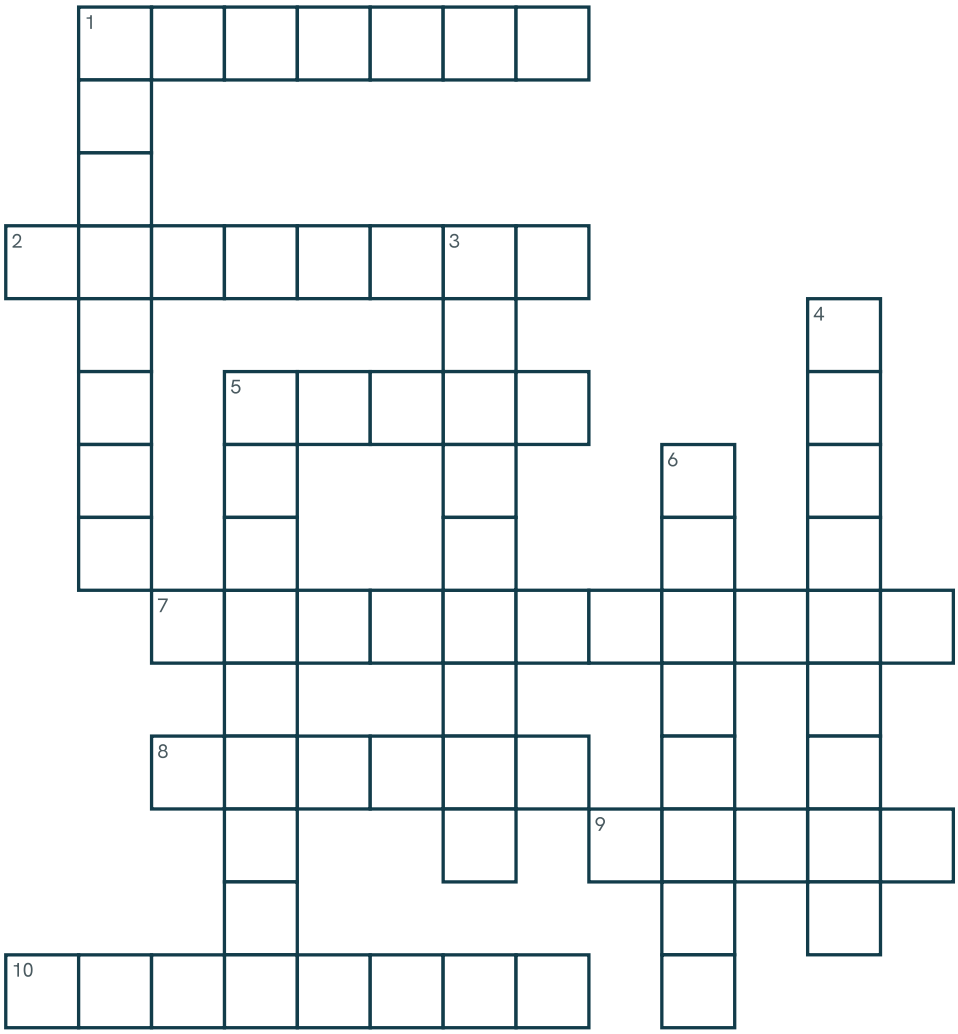
6 Rules on what goes on the can (9)

7 To heat-treat for safety (10)

Crossword 98. Hard Seltzer, lesson 1

Base and fermentation

A clean sugar ferment and the nutrients it needs.



Across

1 Rotten-egg fault from stressed yeast (7)

2 Corn sugar for a seltzer base (8)

5 Ferment with no fruity or sulphur notes (5)

7 Keep it steady to avoid off-flavours (11)

8 Salt added because a sugar wash cannot hold its pH (6)

9 Microbe chosen for a clean ferment (5)

10 Element yeast runs short of in sugar washes (8)

Down

1 Plan of nutrient additions by day (8)

3 ___ additions: nutrient given over several days (9)

4 Base with almost no yeast nutrients (5,4)

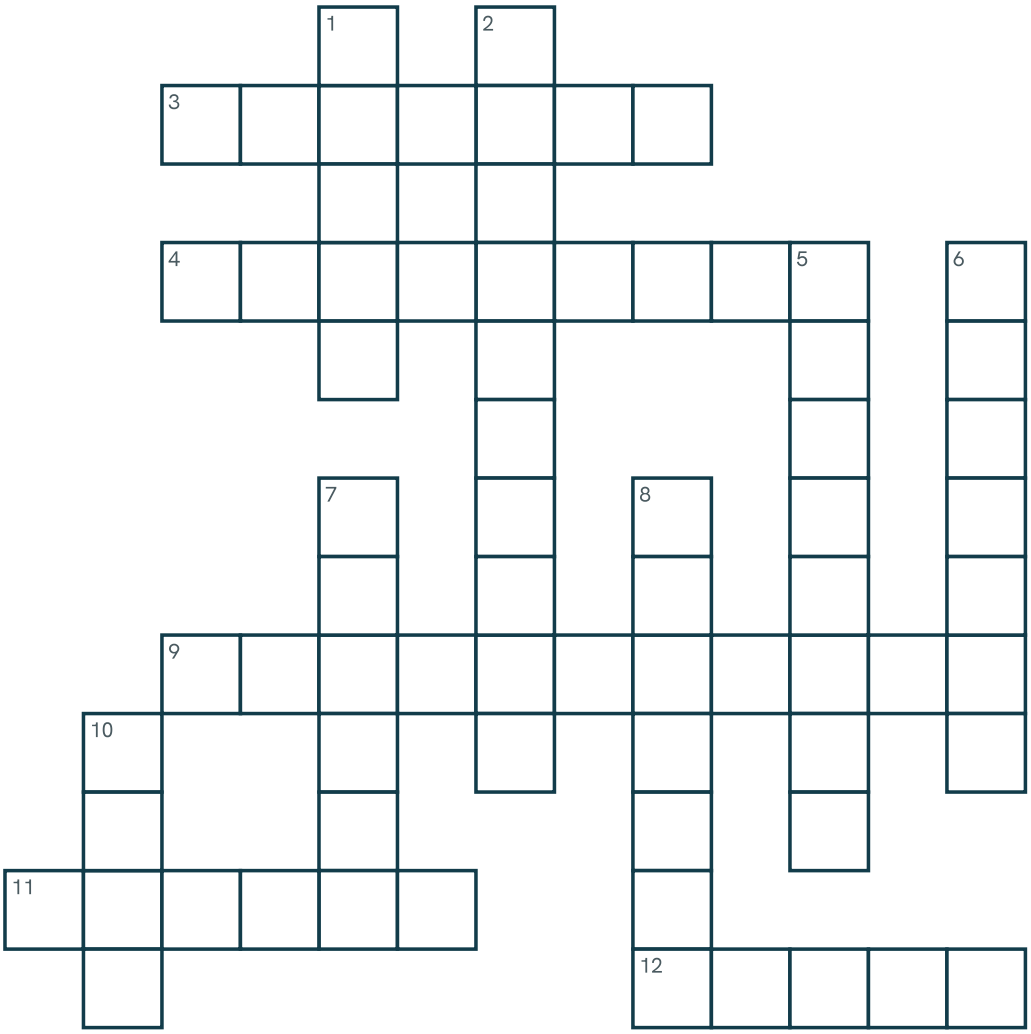
5 Sucrose from sugar cane (4,5)

6 Yeast food added in stages (8)

Crossword 99. Hard Seltzer, lesson 2

Finish and the rules

Clarity, flavour and tax for a seltzer range.



Across

3 Water-white look a seltzer needs (7)

4 Kept to a touch in most seltzers (9)

9 Fizz, usually set high (11)

11 Adding an agent to clear haze (6)

12 Three flavours from one base make one (5)

Down

1 How a seltzer is ___ depends on the state (5)

2 Passing the base through a filter (10)

5 Letting yeast drop out over time (8)

6 How most seltzer reaches the shelf (7)

7 Filter medium that strips off-flavours (6)

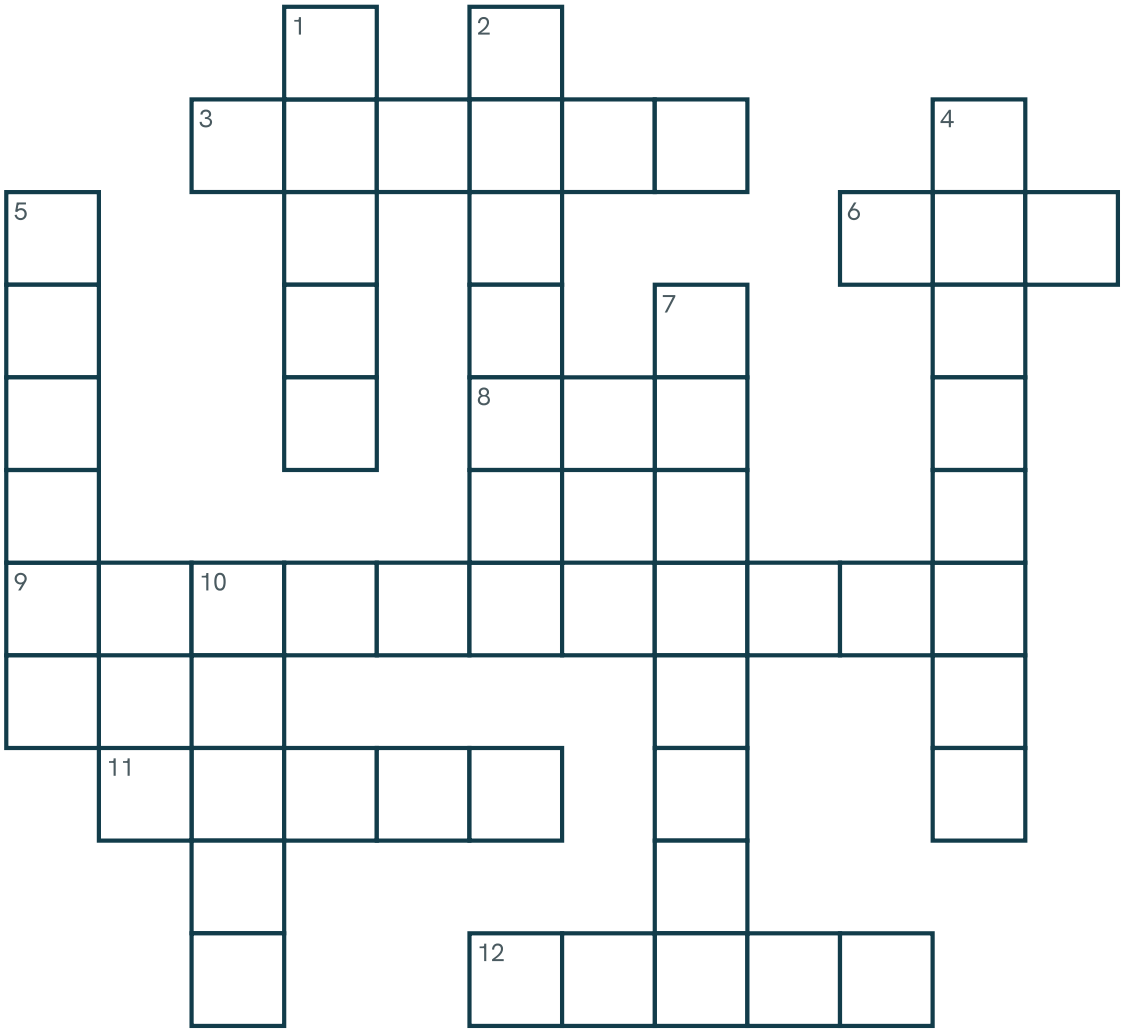
8 Added after fermentation, often from extracts (7)

10 Gives a seltzer its bright finish (4)

Crossword 100. Kombucha, lesson 1

The first ferment

Tea, sugar and a living culture.



Across

3 Acid that gives kombucha its tang (6)

6 Daily record of pH and taste (3)

8 Black or green base for kombucha (3)

9 Warmer runs faster (11)

11 Food for the culture (5)

12 Turns sugar into alcohol for the bacteria (5)

Down

1 Rubbery culture of bacteria and yeast (5)

2 Liquid from a past batch that lowers pH (7)

4 What you taste for each day (8)

5 Why the pH must fall fast (6)

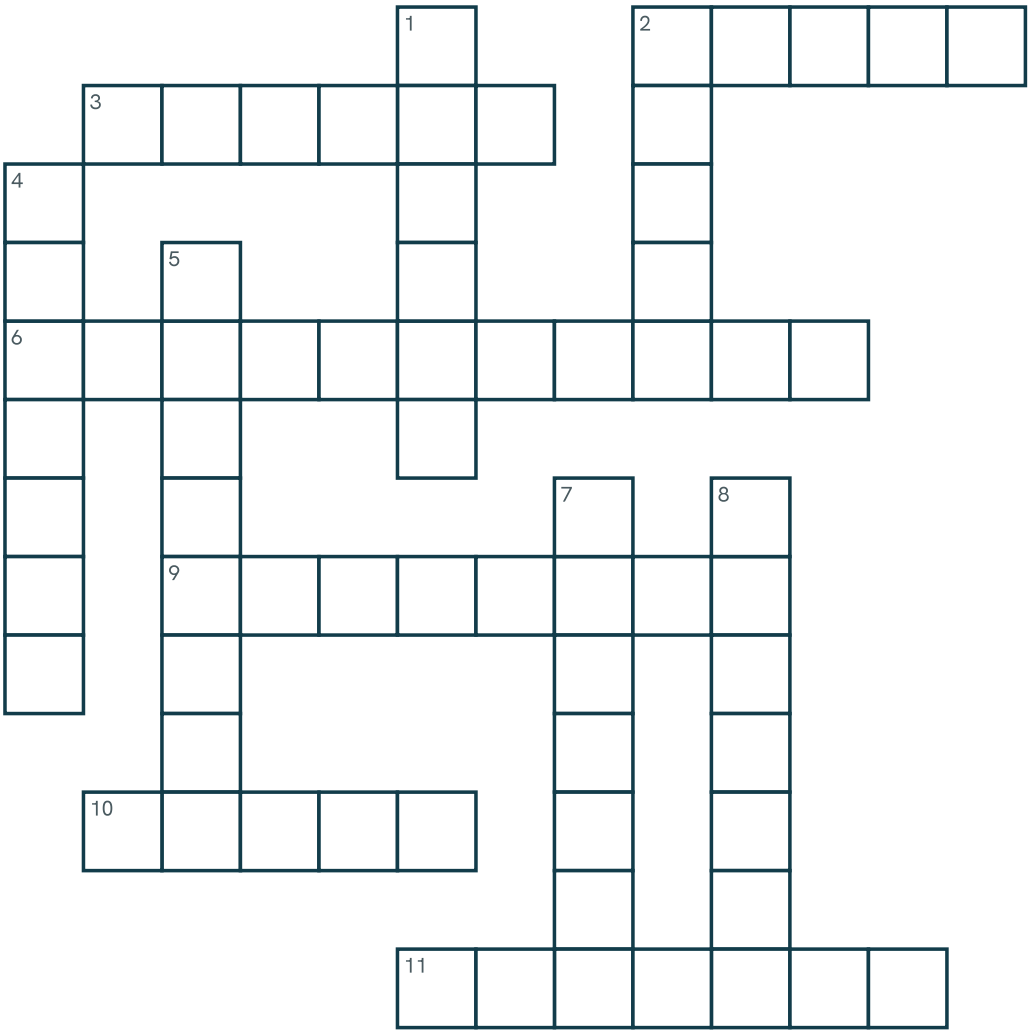
7 Acetic acid makers in the culture (8)

10 Fuzzy, dry spots on top mean bin the batch (5)

Crossword 101. Kombucha, lesson 2

Flavour, fizz and scaling up

Second ferment, safe bottles and a weekly schedule.



Across

2 Mango or berries for a second ferment (5)

3 Ginger or cardamom in the bottle (6)

6 Fizz built up in the bottle (11)

9 Weekly brewing plan (8)

10 Mint or tulsi for flavour (5)

11 Clean kit matters more as you scale (7)

Down

1 ___ ferment: fruit and spices in the bottle (6)

2 Labelling regulator for kombucha, by its letters (5)

4 Can creep up in warm conditions (7)

5 Too much can burst a bottle (8)

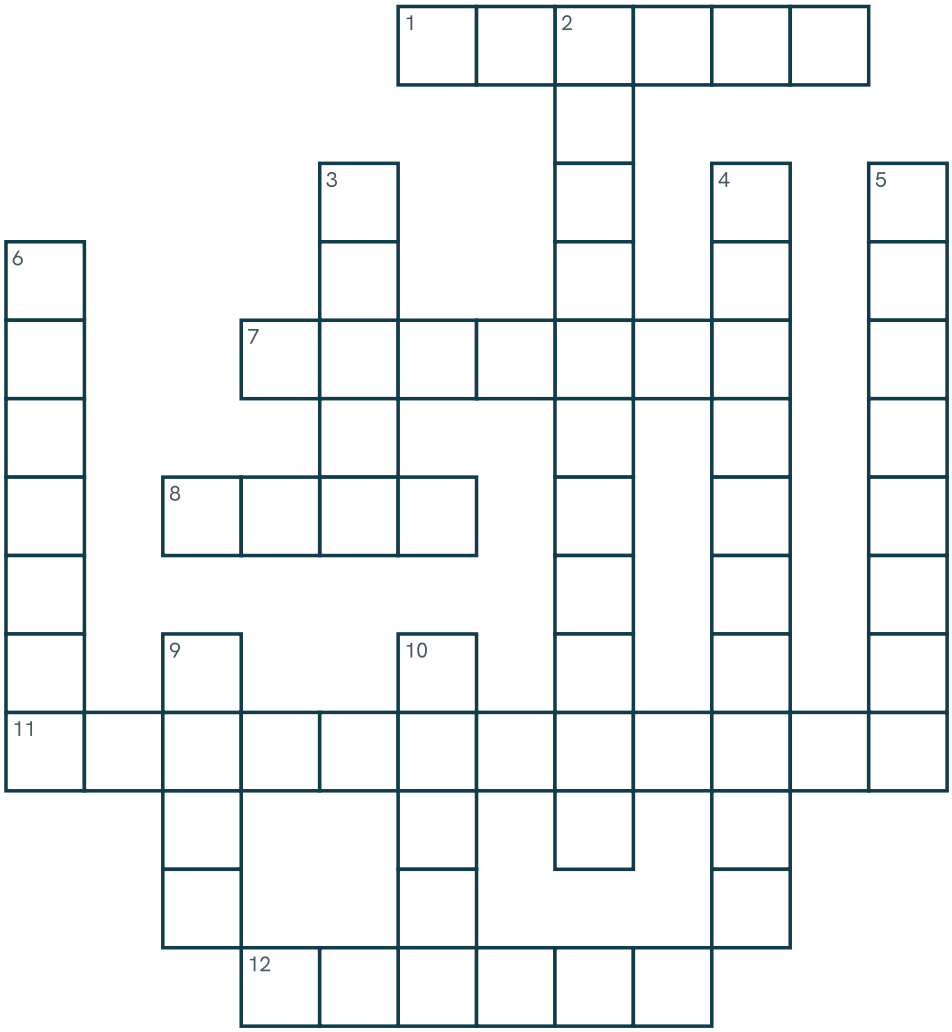
7 Opening bottles to let gas out (7)

8 What to do with alcohol, not guess (7)

Crossword 102. Non-Alcoholic Beer, lesson 1

Making beer without the alcohol

Two routes to beer with almost no alcohol.



Across

1 ___ distillation boils off alcohol at low heat (6)

7 Sugar some special yeasts cannot ferment (7)

8 Grain that adds body (4)

11 To remove alcohol after fermentation (12)

12 Cold hop addition for aroma (3,3)

Down

2 Fizz that lifts thin body (11)

3 Choose one that cannot ferment maltose (5)

4 ___ malts give body and colour (10)

5 Barrier used in reverse osmosis (8)

6 ___ fermentation: stop the alcohol forming (7)

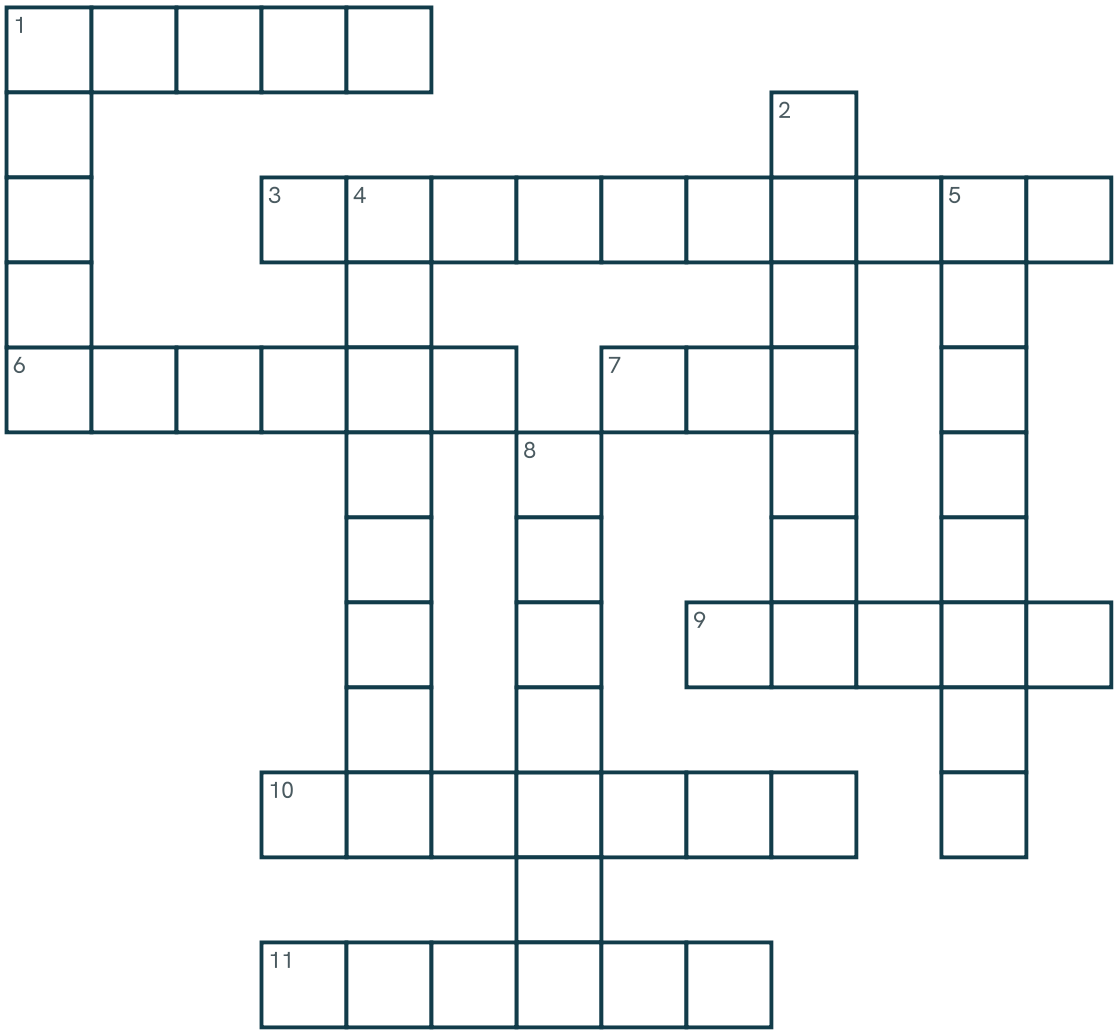
9 Run it hot to leave fewer fermentable sugars (4)

10 Sweet, raw flavour of many first attempts (5)

Crossword 103. Non-Alcoholic Beer, lesson 2

Stability, testing and selling it

Keeping it safe, measuring it right and selling it.



Across

1 ABV ___ for a non-alcoholic claim, which has changed (5)

3 Heat treatment a beer with no alcohol needs (10)

6 First release to market (6)

7 Alcohol by volume, by its letters (3)

9 Regulator that sets the label limit, by its letters (5)

10 Matters more with no alcohol to protect the beer (7)

11 Keep it low to stop staling (6)

Down

1 Must match the current rules (5)

2 One group buying alcohol-free beer (7)

4 Hard to get when measuring tiny amounts of alcohol (8)

5 What unprotected beer is open to (8)

8 Setting what the beer sells for (7)

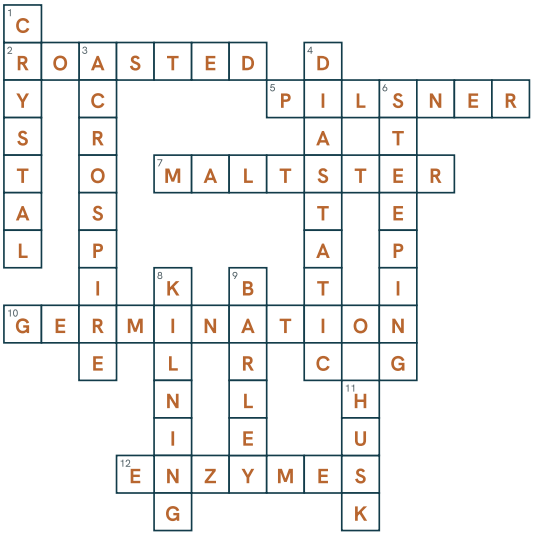


THE BACK OF THE BOOK

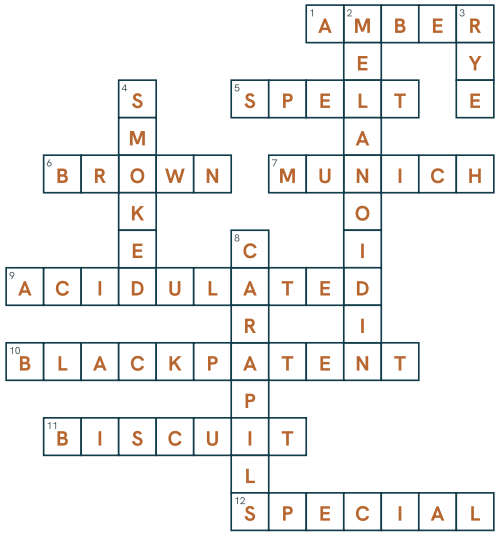
Answers

Every grid, solved. Only for checking, of course.

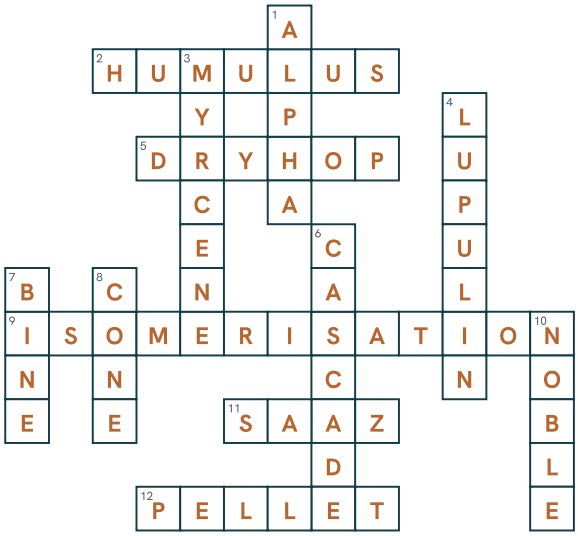
1. Malt basics (page 7)



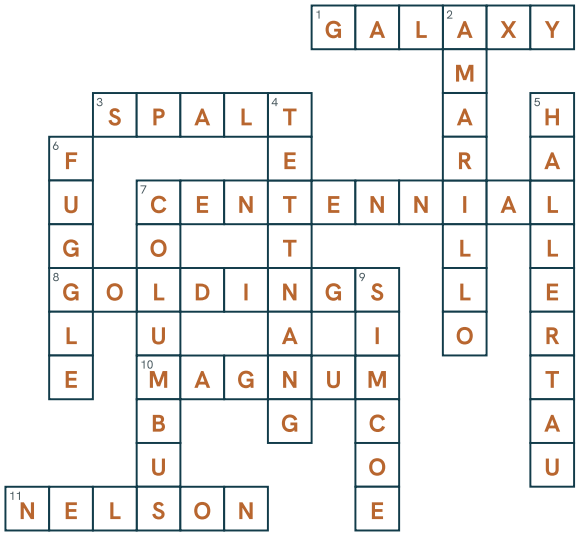
2. Malt varieties (page 8)



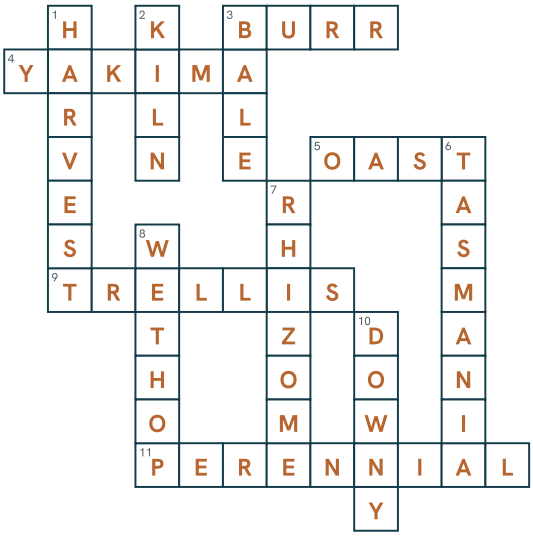
3. Hops (page 9)



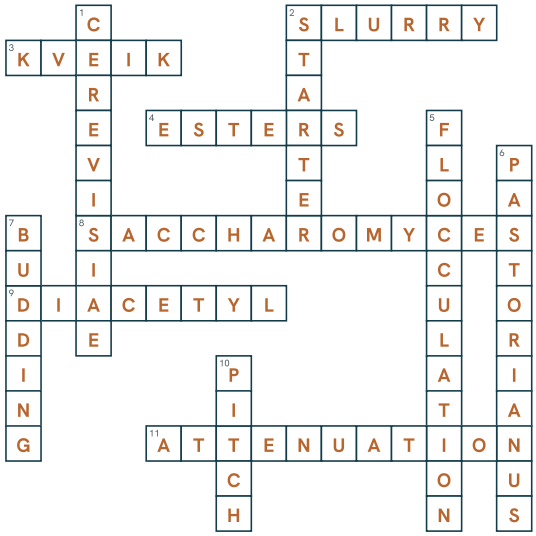
4. Hop varieties (page 10)



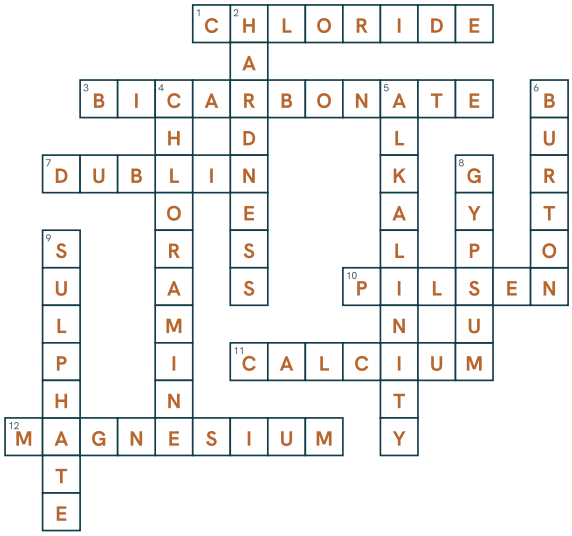
5. Hop farming (page 11)



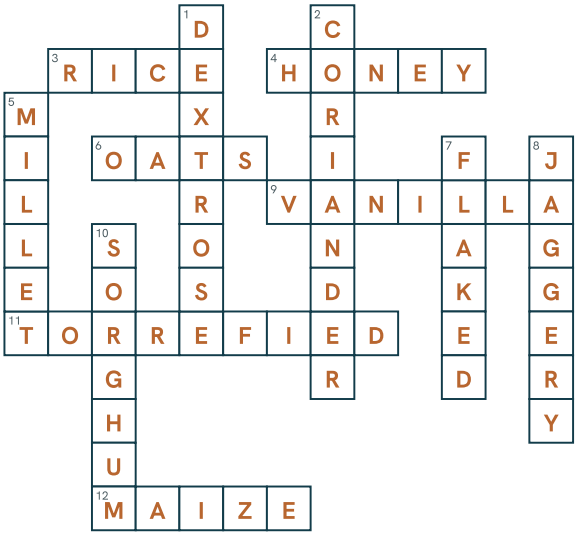
6. Yeast (page 12)



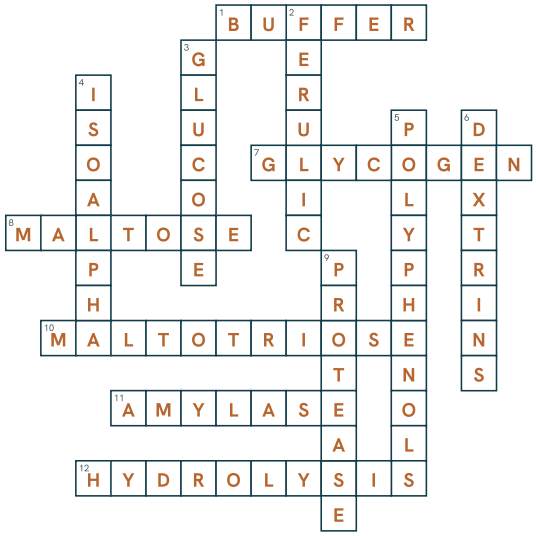
7. Brewing water (page 13)



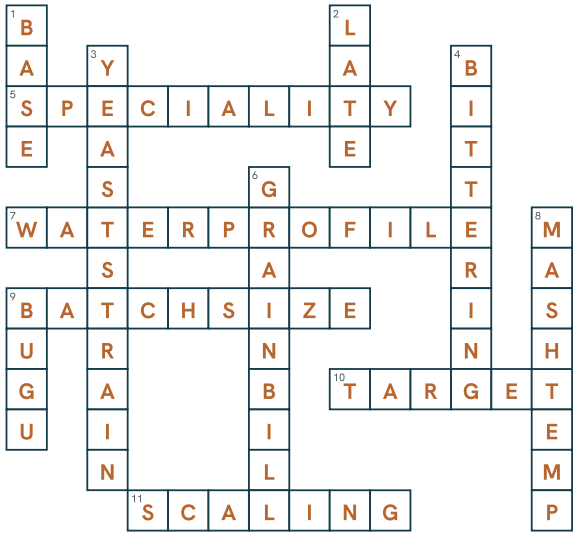
8. Adjuncts (page 14)



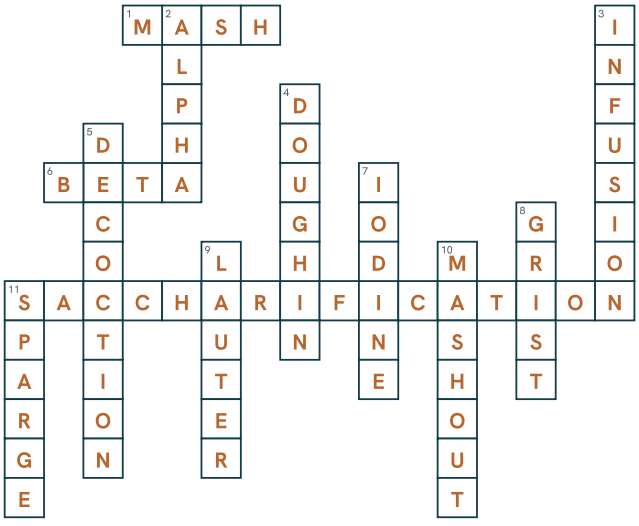
9. Brewing chemistry (page 15)



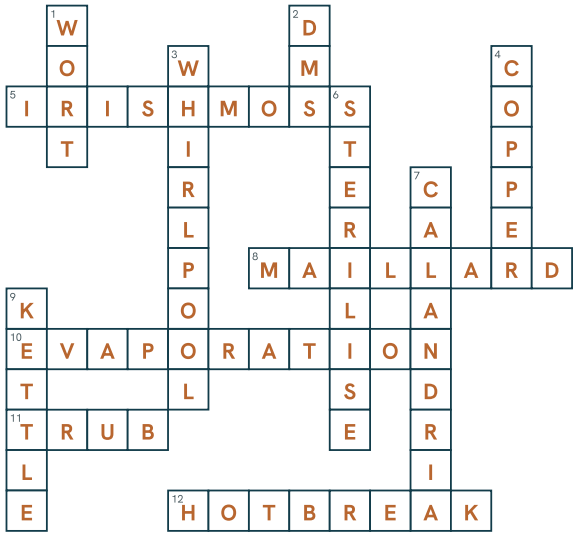
10. Recipe design (page 16)



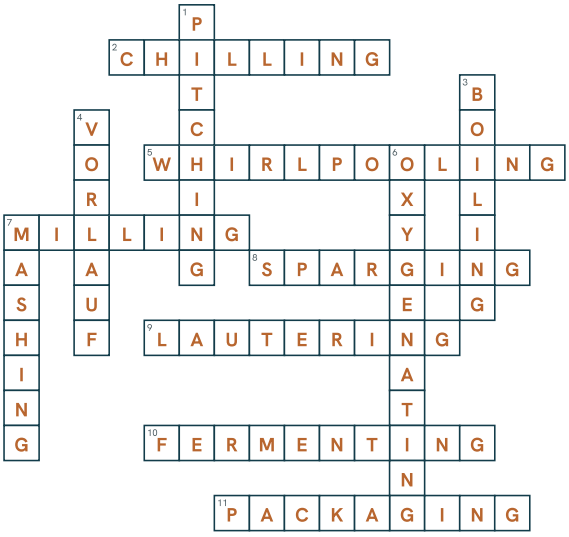
11. Mashing (page 18)



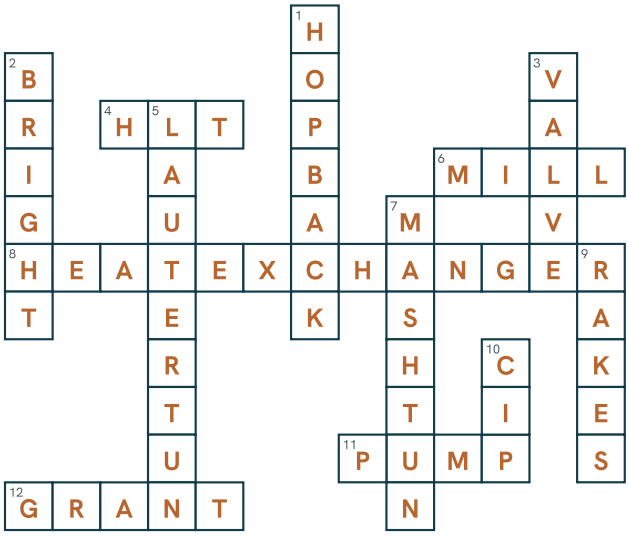
12. The boil (page 19)



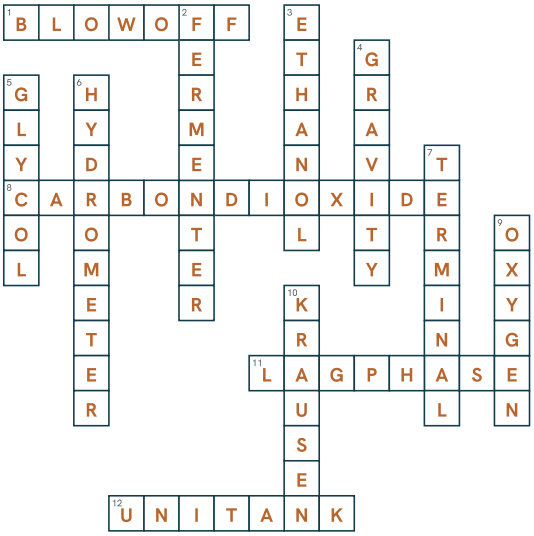
13. Brew day in order (page 20)



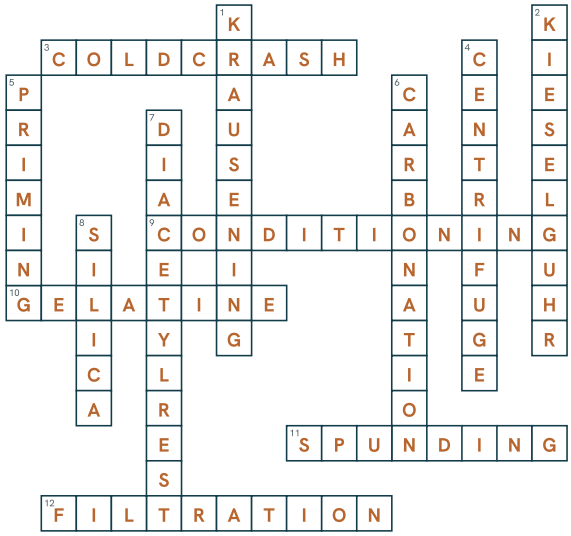
14. Brewhouse kit (page 21)



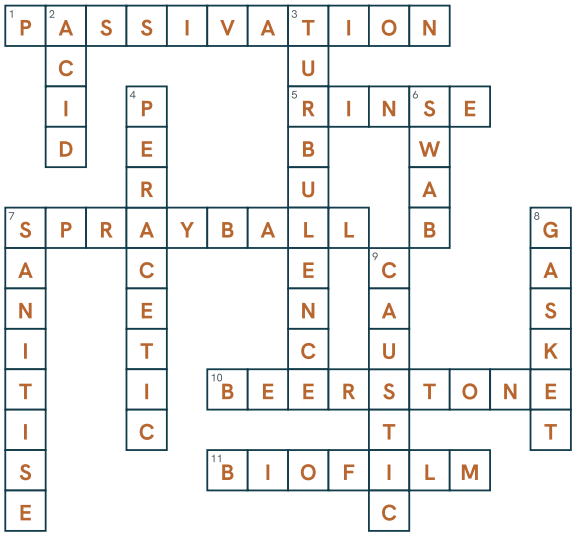
15. Fermentation (page 22)



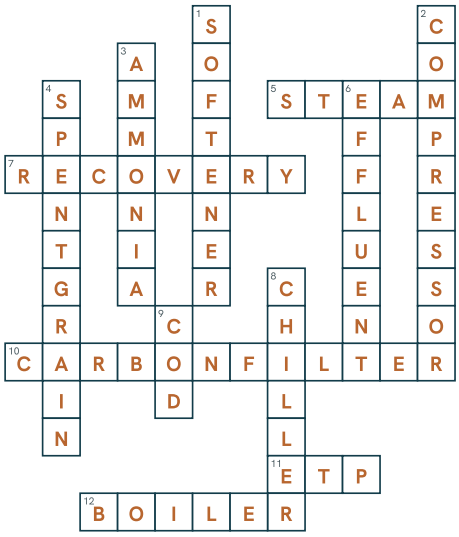
16. Cellar and conditioning (page 23)



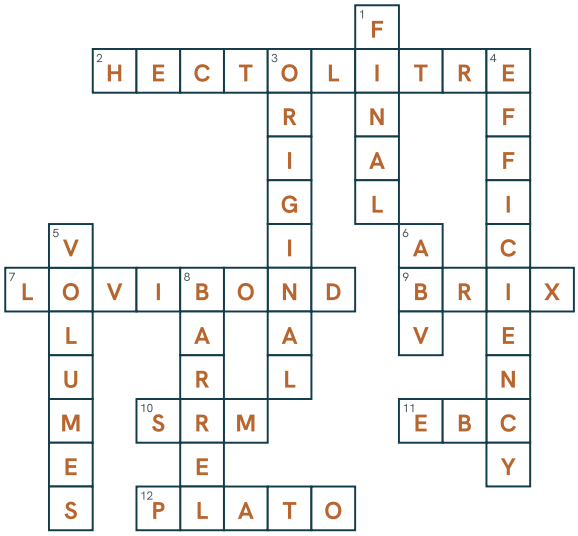
17. Cleaning and hygiene (page 24)



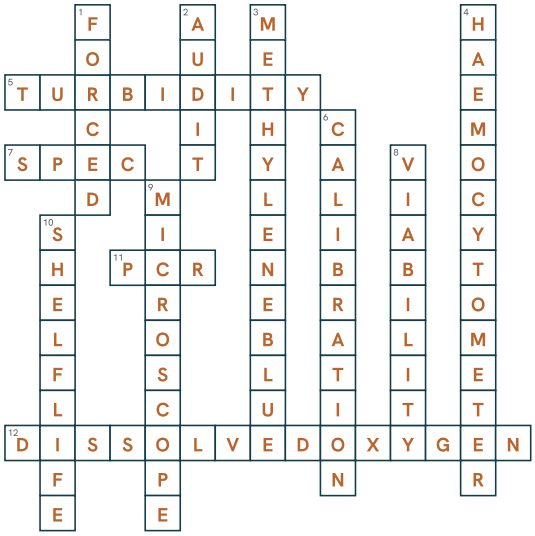
18. Utilities and the plant (page 25)



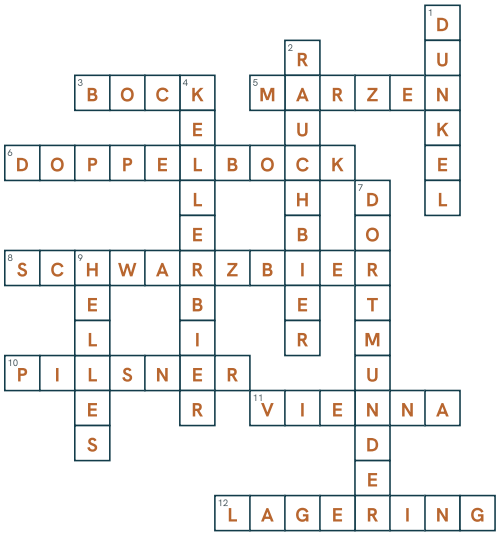
19. Brewing numbers (page 26)



20. Quality control (page 27)



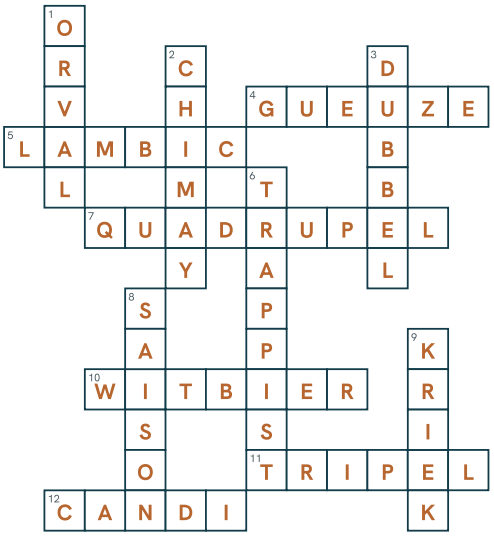
21. Lagers (page 29)



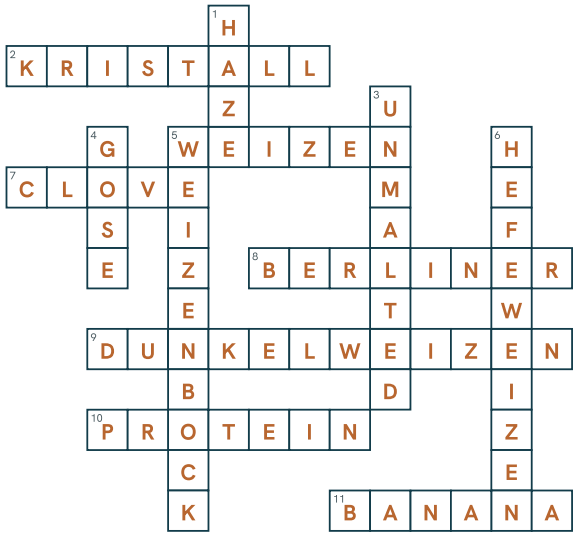
22. British ales (page 30)



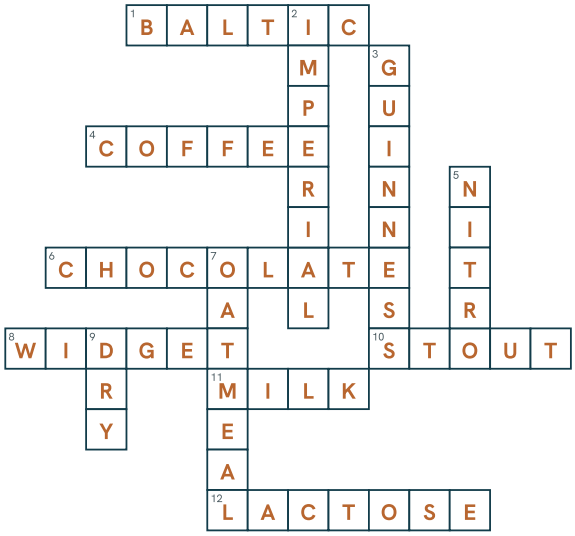
23. Belgian beers (page 31)



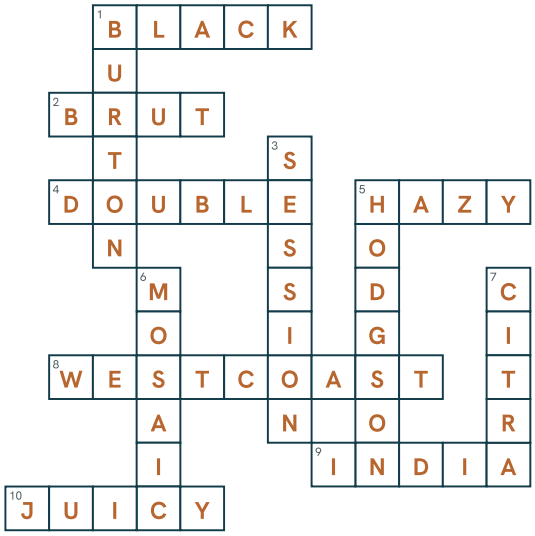
24. Wheat beers (page 32)



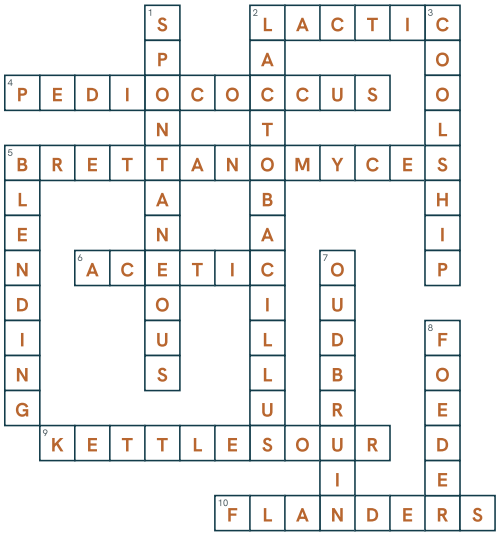
25. Stouts and porters (page 33)



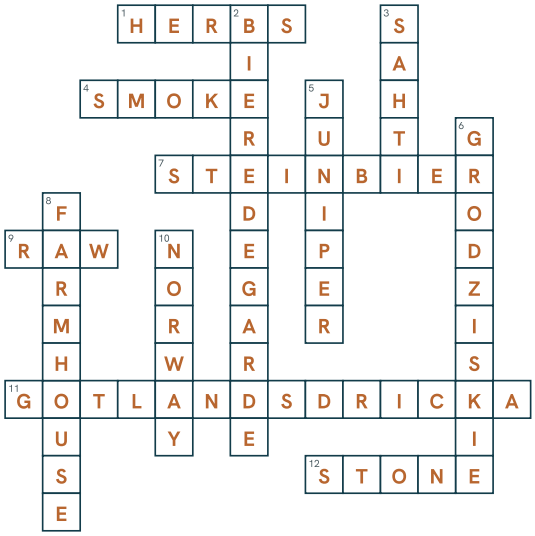
26. India pale ale (page 34)



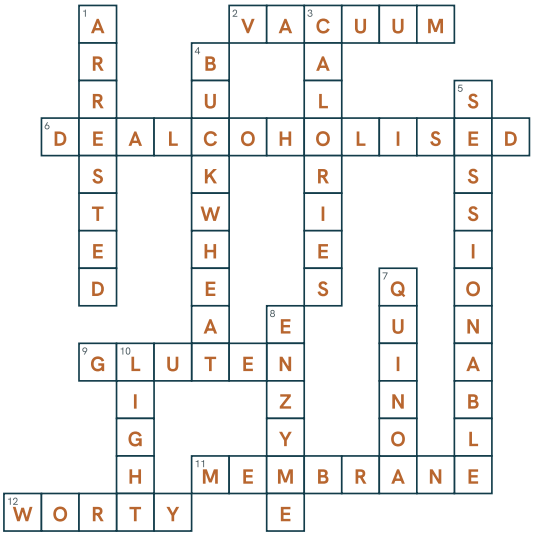
27. Sour beers (page 35)



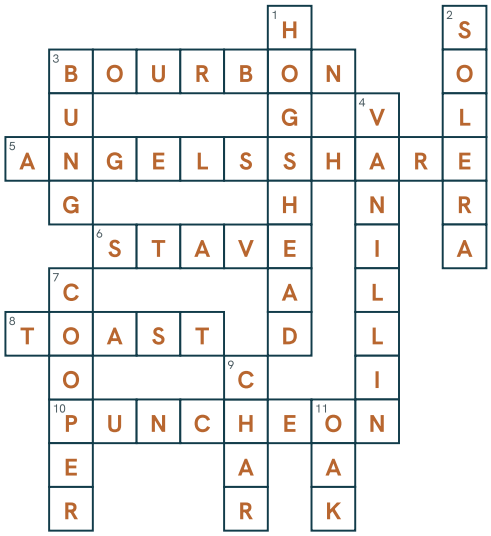
28. Farmhouse traditions (page 36)



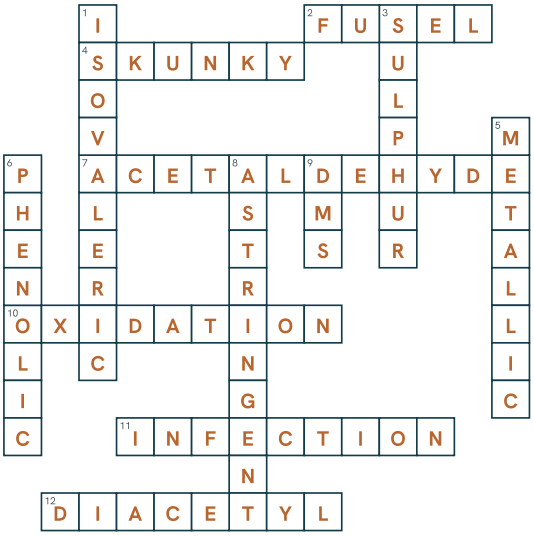
29. Low and no alcohol (page 37)



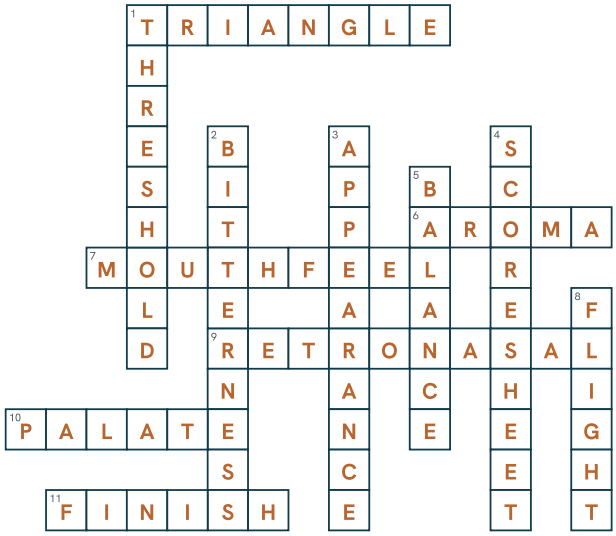
30. Barrel ageing (page 38)



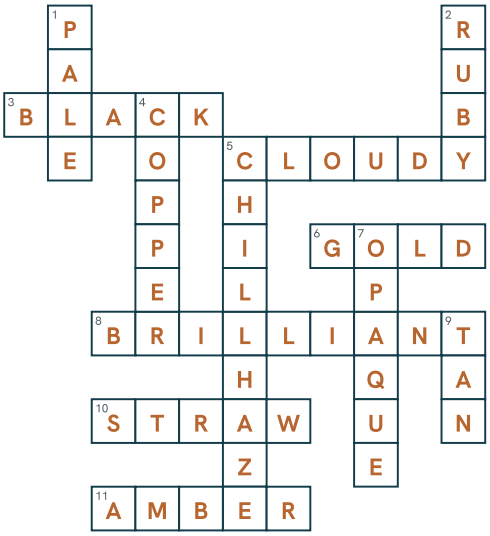
31. Off-flavours (page 40)



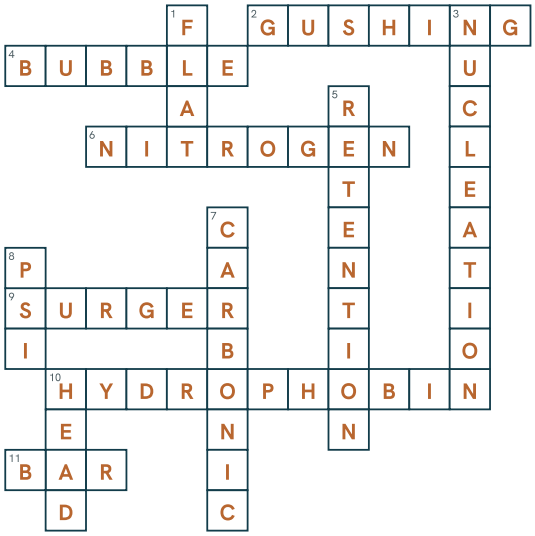
32. Tasting and sensory (page 41)



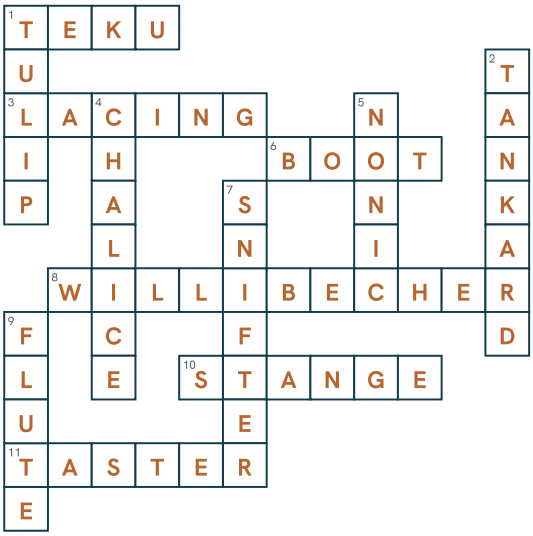
33. Colour and clarity (page 42)



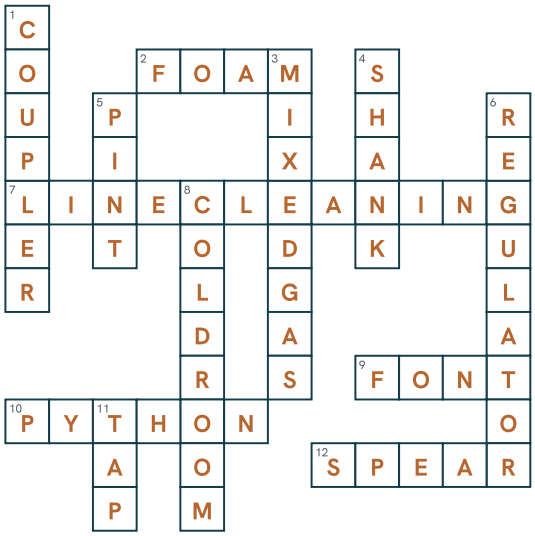
34. Foam and fizz (page 43)



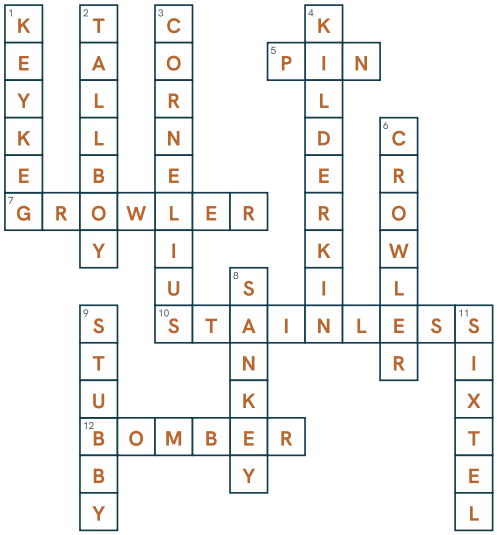
35. Glassware (page 44)



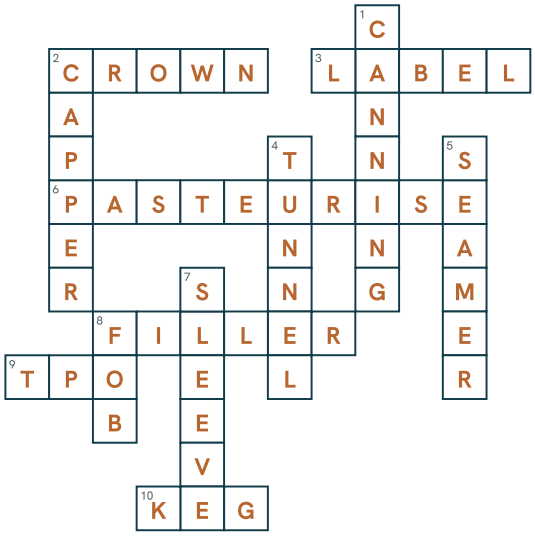
36. Draught and serving (page 45)



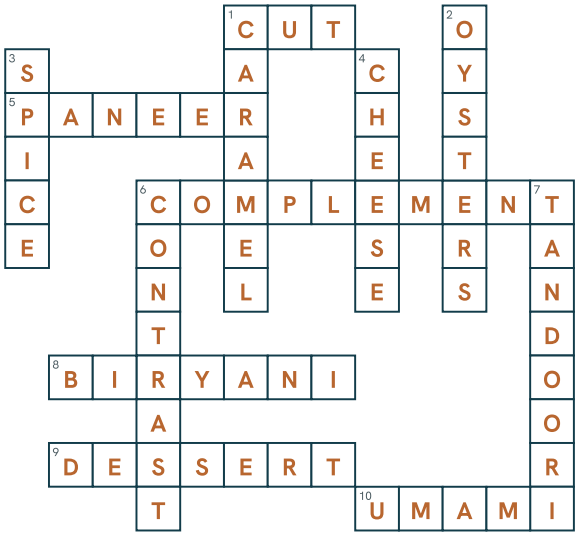
37. Kegs, casks and bottles (page 46)



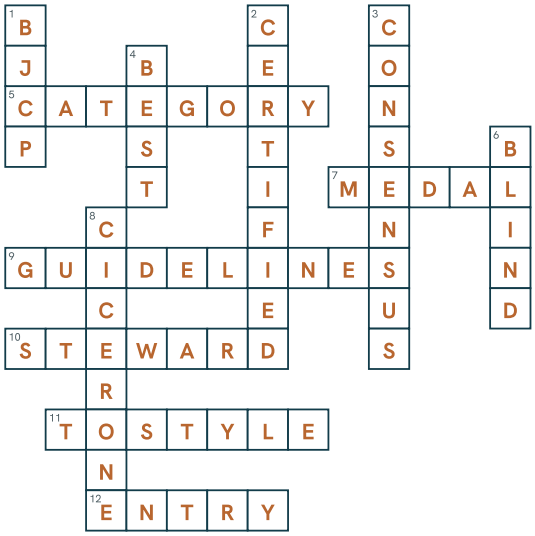
38. Packaging (page 47)



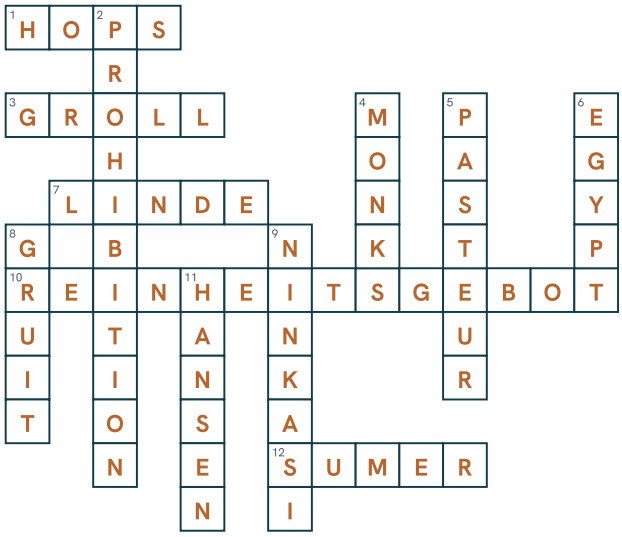
39. Beer and food (page 48)



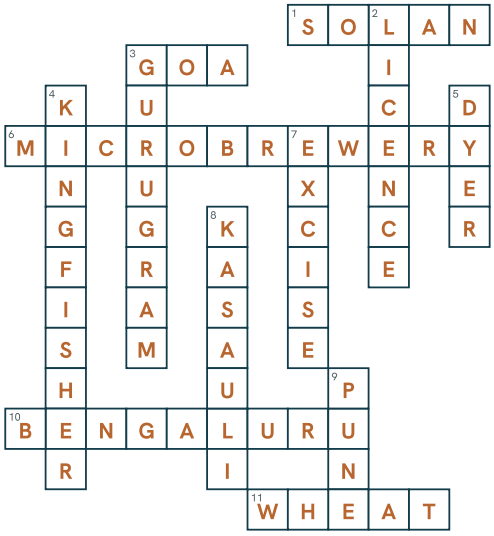
40. Judging and competitions (page 49)



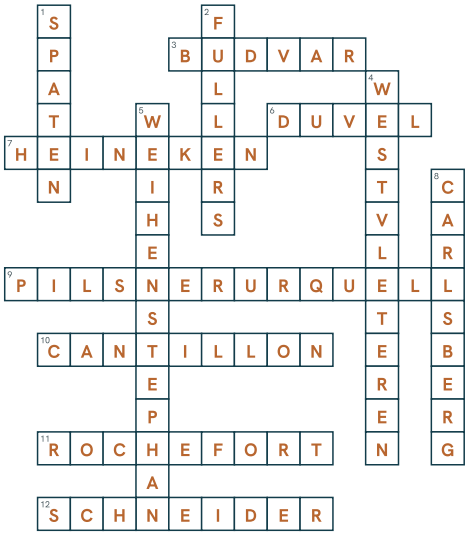
41. Beer history (page 51)



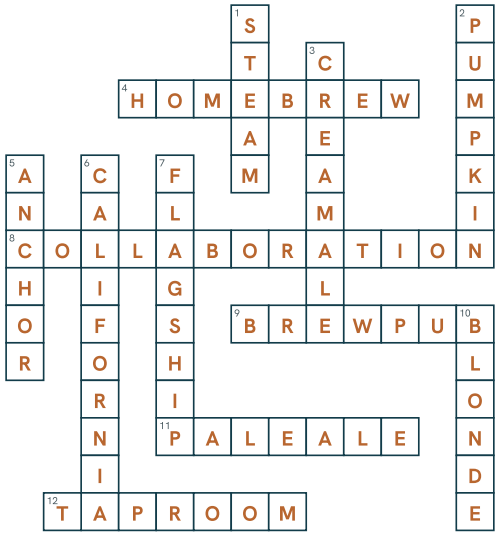
42. Beer in India (page 52)



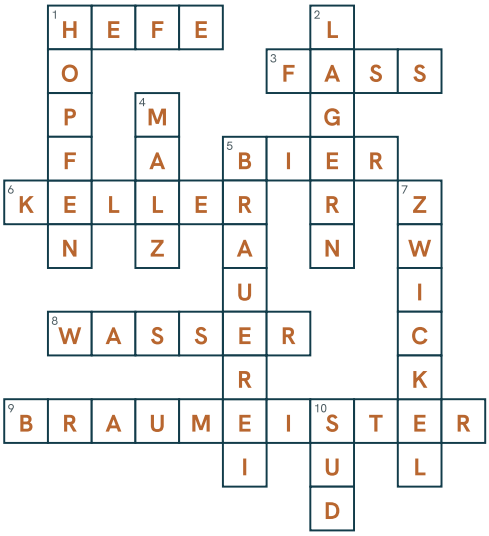
43. Famous breweries (page 53)



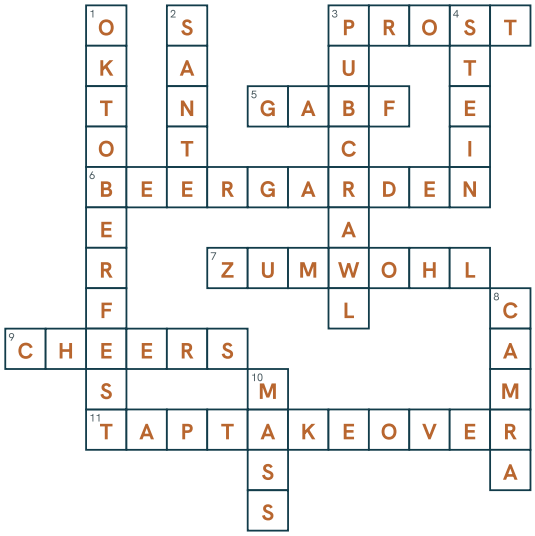
44. American craft (page 54)



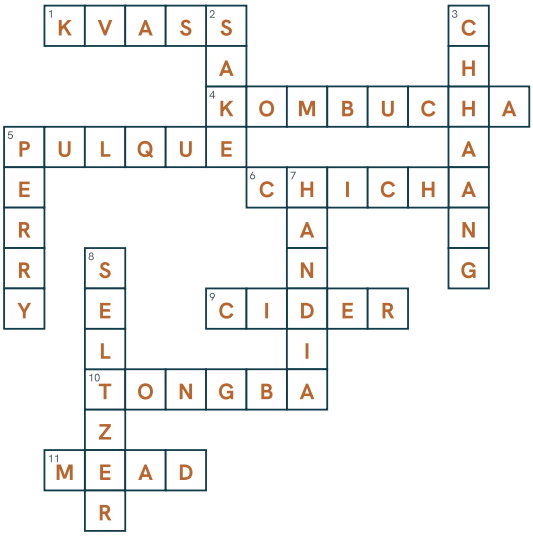
45. German beer words (page 55)



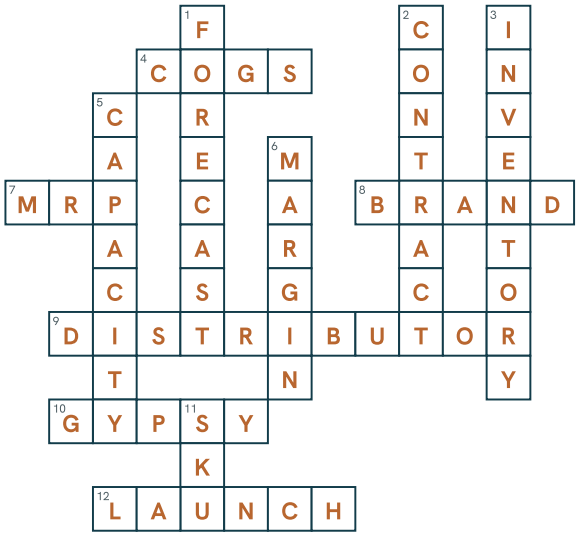
46. Festivals and culture (page 56)



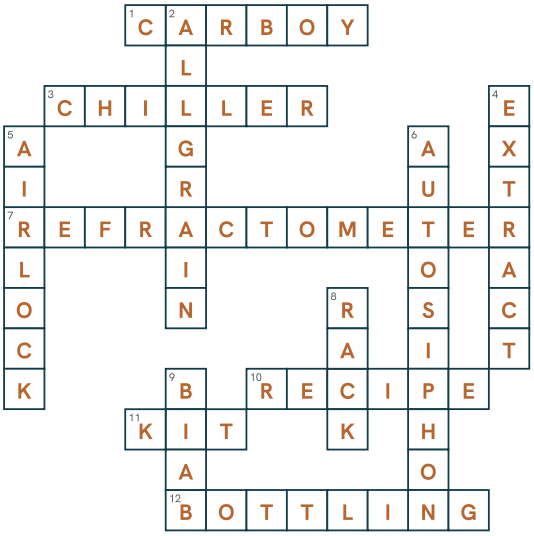
47. Other fermented drinks (page 57)



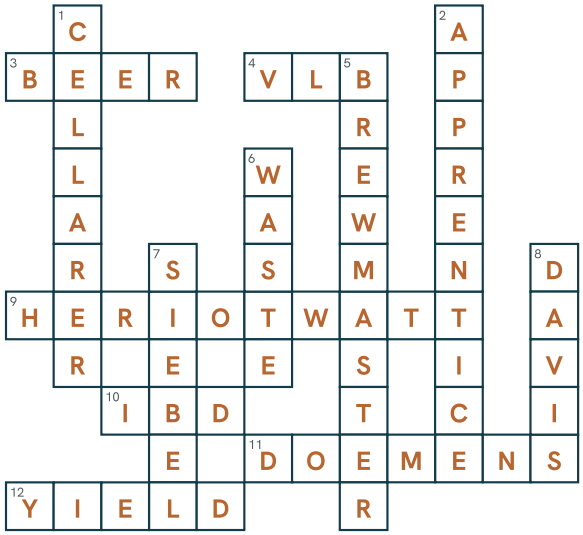
48. The business of beer (page 58)



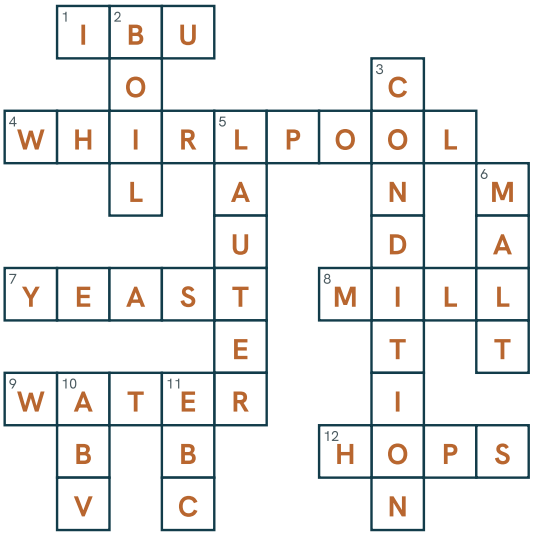
49. Homebrewing (page 59)



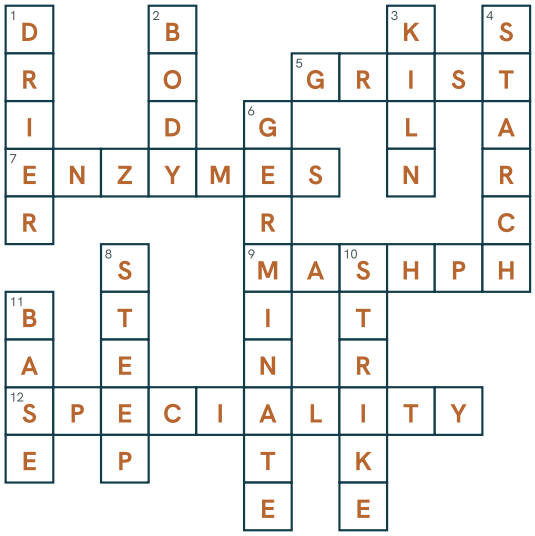
50. Schools and careers (page 60)



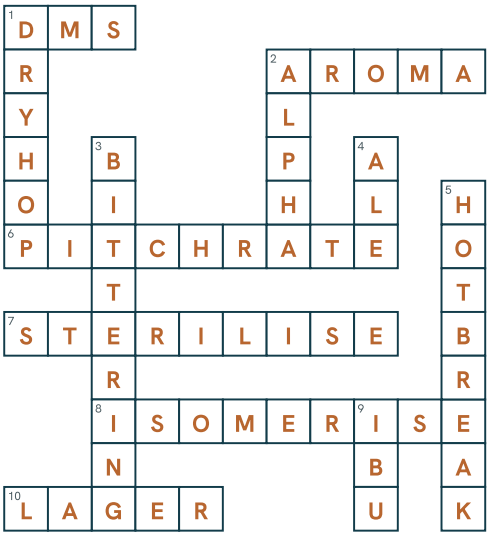
51. Grain to glass (page 62)



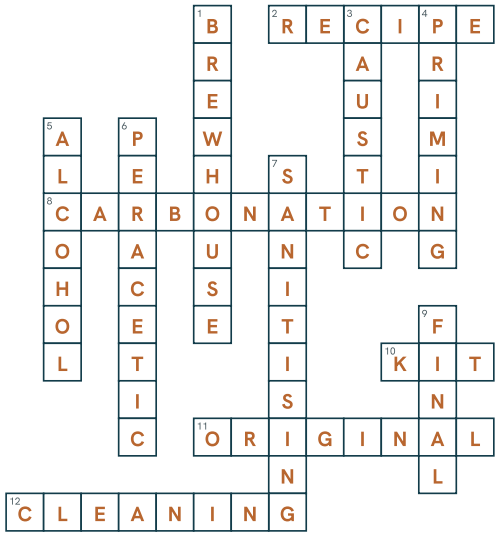
52. Malt, water and the mash (page 63)



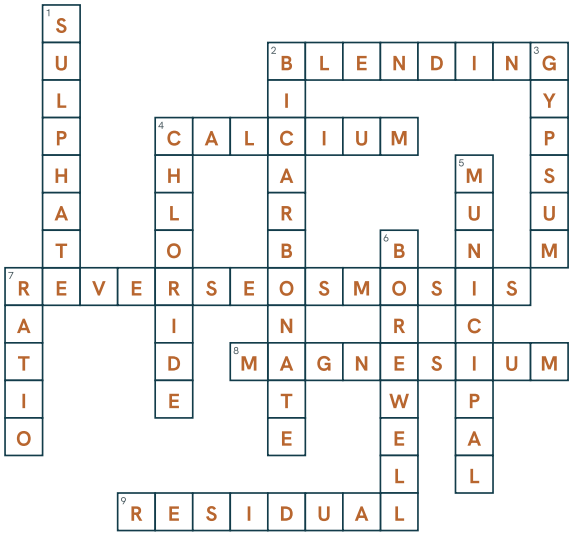
53. Hops, boil and yeast (page 64)



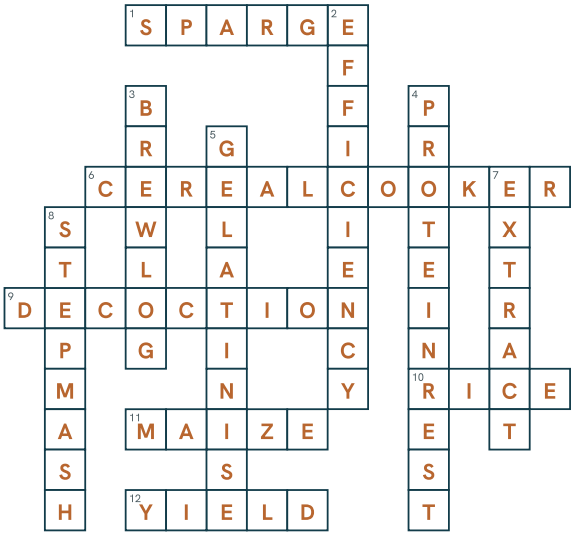
54. Sanitation, kit and your first recipe (page 65)



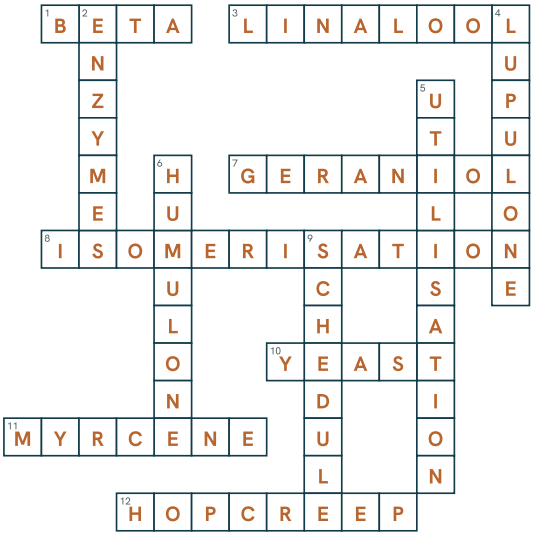
55. Brewing water chemistry (page 66)



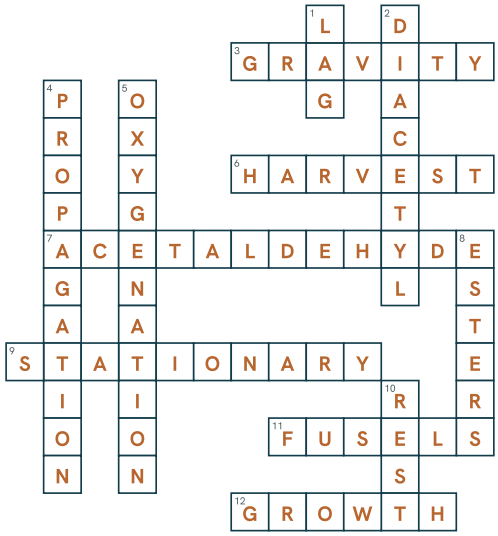
56. Mash chemistry and advanced mashing (page 67)



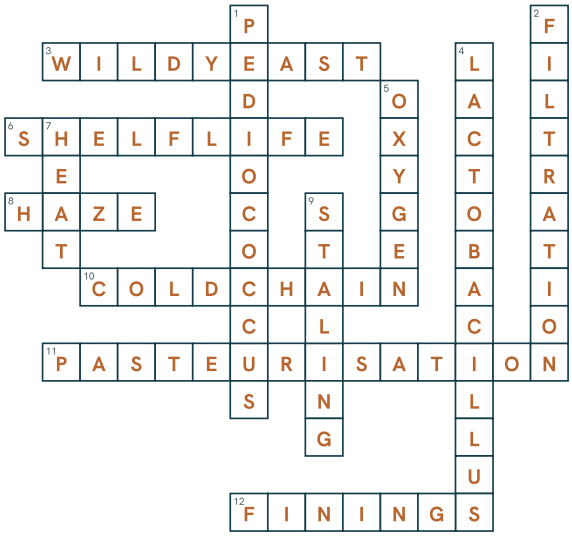
57. Hop chemistry (page 68)



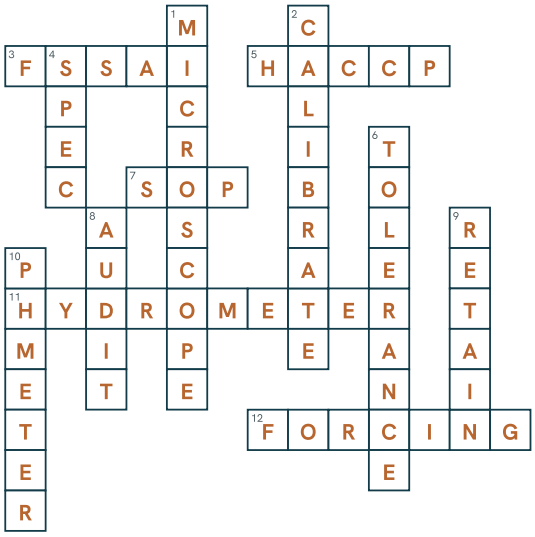
58. Yeast and fermentation microbiology (page 69)



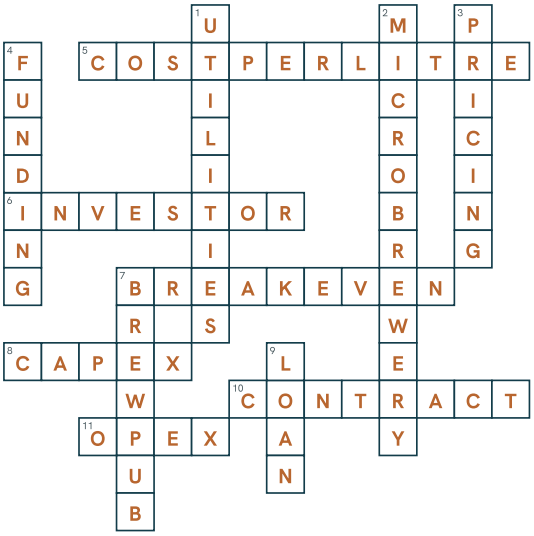
59. Contamination, stability and shelf life (page 70)



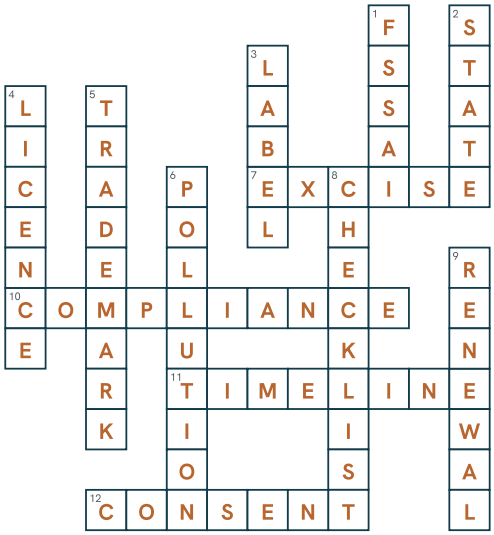
60. Quality assurance and control (page 71)



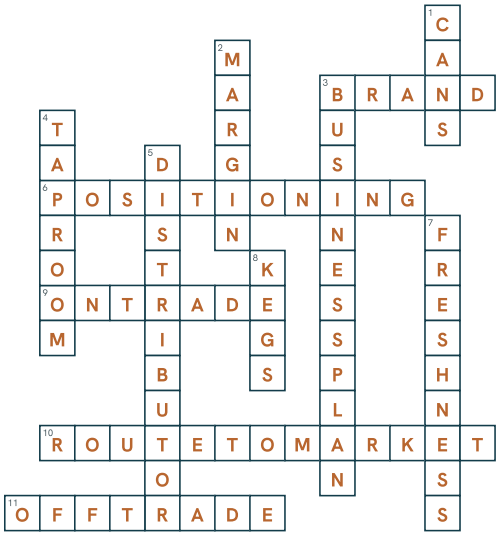
61. Plan and finance (page 72)



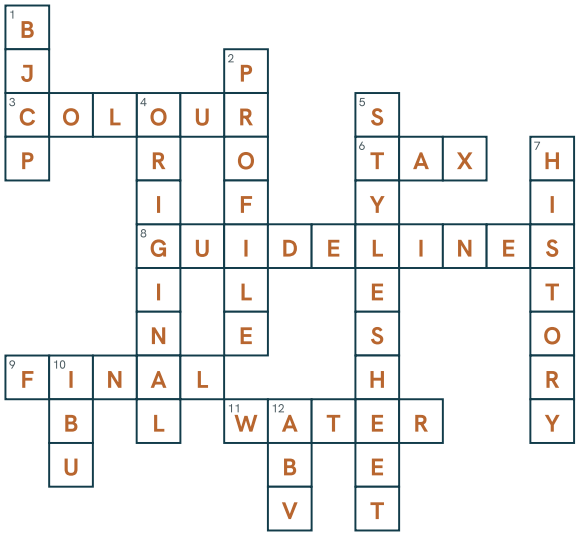
62. Licensing and regulation in India (page 73)



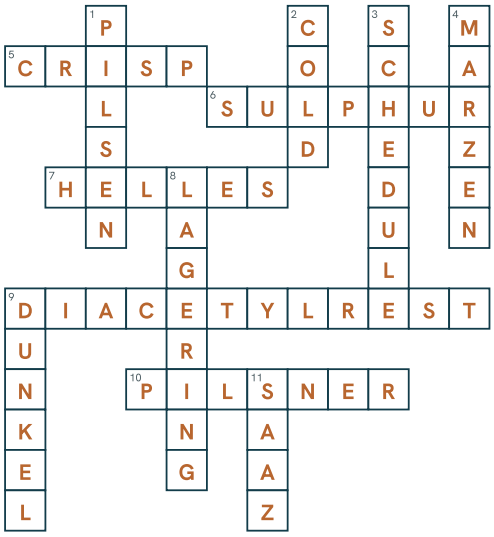
63. Market, brand and distribution (page 74)



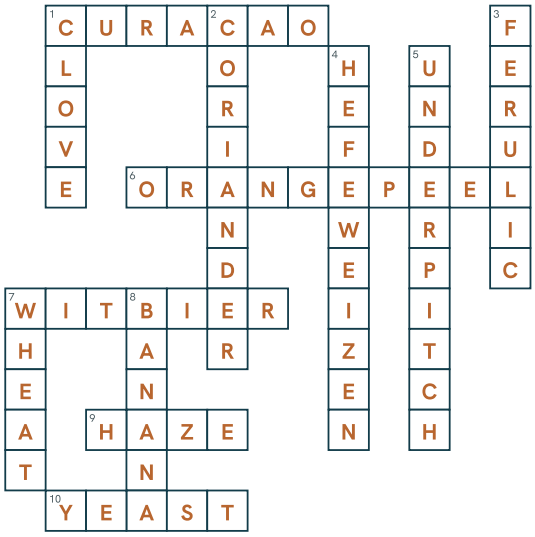
64. How styles work (page 75)



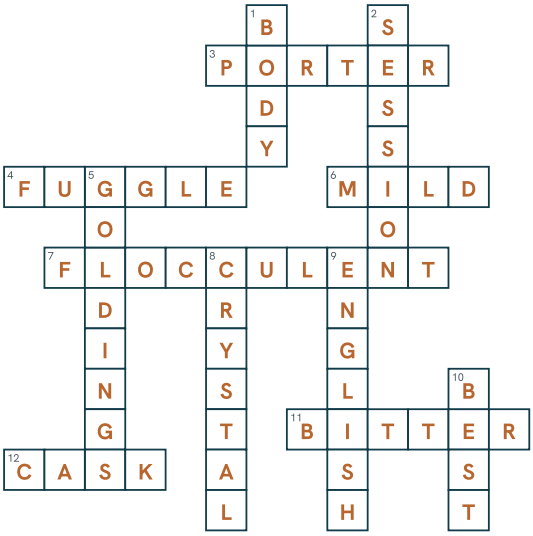
65. Lagers (page 76)



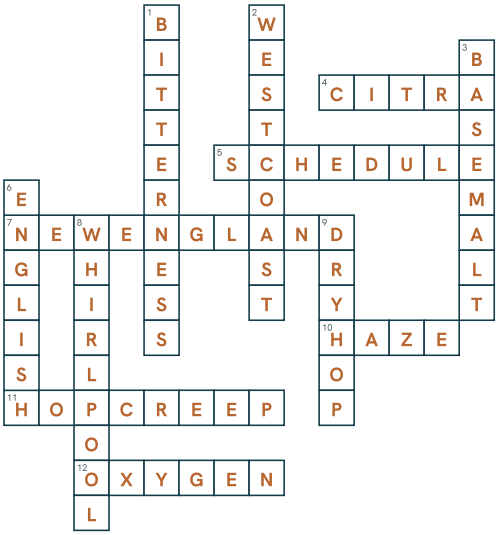
66. Wheat beers (page 77)



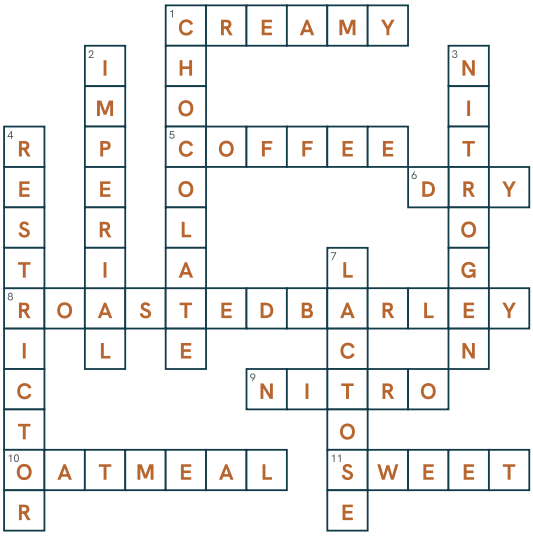
67. British ales (page 78)



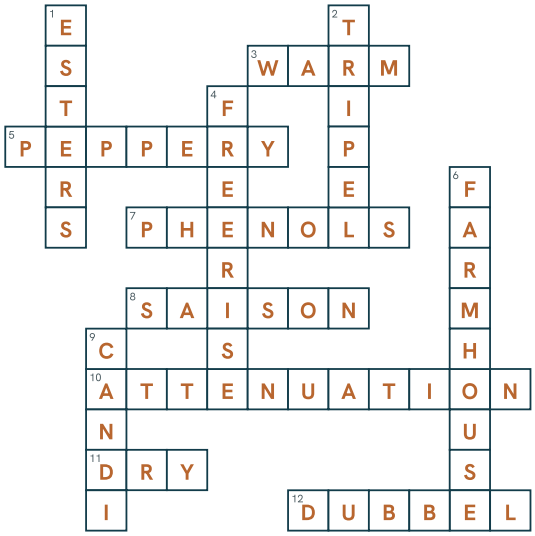
68. IPAs (page 79)



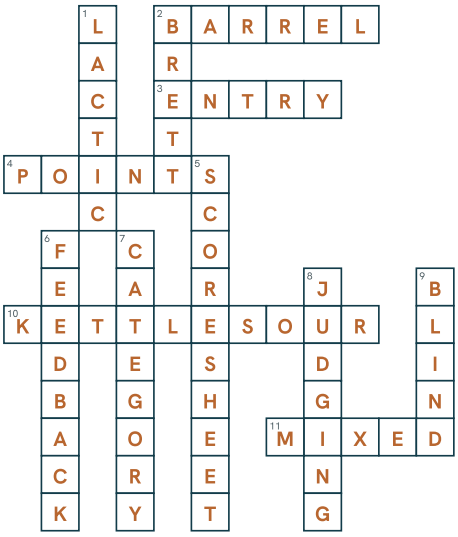
69. Stouts and dark beers (page 80)



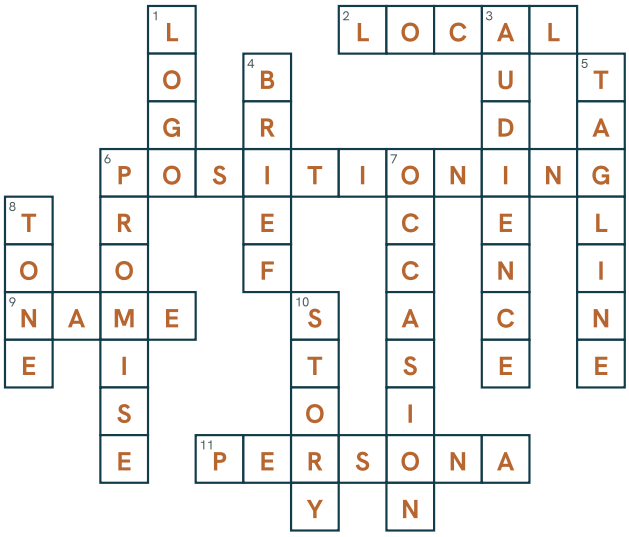
70. Belgian ales (page 81)



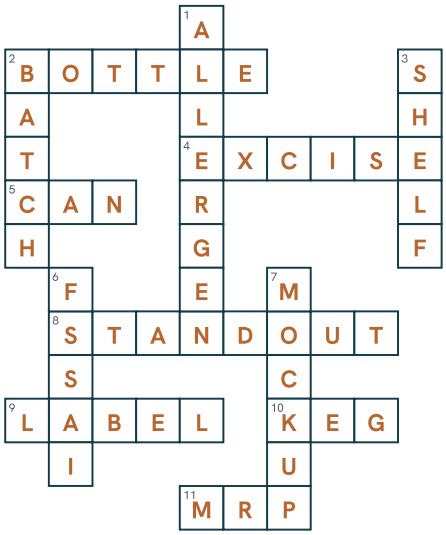
71. Sours and competition brewing (page 82)



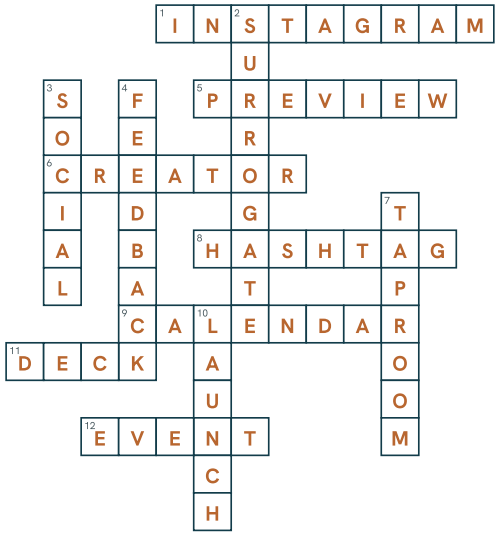
72. Brand identity (page 83)



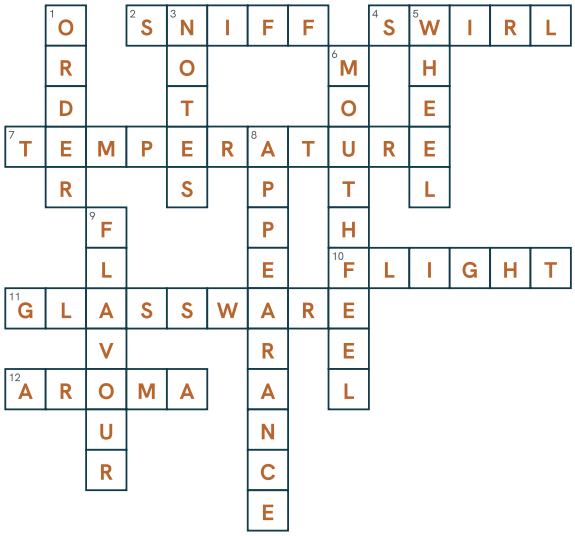
73. Packaging that sells (page 84)



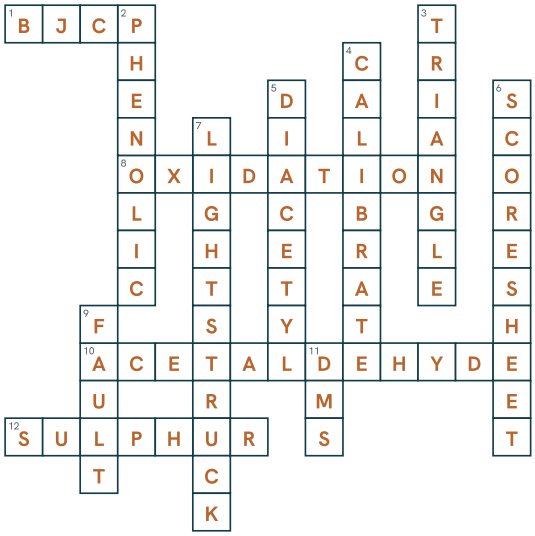
74. Launch and promotion (page 85)



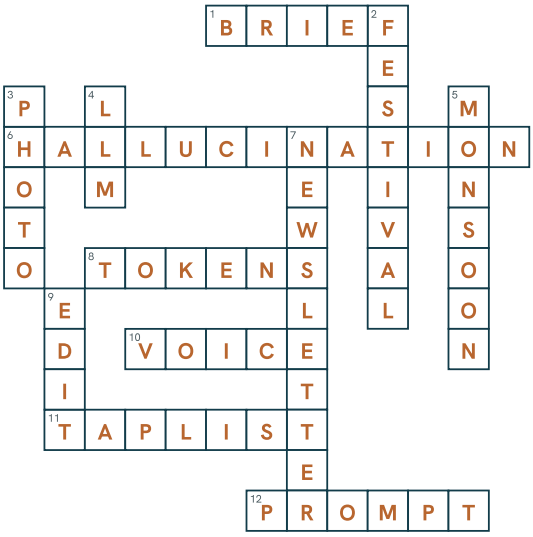
75. How tasting works (page 86)



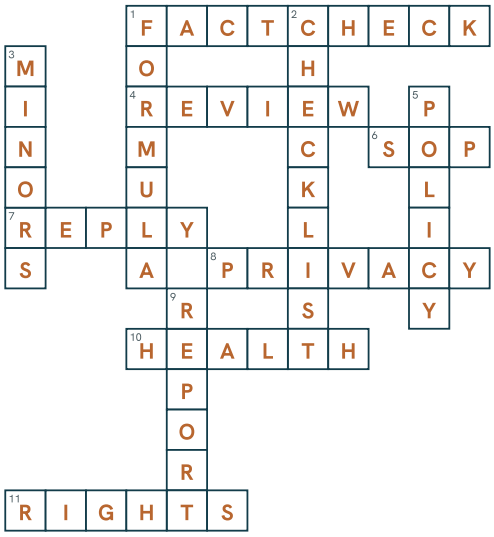
76. Off-flavours and scoring (page 87)



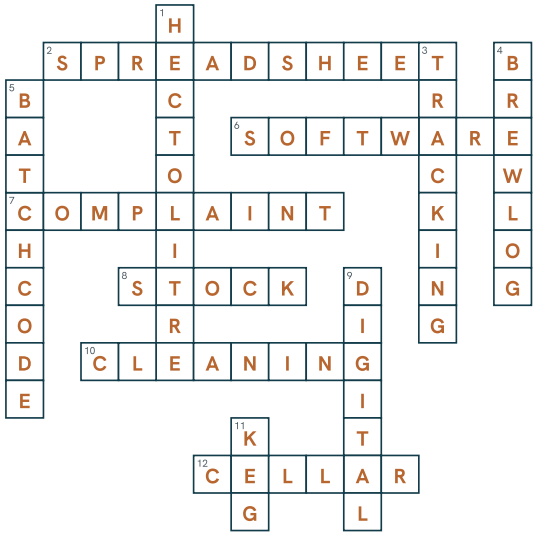
77. GenAI for your brewery's words (page 88)



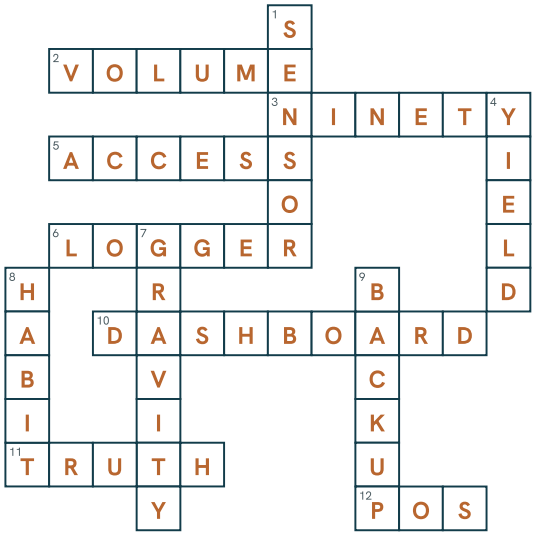
78. Repetitive tasks and responsible AI (page 89)



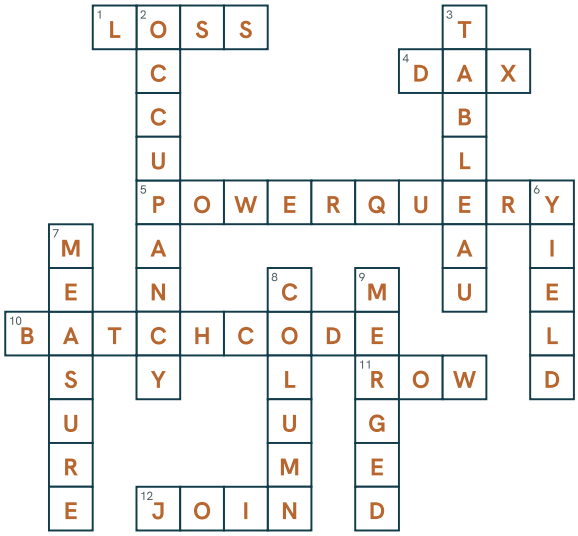
79. From paper to data (page 90)



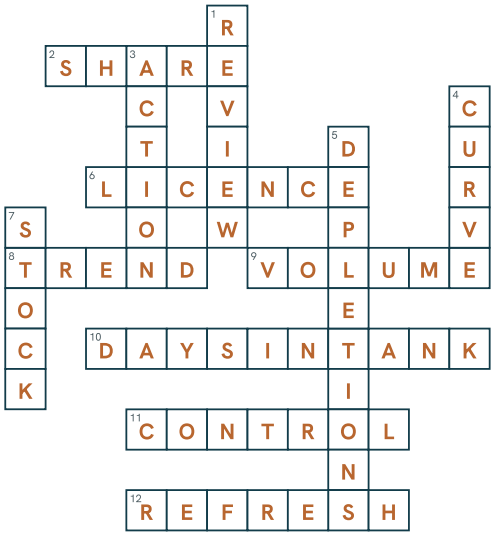
80. Sensors, dashboards and your plan (page 91)



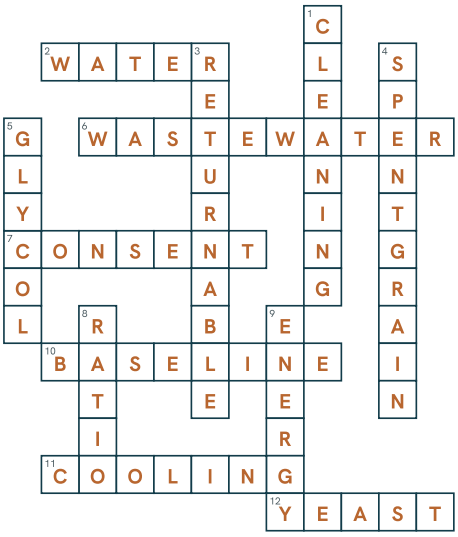
81. The brewery data model (page 92)



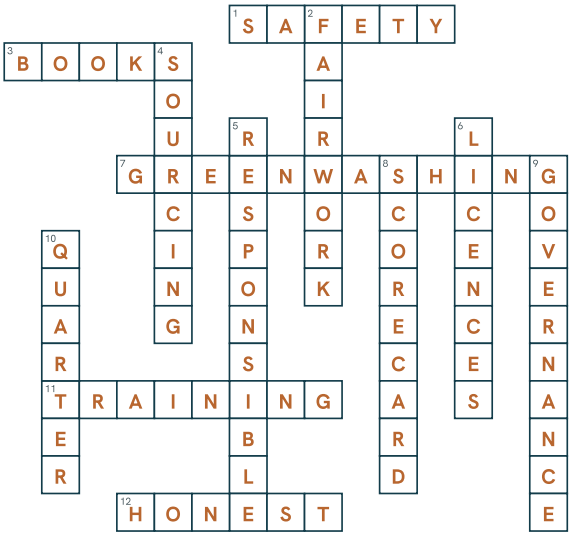
82. Dashboards people open (page 93)



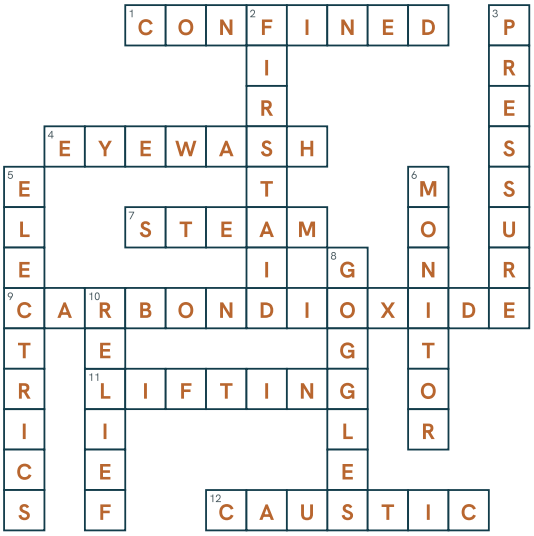
83. The E: water, energy and waste (page 94)



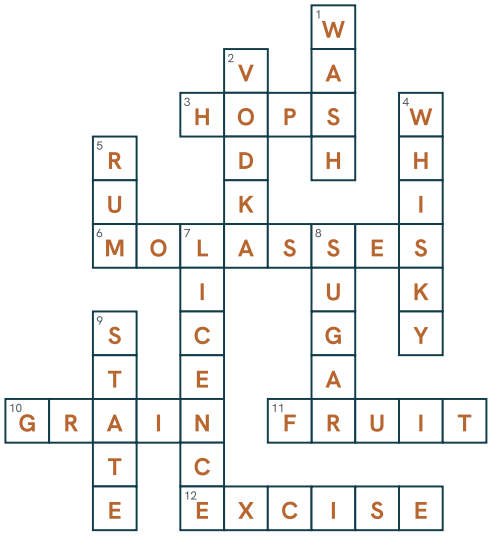
84. The S and the G, and telling the story (page 95)



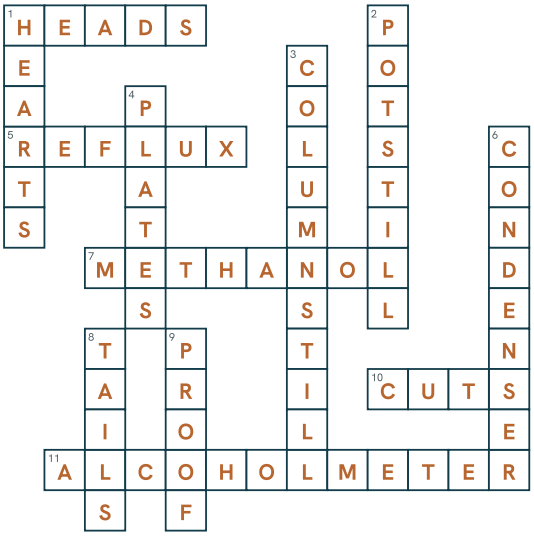
85. The hazards and how to work around them (page 96)



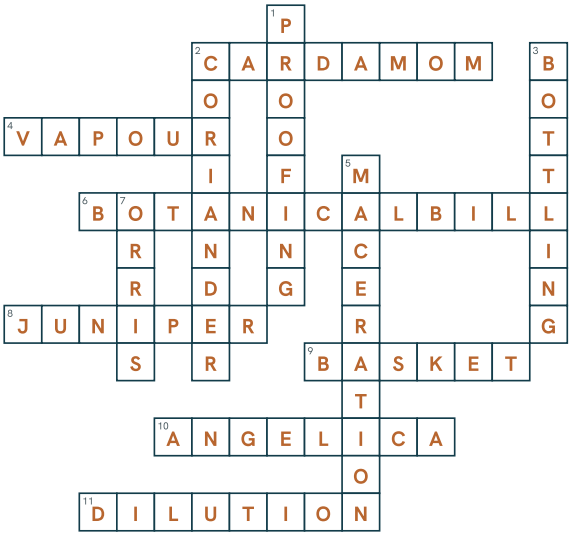
86. Wash, spirits and the law (page 97)



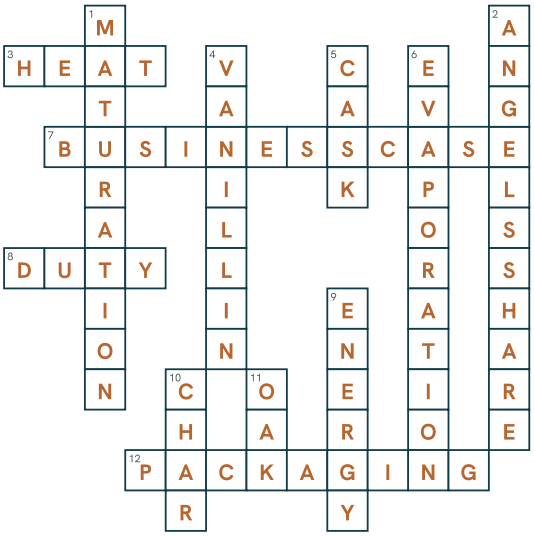
87. Stills and the run (page 98)



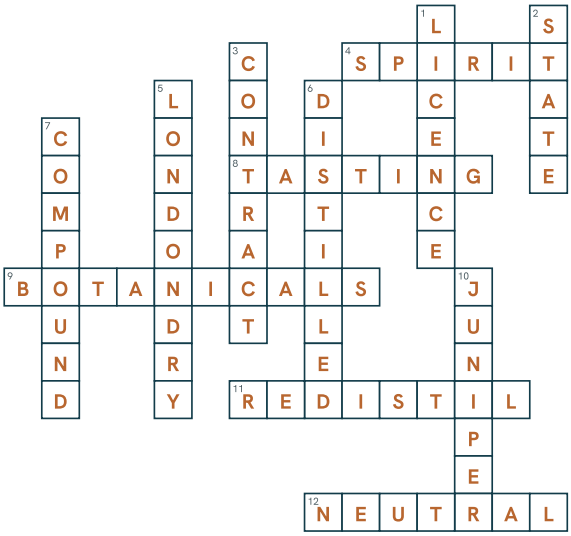
88. Gin and botanicals (page 99)



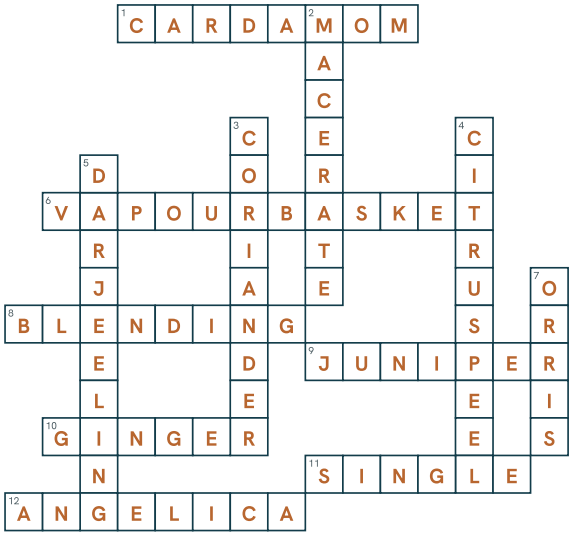
89. Ageing, costs and getting to market (page 100)



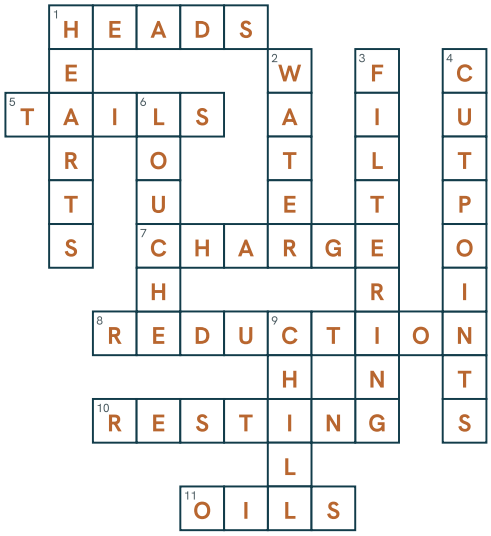
90. What gin is, and the law (page 101)



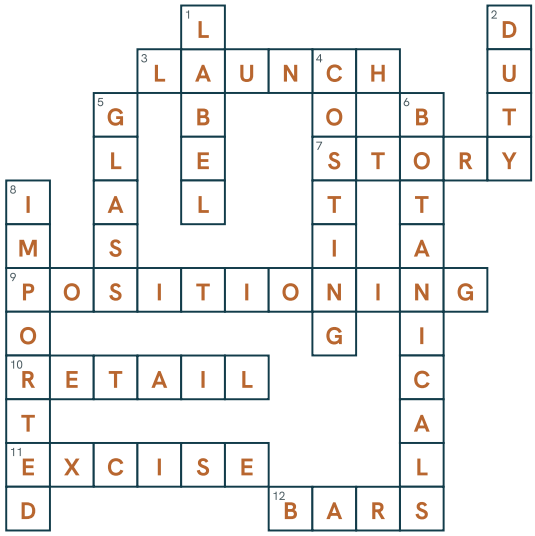
91. Botanicals (page 102)



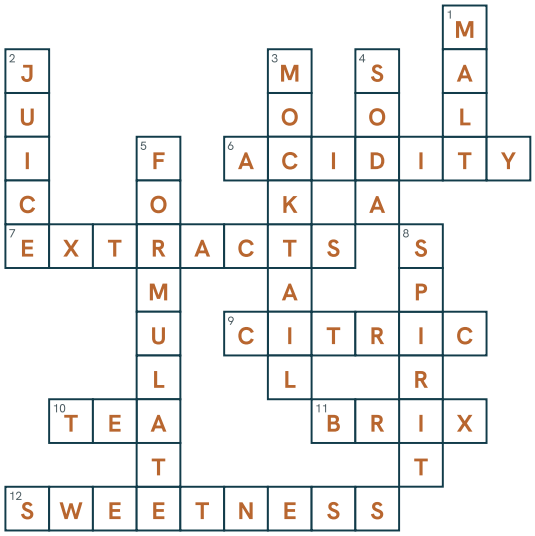
92. The run and the finish (page 103)



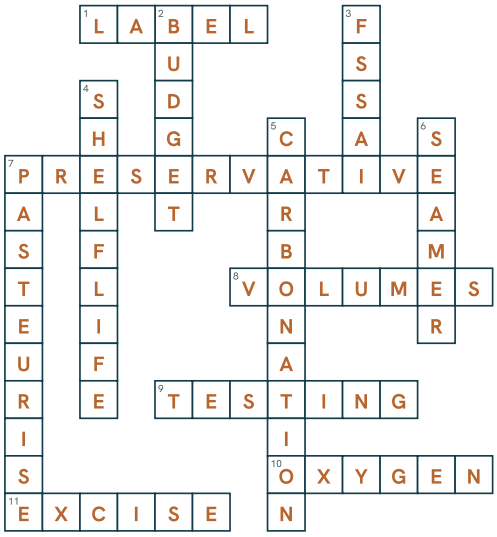
93. Brand, costing and market (page 104)



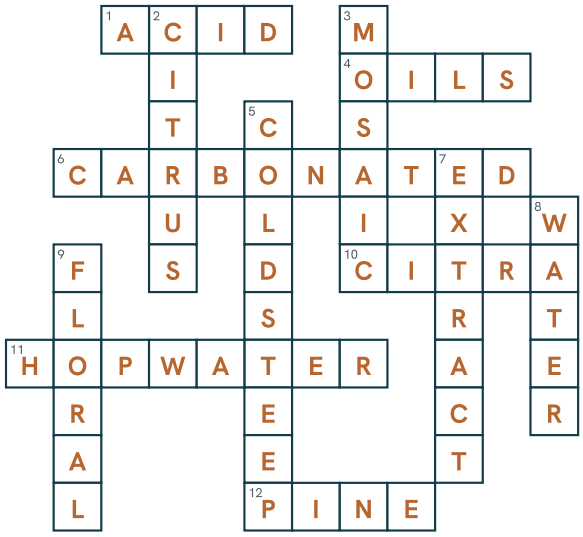
94. Formulation (page 105)



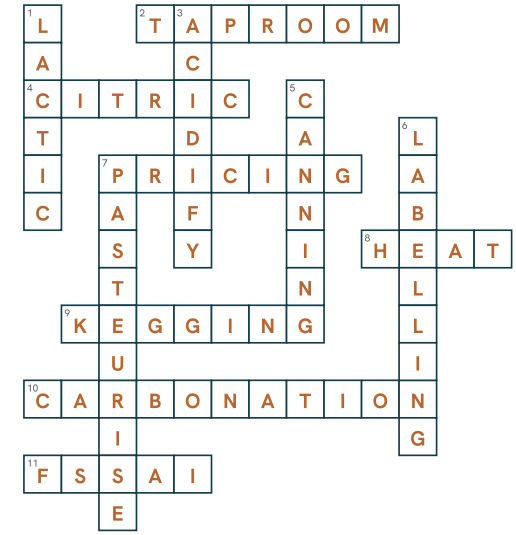
95. Production, shelf life and the rules (page 106)



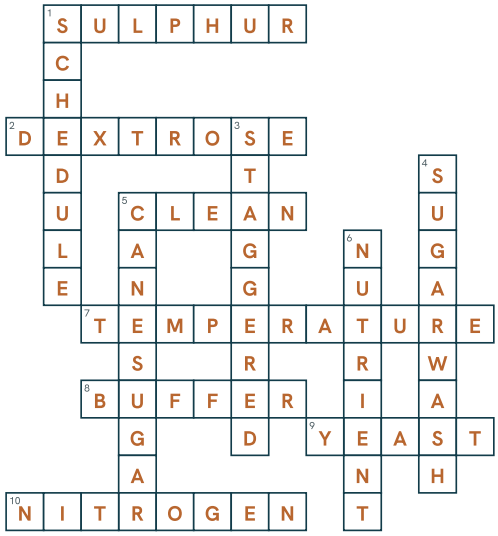
96. Making hop water (page 107)



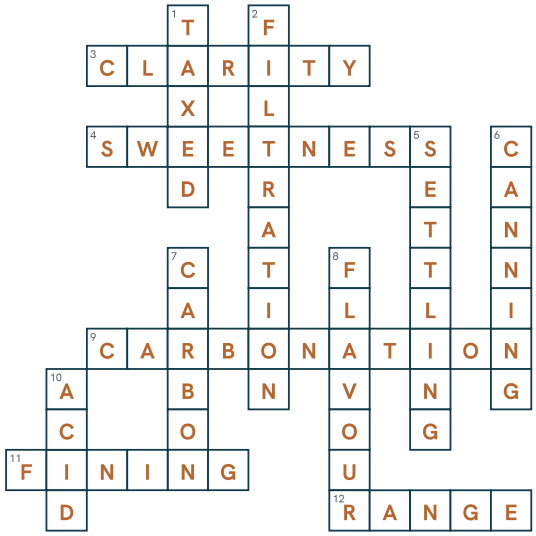
97. Stability and selling it (page 108)



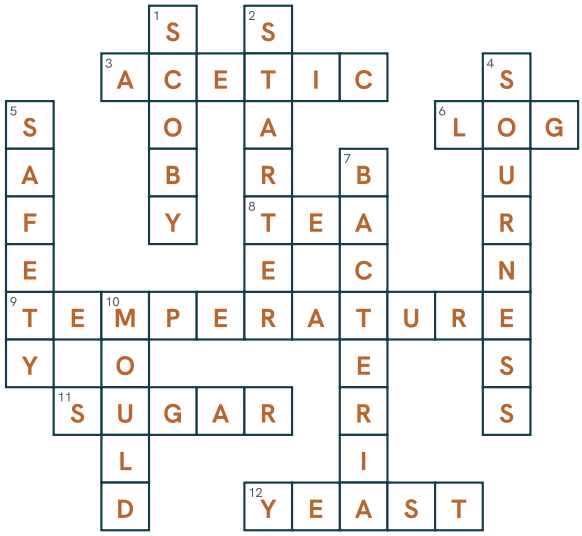
98. Base and fermentation (page 109)



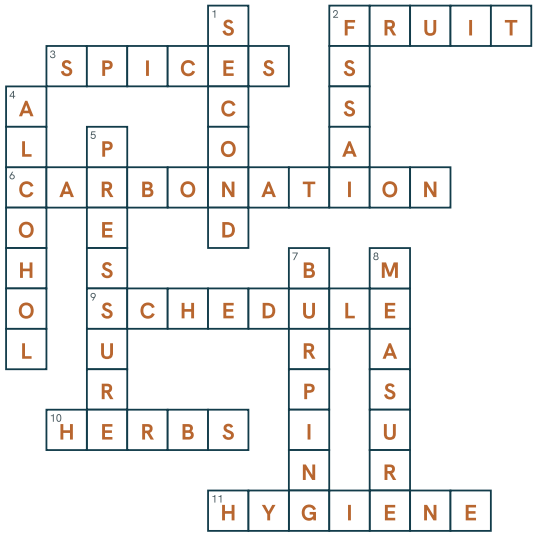
99. Finish and the rules (page 110)



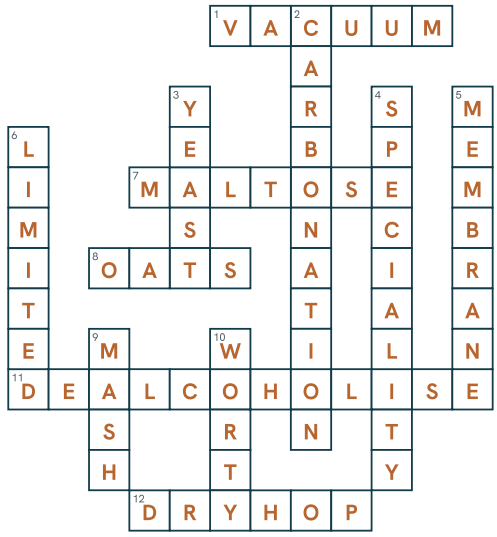
100. The first ferment (page 111)



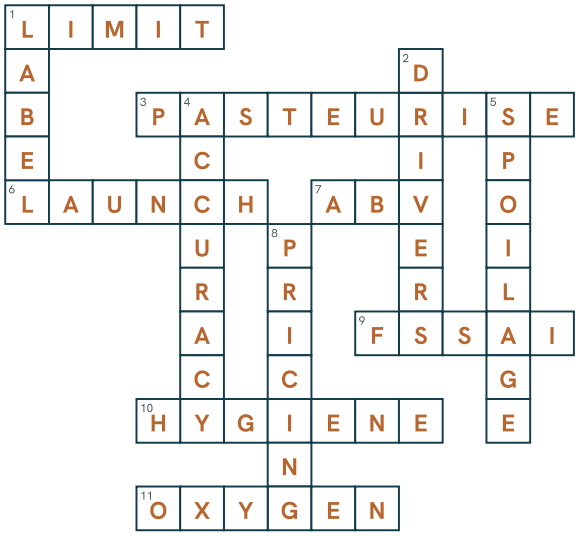
101. Flavour, fizz and scaling up (page 112)



102. Making beer without the alcohol (page 113)



103. Stability, testing and selling it (page 114)



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